

# Large Language Model Enhanced Recommender Systems: Methods, Applications and Trends



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Tutorial Website

# Agenda

## 1 Introduction



Zijian Zhang



## 2 Knowledge Enhancement



Pengyue Jia



## 3 Interaction Enhancement



Ziwei Liu



## 4.1 Model Enhancement 1



Maolin Wang



## 4.2 Model Enhancement 2



Yuhao Wang

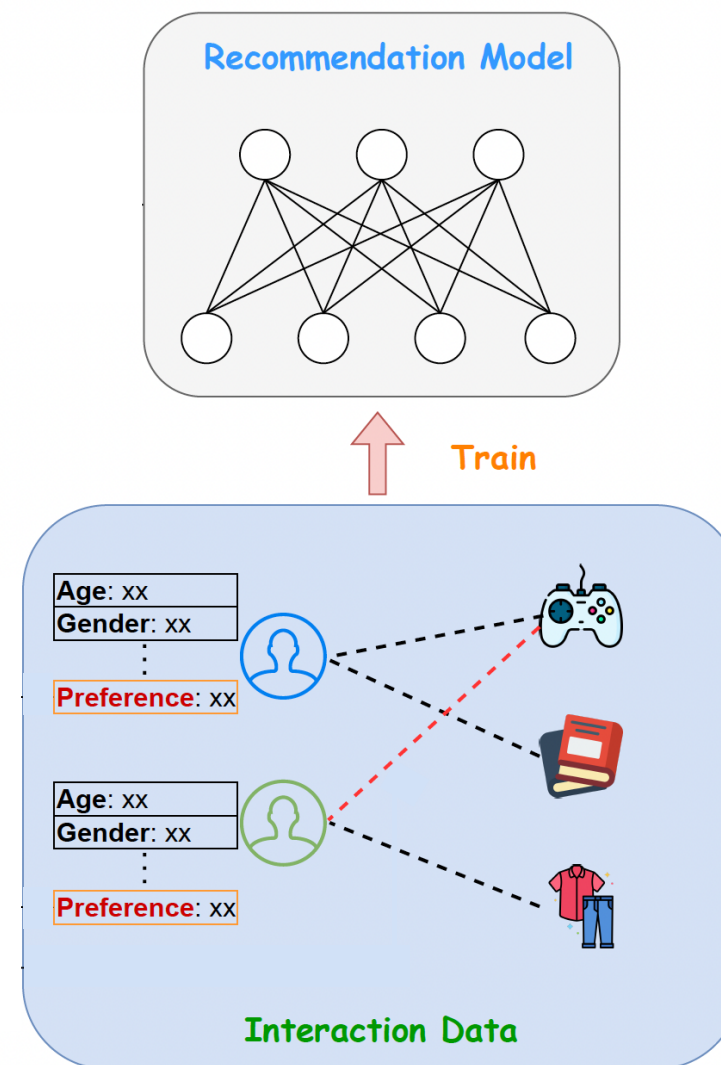
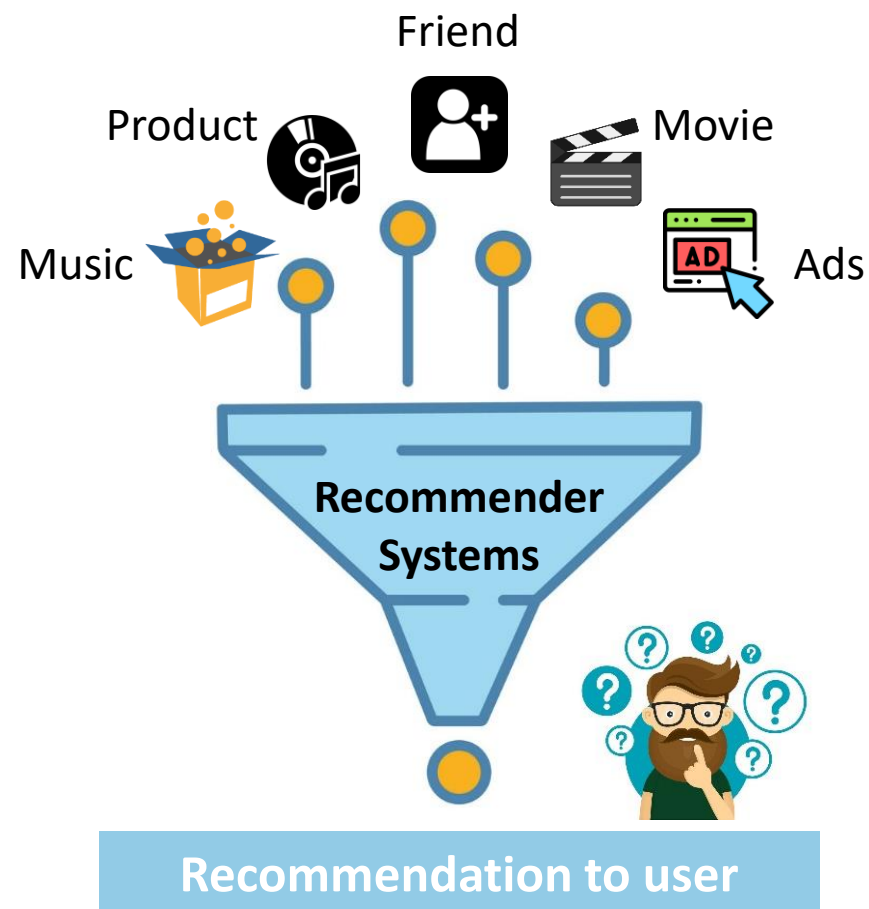


## 5 Conclusion



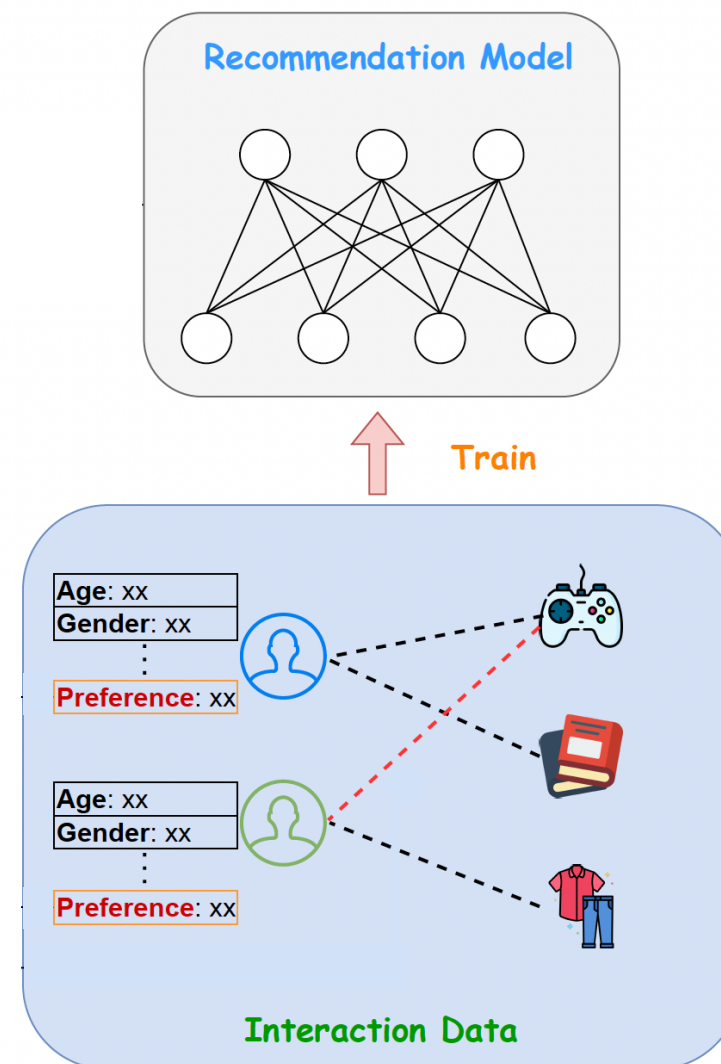
Yejing Wang

## Information overload



## Limitations of RS

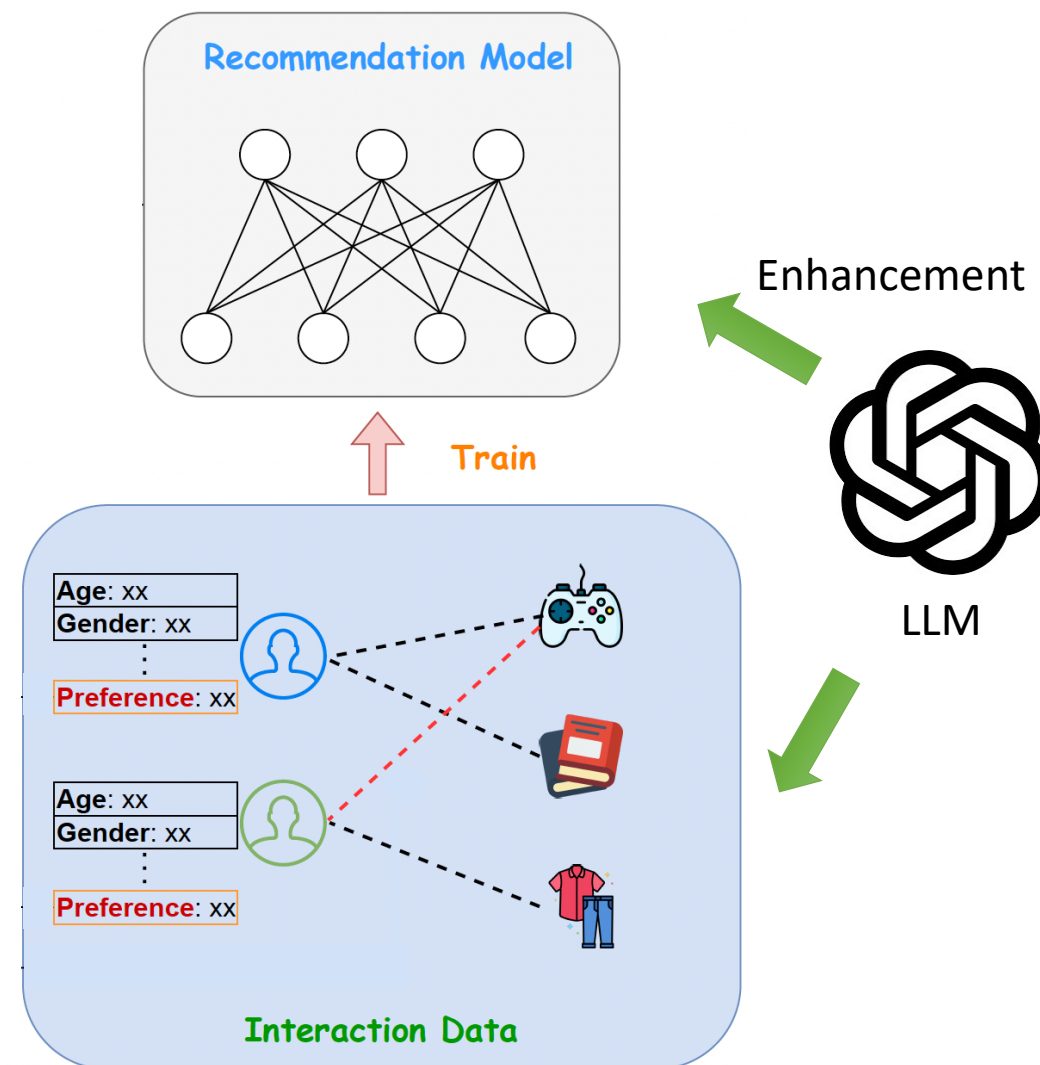
- Feature level: Lack of reasoning and knowledge in feature representations
- Interaction level: Data sparsity in user-item interactions
- Model level: Overlooking semantic understanding





## Advantages of LLM

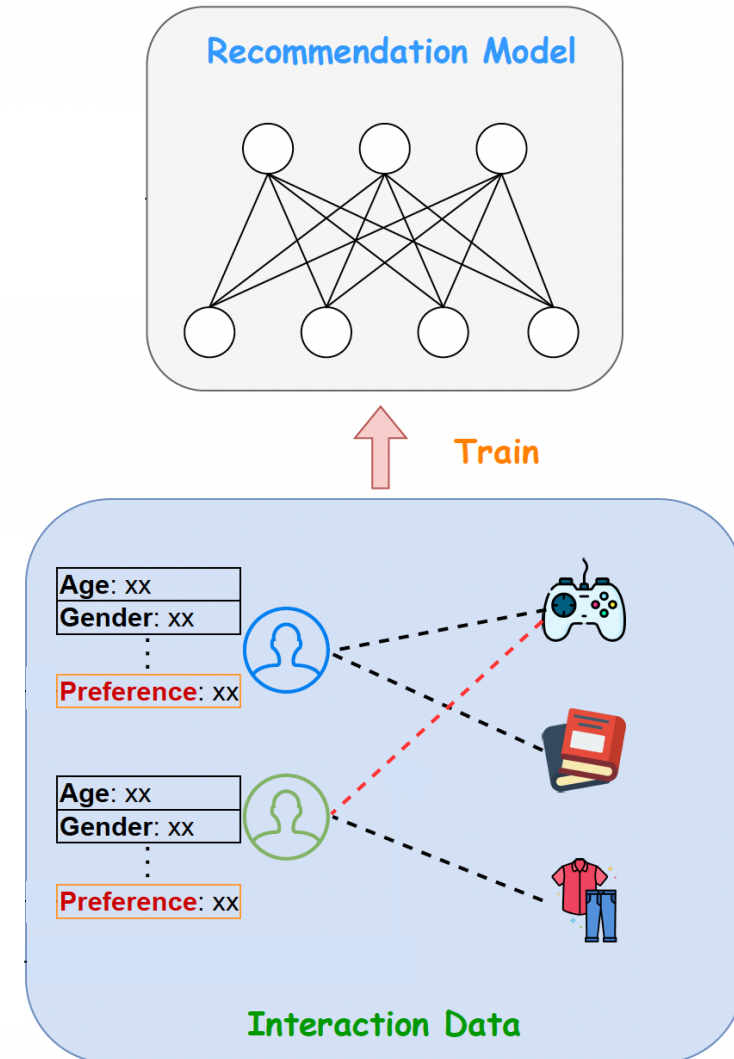
- Feature level: Reasoning abilities and world knowledge to derive textual descriptions
- Interaction level: Approximate users and derive new user-item interactions
- Model level: Analyze interactions from a semantic view



## LLM-enhanced RS (LLMERS)

The conventional recommender systems are enhanced by LLM via assistance in training or supplementary for data, while no need for LLM inference during the service

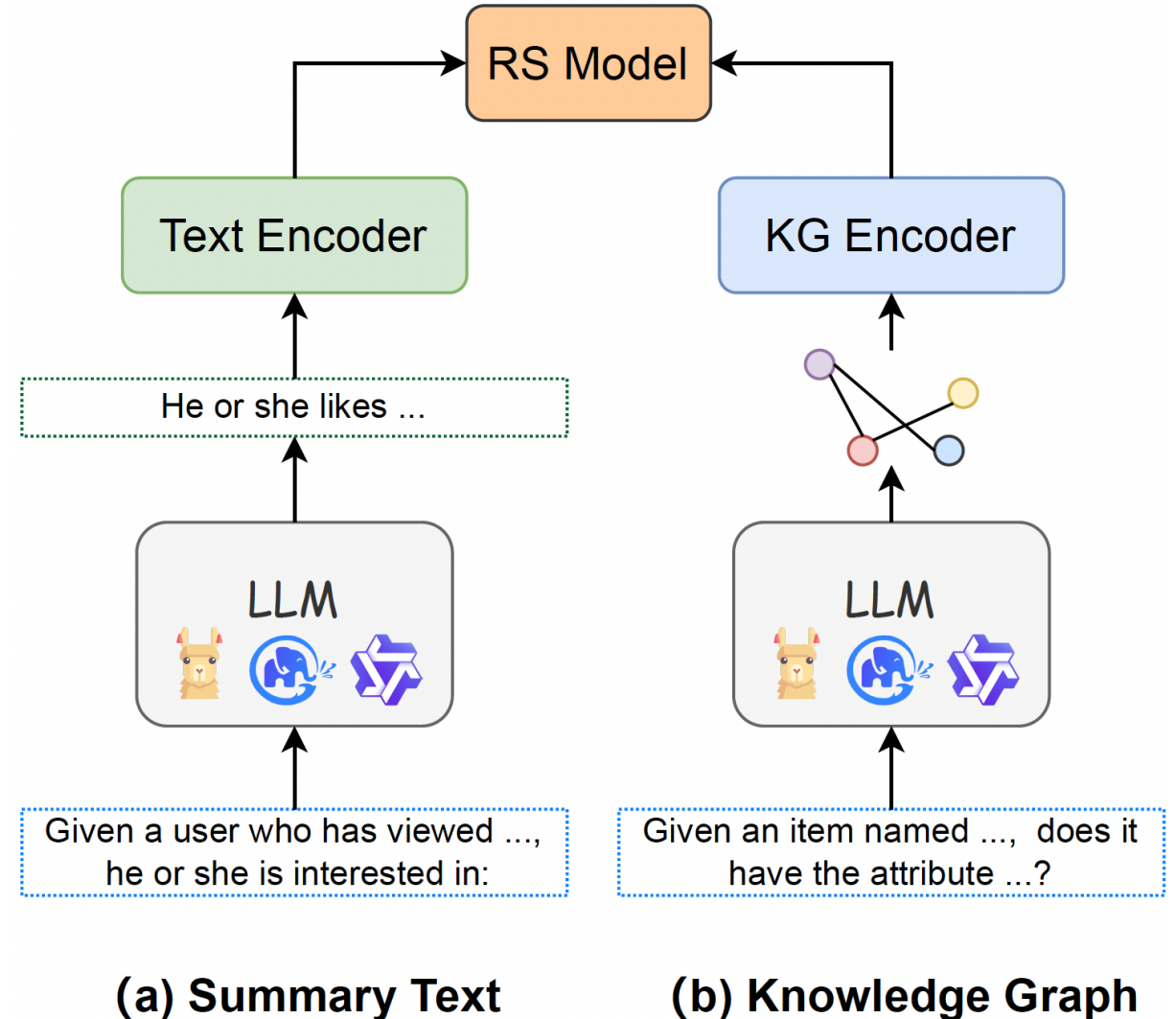
- Knowledge enhancement
- Interaction enhancement
- Model enhancement



# Knowledge enhancement

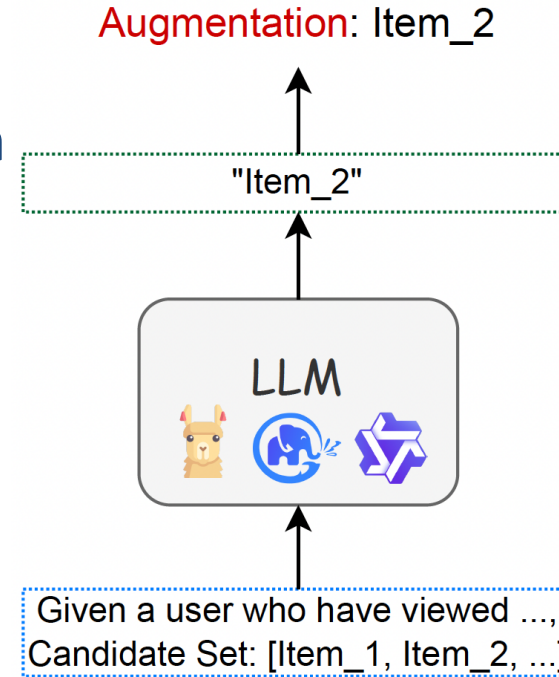
## Knowledge enhancement

- Summary Text: Utilizes LLM to summarize characteristics of items or reason for user preferences.
- Knowledge Graph: Applies LLM to generate or augment structured Knowledge Graphs

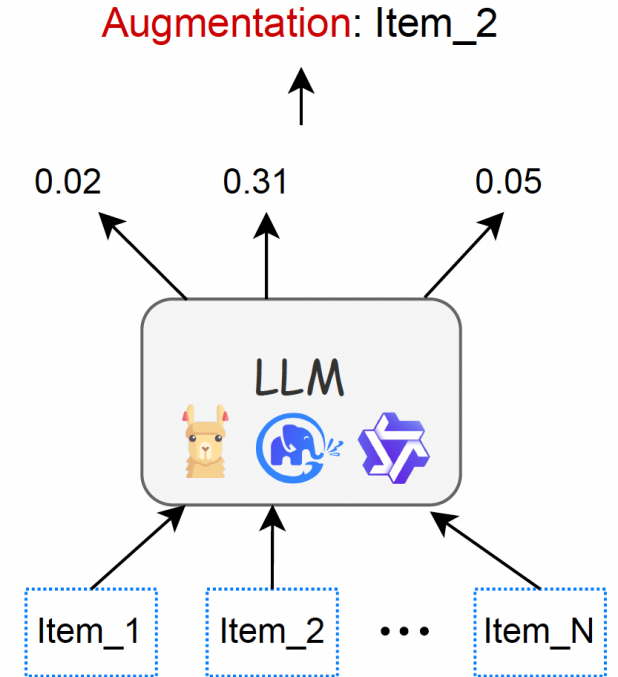


## Interaction enhancement

- Text-based: LLM outputs names of pseudo-interacted items as augmentation
- Score-based: LLM derives probabilities of possible interactions for augmentation



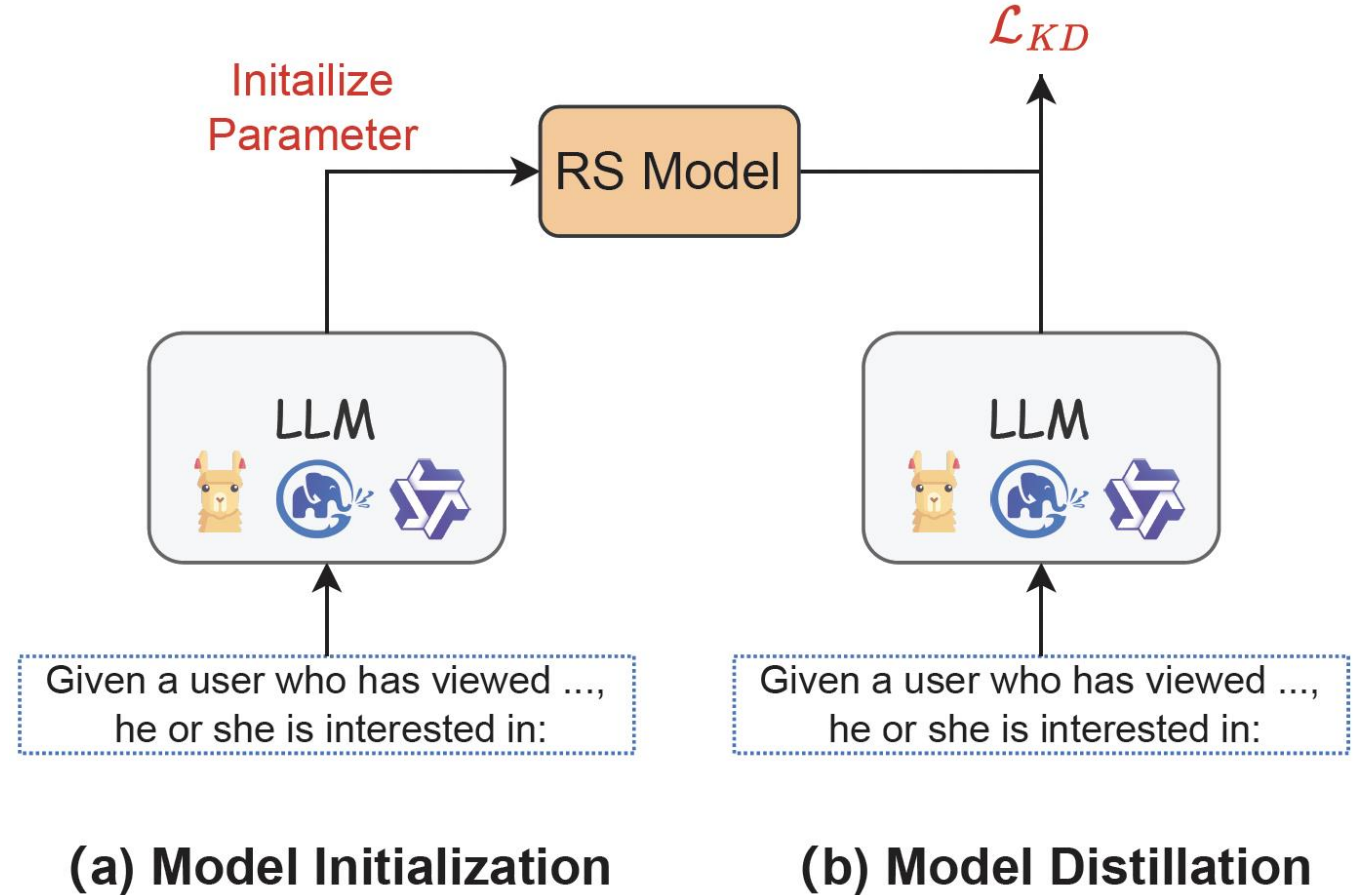
(a) Text-based



(b) Score-based

## Model enhancement

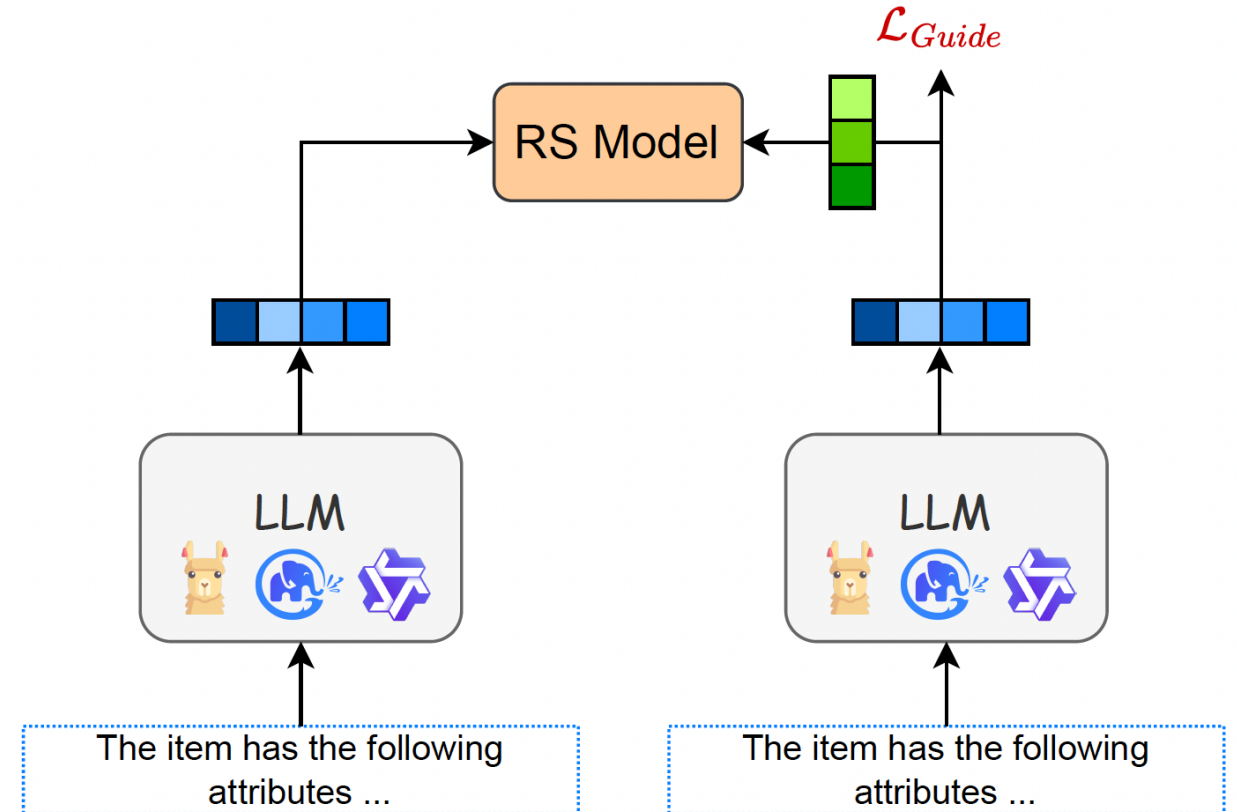
- Model Initialization: Pretrain or initialize RS model weights with LLM's semantics before training
- Model Distillation: Transfer powerful abilities of LLM to smaller RS models





## Model enhancement

- Embedding Utilization: Directly use LLM-derived embeddings as a semantic supplement for RS
- Embedding Guidance: Use LLM embeddings as guidance for training or parameter synthesis of RS models



(c) Embedding Utilization

(d) Embedding Guidance

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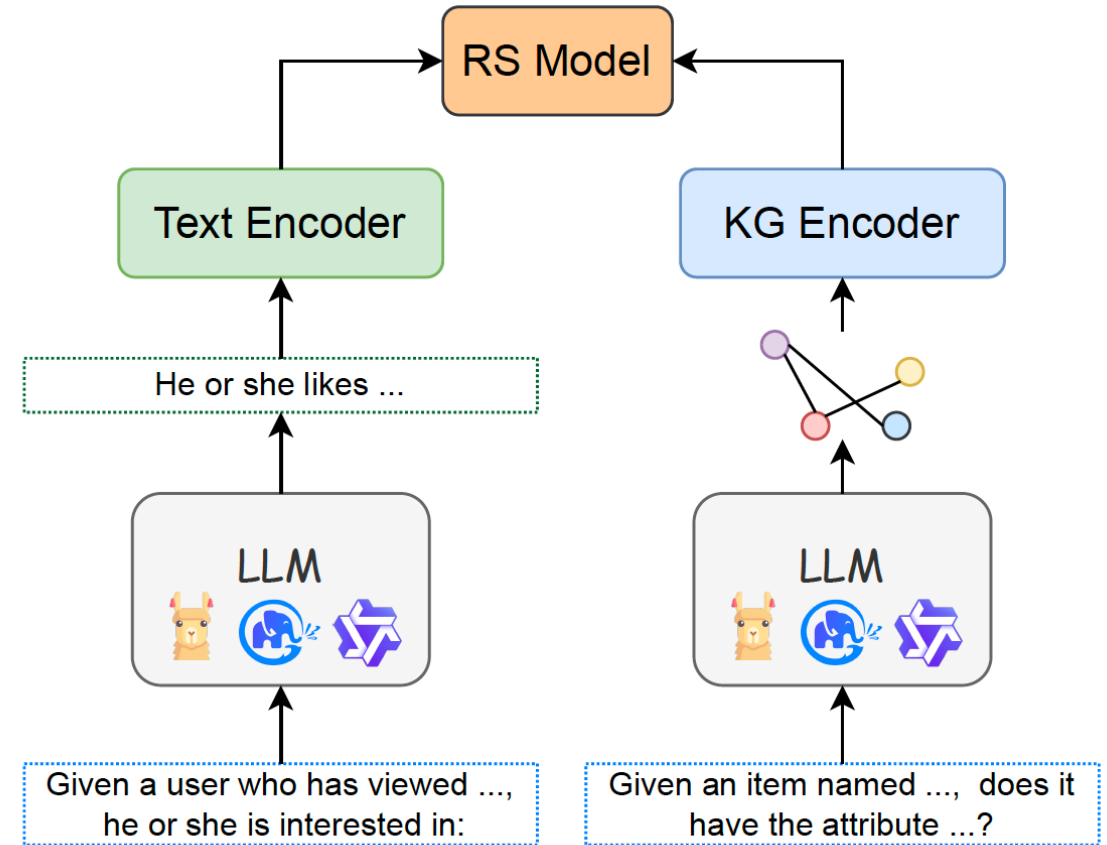
## 5 Conclusion



Yejing Wang

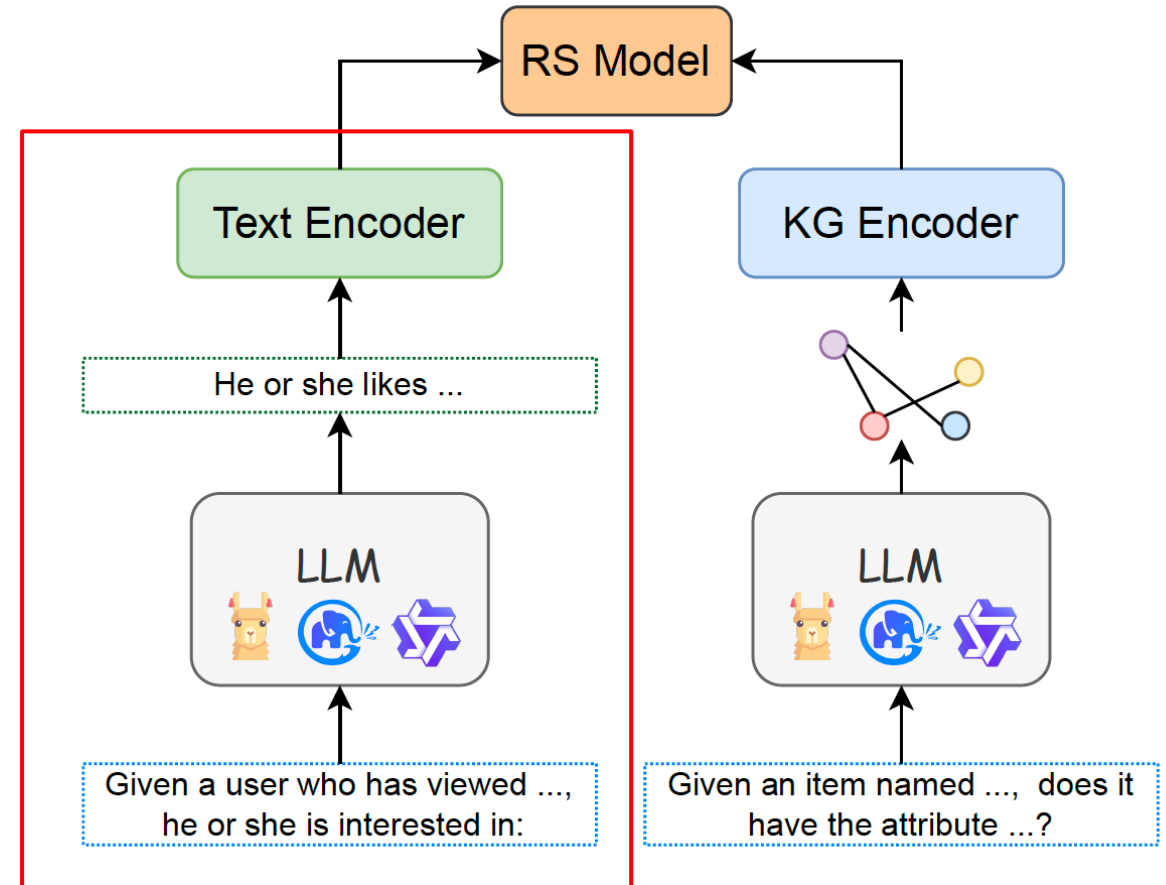
# Knowledge Enhancement

- LLM owns extensive world knowledge and powerful reasoning abilities, which can supplement the RS with external knowledge.
- Categories
  - Summary Text
  - Knowledge Graph
  - Combination



# Summary Text

- This subcategory refers to utilizing LLM to summarize the characteristics of items or to reason for the preference of users.
- For example, the prompt for summarization can be “Given a user who has viewed <Browsing History>, please explain what he or she is interested in: ”.
- Categories
  - User Only
  - Item Only
  - User & Item



- Background
  - The explainability of recommendation systems is crucial for enhancing user trust and satisfaction.
- Motivation
  - Fine-tuning LLM models for recommendation tasks incurs high computational costs and alignment issues with existing systems, limiting the application potential of proven proprietary/closed-source LLM models, such as GPT-4.
- Key Components
  - Semantic embedding
  - User multi-preference extraction using zero-shot prompting
  - Semantic alignment



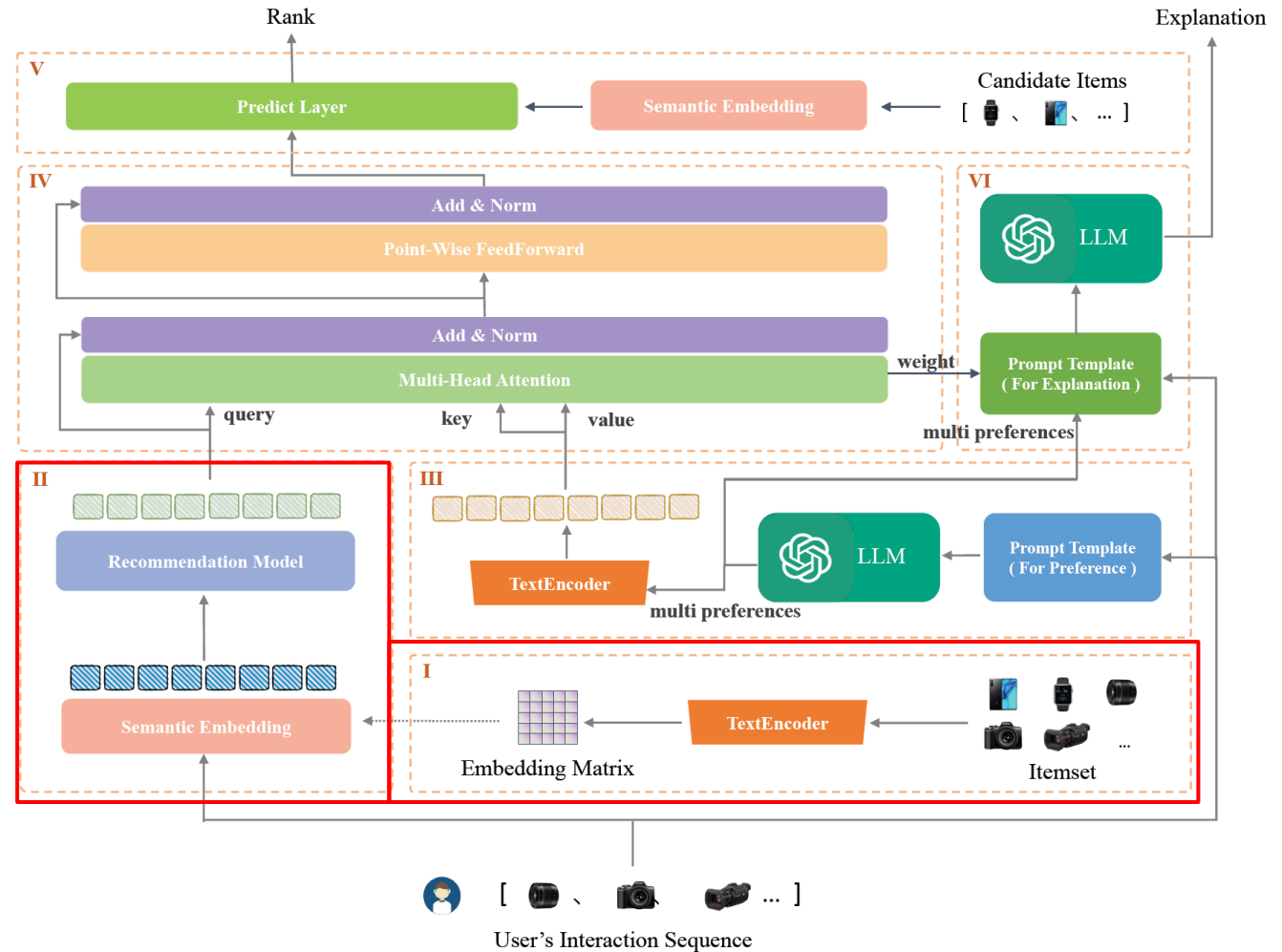
## • Semantic Embedding

- Unlike traditional recommender systems that use item IDs, LANE encodes user history as semantic vectors using a text encoder

$$\mathbf{M} = \begin{bmatrix} \mathbf{m}_1 \\ \mathbf{m}_2 \\ \vdots \\ \mathbf{m}_n \end{bmatrix} = \text{TextEncoder} \left( \begin{bmatrix} i_1 \\ i_2 \\ \vdots \\ i_n \end{bmatrix} \right)$$

## • Integrated Model Module

- LANE incorporates a standard sequential RS, but adapts its embedding layer to use the semantic vectors generated previously.



- Extracting Multi-Preferences using LLM Zero-Shot Prompting

$$\text{Preference}^u = \text{LLM}(\text{prompt}_p(S^u)),$$

$$\mathbf{P}^u = \text{TextEncoder}(\text{Preference}^u)$$

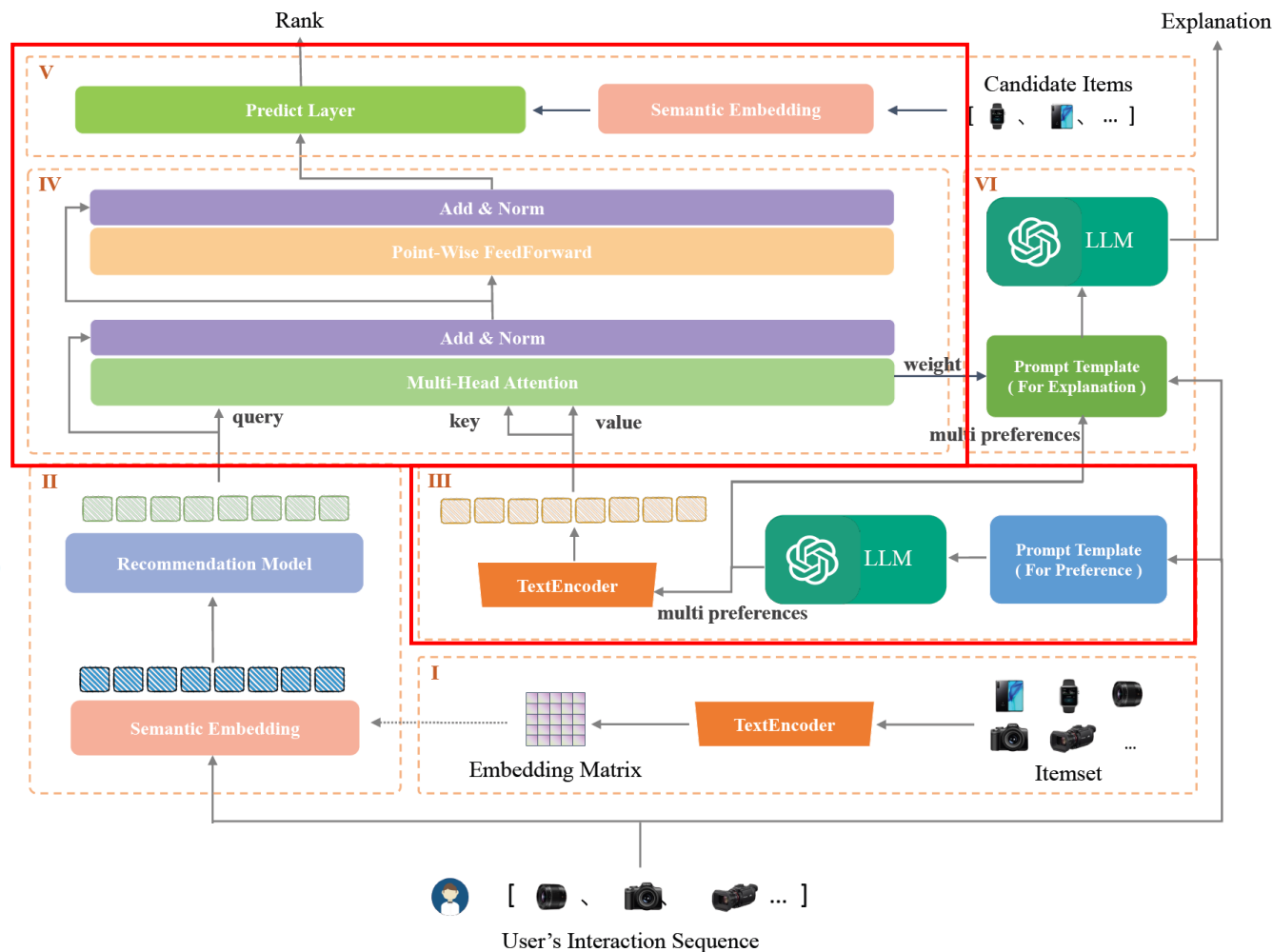
- Semantic Alignment via Multi-Head Attention

$$\text{att}^u = \text{LayerNorm}(\text{Multihead}(\mathbf{Q}^u, \mathbf{P}^u, \mathbf{P}^u)) + \mathbf{Q}^u,$$

$$\mathbf{F}^u = \begin{bmatrix} \mathbf{f}_1^u \\ \mathbf{f}_2^u \\ \vdots \\ \mathbf{f}_n^u \end{bmatrix} = \text{LayerNorm}(\text{FNN}(\text{att}^u)) + \text{att}^u$$

- Prediction

$$\mathbf{r}_{t,i}^u = \mathbf{f}_t^u \cdot \mathbf{m}_i,$$



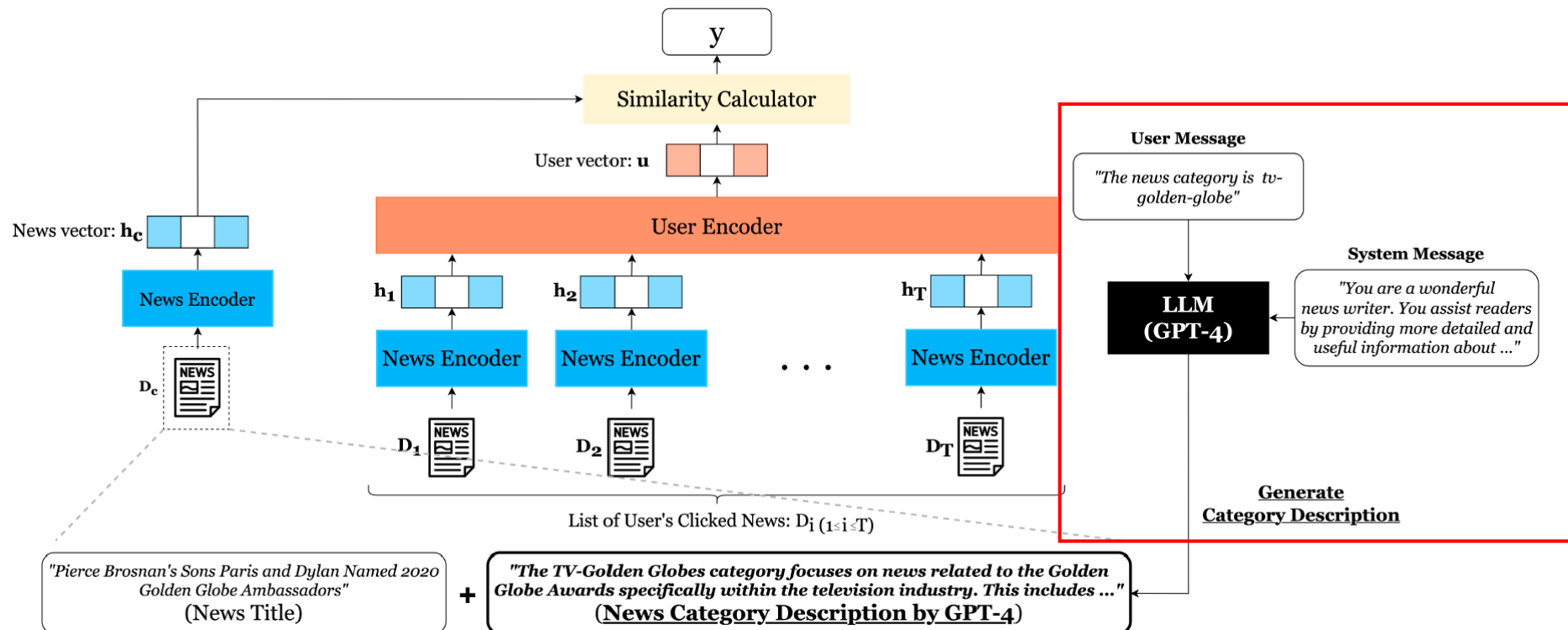
- LANE consistently achieves significant improvements in both NDCG@10 and HR@10 across all datasets and baseline models.

| Dataset        | Beauty        |               | Steam         |               | ML-1M         |              |
|----------------|---------------|---------------|---------------|---------------|---------------|--------------|
|                | NDCG@10       | HR@10         | NDCG@10       | HR@10         | NDCG@10       | HR@10        |
| GRU4Rec        | 0.2853        | 0.4422        | 0.5025        | 0.7492        | 0.5277        | 0.7614       |
| LANE-GRU4Rec   | 0.3238        | 0.4825        | 0.5109        | 0.7598        | 0.5667        | 0.7844       |
| <b>Improv.</b> | <b>13.49%</b> | <b>9.11%</b>  | <b>1.67 %</b> | <b>1.41 %</b> | <b>7.39%</b>  | <b>3.02%</b> |
| BERT4Rec       | 0.224         | 0.3808        | 0.477         | 0.7261        | 0.4611        | 0.7151       |
| LANE-ERT4Rec   | 0.2734        | 0.4526        | 0.4982        | 0.7495        | 0.5111        | 0.7646       |
| <b>Improv.</b> | <b>22.05%</b> | <b>18.86%</b> | <b>4.44%</b>  | <b>3.22%</b>  | <b>10.84%</b> | <b>6.92%</b> |
| SASRec         | 0.2831        | 0.423         | 0.4789        | 0.728         | 0.5701        | 0.7983       |
| LANE-SASRec    | 0.3511        | 0.5172        | 0.5649        | 0.803         | 0.5888        | 0.8106       |
| <b>Improv.</b> | <b>24.02%</b> | <b>22.27%</b> | <b>17.96%</b> | <b>10.30%</b> | <b>3.28%</b>  | <b>1.54%</b> |

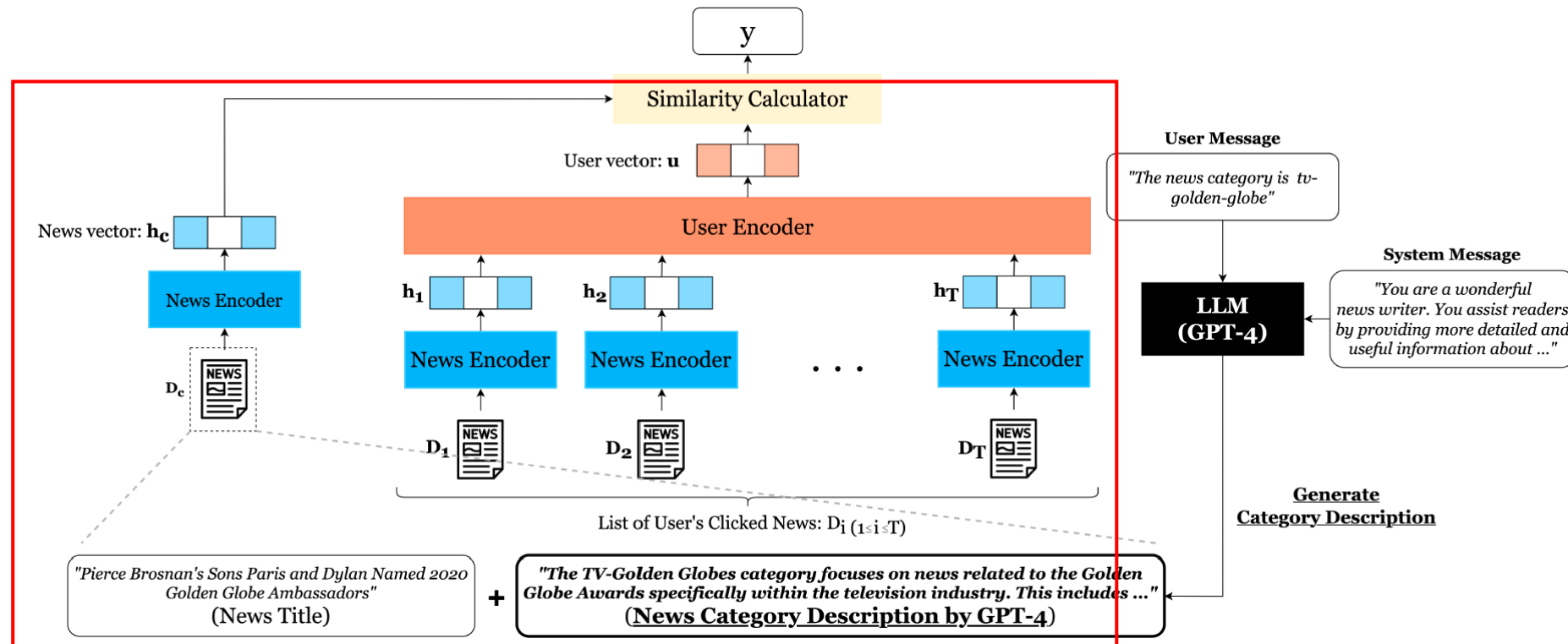
- **Practical Values:** LANE works without fine-tuning LLMs, making it easy to integrate into existing systems while also providing explainable recommendations.

- Background
  - The explainability and informativeness of news metadata—especially category information—play a crucial role in personalized news recommendation.
- Motivation
  - Manually creating detailed descriptions for news categories is labor-intensive.
  - Existing category templates are too generic and provide insufficient semantic guidance.
  - Pretrained language models often fail to interpret short category names effectively without richer context.
- Contribution
  - Generating descriptions with LLMs
  - Incorporating descriptions into recommendation models

- Category Description Generation
  - User Message + System Message -> LLM -> Generated Category Description



- Integration into News Encoder
  - $D_{input} = D_{title} [sep] D_{desc}$ , encoding with bert-style news encoder





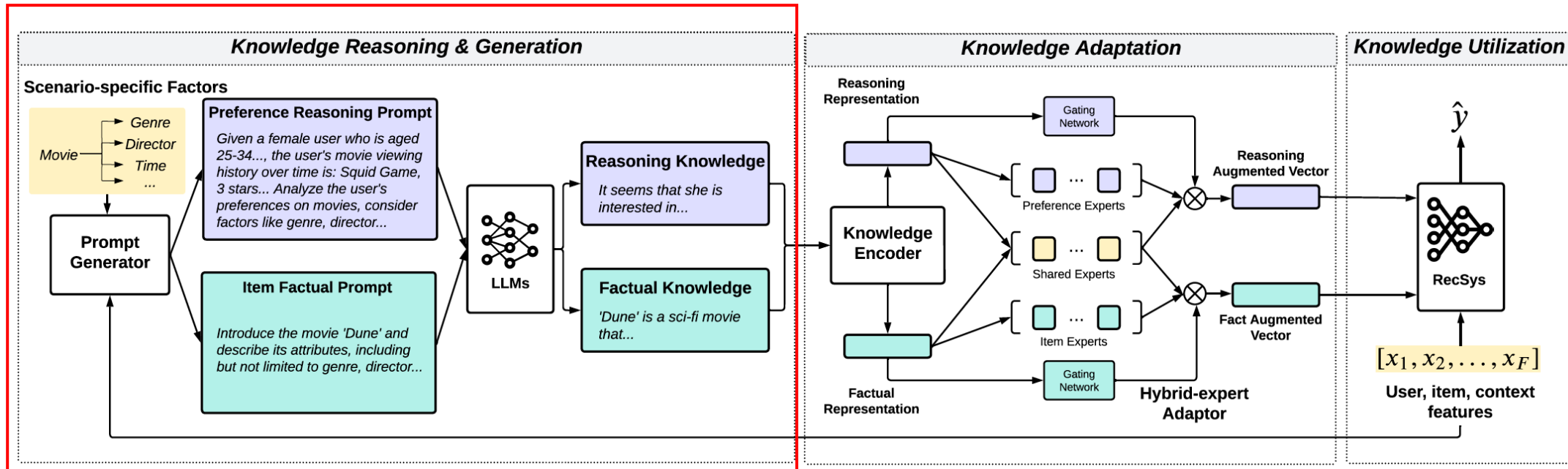
- The proposed method consistently outperforms both the title-only and template-based baselines across all models and settings, achieving up to 5.8% AUC improvement.

| Recommendation Model | PLM        | Method                                     | AUC          | MRR          | nDCG@5       | nDCG@10      |
|----------------------|------------|--------------------------------------------|--------------|--------------|--------------|--------------|
| NAML [19]            | DistilBERT | <i>title only</i>                          | 0.675        | 0.292        | 0.317        | 0.384        |
|                      |            | <i>title + template-based</i>              | 0.690        | 0.295        | 0.327        | 0.393        |
|                      |            | <i>title + generate-description (ours)</i> | <b>0.713</b> | <b>0.326</b> | <b>0.363</b> | <b>0.425</b> |
|                      | BERT       | <i>title only</i>                          | 0.700        | 0.318        | 0.350        | 0.414        |
|                      |            | <i>title + template-based</i>              | 0.696        | 0.308        | 0.340        | 0.405        |
|                      |            | <i>title + generate-description (ours)</i> | <b>0.707</b> | <b>0.322</b> | <b>0.357</b> | <b>0.420</b> |
| NRMS [21]            | DistilBERT | <i>title only</i>                          | 0.674        | 0.297        | 0.322        | 0.387        |
|                      |            | <i>title + template-based</i>              | 0.675        | 0.311        | 0.341        | 0.400        |
|                      |            | <i>title + generate-description (ours)</i> | <b>0.707</b> | <b>0.324</b> | <b>0.359</b> | <b>0.422</b> |
|                      | BERT       | <i>title only</i>                          | 0.689        | 0.306        | 0.336        | 0.400        |
|                      |            | <i>title + template-based</i>              | 0.667        | 0.301        | 0.329        | 0.389        |
|                      |            | <i>title + generate-description (ours)</i> | <b>0.706</b> | <b>0.320</b> | <b>0.355</b> | <b>0.418</b> |
| NPA [20]             | DistilBERT | <i>title only</i>                          | 0.700        | 0.311        | 0.344        | 0.408        |
|                      |            | <i>title + template-based</i>              | 0.698        | 0.309        | 0.342        | 0.407        |
|                      |            | <i>title + generate-description (ours)</i> | <b>0.707</b> | <b>0.319</b> | <b>0.354</b> | <b>0.417</b> |
|                      | BERT       | <i>title only</i>                          | 0.689        | 0.301        | 0.332        | 0.398        |
|                      |            | <i>title + template-based</i>              | 0.694        | 0.314        | 0.345        | 0.410        |
|                      |            | <i>title + generate-description (ours)</i> | <b>0.710</b> | <b>0.324</b> | <b>0.360</b> | <b>0.422</b> |

- Practical Values:** LLM-generated category descriptions offer a simple and generalizable way to enrich news representations, enabling better recommendations without additional training or domain-specific effort.

- Background
  - Classical recommender systems are limited to closed-domain data and lack access to open-world knowledge, constraining their reasoning and generalization abilities.
- Motivation
  - Directly using LLMs as recommenders leads to poor accuracy, high latency, and difficulty handling compositional reasoning.
  - It is challenging to extract useful, aligned, and reliable knowledge from LLMs that is compatible with recommendation models.
- Contribution
  - Propose KAR, a model-agnostic framework that extracts both reasoning knowledge about users and factual knowledge about items from LLMs.
  - Design a hybrid-expert adaptor to transform textual knowledge into dense augmented vectors aligned with recommendation space.

- Knowledge Reasoning and Generation
  - Use LLMs with **factorized prompts** to extract two types of open-world knowledge:
    - Reasoning knowledge  $k_i^{(p)}$  for user  $i$
    - Factual knowledge  $k_i^{(l)}$  for item  $i$



- Knowledge Adaptation and Utilization

- Use a knowledge encoder (e.g., BERT) to encode LLM outputs
- Apply hybrid-expert adaptor with gating networks and shared/dedicated experts to transform to

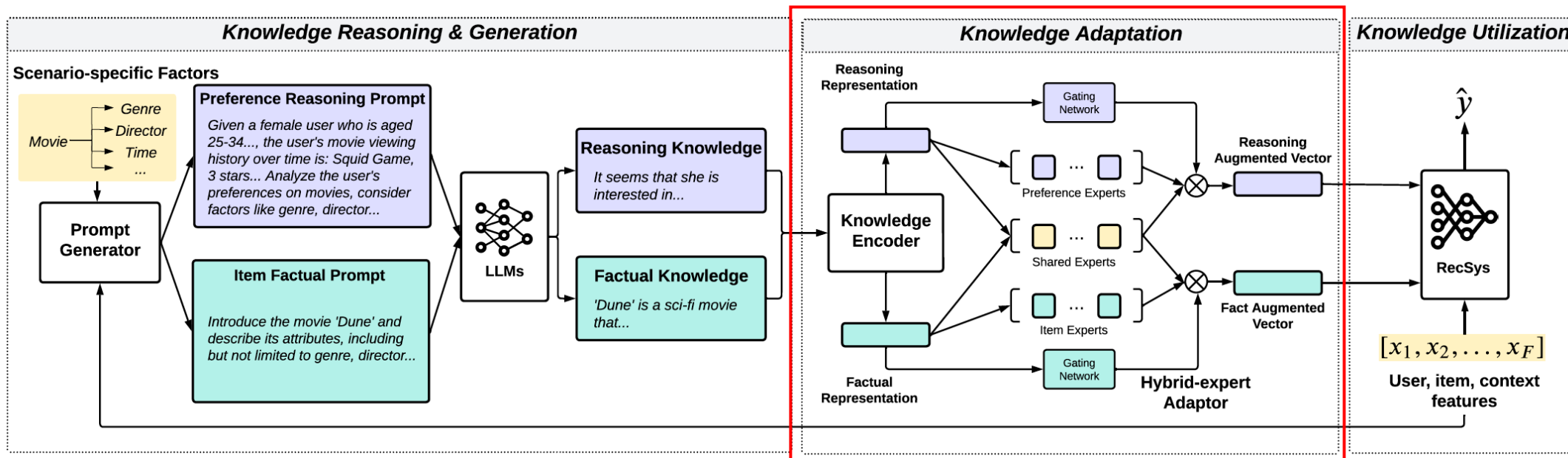
$$\alpha_i^p = \text{Softmax}(g^p(r_i^p)), \quad \alpha_i^l = \text{Softmax}(g^l(r_i^l)),$$

$$r_i^{(p)} = \text{Aggr}(\text{Encoder}(k_i^{(p)}))$$

$$r_i^{(l)} = \text{Aggr}(\text{Encoder}(k_i^{(l)}))$$

$$\hat{r}_i^p = \sum_{e \in S_s} \alpha_{i,e}^p \times e(r_i^p) + \sum_{e \in S_p} \alpha_{i,e}^p \times e(r_i^p),$$

$$\hat{r}_i^l = \sum_{e \in S_s} \alpha_{i,e}^l \times e(r_i^l) + \sum_{e \in S_l} \alpha_{i,e}^l \times e(r_i^l),$$



- KAR consistently improves AUC and logloss across 9 CTR models, with up to **1.6% AUC** and **3.1% logloss gain**

| Backbone Model | AUC    |                |         | Logloss |                |         |
|----------------|--------|----------------|---------|---------|----------------|---------|
|                | base   | KAR            | improv. | base    | KAR            | improv. |
| DCNv2          | 0.7924 | <b>0.8049*</b> | 1.58 %  | 0.5451  | <b>0.5315*</b> | 2.50 %  |
| DCN            | 0.7929 | <b>0.8043*</b> | 1.46 %  | 0.5457  | <b>0.5319*</b> | 2.53 %  |
| DeepFM         | 0.7928 | <b>0.8041*</b> | 1.44 %  | 0.5462  | <b>0.5321*</b> | 2.57 %  |
| FiBiNet        | 0.7925 | <b>0.8051*</b> | 1.59 %  | 0.5450  | <b>0.5310*</b> | 2.56 %  |
| AutoInt        | 0.7934 | <b>0.8060*</b> | 1.59 %  | 0.5440  | <b>0.5297*</b> | 2.65 %  |
| FiGNN          | 0.7944 | <b>0.8054*</b> | 1.39 %  | 0.5424  | <b>0.5307*</b> | 2.16 %  |
| xDeepFM        | 0.7942 | <b>0.8041*</b> | 1.25 %  | 0.5457  | <b>0.5317*</b> | 2.57 %  |
| DIEN           | 0.7960 | <b>0.8059*</b> | 1.25 %  | 0.5469  | <b>0.5298*</b> | 3.13 %  |
| DIN            | 0.7975 | <b>0.8066*</b> | 1.15 %  | 0.5387  | <b>0.5304*</b> | 1.55%   |

\* denotes statistically significant improvement (t-test with  $p$ -value  $< 0.05$ ) over the backbone model.

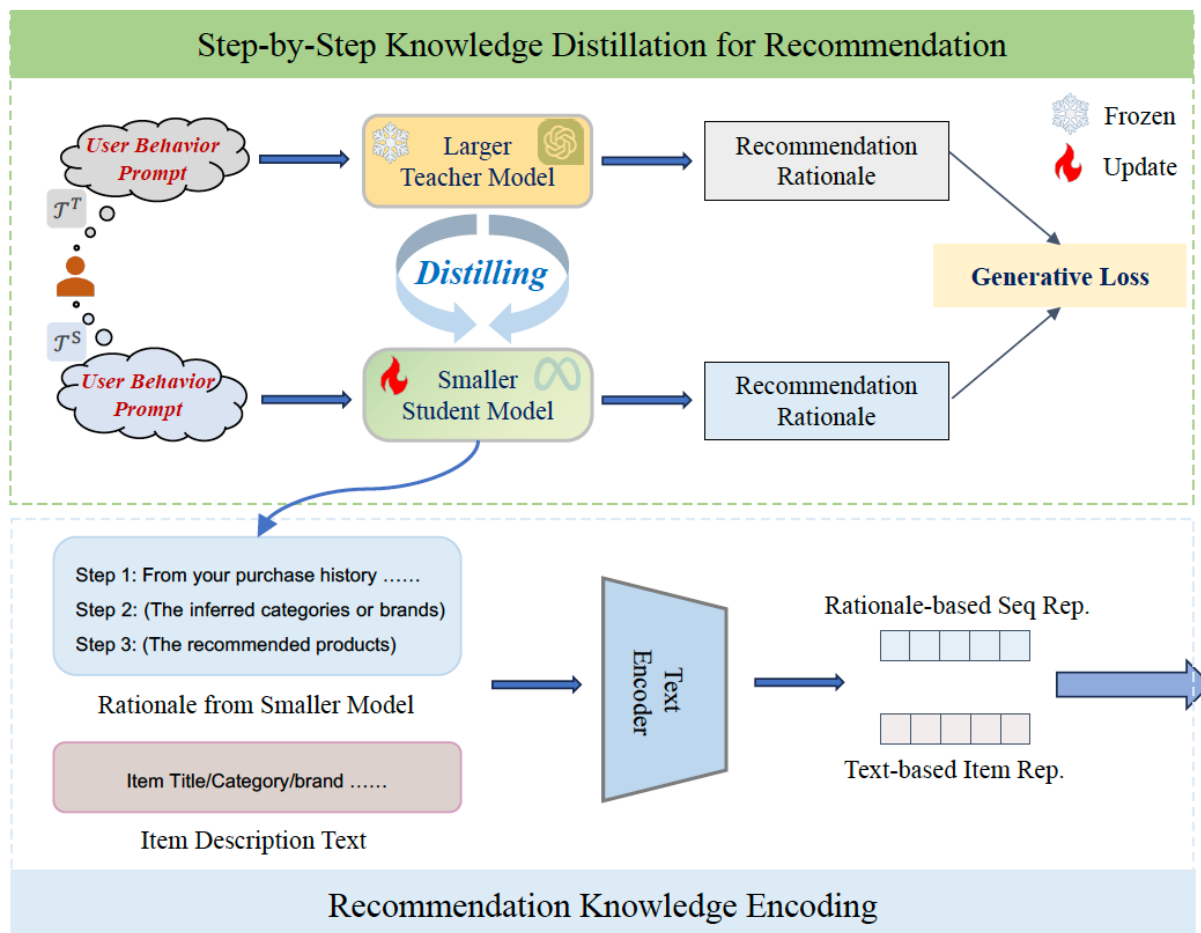
- Practical Values:** KAR offers a practical, modular, and efficient solution to enhance recommenders using LLM-generated knowledge, without modifying their internal architecture.

- Background
  - LLMs show strong reasoning ability and knowledge but are expensive and hard to deploy in real systems.
- Motivation
  - One-shot generation by LLMs often produces irrelevant or incorrect recommendations.
  - The scale and inference cost of LLMs are prohibitive for production use.
- Contribution
  - Propose SLIM, use CoT prompting with LLMs to generate rationales.
  - Distill these rationales into a small LLM for efficient deployment.
  - Enable both ID-based and ID-agnostic recommendation by encoding rationales and item descriptions via a text encoder.



- Step-by-Step Knowledge Distillation
  - Chain-of-Thought Prompting on Teacher Model with user prompt:
    - Summarize preferences
    - Infer categories or brands
    - Recommend specific products
- Fine-tune Student Model with Rationales:
  - The objective is to minimize generative loss

$$\mathcal{L}_{\text{distill}} = \sum_{u \in U'} \sum_{t=1}^{|r'_u|} \log P_{\theta}(r'_{u,t} | p'_u, r'_{u,<t})$$



- Rationale Encoding and Enhanced Recommendation

- Text Encoding:

- Convert student-generated rationales  $r_u$  and item descriptions  $f_i$  into embeddings

$$\mathbf{z}_{\text{text},i} = \text{TextEncoder}(f_i), \quad \mathbf{s}_{\text{text},u} = \text{TextEncoder}(r_u)$$

- ID-Based Integration:

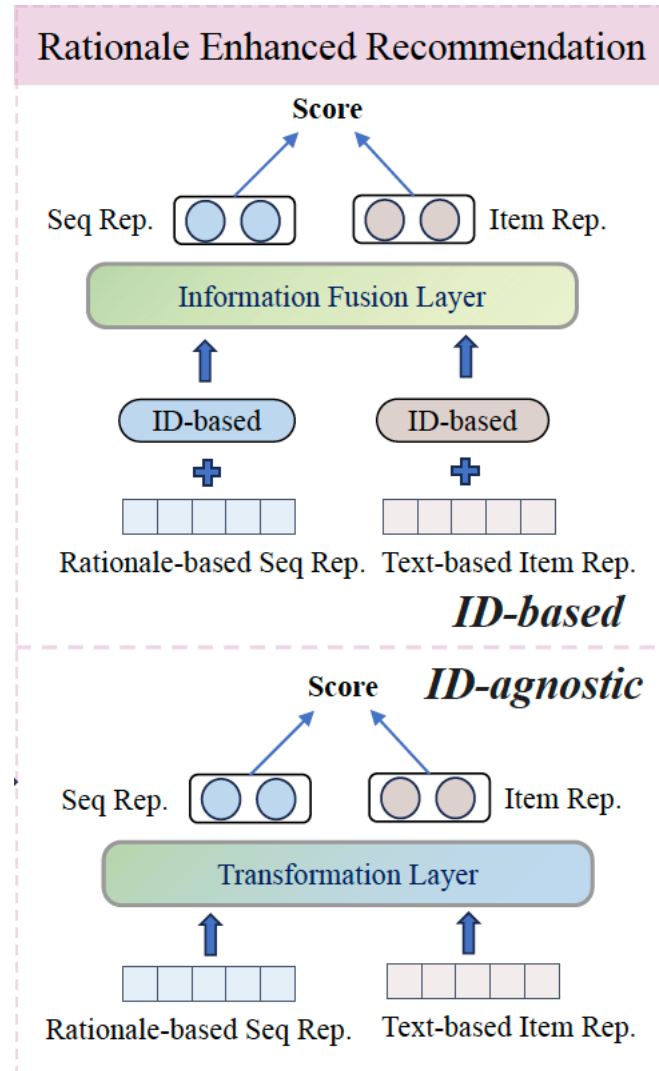
- Use an Information Fusion Layer to combine text and ID embeddings

$$\mathbf{z}_i = g_f([g_l(\mathbf{z}_{\text{text},i}); \mathbf{z}_{\text{id},i}]), \quad \mathbf{s}_u = g_f([g_l(\mathbf{s}_{\text{text},u}); \mathbf{s}_{\text{SeqEnc},u}])$$

- ID-Agnostic Matching:

- Apply a transformation to match rationale and item vectors

$$\mathbf{z}_i = g_t(\mathbf{z}_{\text{text},i}), \quad \mathbf{s}_u = g_t(\mathbf{s}_{\text{text},u})$$

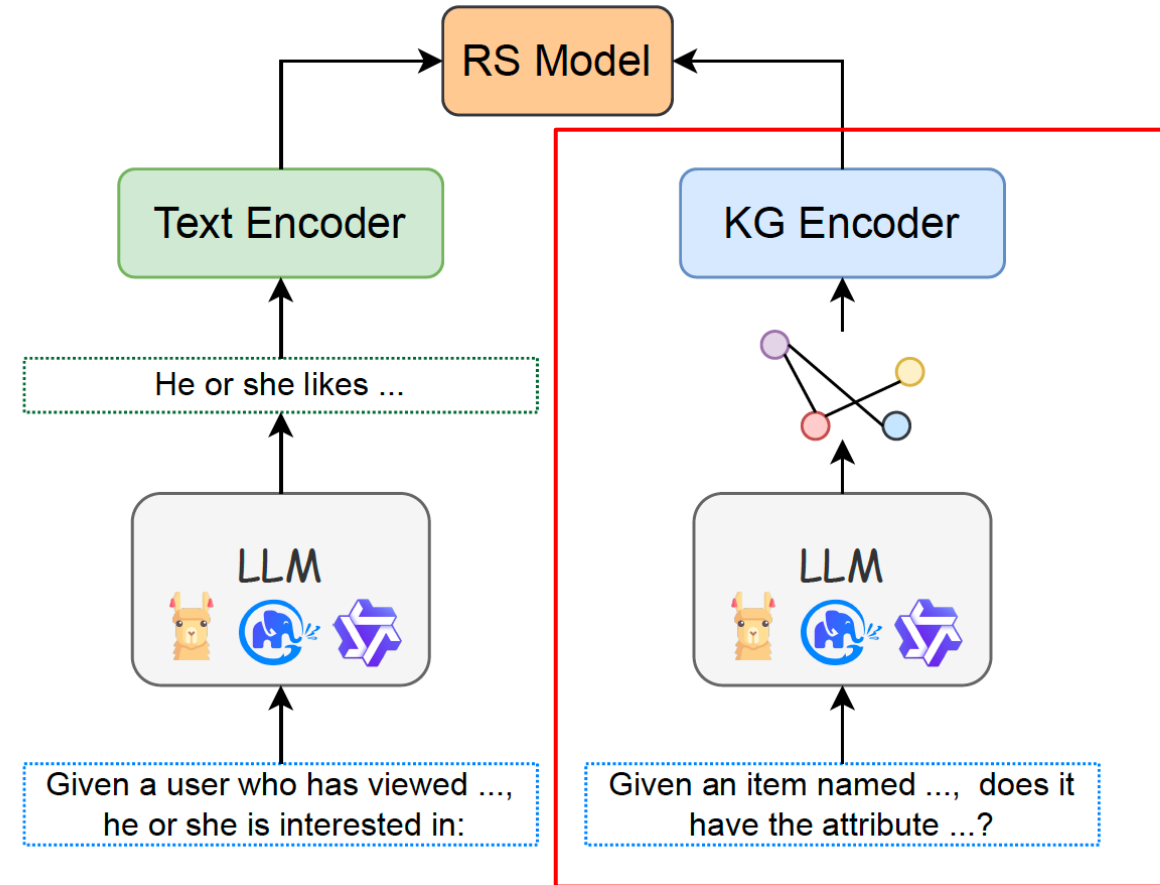


- SLIM consistently improves performance across three datasets and three backbone models (GRU4Rec, SASRec, SRGNN), outperforming both traditional and ChatGPT-augmented baselines.

| Methods              | Games               |                     |                     | Food                |                     |                     | Home                |                     |                     |
|----------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|---------------------|
|                      | NDCG@10             | Hit @10             | Hit @20             | NDCG@10             | Hit @10             | Hit @20             | NDCG@10             | Hit @10             | Hit @20             |
| GRU4Rec              | 17.61 ± 0.18        | 30.87 ± 0.56        | 42.39 ± 0.62        | 9.10 ± 0.30         | 15.27 ± 0.58        | 19.51 ± 0.24        | 2.19 ± 0.21         | 4.17 ± 0.39         | 7.53 ± 0.64         |
| GRU4Rec <sup>+</sup> | 27.33 ± 0.53        | 44.06 ± 0.79        | 56.53 ± 1.24        | 17.75 ± 0.78        | 31.10 ± 1.09        | 45.01 ± 1.46        | 12.19 ± 1.02        | 26.76 ± 2.58        | 49.10 ± 3.82        |
| SLIM <sup>-</sup>    | 27.70 ± 0.47        | 45.13 ± 0.56        | 57.70 ± 0.37        | 17.97 ± 0.70        | 31.78 ± 1.47        | 46.88 ± 2.10        | 13.59 ± 1.05        | 30.30 ± 2.01        | 55.85 ± 3.55        |
| SLIM                 | <b>28.37 ± 0.41</b> | <b>45.68 ± 0.53</b> | <b>58.09 ± 0.58</b> | <b>18.32 ± 0.53</b> | <b>32.56 ± 1.30</b> | <b>46.92 ± 1.82</b> | <b>15.64 ± 0.51</b> | <b>34.33 ± 1.53</b> | <b>62.93 ± 3.46</b> |
| Improv.              | 3.81%               | 3.68%               | 2.76%               | 3.21%               | 4.69%               | 4.24%               | 28.3%               | 28.29%              | 28.17%              |
| SASRec               | 22.73 ± 0.28        | 37.77 ± 0.52        | 51.53 ± 0.39        | 26.78 ± 0.24        | 35.78 ± 0.36        | 43.32 ± 0.55        | 2.66 ± 0.22         | 5.56 ± 0.72         | 14.93 ± 1.53        |
| SASRec <sup>+</sup>  | 27.46 ± 0.19        | 44.88 ± 0.63        | 58.90 ± 0.38        | 30.95 ± 0.38        | 44.98 ± 0.53        | 55.61 ± 1.12        | 5.58 ± 0.10         | 11.09 ± 0.16        | 20.69 ± 0.77        |
| SLIM <sup>-</sup>    | <b>31.58 ± 0.35</b> | 50.83 ± 0.62        | 63.45 ± 0.71        | 32.65 ± 0.15        | 48.01 ± 0.48        | 59.25 ± 0.56        | 5.95 ± 0.32         | 11.83 ± 0.62        | <b>22.43 ± 0.54</b> |
| SLIM                 | 31.43 ± 0.39        | <b>51.11 ± 0.82</b> | <b>64.10 ± 0.26</b> | <b>32.80 ± 0.40</b> | <b>48.27 ± 0.64</b> | <b>59.30 ± 0.89</b> | <b>6.01 ± 0.19</b>  | <b>12.01 ± 0.38</b> | 22.29 ± 0.85        |
| Improv.              | 14.46%              | 13.88%              | 8.83%               | 5.98%               | 7.31%               | 6.64%               | 7.71%               | 8.3%                | 7.73%               |
| SRGNN                | 16.45 ± 0.22        | 29.29 ± 0.14        | 40.99 ± 0.52        | 10.99 ± 2.07        | 20.32 ± 4.30        | 32.14 ± 6.55        | 5.04 ± 0.83         | 13.48 ± 2.23        | 37.22 ± 3.85        |
| SRGNN <sup>+</sup>   | 21.54 ± 0.64        | 36.77 ± 1.05        | 49.11 ± 1.54        | 11.91 ± 0.71        | 21.39 ± 1.91        | 33.63 ± 3.41        | 11.61 ± 1.14        | 25.22 ± 2.46        | 43.85 ± 3.58        |
| SLIM <sup>-</sup>    | 22.35 ± 1.48        | 37.69 ± 1.53        | 51.29 ± 0.59        | <b>12.92 ± 0.78</b> | <b>23.80 ± 1.60</b> | <b>37.22 ± 2.75</b> | 11.25 ± 1.42        | 24.28 ± 3.19        | 44.05 ± 5.97        |
| SLIM                 | <b>23.77 ± 0.20</b> | <b>39.81 ± 0.52</b> | <b>52.34 ± 0.63</b> | 12.38 ± 0.51        | 22.98 ± 1.30        | 36.44 ± 1.72        | <b>12.29 ± 1.39</b> | <b>26.51 ± 2.71</b> | <b>47.01 ± 3.61</b> |
| Improv.              | 10.35%              | 8.27%               | 6.58%               | 3.95%               | 7.43%               | 8.36%               | 5.86%               | 5.11%               | 7.21%               |

- Practical Values:** SLIM enables small language models to act as effective reasoners in sequential recommendation by distilling step-by-step rationales from LLMs at a fraction of the computational cost.

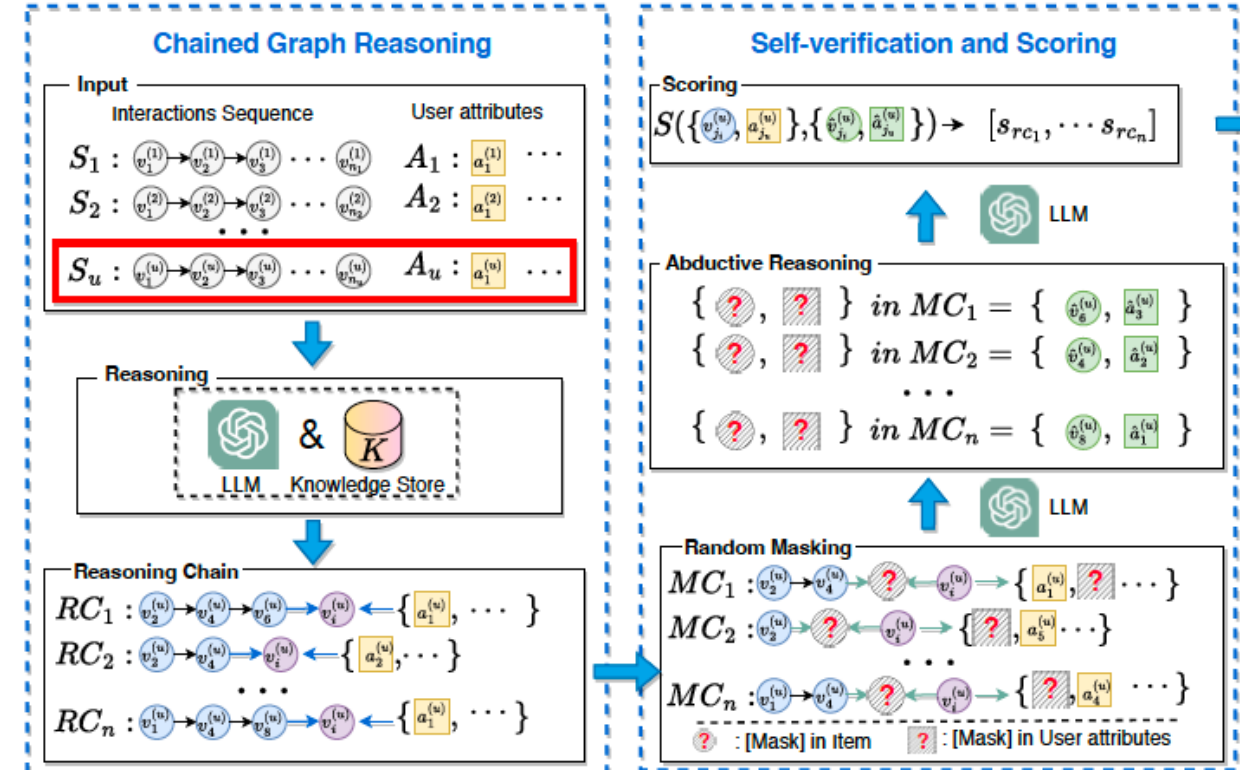
- Summary text is a type of unstructured knowledge, while the **structured Knowledge Graph (KG)** may drive **better integration**.
- Many work explore how to **apply the LLM to generate a KG** or **augment an existing KG** for enhancing RS.
- Categories
  - Generation
  - Completion & Fusion



- Background
  - Recent works use knowledge graphs or behavior graphs to enhance reasoning, but most rely on pre-defined structures and cannot flexibly adapt to personalized needs.
  - LLMs exhibit powerful reasoning and abductive inference ability.
- Motivation
  - Can LLMs generate personalized multi-hop reasoning chains for sequential RS?
  - How can we verify, score, and incorporate these LLM-generated chains into real-world recommenders?
  - Can this be done without relying on explicit external knowledge graphs?
- Contributions
  - Adaptive Reasoning Module

- Chain Construction and Verification
  - Chained Graph Reasoning
    - LLM builds reasoning chains  $RC$  from interaction sequence  $S_u$  and attribute  $A_u$
  - Random Masking & Abductive Reasoning
    - Each chain is perturbed into masked candidates  $MC_i$ .
    - The LLM performs abductive reasoning to infer whether masked parts still lead to the correct reasoning path.
- Scoring

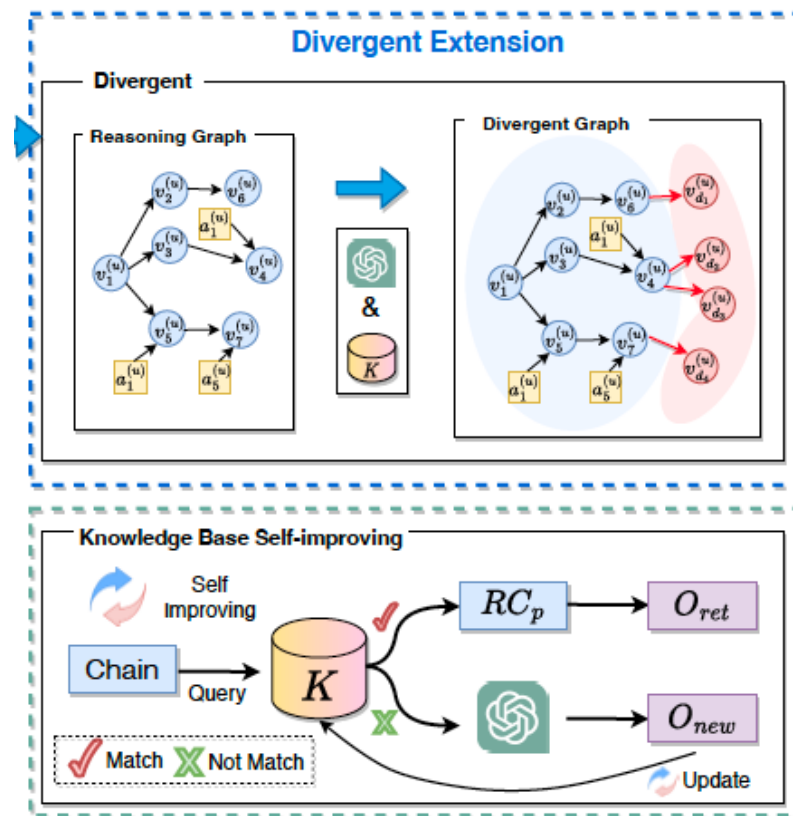
$$S(RC_i, A_u, S_u) \rightarrow [s_{RC_1}, \dots, s_{RC_n}]$$





- Reasoning Enhancement and Model Fusion
  - Divergent Extension:
    - From the reasoning graph, LLM identifies alternative paths by querying unexplored nodes, forming divergent reasoning graphs.
  - Self-Improving Knowledge Base:
    - Mismatches between retrieved reasoning and newly inferred reasoning are used to update the knowledge base.
  - Final Fusion:

$$E_{\text{fusion}} = \text{Fusion}(E_{\text{base}}, E_{\text{ori}}, E_{\text{div}})$$



- LLMRG improves NDCG and HR by over 25–30% on ML-1M, Amazon Beauty, and Amazon Clothing, outperforming all baselines.

| Dataset         | Metric  | FDSA     |           |           | BERT4Rec |           |           | CL4SRec  |           |           | DuoRec   |           |           |
|-----------------|---------|----------|-----------|-----------|----------|-----------|-----------|----------|-----------|-----------|----------|-----------|-----------|
|                 |         | Original | GPT3.5    | GPT4      | Original | GPT3.5    | GPT4      | Original | GPT3.5    | GPT4      | Original | GPT3.5    | GPT4      |
| ML-1M           | HR@5    | 0.0909   | + 20.70%  | + 25.79%  | 0.1124   | + 26.67%  | + 32.56%  | 0.1141   | + 19.98%  | + 21.02%  | 0.2011   | + 12.87%  | + 14.76%  |
|                 | HR@10   | 0.1631   | + 17.93 % | + 22.87%  | 0.1910   | + 13.52 % | + 16.49 % | 0.1866   | + 17.30 % | + 19.31 % | 0.2837   | + 14.10 % | + 15.53 % |
|                 | NDCG@5  | 0.0599   | + 21.33%  | + 30.27 % | 0.0713   | + 25.74 % | + 32.82 % | 0.0721   | + 14.97 % | + 16.78 % | 0.1265   | + 23.55 % | + 26.01 % |
|                 | NDCG@10 | 0.0878   | + 21.78%  | + 28.25%  | 0.0980   | + 23.34%  | + 28.06%  | 0.1013   | + 17.67%  | + 20.42%  | 0.1663   | + 12.86%  | + 13.77%  |
| Amazon Beauty   | HR@5    | 0.0237   | + 13.89 % | + 17.53 % | 0.0201   | + 19.17 % | + 23.22 % | 0.0398   | + 11.15 % | + 14.15 % | 0.0552   | + 9.31 %  | + 11.93 % |
|                 | HR@10   | 0.0418   | + 15.02 % | + 17.78 % | 0.0413   | + 17.79 % | + 22.14 % | 0.0664   | + 10.22 % | + 11.32 % | 0.0839   | + 5.14 %  | + 6.61 %  |
|                 | NDCG@5  | 0.0195   | + 16.20 % | + 18.64 % | 0.0192   | + 14.21 % | + 17.63 % | 0.0221   | + 8.45 %  | + 10.18 % | 0.0350   | + 7.42 %  | + 9.24 %  |
|                 | NDCG@10 | 0.0275   | + 14.78 % | + 17.64 % | 0.0263   | + 11.53 % | + 14.76 % | 0.0322   | + 8.17 %  | + 9.68 %  | 0.0447   | + 6.67 %  | + 7.95 %  |
| Amazon Clothing | HR@5    | 0.0119   | + 20.67 % | + 23.92 % | 0.0128   | + 16.09 % | + 19.10 % | 0.0166   | + 7.90 %  | + 10.92 % | 0.0190   | + 9.98 %  | + 11.40 % |
|                 | HR@10   | 0.0197   | + 14.45 % | + 17.88 % | 0.0202   | + 10.52 % | + 13.72 % | 0.0273   | + 11.21 % | + 14.99 % | 0.0311   | + 7.65 %  | + 9.48 %  |
|                 | NDCG@5  | 0.0073   | + 8.16 %  | + 10.86 % | 0.0081   | + 7.39 %  | + 10.39 % | 0.0093   | + 6.02 %  | + 9.09 %  | 0.0118   | + 6.74 %  | + 9.19 %  |
|                 | NDCG@10 | 0.0109   | + 6.01 %  | + 8.13 %  | 0.0113   | + 5.21 %  | + 5.94 %  | 0.0125   | + 4.32 %  | + 8.07 %  | 0.0155   | + 7.89 %  | + 9.29 %  |

- Practical Values:** LLMRG enables LLM-based, adaptive, multi-hop reasoning without external knowledge graphs, improving recommendation via personalized and verifiable chains.

- Background
  - Existing KG-based recommenders often convert textual information into IDs, ignore semantic connections, and struggle to capture high-order relations due to GNNs' limitations.
- Motivation
  - KGs are incomplete or too sparse
  - Converting entities into IDs discards rich semantic information, and GNNs fail to effectively propagate information over long distances in the KG.
- Contributions
  - Item-item semantic graph
  - Representation alignment and neighbor augmentation modules

# Completion & Fusion - CoLaKG

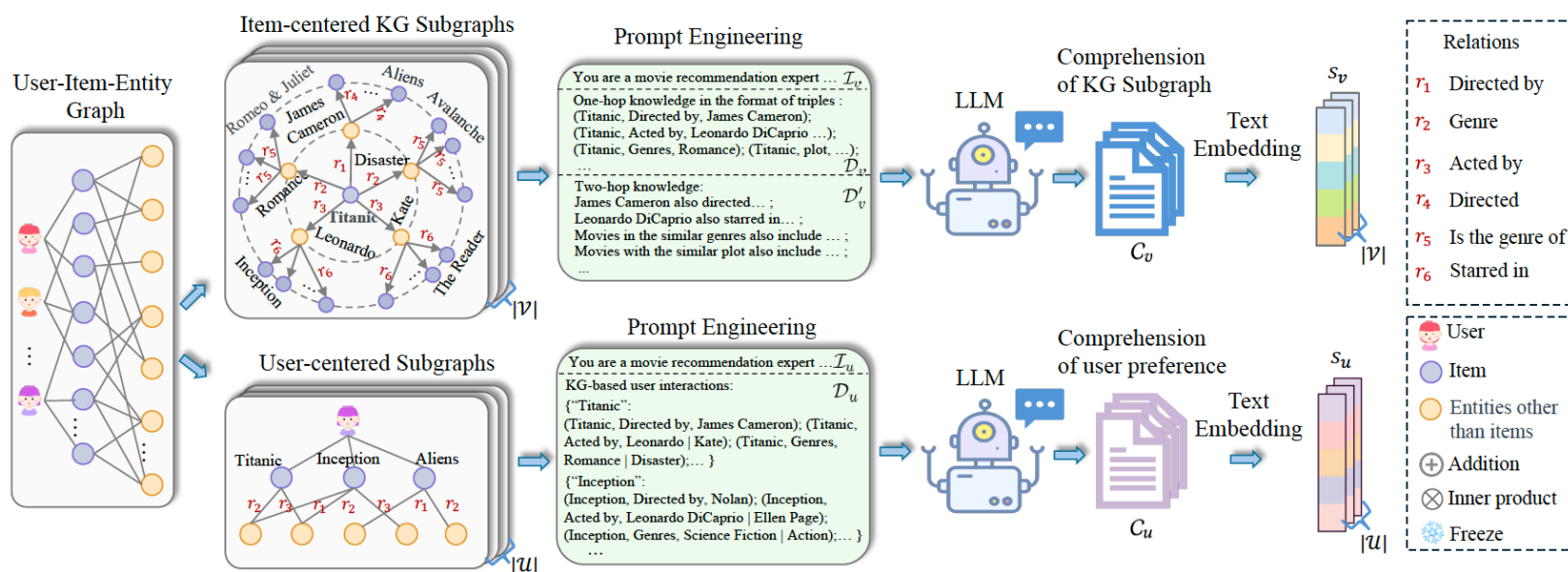
- KG Comprehension via LLM

- Item-centered KG Subgraph Construction
- Text Conversion & Prompting
- Embedding Generation
  - Use pre-trained model  $P$  to obtain semantic embedding

$$T_v = \{(v, r, e)\}$$

$$C_v = \text{LLMs}(I_v, D_v, D'_v)$$

$$s_v = P(C_v)$$



- Semantic Graph Construction & Integration
  - Semantic Item-Item Graph
    - Use cosine similarity to define edge weights
    - Construct graph
  - Representation Integration
    - Align semantic embeddings to ID space
    - Fuse with ID embeddings
  - Neighbor Augmentation

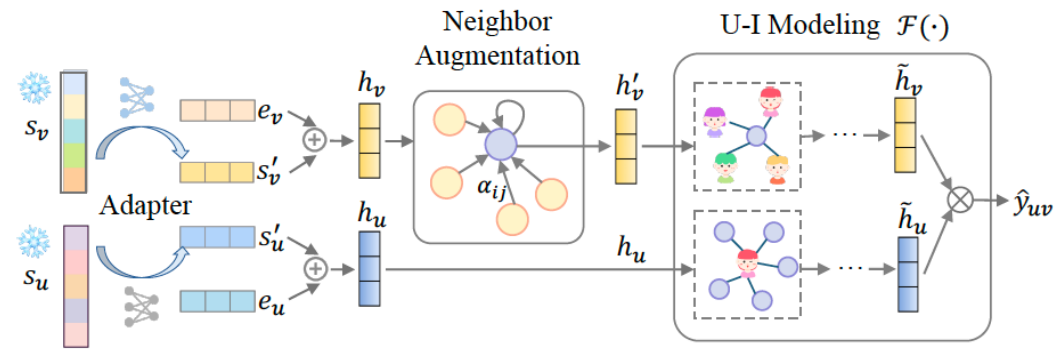
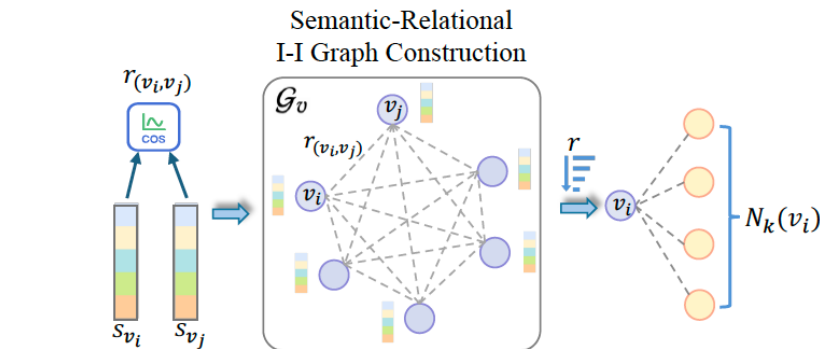
$$r(v_i, v_j) = \text{sim}(s_{v_i}, s_{v_j})$$

$$\mathcal{G}_v = \{(v_i, r(v_i, v_j), v_j)\}$$

$$s'_v = \sigma(W_1 s_v), \quad s'_u = \sigma(W_2 s_u)$$

$$h_v = \frac{1}{2}(e_v + s'_v), \quad h_u = \frac{1}{2}(e_u + s'_u)$$

$$\mathbf{h}'_{v_i} = \sigma \left( \frac{1}{2} \left( \mathbf{h}_{v_i} + \sum_{j \in \mathcal{N}_k(v_i)} \alpha_{ij} \mathbf{h}_{v_j} \right) \right)$$





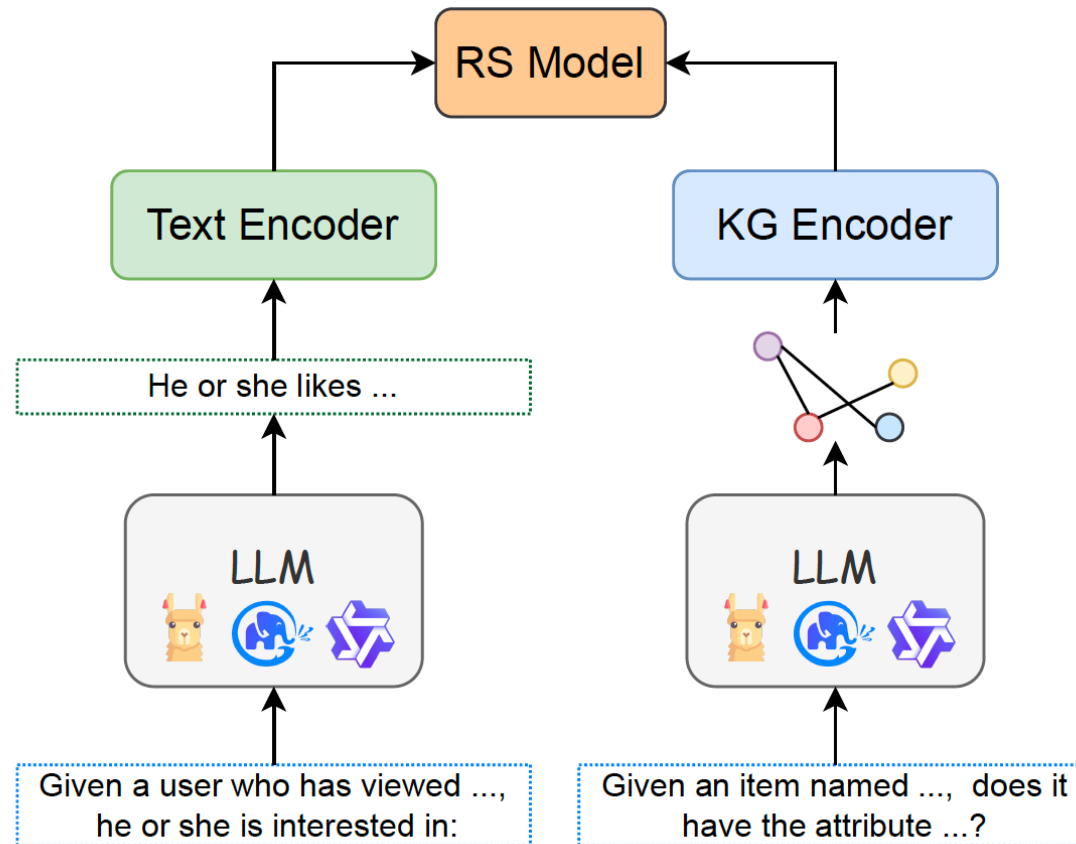
- CoLaKG achieves the best performance across all four datasets, demonstrating strong generalization and semantic reasoning ability.

| Model     | MovieLens     |               |               |               | Last-FM       |               |               |               | MIND          |               |               |               | Funds         |               |               |               |
|-----------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|           | R@10          | N@10          | R@20          | N@20          | R@10          | N@10          | R@20          | N@20          | R@10          | N@10          | R@20          | N@20          | R@10          | N@10          | R@20          | N@20          |
| BPR-MF    | 0.1257        | 0.3100        | 0.2048        | 0.3062        | 0.1307        | 0.1352        | 0.1971        | 0.1685        | 0.0315        | 0.0238        | 0.0537        | 0.0310        | 0.4514        | 0.3402        | 0.5806        | 0.3809        |
| NFM       | 0.1346        | 0.3558        | 0.2129        | 0.3379        | 0.2246        | 0.2327        | 0.3273        | 0.2830        | 0.0495        | 0.0356        | 0.0802        | 0.0458        | 0.4388        | 0.3187        | 0.5756        | 0.3651        |
| LightGCN  | 0.1598        | 0.3901        | 0.2512        | 0.3769        | 0.2589        | 0.2799        | 0.3642        | 0.3321        | 0.0624        | 0.0492        | 0.0998        | 0.0609        | 0.4992        | 0.3778        | 0.6353        | 0.4204        |
| CKE       | 0.1524        | 0.3783        | 0.2373        | 0.3609        | 0.2342        | 0.2545        | 0.3266        | 0.3001        | 0.0526        | 0.0417        | 0.0822        | 0.0510        | 0.4926        | 0.3702        | 0.6294        | 0.4130        |
| RippleNet | 0.1415        | 0.3669        | 0.2201        | 0.3423        | 0.2267        | 0.2341        | 0.3248        | 0.2861        | 0.0472        | 0.0364        | 0.0785        | 0.0451        | 0.4764        | 0.3591        | 0.6124        | 0.4003        |
| KGAT      | 0.1536        | 0.3782        | 0.2451        | 0.3661        | 0.2470        | 0.2595        | 0.3433        | 0.3075        | 0.0594        | 0.0456        | 0.0955        | 0.0571        | 0.5037        | 0.3751        | 0.6418        | 0.4182        |
| KGIN      | 0.1631        | 0.3959        | 0.2562        | 0.3831        | 0.2562        | 0.2742        | 0.3611        | 0.3215        | 0.0640        | 0.0518        | 0.1022        | 0.0639        | 0.5079        | 0.3857        | 0.6428        | 0.4259        |
| KGCL      | 0.1554        | 0.3797        | 0.2465        | 0.3677        | <u>0.2599</u> | 0.2763        | <u>0.3652</u> | 0.3284        | <u>0.0671</u> | <u>0.0543</u> | <u>0.1059</u> | <u>0.0670</u> | 0.5071        | 0.3877        | 0.6355        | 0.4273        |
| KGRec     | <u>0.1640</u> | <u>0.3968</u> | <u>0.2571</u> | <u>0.3842</u> | 0.2571        | 0.2748        | 0.3617        | 0.3251        | 0.0627        | 0.0506        | 0.1003        | 0.0625        | <u>0.5104</u> | <u>0.3913</u> | <u>0.6467</u> | <u>0.4304</u> |
| RLMRec    | 0.1613        | 0.3920        | 0.2524        | 0.3787        | 0.2597        | <u>0.2812</u> | 0.3651        | <u>0.3335</u> | 0.0619        | 0.0486        | 0.0990        | 0.0602        | 0.4988        | 0.3784        | 0.6351        | 0.4210        |
| CoLaKG    | <b>0.1699</b> | <b>0.4130</b> | <b>0.2642</b> | <b>0.3974</b> | <b>0.2738</b> | <b>0.2948</b> | <b>0.3803</b> | <b>0.3471</b> | <b>0.0698</b> | <b>0.0562</b> | <b>0.1087</b> | <b>0.0684</b> | <b>0.5273</b> | <b>0.4012</b> | <b>0.6524</b> | <b>0.4392</b> |

- Practical Values:** By leveraging LLMs to comprehend and augment KG information, CoLaKG bridges the gap between structured knowledge and collaborative filtering, resulting in more robust and semantically aligned recommendation

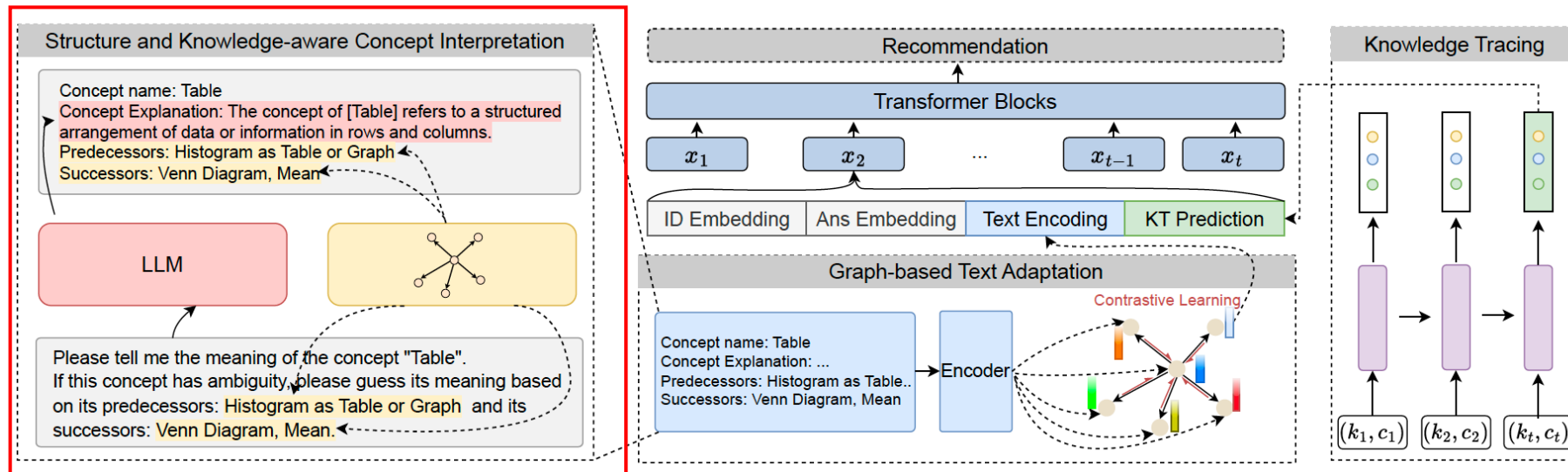


- In consideration of the effectiveness of Summary text and Knowledge graph, some work also resort to combine both of them.

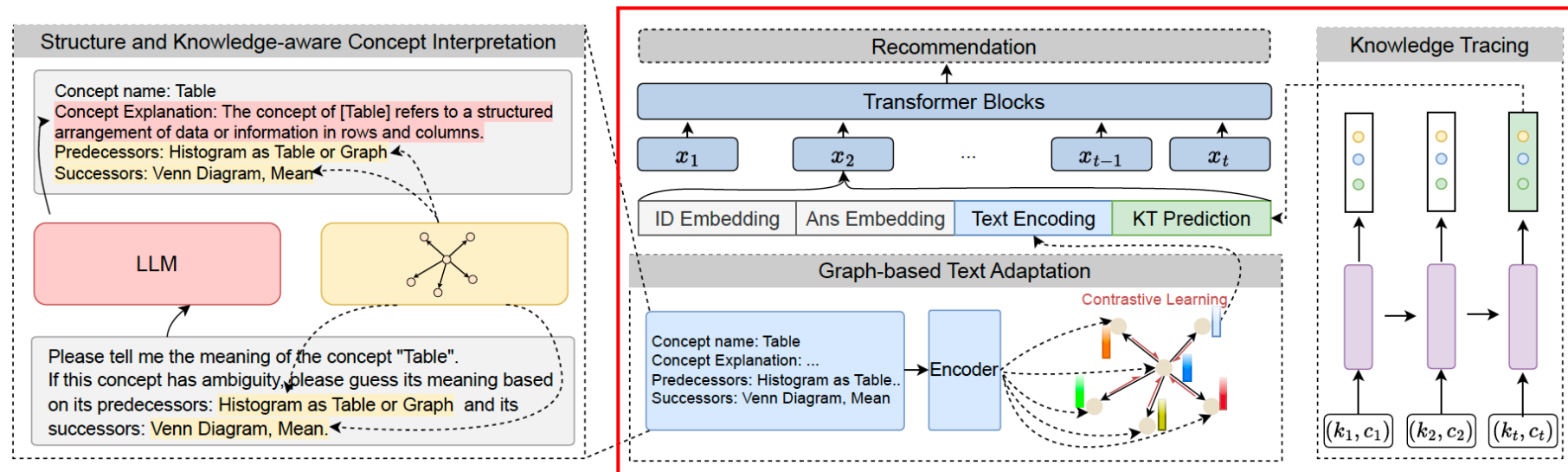


- Background
  - Existing KT and recommendation models often rely on ID-based concept representations, which fail to generalize and lack interpretability.
- Motivation
  - Prior methods overlook the textual and relational information between concepts, which leads to limited generalization, especially when dealing with unseen or ambiguous concepts.
- Contributions
  - Structure and Knowledge-aware Concept Interpretation
  - Graph-based Text Adaptation

- Structure and Knowledge-aware Concept Interpretation
  - For a given concept  $c$ , SKarRec constructs prompt with (1) name and explanation (2) its predecessors and successors in the concept graph
  - Prompt is sent to LLM to generate rich interpretation
  - The interpreted concept text is used as semantic anchor for downstream tasks



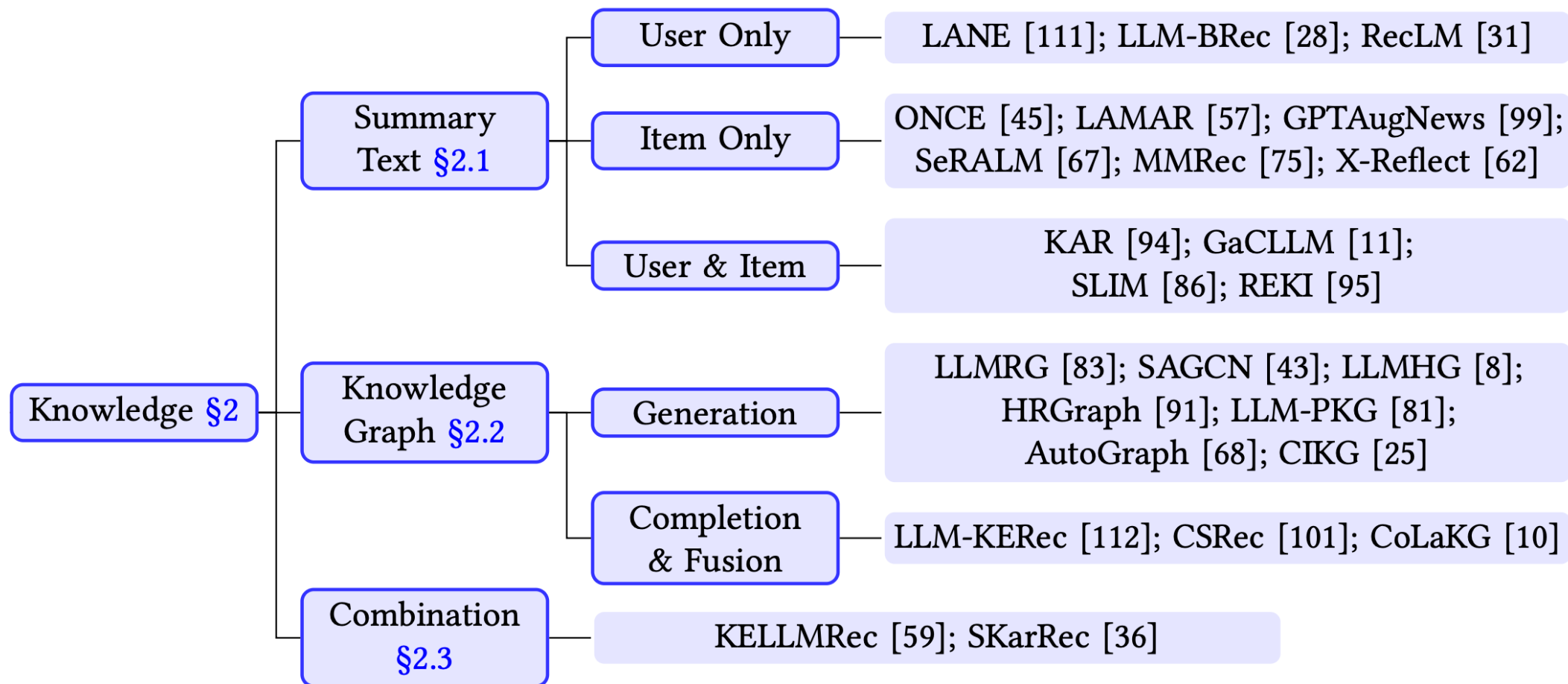
- Graph-based Text Adaptation for Recommendation and Tracing
  - Interpreted concept texts are encoded into embeddings using a transformer encoder.
  - Graph  $G$  over concepts is built using their relations
  - Contrastive learning: positive pair (connected nodes), negative pair (unconnected)



- SKarRec achieves the best performance on ASSIST09, ASSIST12, and Junyi across all three metrics, significantly outperforming prior graph-based, text-only, and ID-only methods.

| Dataset  | Metric | Graph-based Methods |        | ID-Only Methods |          | Text-Only Methods |           | ID-Text Methods |               |         |                |
|----------|--------|---------------------|--------|-----------------|----------|-------------------|-----------|-----------------|---------------|---------|----------------|
|          |        | ACKRec              | GCARec | SASRec          | BERT4Rec | ZESRec            | RECFORMER | FDSA            | S3-Rec        | UniSRec | SKarRec        |
| ASSIST09 | HR@1   | 0.2408              | 0.2513 | 0.7649          | 0.1935   | 0.5809            | 0.6072    | 0.7627          | <u>0.7661</u> | 0.7597  | <b>0.7922*</b> |
|          | NDCG@5 | 0.3787              | 0.3904 | <u>0.8736</u>   | 0.2203   | 0.7269            | 0.6540    | 0.8722          | 0.8731        | 0.8647  | <b>0.8838*</b> |
|          | MRR    | 0.3627              | 0.3756 | 0.8503          | 0.2342   | 0.6998            | 0.6711    | 0.8484          | <u>0.8505</u> | 0.8439  | <b>0.8646*</b> |
| ASSIST12 | HR@1   | 0.1976              | 0.2119 | 0.2838          | 0.2540   | 0.2596            | 0.1990    | 0.2861          | <u>0.2904</u> | 0.2888  | <b>0.2933*</b> |
|          | NDCG@5 | 0.3078              | 0.3169 | 0.4895          | 0.4247   | 0.4649            | 0.2055    | 0.4886          | <b>0.4957</b> | 0.4934  | <u>0.4954</u>  |
|          | MRR    | 0.3064              | 0.3182 | 0.4574          | 0.4047   | 0.4343            | 0.2536    | 0.4579          | <u>0.4633</u> | 0.4617  | <b>0.4645*</b> |
| Junyi    | HR@1   | 0.1447              | 0.1284 | 0.8469          | 0.3550   | 0.8546            | 0.8279    | 0.8492          | <u>0.8608</u> | 0.8407  | <b>0.8730*</b> |
|          | NDCG@5 | 0.1872              | 0.1650 | 0.8907          | 0.4826   | 0.8930            | 0.7625    | 0.8917          | <u>0.9022</u> | 0.8890  | <b>0.9076*</b> |
|          | MRR    | 0.1918              | 0.1740 | 0.8825          | 0.4661   | 0.8860            | 0.8497    | 0.8840          | <u>0.8942</u> | 0.8799  | <b>0.9014*</b> |

- Practical Values:** By integrating structural graph knowledge and language-model-based semantic reasoning, SKarRec offers a robust framework for educational recommendation and knowledge tracing with superior generalization and interpretability





# Agenda

## 1 Introduction



Zijian Zhang



## 2 Knowledge Enhancement



Pengyue Jia



## 3 Interaction Enhancement



Ziwei Liu



## 4.1 Model Enhancement 1



Maolin Wang



## 4.2 Model Enhancement 2



Yuhao Wang



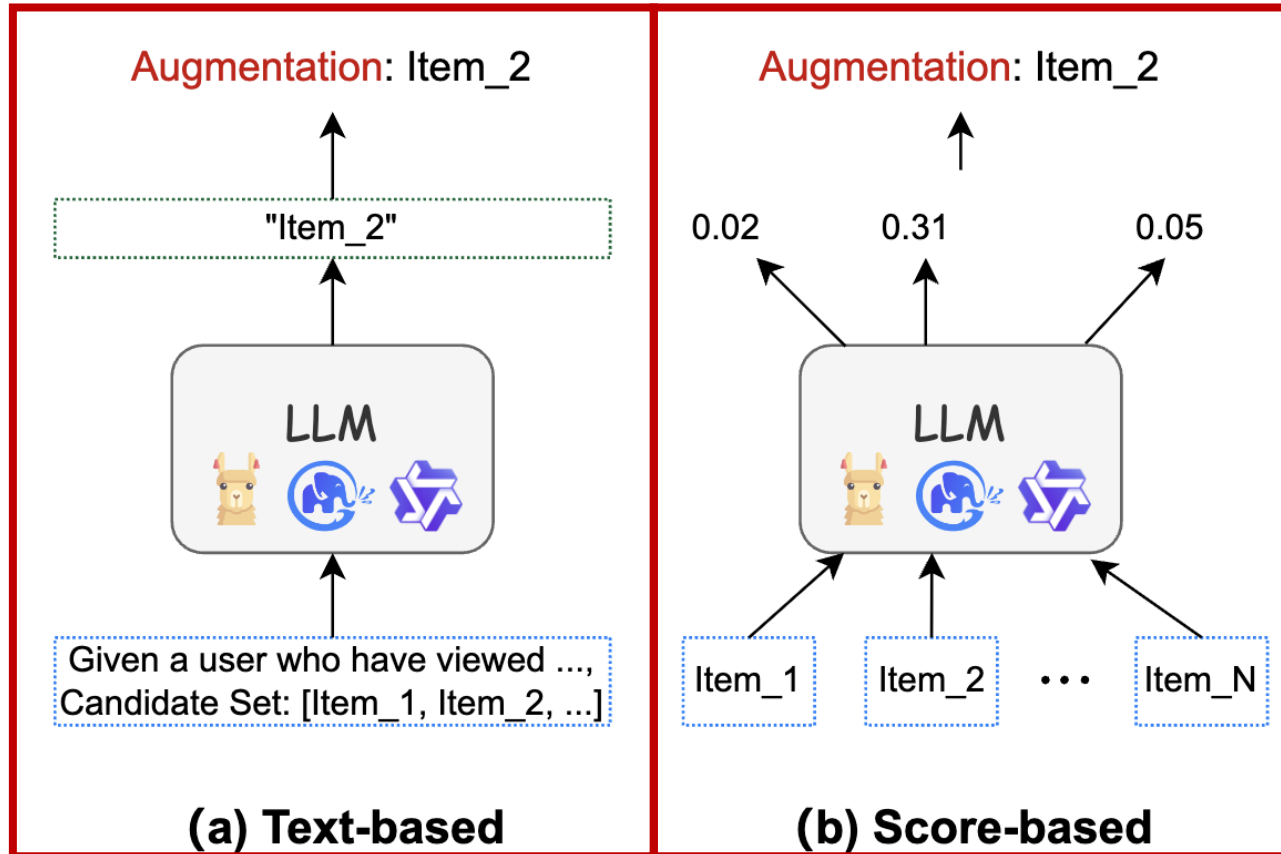
## 5 Conclusion



Yejing Wang

# Interaction Enhancement

We basically divide the categories of LLM for interaction enhancement in **two aspects**:



## □ Text-based:

As shown in Figure (a), test-based methods first conclude a **candidate set** and then utilize LLM to **conduct the possible next interacted item** to achieve augmentation.

## □ Score-based:

For comparison (shown in Figure (b)), score-based methods utilize LLM to **rank the importance of each item**.

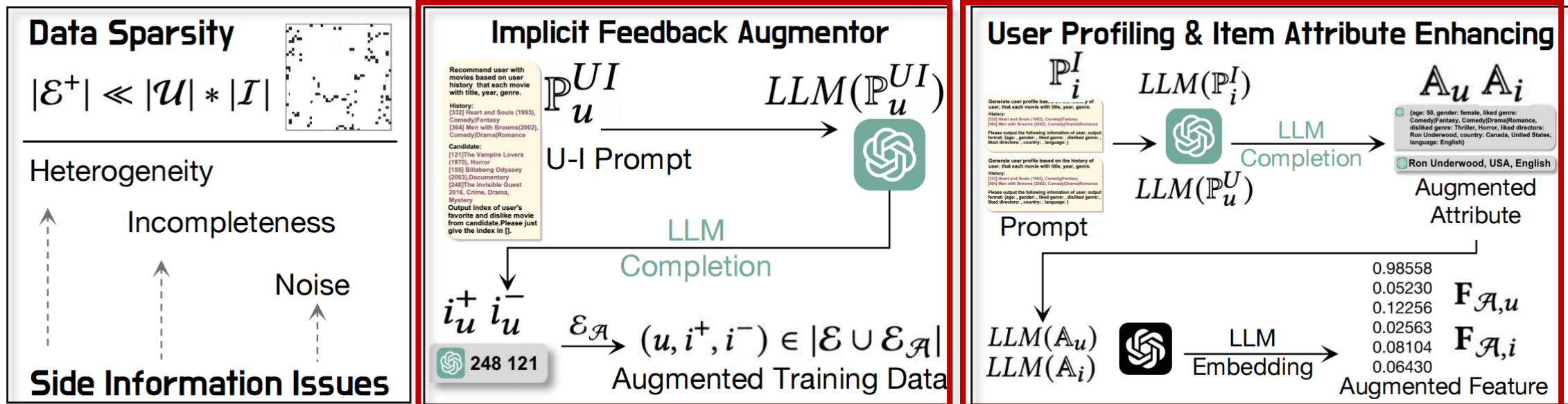
**Low-quality side information & Data Sparsity** often contains bad affects, which severely affects the accuracy of recommendation.

**Large Language Model (LLM)** is a promising technique to predict the user-item interaction and augment the side information of users/items. However, there are still two severe problems that hinder it work as a powerful data augmentor:

- **Incomplete input information**: It is hard to enable LLM to reason over full user-item interaction patterns due to the limited input token. (**P1**)
- **Low-quality of generated data**: LLM generators often introduce low-quality content that may compromise the results because of the noise and the hallucination of the model. (**P2**)

# Text-based: LLMRec

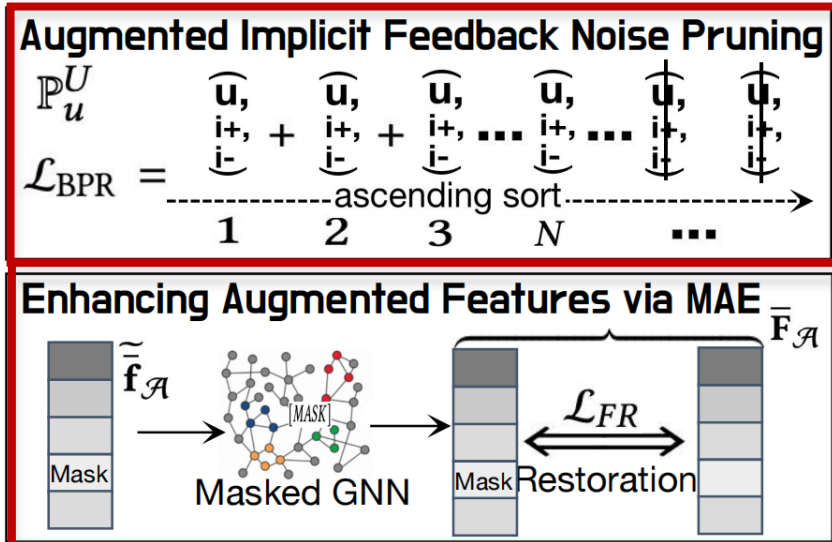
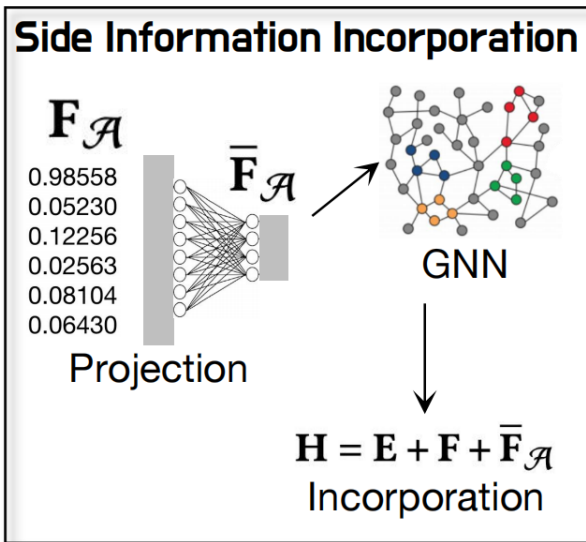
Specifically, for the first problem '**P1: Incomplete input information**', The authors design two individual modules to from a natural language perspective.



- **Implicit Feedback Augmentor**: Feeding the side feature of items and an item candidates pool (acquired by a basic Rec model) into LLM.
- **User Profiling & Item Attribute Enhancing**: Leveraging LLM's reasoning ability to summarize users' interactions and unify the item attributes.

# Text-based: LLMRec

For the ‘**P2: Low-quality of generated data**’, The authors propose the **Augmented Implicit Feedback Noise Pruning and Enhancing Augmented Features via MAE**.



## Augmented Implicit Feedback Noise Pruning:

- Leveraging **ascending-sort mechanism** with a BPR loss to filter values and selects the **top-N score**.

$$\mathcal{L}_{BPR} = \sum_{(u, i^+, i^-)}^{|\mathcal{E} \cup \mathcal{E}_{\mathcal{A}}|} -\log(\sigma(\hat{y}_{u, i^+} - \hat{y}_{u, i^-})) + \omega_2 \cdot \|\Theta\|^2$$

$$\mathcal{E}_{\mathcal{A}} \subseteq \{LLM(\mathbb{P}_u) | u \in \mathcal{U}\}, \quad |\mathcal{E}_{\mathcal{A}}| = \omega_3 * B$$

$$\sum_{(u, i^+, i^-)}^{(1-\omega_4) * |\mathcal{E} \cup \mathcal{E}_{\mathcal{A}}|} -SortAscend(\log(\sigma(\hat{y}_{u, i^+} - \hat{y}_{u, i^-}))) [0 : N] + \omega_2 \cdot \|\Theta\|^2$$

- Enhancing Augmented Semantic Features via MAE:** 1. Masking features of selected subset of the node; 2. Leveraging **feature restoration loss** to compare the **masked matrix** with the **original one**.

$$\tilde{\mathbf{f}}_{\mathcal{A}} = \begin{cases} \mathbf{f}_{[MASK]} & v \in \tilde{\mathcal{V}} \\ \mathbf{f}_{\mathcal{A}} & v \notin \tilde{\mathcal{V}} \end{cases} \quad \mathcal{L}_{FR} = \frac{1}{|\tilde{\mathcal{V}}|} \sum_{v \in \tilde{\mathcal{V}}} \left(1 - \frac{\tilde{\mathbf{f}}_{\mathcal{A}} \cdot \mathbf{f}_{\mathcal{A}}}{\|\tilde{\mathbf{f}}_{\mathcal{A}}\| \cdot \|\mathbf{f}_{\mathcal{A}}\|}\right)^{\gamma}$$



# Text-based: LLMRec

For the experimental results, LLMRec achieves remarkable results under two public datasets and beats all the compared baselines with significant improvement.

| Baseline                                | Netflix       |               |               |               |               |               |               | MovieLens     |               |               |               |               |               |               |
|-----------------------------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|                                         | R@10          | N@10          | R@20          | N@20          | R@50          | N@50          | P@20          | R@10          | N@10          | R@20          | N@20          | R@50          | N@50          | P@20          |
| General Collaborative Filtering Methods |               |               |               |               |               |               |               |               |               |               |               |               |               |               |
| MF-BPR                                  | 0.0282        | 0.0140        | 0.0542        | 0.0205        | 0.0932        | 0.0281        | 0.0027        | 0.1890        | 0.0815        | 0.2564        | 0.0985        | 0.3442        | 0.1161        | 0.0128        |
| NGCF                                    | 0.0347        | 0.0161        | 0.0699        | 0.0235        | 0.1092        | 0.0336        | 0.0032        | 0.2084        | 0.0886        | 0.2926        | 0.1100        | 0.4262        | 0.1362        | 0.0146        |
| LightGCN                                | 0.0352        | 0.0160        | 0.0701        | 0.0238        | 0.1125        | 0.0339        | 0.0032        | 0.1994        | 0.0837        | 0.2660        | 0.1005        | 0.3692        | 0.1209        | 0.0133        |
| Recommenders with Side Information      |               |               |               |               |               |               |               |               |               |               |               |               |               |               |
| VBPR                                    | 0.0325        | 0.0142        | 0.0553        | 0.0199        | 0.1024        | 0.0291        | 0.0028        | 0.2144        | 0.0929        | 0.2980        | 0.1142        | 0.4076        | 0.1361        | 0.0149        |
| MMGCN                                   | 0.0363        | 0.0174        | 0.0699        | 0.0249        | 0.1164        | 0.0342        | 0.0033        | 0.2314        | 0.1097        | 0.2856        | 0.1233        | 0.4282        | 0.1514        | 0.0147        |
| GRCN                                    | 0.0379        | 0.0192        | 0.0706        | 0.0257        | 0.1148        | 0.0358        | 0.0035        | 0.2384        | 0.1040        | 0.3130        | 0.1236        | 0.4532        | 0.1516        | 0.0150        |
| Data Augmentation Methods               |               |               |               |               |               |               |               |               |               |               |               |               |               |               |
| LATTICE                                 | 0.0433        | 0.0181        | 0.0737        | 0.0259        | 0.1301        | 0.0370        | 0.0036        | 0.2116        | 0.0955        | 0.3454        | 0.1268        | 0.4667        | 0.1479        | 0.0167        |
| MICRO                                   | <u>0.0466</u> | 0.0196        | <u>0.0764</u> | 0.0271        | <u>0.1306</u> | 0.0378        | <u>0.0038</u> | 0.2150        | <u>0.1131</u> | <u>0.3461</u> | <u>0.1468</u> | <u>0.4898</u> | <u>0.1743</u> | <u>0.0175</u> |
| Self-supervised Methods                 |               |               |               |               |               |               |               |               |               |               |               |               |               |               |
| CLCRec                                  | 0.0428        | 0.0217        | 0.0607        | 0.0262        | 0.0981        | 0.0335        | 0.0030        | 0.2266        | 0.0971        | 0.3164        | 0.1198        | 0.4488        | 0.1459        | 0.0158        |
| MMSSL                                   | 0.0455        | <u>0.0224</u> | 0.0743        | <u>0.0287</u> | 0.1257        | <u>0.0383</u> | 0.0037        | <u>0.2482</u> | 0.1113        | 0.3354        | 0.1310        | 0.4814        | 0.1616        | 0.0170        |
| LLMRec                                  | <b>0.0531</b> | <b>0.0272</b> | <b>0.0829</b> | <b>0.0347</b> | <b>0.1382</b> | <b>0.0456</b> | <b>0.0041</b> | <b>0.2603</b> | <b>0.1250</b> | <b>0.3643</b> | <b>0.1628</b> | <b>0.5281</b> | <b>0.1901</b> | <b>0.0186</b> |
| $p$ -value                              | $2.9e^{-4}$   | $3.0e^{-3}$   | $9.4e^{-5}$   | $1.5e^{-3}$   | $2.8e^{-5}$   | $2.2e^{-3}$   | $3.4e^{-5}$   | $2.8e^{-5}$   | $1.6e^{-2}$   | $3.1e^{-3}$   | $4.1e^{-4}$   | $1.9e^{-3}$   | $1.3e^{-2}$   | $1.8e^{-3}$   |
| Improv.                                 | 13.95%        | 21.43%        | 8.51%         | 20.91%        | 5.82%         | 19.06%        | 7.89%         | 4.88%         | 10.52%        | 5.26%         | 10.90%        | 7.82%         | 9.06%         | 6.29%         |

## Overall Performance Results



# Text-based: LlamaRec

While **Conventional methods** excel at mining collaborative information and modeling sequential behavior, they struggle with **data sparsity** and **long-tail problem**.

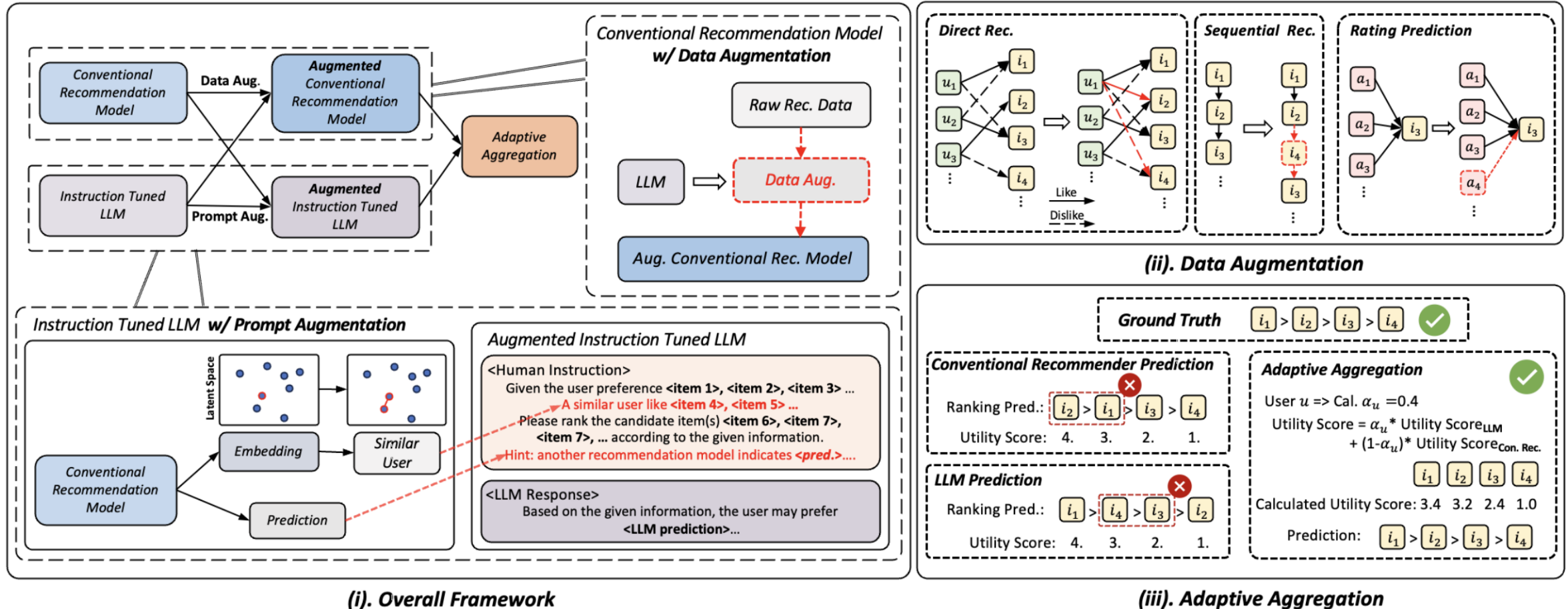
↑  
**Significant Gap**  
↓

**LLM-based methods** are proficient at utilizing rich textual contexts, but face challenges in utilizing **collaborative or sequential information**.

- **Q1**: How to leverage LLM's reasoning ability in conventional methods?
- **Q2**: How to introduce collaborative/sequential information in LLM-based methods?
- **Q3**: How to aggregate the LLM-based method and conventional methods properly?

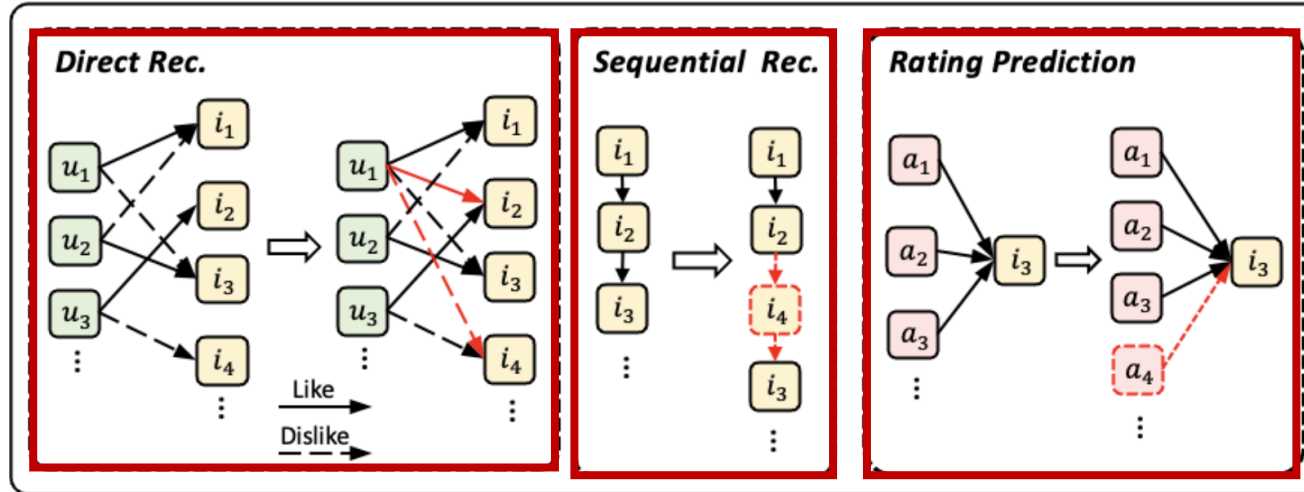
# Text-based: LlamaRec

To overcome the above problems, the authors propose LlamaRec, which allows the conventional & LLM-based augmentation to **mutually augment each other** and **adaptively aggregates the results**.



# Text-based: LlamaRec

Specifically, for the first problem ‘**Q1**: How to leverage LLM’s reasoning ability in conventional methods?’, The authors designed the **Data Augmentation** module.



## • Data Augmentation:

### □ Direct Recommendation:

Leveraging LLM to rank each item pair and form the corresponding BPR loss:

$$\mathcal{L}'_{BPR} = - \sum_{(u,i,j) \in \mathcal{D}'} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj})$$

### □ Sequential Recommendation:

Leveraging LLM to predict the user preferred items in the un-interacted list, and then inserting them in the interaction sequence.

### □ Rating Prediction:

Leveraging LLM’s world knowledge to provide side information of item features.

# Text-based: LlamaRec

For the second problem ‘**Q2**: How to introduce collaborative/sequential information in LLM-based methods?’, The authors design the two **Prompt Augmentation** strategies.

## Top-k Recommendation Prompt Example:

**Instruction:** Rank the candidate movies based on user historical interactions and make the top  $k$  recommendations.

**Interaction History:** Beyond Rangoon (1995); Alien (1979); Hollow Reed (1996); Primary Colors (1998); ...; Birds, The (1963)

**Candidate Items:** Last Dance (1996); Remains of the Day, The (1993); Assassins (1995); ...; Fatal Instinct (1993)

**Similar User Interaction History:** L.A. Confidential (1997); Apt Pupil (1998); Kolya (1996); ...; Star Wars (1977)

**Conventional Model Prediction:** Remains of the Day, The (1993); Addiction, The (1995); ...; Fugitive, The (1993)

**Output:** Fugitive, The (1993); Angel Baby (1995); ...; Remains of the Day, The (1993)

## Rating Prediction Prompt Example:

**Instruction:** Predict the rating of a target movie based on the user's historical movie ratings.

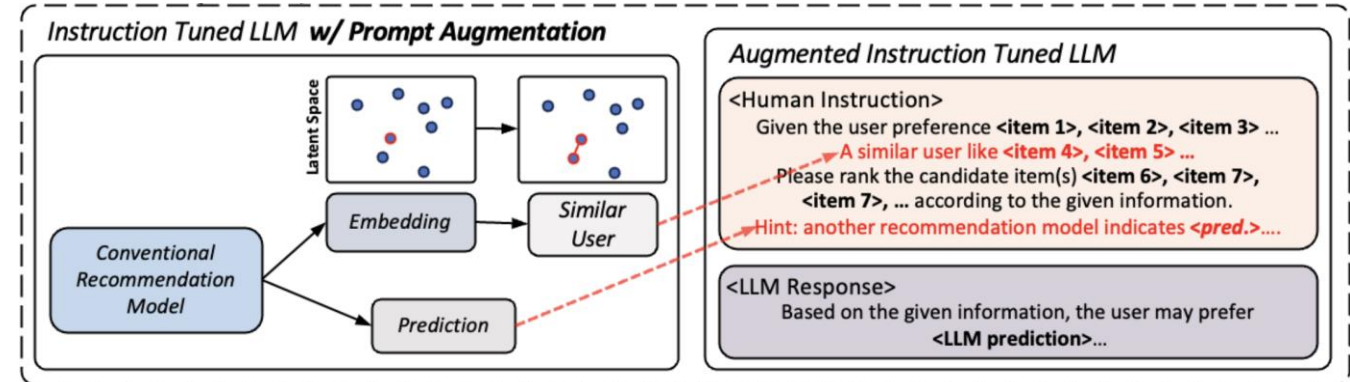
**Rating History:** Independence Day (1996): 3; Grosse Fatigue (1994): 3; Face/Off (1997): 4; ...; Shall We Dance? (1996): 3

**Candidate Item:** Pink Floyd - The Wall (1982)

**Similar User Rating History:** L.A. Confidential (1997): 3; Apt Pupil (1998): 4; ...; English Patient, The (1996): 3

**Conventional Model Prediction:** 3.2

**Output:** 3



## • Information from Similar User:

Regarding the interacted items of similar users as collaborative information to enrich the prompt for LLM Instruction-tuning.

## • Knowledge from Conventional Rec Model:

Combining the prediction results from conventional models with the prompt.



# Text-based: LlamaRec

Moreover, for ‘**Q3**: How to aggregate the LLM-based method and conventional methods properly?’, LlamaRec introduce the **Adaptive Aggregation**.

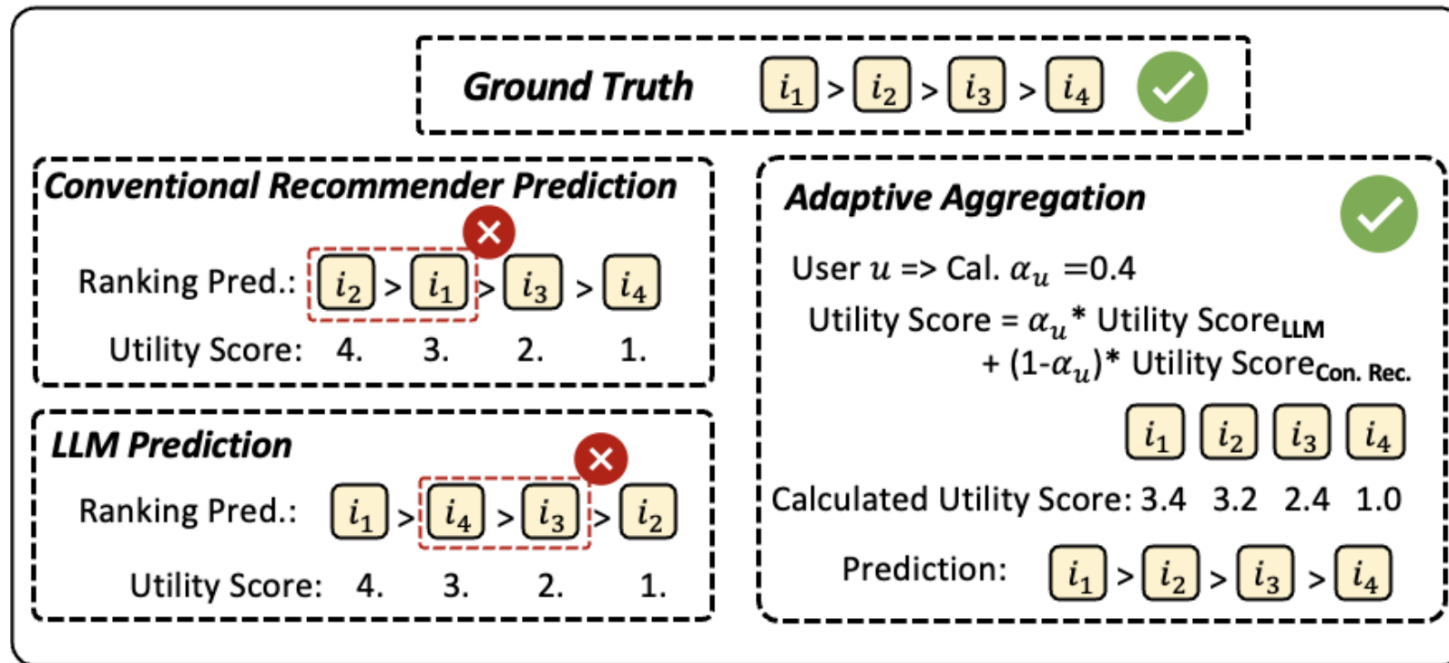
Long-tail coefficient:

$$\ell_u = \log(N(u) + 1)$$

The Utility Score of **Rating Prediction** task can be formulize as:

$$\alpha_u = \max\left(\frac{\ell_{max} - \ell_u}{\ell_{max} - \ell_{min}}, \alpha_2\right) \cdot \alpha_1$$

$$U_u = \alpha_u U_{LLM} + (1 - \alpha_u) U_{Rec}$$



For the **top- $k$  recommendation** task, the LLM is employed to rerank the item list and the **Utility Score** can be calculated as the same.

# Text-based: LlamaRec

For the experimental results, LlamaRec achieves optimal results under **three public datasets** and **three recommendation tasks**.

| Backbone | Method    | ML-100K        |                |                |                | ML-1M          |                |                |                | BookCrossing   |                |                |                |
|----------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|          |           | H@3 ↑          | N@3 ↑          | H@5 ↑          | N@5 ↑          | H@3 ↑          | N@3 ↑          | H@5 ↑          | N@5 ↑          | H@3 ↑          | N@3 ↑          | H@5 ↑          | N@5 ↑          |
| MF       | Base      | 0.0455         | 0.0325         | 0.0690         | 0.0420         | 0.0255         | 0.0187         | 0.0403         | 0.0248         | 0.0294         | 0.0227         | 0.0394         | 0.0269         |
|          | IFT       | 0.0546         | 0.0388         | 0.0790         | 0.0488         | 0.0242         | 0.0175         | 0.0410         | 0.0244         | 0.0247         | 0.0177         | 0.0377         | 0.0230         |
|          | Llama4Rec | <b>0.0645*</b> | <b>0.0474*</b> | <b>0.0919*</b> | <b>0.0588*</b> | <b>0.0281*</b> | <b>0.0203*</b> | <b>0.0433*</b> | <b>0.0265*</b> | <b>0.0365*</b> | <b>0.0284*</b> | <b>0.0462*</b> | <b>0.0324*</b> |
|          | Impro.    | 18.13%         | 22.16%         | 16.33%         | 20.49%         | 10.20%         | 8.56%          | 5.61%          | 6.85%          | 24.15%         | 25.55%         | 17.26%         | 20.45%         |
| LightGCN | Base      | 0.0492         | 0.0343         | 0.0744         | 0.0447         | 0.0283         | 0.0203         | 0.0432         | 0.0264         | 0.0358         | 0.0272         | 0.0480         | 0.0322         |
|          | IFT       | 0.0537         | 0.0381         | 0.0846         | 0.0507         | 0.0268         | 0.0193         | 0.0441         | 0.0263         | 0.0287         | 0.0202         | 0.0448         | 0.0268         |
|          | Llama4Rec | <b>0.0647*</b> | <b>0.0476*</b> | <b>0.0967*</b> | <b>0.0608*</b> | <b>0.0304*</b> | <b>0.0222*</b> | <b>0.0461*</b> | <b>0.0286*</b> | <b>0.0434*</b> | <b>0.0338*</b> | <b>0.057*</b>  | <b>0.0394*</b> |
|          | Impro.    | 20.48%         | 24.93%         | 14.30%         | 19.92%         | 7.42%          | 9.36%          | 4.54%          | 8.33%          | 21.23%         | 24.26%         | 18.75%         | 22.36%         |
| MixGCF   | Base      | 0.0526         | 0.0401         | 0.0757         | 0.0496         | 0.0159         | 0.0115         | 0.0238         | 0.0147         | 0.0426         | 0.0330         | 0.0556         | 0.0384         |
|          | IFT       | 0.0617         | 0.0452         | 0.0906         | 0.0570         | 0.0162         | 0.0114         | <b>0.0259</b>  | 0.0154         | 0.0337         | 0.0243         | 0.0506         | 0.0312         |
|          | Llama4Rec | <b>0.0690*</b> | <b>0.0515*</b> | <b>0.0949*</b> | <b>0.0621*</b> | <b>0.0174*</b> | <b>0.0128*</b> | <b>0.0259</b>  | <b>0.0162*</b> | <b>0.0495*</b> | <b>0.0384*</b> | <b>0.0635*</b> | <b>0.0441*</b> |
|          | Impro.    | 11.83%         | 13.94%         | 4.75%          | 8.95%          | 7.41%          | 11.30%         | 0.00%          | 5.19%          | 16.20%         | 16.36%         | 14.21%         | 14.84%         |
| SGL      | Base      | 0.0505         | 0.0380         | 0.0729         | 0.0472         | 0.0284         | 0.0206         | 0.0434         | 0.0267         | 0.0419         | 0.0319         | 0.0566         | 0.0380         |
|          | IFT       | 0.0520         | 0.0392         | 0.0792         | 0.0503         | 0.0275         | 0.0202         | 0.0438         | 0.0269         | 0.0326         | 0.0237         | 0.0499         | 0.0307         |
|          | Llama4Rec | <b>0.0632*</b> | <b>0.0479*</b> | <b>0.0917*</b> | <b>0.0596*</b> | <b>0.0308*</b> | <b>0.0224*</b> | <b>0.0480*</b> | <b>0.0294*</b> | <b>0.0501*</b> | <b>0.0393*</b> | <b>0.0634*</b> | <b>0.0448*</b> |
|          | Impro.    | 21.54%         | 22.19%         | 15.78%         | 18.49%         | 8.45%          | 8.74%          | 9.59%          | 9.29%          | 19.57%         | 23.20%         | 12.01%         | 17.89%         |

Overall Performance achieved by Direct Recommendation Models



# Text-based: LlamaRec

For the experimental results, LlamaRec achieves optimal results under **three public datasets** and **three recommendation tasks**.

| Backbone | Method    | ML-100K        |                |                |                | ML-1M          |                |               |                | BookCrossing   |                |                |                |
|----------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|----------------|----------------|----------------|----------------|----------------|
|          |           | H@3 ↑          | N@3 ↑          | H@5 ↑          | N@5 ↑          | H@3 ↑          | N@3 ↑          | H@5 ↑         | N@5 ↑          | H@3 ↑          | N@3 ↑          | H@5 ↑          | N@5 ↑          |
| SASRec   | Base      | 0.0187         | 0.0125         | 0.0385         | 0.0205         | 0.0277         | 0.0165         | 0.0502        | 0.0257         | 0.0086         | 0.0049         | 0.0163         | 0.0081         |
|          | IFT       | 0.0204         | 0.0136         | 0.0379         | 0.0207         | 0.0241         | 0.0159         | 0.0473        | 0.0254         | 0.0124         | 0.0086         | 0.0185         | 0.0111         |
|          | Llama4Rec | <b>0.0238*</b> | <b>0.0155*</b> | <b>0.0449*</b> | <b>0.0240*</b> | <b>0.0293*</b> | <b>0.0201*</b> | <b>0.0504</b> | <b>0.0287*</b> | <b>0.0142*</b> | <b>0.0098*</b> | <b>0.0227*</b> | <b>0.0131*</b> |
|          | Impro.    | 16.67%         | 13.97%         | 16.62%         | 15.94%         | 5.78%          | 21.82%         | 0.40%         | 11.67%         | 14.52%         | 13.95%         | 22.70%         | 18.02%         |
| BERT4Rec | Base      | 0.0153         | 0.0104         | 0.0294         | 0.0161         | 0.0107         | 0.0069         | <b>0.0211</b> | 0.0112         | 0.0088         | 0.0058         | 0.0161         | 0.0088         |
|          | IFT       | 0.0174         | 0.0119         | 0.0326         | 0.0100         | 0.0106         | 0.0071         | 0.0188        | 0.0104         | 0.0127         | 0.0092         | 0.0180         | 0.0113         |
|          | Llama4Rec | <b>0.0198*</b> | <b>0.0134*</b> | <b>0.0332</b>  | <b>0.0189*</b> | <b>0.0115*</b> | <b>0.0078*</b> | 0.0206        | <b>0.0115*</b> | <b>0.0154*</b> | <b>0.0108*</b> | <b>0.023*</b>  | <b>0.0139*</b> |
|          | Impro.    | 13.79%         | 12.61%         | 1.84%          | 17.39%         | 7.48%          | 9.86%          | -2.37%        | 2.68%          | 21.26%         | 17.39%         | 27.78%         | 23.01%         |
| CL4SRec  | Base      | 0.0243         | 0.0143         | 0.0436         | 0.0222         | 0.0259         | 0.0153         | <b>0.0492</b> | 0.0248         | 0.0083         | 0.0048         | 0.0165         | 0.0082         |
|          | IFT       | 0.0230         | 0.0149         | 0.0428         | 0.0230         | 0.0234         | 0.0155         | 0.0447        | 0.0241         | 0.0102         | 0.0071         | 0.0177         | 0.0102         |
|          | Llama4Rec | <b>0.0255*</b> | <b>0.0182*</b> | <b>0.0440</b>  | <b>0.0255*</b> | <b>0.0278*</b> | <b>0.0185*</b> | 0.0482        | <b>0.0268*</b> | <b>0.0138*</b> | <b>0.0093*</b> | <b>0.0220*</b> | <b>0.0127*</b> |
|          | Impro.    | 4.94%          | 22.15%         | 0.92%          | 10.87%         | 7.34%          | 19.35%         | -2.03%        | 8.06%          | 35.29%         | 30.99%         | 24.29%         | 24.51%         |

Overall Performance achieved by Sequential Recommendation Models

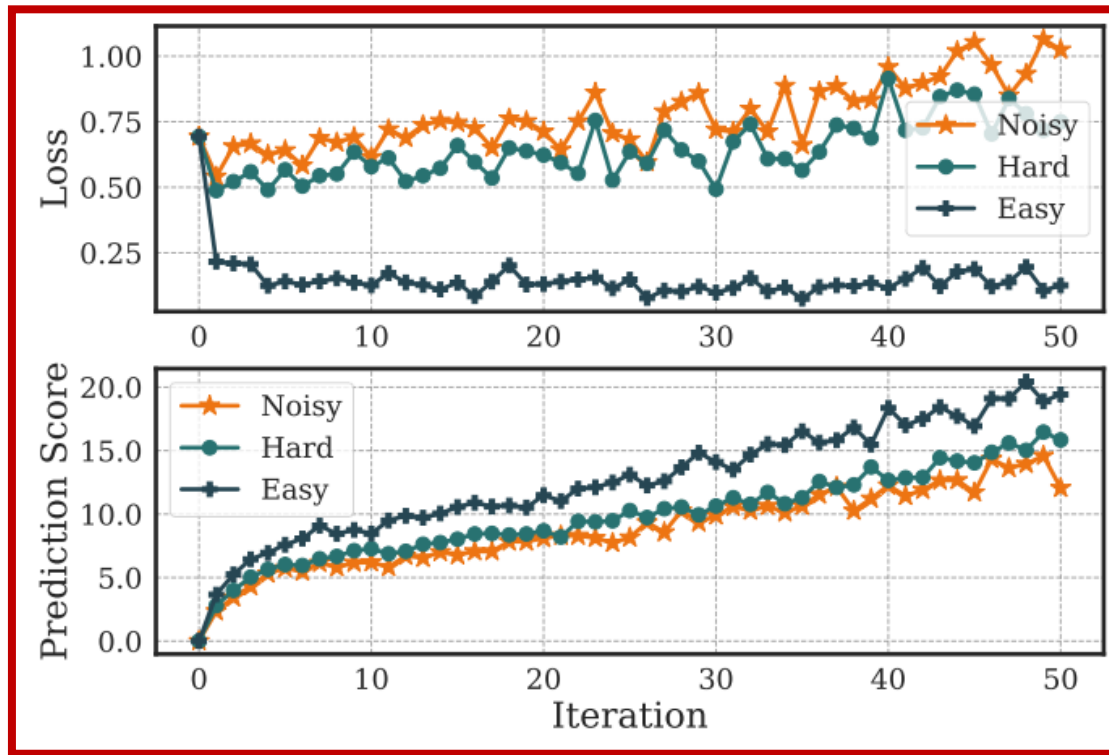
# Text-based: LlamaRec

For the experimental results, LlamaRec achieves optimal results under **three public datasets** and **three recommendation tasks**.

| Backbone | Method    | ML-100K        |                | ML-1M          |                | BookCrossing   |                |
|----------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|
|          |           | RMSE ↓         | MAE ↓          | RMSE ↓         | MAE ↓          | RMSE ↓         | MAE ↓          |
| LLaMA    | IFT       | 1.2792         | 0.8940         | 1.2302         | 0.8770         | 2.0152         | 1.3782         |
| DeepFM   | Base      | 1.0487         | 0.8082         | 0.9455         | 0.7409         | 1.7738         | 1.3554         |
|          | Llama4Rec | <b>1.0306*</b> | <b>0.7987*</b> | <b>0.9360*</b> | <b>0.7321*</b> | <b>1.6958*</b> | <b>1.2843*</b> |
|          | Impro.    | 1.73%          | 1.18%          | 1.00%          | 1.19%          | 4.40%          | 5.25%          |
| NFM      | Base      | 1.0284         | 0.8005         | 0.9438         | 0.7364         | 2.121          | 1.5984         |
|          | Llama4Rec | <b>1.0189*</b> | <b>0.7961</b>  | <b>0.9369*</b> | <b>0.7303*</b> | <b>1.9253*</b> | <b>1.4473*</b> |
|          | Impro.    | 0.92%          | 0.55%          | 0.73%          | 0.83%          | 9.23%          | 9.45%          |
| DCN      | Base      | 1.0478         | 0.8063         | 0.9426         | 0.7342         | 2.0216         | 1.4622         |
|          | Llama4Rec | <b>1.0367*</b> | <b>0.8033</b>  | <b>0.9345*</b> | <b>0.7272*</b> | <b>1.8518*</b> | <b>1.3566*</b> |
|          | Impro.    | 1.06%          | 0.37%          | 0.86%          | 0.95%          | 8.40%          | 7.22%          |
| AFM      | Base      | 1.0471         | 0.8035         | 0.9508         | 0.7464         | 1.6516         | 1.2614         |
|          | Llama4Rec | <b>1.0340*</b> | <b>0.7996</b>  | <b>0.9426*</b> | <b>0.7394*</b> | <b>1.6244*</b> | <b>1.2259*</b> |
|          | Impro.    | 1.25%          | 0.49%          | 0.86%          | 0.94%          | 1.65%          | 2.81%          |
| xDeepFM  | Base      | 1.1472         | 0.8836         | 0.9519         | 0.7428         | 2.1756         | 1.6461         |
|          | Llama4Rec | <b>1.0947*</b> | <b>0.8483*</b> | <b>0.9401*</b> | <b>0.7336*</b> | <b>1.9610*</b> | <b>1.4833*</b> |
|          | Impro.    | 4.58%          | 4.00%          | 1.24%          | 1.24%          | 9.86%          | 9.89%          |
| AutoInt  | Base      | 1.0500         | 0.8120         | 0.9471         | 0.7404         | 1.9148         | 1.4501         |
|          | Llama4Rec | <b>1.0369*</b> | <b>0.8059*</b> | <b>0.9382*</b> | <b>0.7326*</b> | <b>1.7917*</b> | <b>1.3492*</b> |
|          | Impro.    | 1.25%          | 0.75%          | 0.94%          | 1.05%          | 6.43%          | 6.96%          |

Overall Performance achieved by Rating Prediction Models

Although **Large Language Models (LLM)** show potential in denoising recommendation, there are still some challenges that hinders the directly application of LLM:



- **C1: High Cost of LLM Summarization:**

Assessing the preferences of all users across all items is **computationally intensive**.

- **C2: Misidentifying of Hard Sample & Noise:**

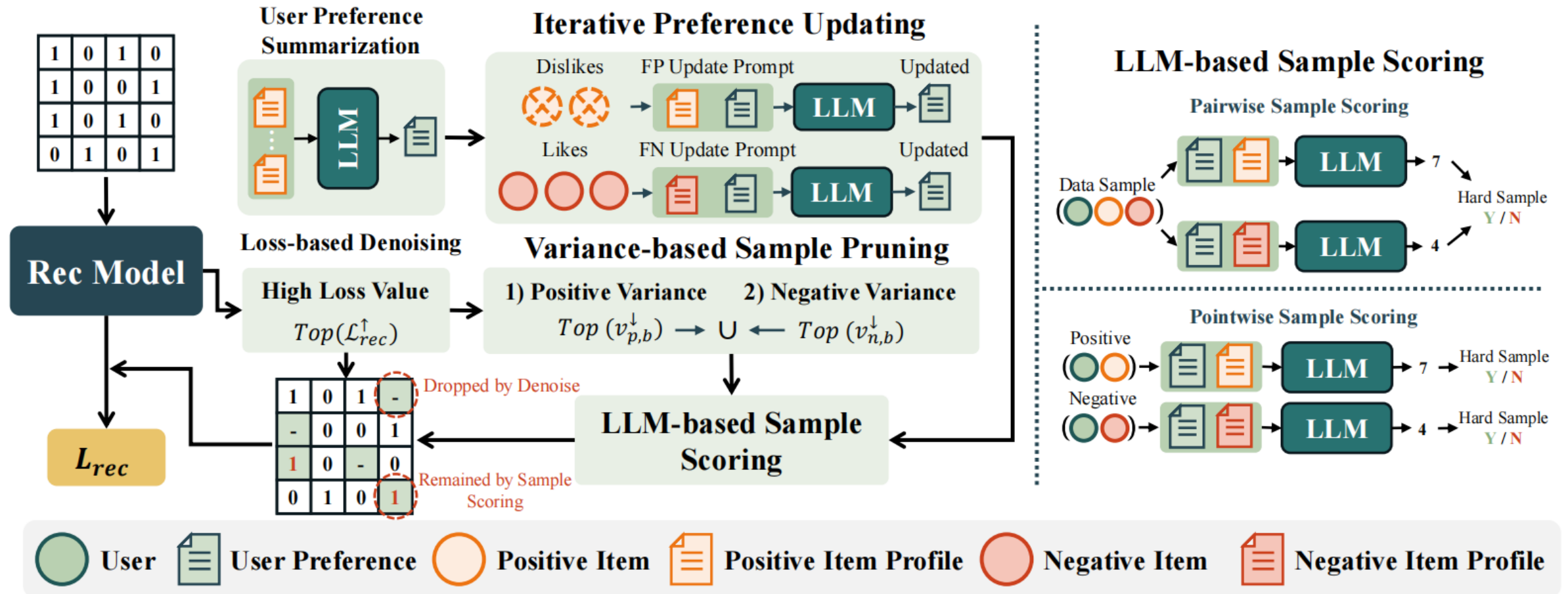
It is challenging to distinguish the difference between the **Noisy** and **Hard Example**.

- **C3: Biased User Preference:**

**False-positive items** can lead to **biased user preference summarization**.

# Text-based: LLMHD

To address the above challenges, the authors proposed LLMHD, featuring with **Variance-based Sample Pruning**, **LLM-based Sample Scoring**, and **Iterative Preference Updating** to fully explore the capabilities of LLM in denoising recommendation.



To address **C1: High Cost of LLM Summarization**, LLMHD introduce the **Variance-based Sample Pruning module**.

1) Positive Variance      2) Negative Variance

$$Top(v_{p,b}^\downarrow) \rightarrow \cup \leftarrow Top(v_{n,b}^\downarrow)$$

Following the observation of previous work: hard samples exhibit relatively higher prediction score variance compared to noisy samples, LLMHD calculate and sort the prediction score of positive/negative items as follows:

$$v_{p,1}^\downarrow > v_{p,2}^\downarrow > \dots > v_{p,b}^\downarrow > \dots > v_{p,|\mathcal{B}_N^p|}^\downarrow, b \in \mathcal{B}_N,$$

$$v_{n,1}^\downarrow > v_{n,2}^\downarrow > \dots > v_{n,b}^\downarrow > \dots > v_{n,|\mathcal{B}_N^n|}^\downarrow, b \in \mathcal{B}_N,$$

So the hard samples can then be defined as:

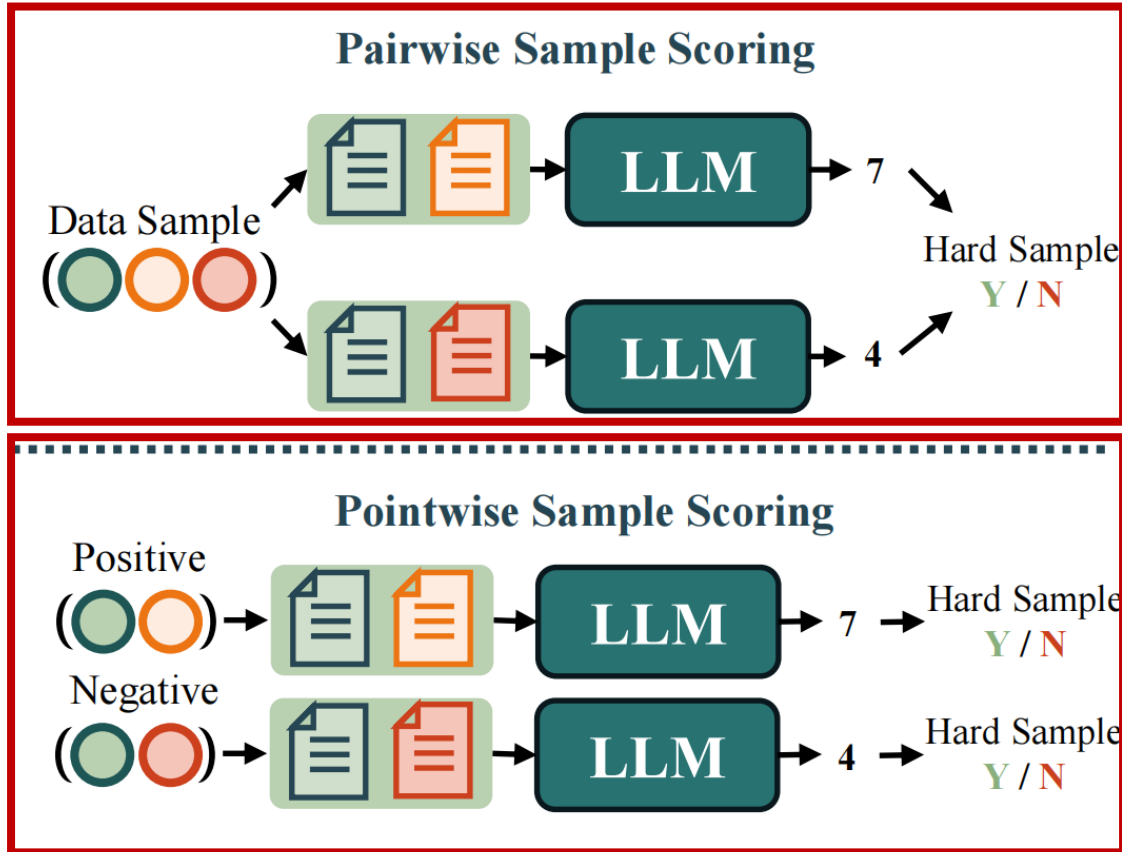
$$\mathcal{B}_{HC} = \{b_j | v_{p,j}^\downarrow \geq v_{p,\varepsilon_v|\mathcal{B}_N^p|}^\downarrow\} \cup \{b_j | v_{n,b_j}^\downarrow \geq v_{n,\varepsilon_v|\mathcal{B}_N^n|}^\downarrow\}$$

Preventing to feed all identified noisy samples to the LLMs for scoring, significantly reduce the computation cost of LLM.



# Text-based: LLMHD

For **C2: Misidentifying of Hard Sample & Noise**, LLMHD introduce a novel **Sample Scoring module** that leverage LLM to **provide auxiliary information** for evaluating the sample hardness.



## Pointwise Sample Scoring:

Considering positive pairs and negative pairs, the positive difference threshold significantly surpasses the negative, hard samples are identified through the indicator function:

$$\mathbb{I}_{\text{point}}(u, i) = \begin{cases} 1, & \text{if } s_{u,i} < \varepsilon_{\text{pos}} \text{ and } y_{u,i} = 1 \\ 1, & \text{if } s_{u,i} > \varepsilon_{\text{neg}} \text{ and } y_{u,i} = 0 \end{cases}$$

$$\mathbb{I}_{\text{pair}}(s_{u,i_p} - s_{u,i_n} > \varepsilon_{\text{pair}})^{\text{wise}},$$

Also for achieving the generalization ability, And smoothly changing each threshold during each training iteration  $T$  to achieve the hardness by the number of iteration  $T$

$$\varepsilon_{\text{pair}} = \max(\varepsilon_{\text{pair}}^{\text{max}} - \frac{1}{\alpha} T, \varepsilon_{\text{pair}}^{\text{min}})$$

$$\varepsilon_{\text{neg}} = \min(\varepsilon_{\text{neg}} + \frac{1}{\alpha} T, \varepsilon_{\text{neg}})$$



# Text-based: LLMHD

To address **C3: Biased User Preference Summarization**, LLMHD introduce the **Iterative Preference Updating module**.

Specifically, the authors refine user preferences iteratively by excluding dislikes (**false-positive**) and incorporating likes (**false-negative**).

The confident items' selection can then be formularize as:

$$v_d = \frac{1}{m} \sum_{j=t-m+1}^t \left( \hat{y}_d^j - \frac{\sum_{j=t-m+1}^t \hat{y}_d^j}{m} \right)^2 \begin{cases} \text{Positive} & \rightarrow v_{d_1^p}^{\uparrow} < v_{d_2^p}^{\uparrow} < \dots < v_{d_{|\mathcal{D}_{pos}|}^p}^{\uparrow}, d_k^p \in \mathcal{D}_{pos} \rightarrow \mathbb{I}_{\text{FP}}\left(\sum_{j=0}^t \mathbb{I}(d_k^p) \geq \varepsilon_{\gamma}\right) \\ \text{Negative} & \rightarrow v_{d_1^n}^{\uparrow} < v_{d_2^n}^{\uparrow} < \dots < v_{d_{|\mathcal{D}_{neg}|}^n}^{\uparrow}, d_k^n \in \mathcal{D}_{neg} \rightarrow \mathbb{I}_{\text{FN}}\left(\sum_{j=0}^t \mathbb{I}(d_k^n) \geq \varepsilon_{\gamma}\right) \end{cases}$$

At last, LLMHD can refine the original preference based on **identified false-positives and false-negatives** to construct the unbiased user profiles.

$$\mathcal{P}_u^* = LLM(T_{\text{FP}}(\mathcal{P}_u, \mathcal{P}_{i_{\text{fp}}}))$$

$$\mathcal{P}_u^* = LLM(T_{\text{FN}}(\mathcal{P}_u, \mathcal{P}_{i_{\text{fn}}}))$$

# Text-based: LLMHD



August 3-7, 2025



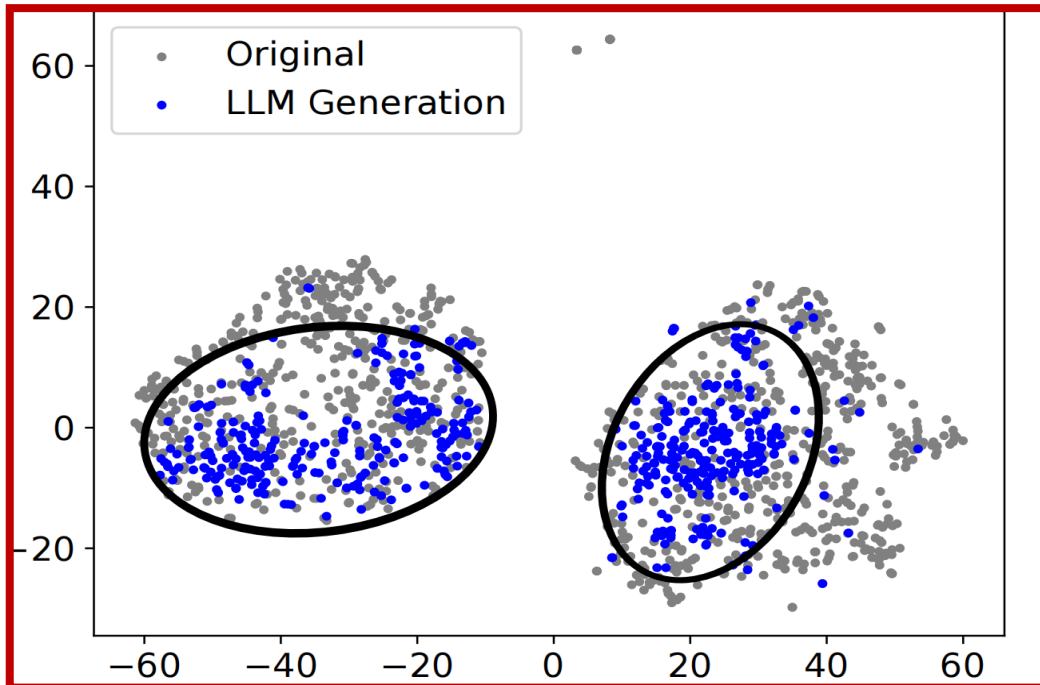
| Dataset  |                      | Amazon-book   |               |               |               | Yelp          |               |               |               | Steam         |               |               |               |
|----------|----------------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| Backbone | Method               | R@5           | R@10          | N@5           | N@10          | R@5           | R@10          | N@5           | N@10          | R@5           | R@10          | N@5           | N@10          |
| NGCF     | BCE                  | 0.0353        | 0.0570        | 0.0365        | 0.0438        | 0.0236        | 0.0431        | 0.0283        | 0.0350        | 0.0223        | 0.0405        | 0.0236        | 0.0305        |
|          | BPR                  | 0.0389        | 0.0651        | 0.0406        | 0.0494        | 0.0280        | 0.0495        | 0.0338        | 0.0405        | 0.0381        | 0.0629        | 0.0453        | 0.0525        |
|          | T-CE                 | 0.0393        | 0.0650        | 0.0402        | 0.0489        | 0.0259        | 0.0450        | 0.0313        | 0.0373        | 0.0257        | 0.0448        | 0.0288        | 0.0354        |
|          | R-CE                 | 0.0366        | 0.0587        | 0.0369        | 0.0444        | 0.0254        | 0.0438        | 0.0302        | 0.0360        | 0.0236        | 0.0435        | 0.0254        | 0.0328        |
|          | RGCF                 | <u>0.0415</u> | <u>0.0658</u> | <u>0.0422</u> | <u>0.0502</u> | <u>0.0287</u> | <u>0.0485</u> | <u>0.0344</u> | <u>0.0406</u> | <u>0.0401</u> | <u>0.0644</u> | <u>0.0472</u> | <u>0.0543</u> |
|          | DCF                  | 0.0398        | 0.0617        | 0.0399        | 0.0472        | 0.0281        | 0.0488        | 0.0353        | 0.0414        | 0.0264        | 0.0446        | 0.0308        | 0.0365        |
|          | LLMHD <sub>BCE</sub> | 0.0406        | 0.0668        | 0.0413        | 0.0503        | 0.0276        | 0.0477        | 0.0329        | 0.0386        | 0.0267        | 0.0459        | 0.0297        | 0.0364        |
|          | LLMHD <sub>BPR</sub> | <b>0.0455</b> | <b>0.0743</b> | <b>0.0455</b> | <b>0.0552</b> | <b>0.0338</b> | <b>0.0579</b> | <b>0.0398</b> | <b>0.0474</b> | <b>0.0418</b> | <b>0.0696</b> | <b>0.0496</b> | <b>0.0579</b> |
| LightGCN | BCE                  | 0.0558        | 0.0849        | 0.0565        | 0.0665        | 0.0390        | 0.0660        | 0.0481        | 0.0557        | 0.0448        | 0.0732        | 0.0529        | 0.0612        |
|          | BPR                  | 0.0587        | 0.0904        | 0.0598        | 0.0704        | 0.0359        | 0.0609        | 0.0446        | 0.0516        | 0.0510        | 0.0828        | 0.0597        | 0.0693        |
|          | T-CE                 | 0.0590        | 0.0895        | 0.0592        | 0.0697        | 0.0401        | 0.0677        | 0.0504        | 0.0580        | 0.0463        | 0.0758        | 0.0555        | 0.0640        |
|          | R-CE                 | 0.0557        | 0.0834        | 0.0566        | 0.0658        | 0.0389        | 0.0650        | 0.0474        | 0.0550        | 0.0461        | 0.0757        | 0.0543        | 0.0630        |
|          | RGCF                 | <u>0.0619</u> | <u>0.0956</u> | <u>0.0644</u> | <u>0.0753</u> | <u>0.0420</u> | <u>0.0693</u> | <u>0.0501</u> | <u>0.0579</u> | <u>0.0519</u> | <u>0.0849</u> | <u>0.0599</u> | <u>0.0702</u> |
|          | DCF                  | 0.0590        | 0.0898        | 0.0596        | 0.0701        | 0.0403        | 0.0680        | 0.0503        | 0.0579        | 0.0477        | 0.0778        | 0.0562        | 0.0650        |
|          | LLMHD <sub>BCE</sub> | 0.0607        | 0.0921        | 0.0607        | 0.0711        | 0.0408        | 0.0689        | 0.0514        | 0.0589        | 0.0469        | 0.0767        | 0.0563        | 0.0647        |
|          | LLMHD <sub>BPR</sub> | <b>0.0652</b> | <b>0.0999</b> | <b>0.0655</b> | <b>0.0767</b> | <b>0.0427</b> | <b>0.0731</b> | <b>0.0518</b> | <b>0.0611</b> | <b>0.0536</b> | <b>0.0867</b> | <b>0.0624</b> | <b>0.0722</b> |
| SGL      | BCE                  | 0.0589        | 0.0902        | 0.0604        | 0.0707        | 0.0377        | 0.0655        | 0.0470        | 0.0548        | 0.0433        | 0.0682        | 0.0505        | 0.0676        |
|          | BPR                  | 0.0608        | 0.0956        | 0.0621        | 0.0736        | 0.0373        | 0.0629        | 0.0465        | 0.0538        | 0.0529        | 0.0838        | 0.0613        | 0.0704        |
|          | T-CE                 | 0.0602        | 0.0909        | 0.0622        | 0.0720        | 0.0408        | 0.0697        | 0.0502        | 0.0587        | 0.0449        | 0.0720        | 0.0532        | 0.0609        |
|          | R-CE                 | 0.0591        | 0.0901        | 0.0601        | 0.0702        | 0.0386        | 0.0645        | 0.0476        | 0.0550        | 0.0456        | 0.0732        | 0.0538        | 0.0618        |
|          | RGCF                 | <u>0.0675</u> | <u>0.1049</u> | <u>0.0681</u> | <u>0.0808</u> | <u>0.0416</u> | <u>0.0715</u> | <u>0.0512</u> | <u>0.0606</u> | <b>0.0552</b> | 0.0881        | 0.0639        | 0.0736        |
|          | DCF                  | 0.0626        | 0.0933        | 0.0641        | 0.0740        | 0.0413        | 0.0683        | 0.0506        | 0.0583        | 0.0455        | 0.0727        | 0.0536        | 0.0615        |
|          | LLMHD <sub>BCE</sub> | 0.0615        | 0.0931        | 0.0640        | 0.0739        | 0.0414        | 0.0708        | 0.0509        | 0.0596        | 0.0462        | 0.0742        | 0.0543        | 0.0619        |
|          | LLMHD <sub>BPR</sub> | <b>0.0693</b> | <b>0.1051</b> | <b>0.0717</b> | <b>0.0837</b> | <b>0.0426</b> | <b>0.0718</b> | <b>0.0523</b> | <b>0.0619</b> | <u>0.0546</u> | <b>0.0887</b> | <b>0.0641</b> | <b>0.0739</b> |
| NCL      | BCE                  | 0.0574        | 0.0871        | 0.0598        | 0.0694        | 0.0391        | 0.0647        | 0.0477        | 0.0548        | 0.0450        | 0.0731        | 0.0529        | 0.0612        |
|          | BPR                  | 0.0605        | 0.0942        | 0.0628        | 0.0740        | 0.0369        | 0.0609        | 0.0451        | 0.0515        | 0.0511        | 0.0835        | 0.0602        | 0.0698        |
|          | T-CE                 | 0.0599        | 0.0898        | 0.0619        | 0.0719        | 0.0411        | 0.0679        | 0.0507        | 0.0582        | 0.0461        | 0.0751        | 0.0543        | 0.0627        |
|          | R-CE                 | 0.0585        | 0.0874        | 0.0604        | 0.0696        | 0.0399        | 0.0655        | 0.0487        | 0.0558        | 0.0459        | 0.0750        | 0.0540        | 0.0625        |
|          | RGCF                 | <u>0.0694</u> | <u>0.1045</u> | <u>0.0706</u> | <u>0.0819</u> | 0.0396        | 0.0660        | 0.0480        | 0.0560        | <u>0.0534</u> | <b>0.0863</b> | 0.0621        | <b>0.0718</b> |
|          | DCF                  | 0.0619        | 0.0929        | 0.0624        | 0.0727        | 0.0424        | 0.0696        | 0.0513        | 0.0589        | 0.0465        | 0.0759        | 0.0550        | 0.0635        |
|          | LLMHD <sub>BCE</sub> | 0.0609        | 0.0915        | 0.0629        | 0.0730        | 0.0417        | 0.0690        | 0.0517        | 0.0591        | 0.0468        | 0.0762        | 0.0551        | 0.0633        |
|          | LLMHD <sub>BPR</sub> | <b>0.0719</b> | <b>0.1053</b> | <b>0.0741</b> | <b>0.0846</b> | <b>0.0432</b> | <b>0.0716</b> | <b>0.0542</b> | <b>0.0620</b> | <b>0.0540</b> | <u>0.0861</u> | <b>0.0624</b> | 0.0717        |

For the experimental results, LLMHD achieves optimal results on three public datasets, validating the effectiveness of proposed methods.

# Text-based: SampleLLM

Existing methods in **Tabular data synthesis** often fall short in RS because of the difficulty in understanding the complicated semantic feature relations.

**Large Language Models (LLMs)** have shown potential in generating synthetic data. However, directly applying LLM to data synthesis still face severe problems:



- **P1: Inconsistent Distribution & Diversity:**

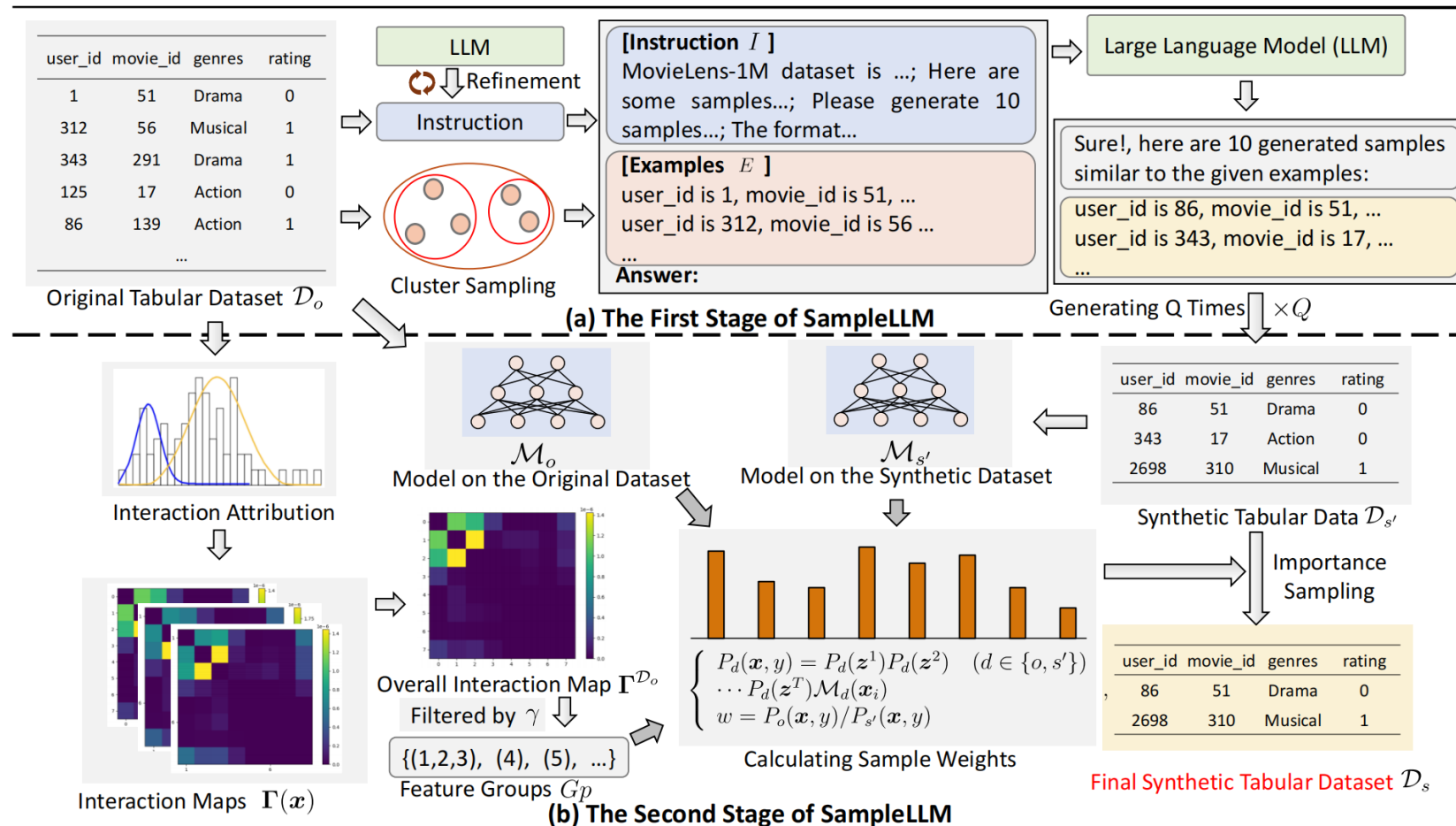
The **distribution mismatch** between LLM's inherent knowledge and the target datasets can lead to reduced output diversity.

- **P2: Distribution Difference Caused by LLM:**

Inherent difference caused by **LLM's input-output processing** still result in a **distribution gap**.

# Text-based: SampleLLM

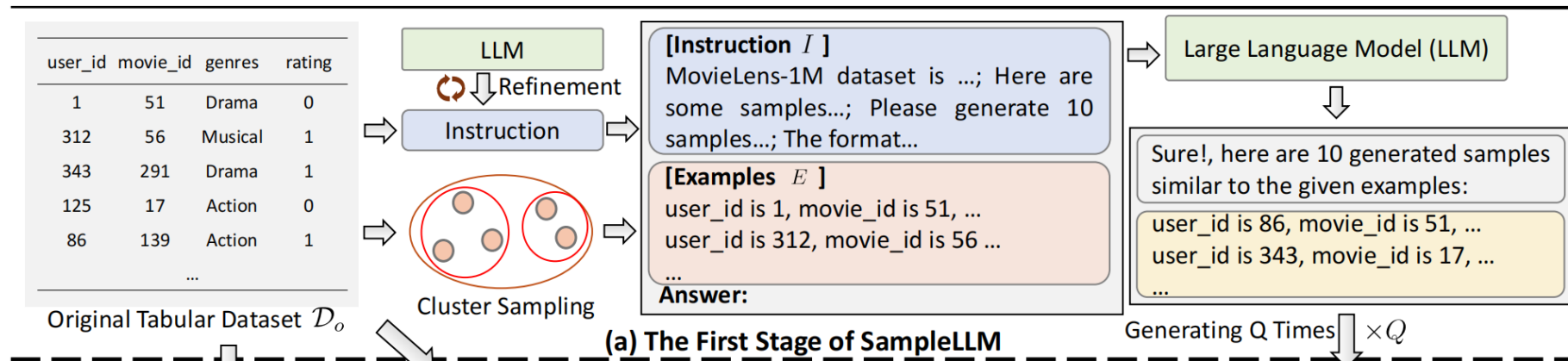
To address the above problems, the authors propose SampleLLM, a **two-stage framework** designed to enhance the quality of LLM-based tabular data synthesis in recommendation tasks.





# Text-based: SampleLLM

To address **P1: Inconsistent Distribution & Diversity**, in the First Stage, the designed instruction and sampled exemplars from the original dataset are selected to serve as the input for the LLM, producing the initial synthetic tabular data.



- **Example Selection**

Leveraging **K-Means** method, SampleLLM can ensures that each group of exemplars **contains samples from diverse regions of the original dataset's distribution** by only using **clusters** for LLM's few-shot learning.

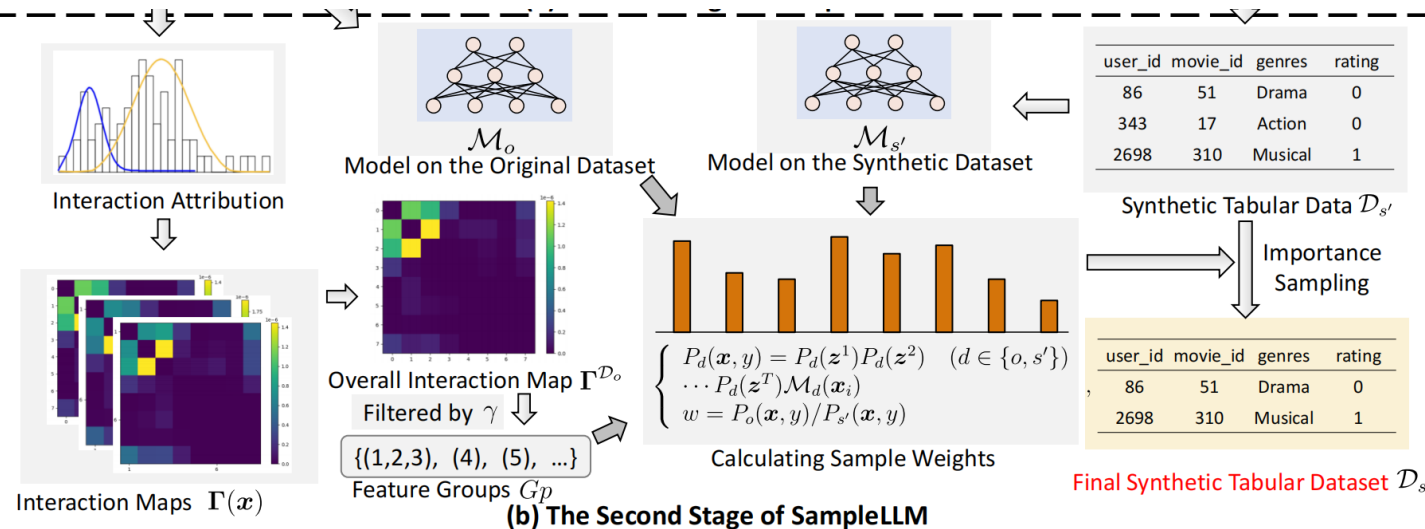
The whole approach can be formulized as follows:

$$\begin{cases} Clu = h(\mathcal{D}_o, a) = \{clu_1, clu_2, \dots, clu_a\} \\ E = \{\mathbf{x}_i | \mathbf{x}_i = g(clu_i); i \in \{1, 2, \dots, a\}\} \end{cases}$$



# Text-based: SampleLLM

In the second stage, a **feature attribution-based importance sampling operation** is performed to address '**P2: Distribution Difference Caused by LLM**'.



## Feature Attribution-based Importance Sampling:

- Current Assumption:

$$P(x_i) = P(x_i^1)P(x_i^2) \dots P(x_i^K)P(y_i | x_i^1, x_i^2, \dots, x_i^K)$$

- Semi-independence assumption:

$$P(x_i) = P(x_i^1, x_i^8)P(x_i^2) \dots P(x_i^K)P(y_i | x_i^1, x_i^2, \dots, x_i^K)$$

Still challenging to identify the significance. -> **Feature Interaction Extraction**

Approximating performance change after setting an informative feature to the non-informative one:

$$EG_i(x) := \mathbb{E}_{x' \sim \mathcal{D}^b} \left[ (x_i - x_i^b) \frac{\delta f(x)}{\delta x_i} \right] \longrightarrow \Gamma_{i,j}(x) = \mathbb{E}_{x \sim \mathcal{D}^b} \left[ (x_i - x_i^b) (x_j - x_j^b) \frac{\partial^2 f(x^b)}{\partial x_i \partial x_j} \right]$$

# Text-based: SampleLLM

In the second stage, a **feature attribution-based importance sampling operation** is performed to address '**P2: Distribution Difference Caused by LLM**'.

## • Calculating Sample Weight

The import  $EG_i(\mathbf{x}) := \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}^b} \left[ \left( x_i - x_i^b \right) \frac{\delta f(\mathbf{x})}{\delta x_i} \right]$  **the derivative**  $\Gamma_{i,j}(\mathbf{x}) = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}^b} \left[ \left( x_i - x_i^b \right) \left( x_j - x_j^b \right) \frac{\partial^2 f(\mathbf{x}^b)}{\partial x_i \partial x_j} \right]$  a:

The sum  $\left\{ \begin{array}{l} F_Y = \gamma \cdot \max(\Gamma^{\mathcal{D}_o}) \\ Pairs = \text{argwhere}_{\Gamma^{\mathcal{D}_o}} (\Gamma^{\mathcal{D}_o} > F_Y) \\ Gp = \text{Merge}(Pairs) \end{array} \right.$  **tion can highlight the significance in the whole dataset**  
 and ca **ment refine**  $\left\{ \begin{array}{l} F_Y = \left\{ \begin{array}{l} P_d(\mathbf{x}, y) = P_d(z^1)P_d(z^2) \quad (d \in \{o, s'\}) \\ \dots P_d(z^T) \mathcal{M}_d(\mathbf{x}_i) \end{array} \right. \\ Pairs \left\{ \begin{array}{l} w = P_o(\mathbf{x}, y) / P_{s'}(\mathbf{x}, y) \end{array} \right. \\ Gp = \text{Merge}(Pairs) \end{array} \right.$

## • Obtaining Discriminant Probability

Then, the label of each tabular sample can be obtained by training a task-specific model:

$$\mathcal{M}(\mathbf{x}_i) = P(y_i | x_i^1, x_i^2, \dots, x_i^K)$$

# Text-based: SampleLLM

The experimental results have shown that SampleLLM achieves remarkable results on **five public datasets**, validating the effectiveness of proposed methods.

They also validate the utility of **synthesizing the real data with synthetic data** for training of different methods after injecting synthetic data into the original data (**Augmentation Utility**).

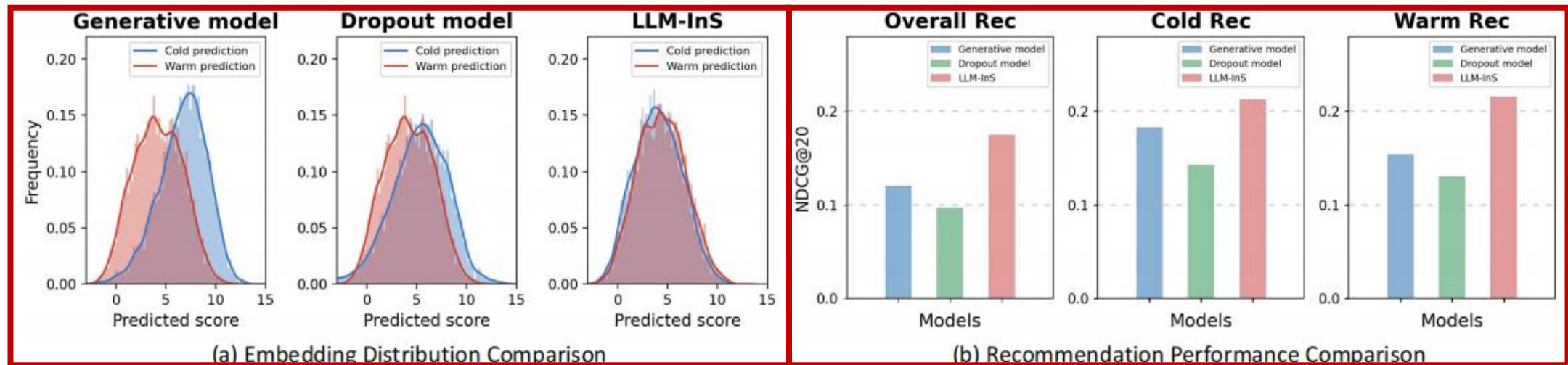
| Approach      | ML-1M          |                | Amazon         |                | Douban         |                | HELOC          |                | CoverType      |               |                |
|---------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|---------------|----------------|
|               | AUC ↑          | Logloss ↓      | AUC ↑          | Logloss ↓      | AUC ↑          | Logloss ↓      | AUC ↑          | Logloss ↓      | Precision ↑    | Recall ↑      | F1 ↑           |
| Original      | 0.8173         | 0.5163         | 0.7075         | 0.4602         | 0.8016         | 0.5158         | 0.7638         | 0.6445         | 0.7395         | 0.7339        | 0.7250         |
| PATE-GAN      | 0.8154         | <u>0.5153</u>  | 0.7022         | 0.4620         | 0.8010         | 0.5162         | 0.7648         | 0.6461         | 0.7388         | 0.7350        | 0.7258         |
| ADS-GAN       | 0.8147         | 0.5157         | 0.7026         | 0.4619         | 0.8016         | 0.5151         | 0.7672         | 0.6451         | 0.7414         | 0.7413        | 0.7312         |
| CTGAN         | 0.8148         | 0.5156         | 0.7027         | 0.4631         | 0.8016         | 0.5150         | 0.7677         | 0.6419         | 0.7417         | 0.7366        | 0.7295         |
| TVAE          | 0.8143         | 0.5162         | 0.7031         | 0.4616         | 0.8019         | 0.5146         | 0.7701         | 0.6455         | 0.7422         | 0.7378        | 0.7305         |
| TabDDPM       | 0.8141         | 0.5159         | 0.7036         | 0.4619         | 0.8020         | 0.5150         | 0.7704         | 0.6424         | 0.7418         | 0.7388        | 0.7299         |
| GReaT         | 0.8153         | 0.5198         | 0.7040         | 0.4616         | 0.8021         | 0.5147         | 0.7703         | 0.6444         | 0.7423         | <b>0.7440</b> | <u>0.7317</u>  |
| REaLTabFormer | 0.8156         | 0.5182         | 0.7041         | 0.4611         | 0.8022         | 0.5145         | 0.7707         | 0.6417         | 0.7428         | 0.7351        | 0.7264         |
| SampleLLM     | <b>0.8180*</b> | <b>0.5140*</b> | <b>0.7082*</b> | <b>0.4601*</b> | <b>0.8027*</b> | <b>0.5139*</b> | <b>0.7732*</b> | <b>0.6403*</b> | <b>0.7445*</b> | <u>0.7437</u> | <b>0.7364*</b> |

Augmentation Utility Perfor (10% synthetic data)  
Perfor (10% synthetic data)

# Score-based: LLM-Ins

**Cold-Start Items** are common in practice and severely affect the performance.

Current Methods adopt **mapping function** to generate fake embedding from cold-start items' content feature, but still face some challenges:

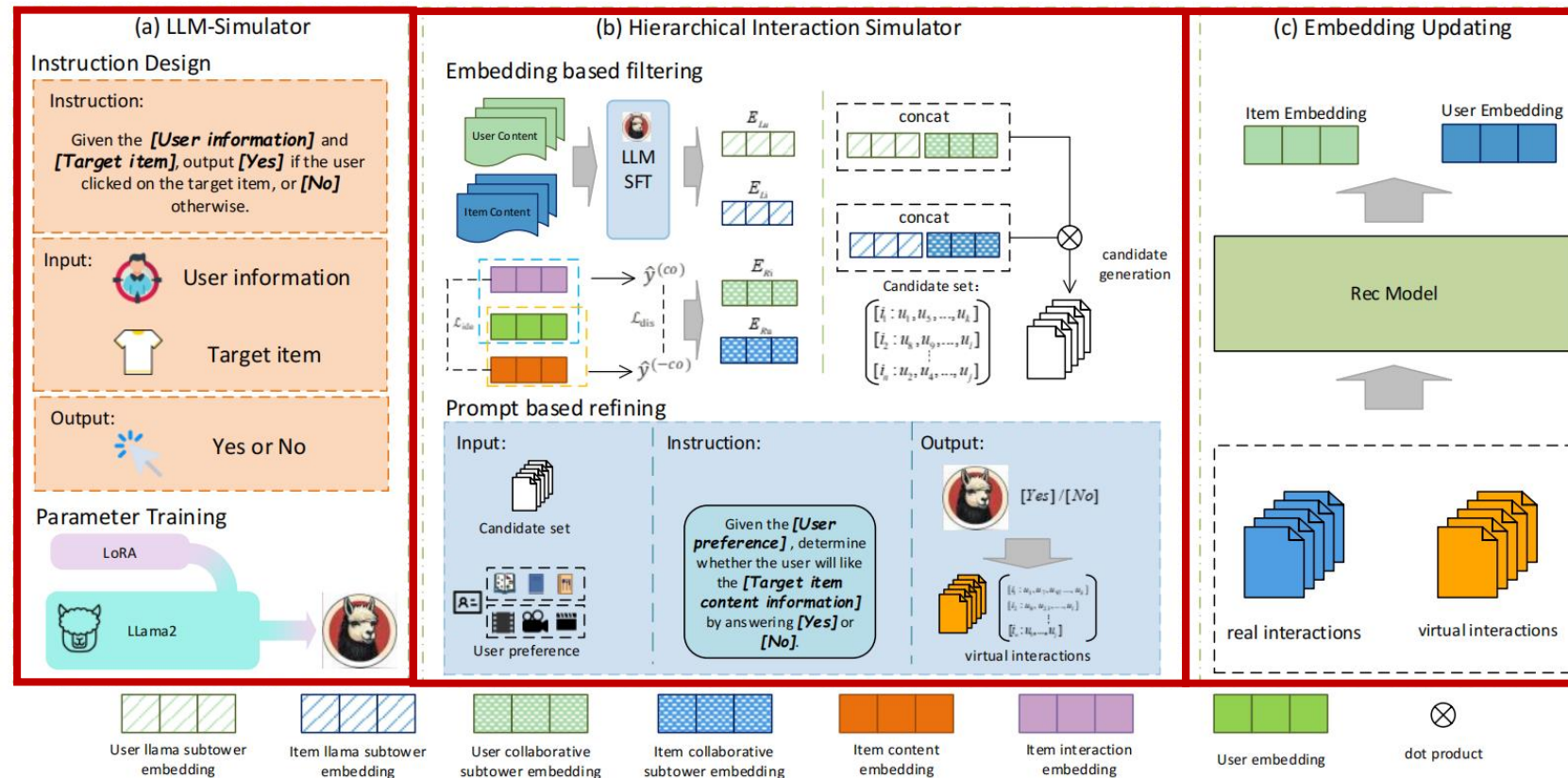


- **C1: Distribution Gap.** The generated embeddings are created from **content features**, while the behavioral embeddings are trained from **sequences**.
- **C2: LLM's Inadaptability.** Applying LLM to interaction generation need **further alignment & unify**.



# Score-based: LLM-Ins

To address the challenges, the authors propose LLM-Ins to model interactions based on content for each cold item (**C1**) and transform them from cold to warm items (**C2**).



## (a) LLM-Simulator:

Generating possible user set for cold items.

## (b) Interaction Simulator:

Mimicing user interactions for each cold item and transform them to warm.

## (c) Embedding Updating :

Adaptively combining the cold items and warm items.

We'll then detail the procedure of **Hierarchical Interaction Simulator** (main procedure).

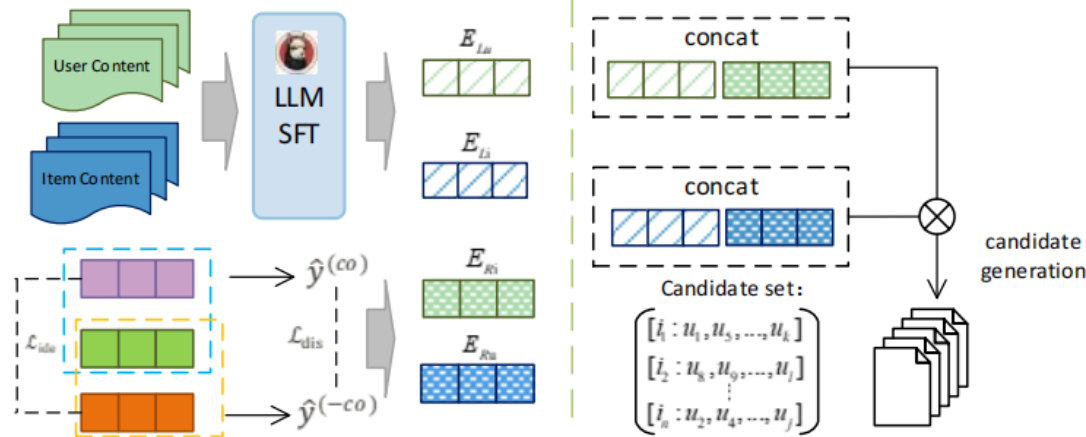


# Score-based: LLM-Ins

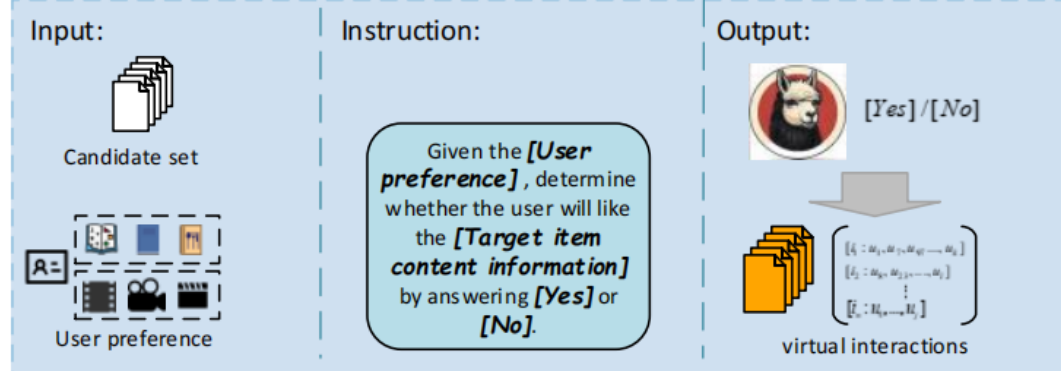
We'll detail the procedure of **Hierarchical Interaction Simulator** (main procedure).

(b) Hierarchical Interaction Simulator

## Embedding based filtering



## Prompt based refining



## Embedding based filtering :

Ensuring Collaborative Subordinate selection aligns closely with real world scenarios.  
Main loss (BPR): Narrowing the gap between (two sides semantic & collaborative)

Auxiliary Subloss (dis): Collaborative/non-collaborative side:  
Featuring with a **user tower** and a **item tower**,

$$\mathcal{L}_{dis} = \frac{1}{|\mathcal{B}|} \sum_{(u,i,j) \in \mathcal{B}} \left( \left| \hat{y}_{ui}^{(co)} - \hat{y}_{ui}^{(-co)} \right|^2 + \left| \hat{y}_{uj}^{(co)} - \hat{y}_{uj}^{(-co)} \right|^2 \right),$$

Auxillary Loss (ide): Embedding distance side.

$$\mathcal{L}_{bce} = - \sum_{i \in \mathcal{B}_I} u_{i..} \log(\hat{u}_{i..}) + (1 - u_{i..}) \log(1 - \hat{u}_{i..})$$

$$\mathcal{L}_{ide} = - \sum_{i \in \mathcal{B}_I} (\bar{d}_i^{(co)} \ln \bar{d}_i^{(-co)} + (1 - \bar{d}_i^{(co)}) \ln(1 - \bar{d}_i^{(-co)})),$$

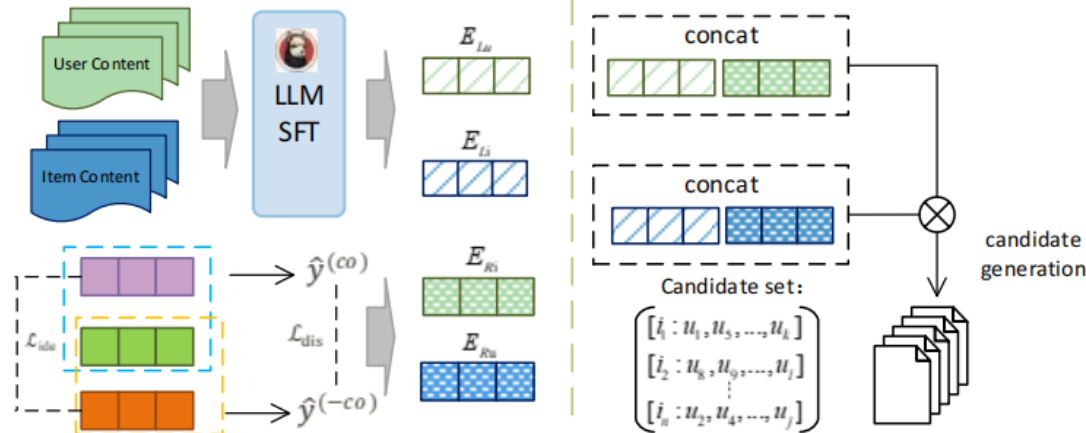
$$\bar{d}_i^{(co)} = \sigma(\mathbf{e}_i^\top \cdot (\mathbf{e}_i - \frac{1}{|\mathcal{B}_I|} \sum_{j \in \mathcal{B}_I} \mathbf{e}_j)),$$

# Score-based: LLM-Ins

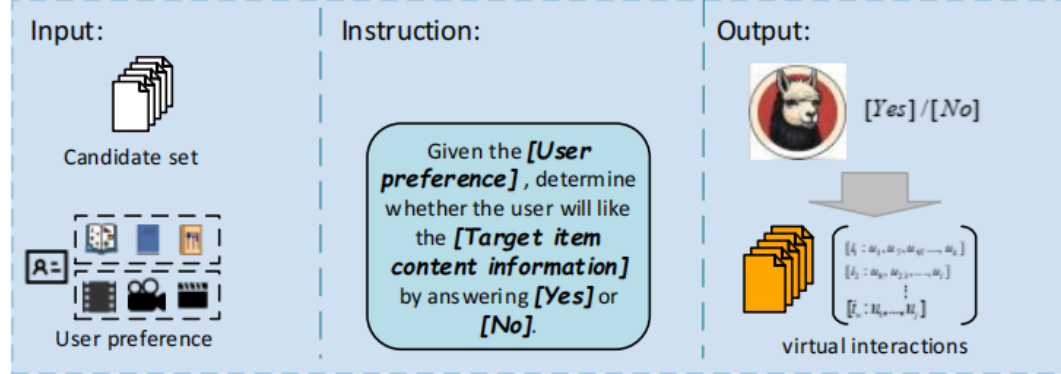
We'll detail the procedure of **Hierarchical Interaction Simulator** (main procedure).

(b) Hierarchical Interaction Simulator

Embedding based filtering



Prompt based refining



## □ Prompt based refining :

The authors concat these two embeddings from **Llama Tower** and **Collaborative Tower**:

More  $E_{Lti} = E_{Li} \parallel E_{Ci}$ ,  $E_{LTu} = E_{Lu} \parallel E_{Cu}$ . LLM to further refine the generated interactions to filter the low-quality interactions:

$$prompt = \mathcal{G}_{u,i}(\mathcal{H}_u, I_t | I_t \in \mathcal{I}_c),$$

$$C_f = L_t(prompt | A_{ns} = yes),$$

# Score-based: LLM-Ins

For the experimental results, LLM-Ins achieves optimal results under **three backbone** MF, NGCF, LightGCN and presents remarkable performance in comparison with **other LLM-based Recommendation model**:

| Method         | Overall Recommendation |               |               |               | Cold Recommendation |               |               |               | Warm Recommendation |               |               |               |
|----------------|------------------------|---------------|---------------|---------------|---------------------|---------------|---------------|---------------|---------------------|---------------|---------------|---------------|
|                | CiteULike              |               | MovieLens     |               | CiteULike           |               | MovieLens     |               | CiteULike           |               | MovieLens     |               |
|                | Recall                 | NDCG          | Recall        | NDCG          | Recall              | NDCG          | Recall        | NDCG          | Recall              | NDCG          | Recall        | NDCG          |
| Backbone       | 0.0812                 | 0.0622        | 0.1418        | <u>0.2330</u> | 0.0041              | 0.0019        | 0.0177        | 0.0190        | 0.2528              | 0.1541        | <u>0.3130</u> | <u>0.3008</u> |
| DropoutNet     | 0.0883                 | 0.0639        | 0.1165        | 0.1978        | 0.2309              | 0.1312        | 0.0340        | 0.0373        | 0.1175              | 0.0692        | 0.2560        | 0.2518        |
| MTPR           | 0.1001                 | 0.0753        | 0.1011        | 0.1551        | 0.2585              | 0.1454        | 0.0779        | 0.0802        | 0.1753              | 0.0697        | 0.2247        | 0.2009        |
| CLCRec         | 0.1293                 | 0.0965        | 0.1253        | 0.2037        | 0.2435              | 0.1425        | 0.0677        | 0.0816        | 0.2149              | 0.1302        | 0.2764        | 0.2612        |
| DeepMusic      | 0.0985                 | 0.0745        | 0.1418        | <u>0.2330</u> | 0.2239              | 0.1259        | 0.0635        | 0.0719        | 0.2528              | 0.1541        | <u>0.3130</u> | <u>0.3008</u> |
| MetaEmb        | 0.0924                 | 0.0714        | 0.1418        | <u>0.2330</u> | 0.2252              | 0.1295        | 0.0248        | 0.0244        | 0.2528              | 0.1541        | <u>0.3130</u> | <u>0.3008</u> |
| GPatch         | 0.1609                 | 0.1197        | 0.1293        | 0.2106        | 0.2606              | 0.1532        | 0.0771        | 0.0760        | 0.2528              | 0.1541        | <u>0.3130</u> | <u>0.3008</u> |
| GAR            | 0.1357                 | 0.1062        | 0.0106        | 0.0195        | 0.2539              | 0.1489        | 0.0110        | 0.0130        | 0.2339              | 0.1455        | 0.2873        | 0.2794        |
| UCC            | 0.1374                 | <u>0.1260</u> | 0.1277        | 0.2020        | 0.0020              | 0.0011        | 0.0063        | 0.0073        | <u>0.3002</u>       | <u>0.2010</u> | 0.2830        | 0.2641        |
| ALDI           | <u>0.1626</u>          | 0.1201        | <u>0.1428</u> | 0.2316        | <u>0.2692</u>       | <u>0.1539</u> | <u>0.1229</u> | <u>0.1295</u> | 0.2528              | 0.1541        | <u>0.3130</u> | <u>0.3008</u> |
| <b>LLM-Ins</b> | <b>0.2285</b>          | <b>0.1747</b> | <b>0.1506</b> | <b>0.2468</b> | <b>0.3601</b>       | <b>0.2126</b> | <b>0.1759</b> | <b>0.1762</b> | <b>0.3252</b>       | <b>0.2156</b> | <b>0.3314</b> | <b>0.3186</b> |
| %Improv.       | 40.52%                 | 38.65%        | 5.46%         | 5.92%         | 33.76%              | 38.14%        | 43.12%        | 36.06%        | 8.33%               | 7.26%         | 5.97%         | 5.92%         |

LightGCN

| Task    | LLM            | CiteULike     |               | MovieLens     |               |
|---------|----------------|---------------|---------------|---------------|---------------|
|         |                | Recall        | NDCG          | Recall        | NDCG          |
| Overall | ChatGPT        | 0.2054        | 0.1641        | 0.1396        | 0.2267        |
|         | TALLRec        | <u>0.2141</u> | <u>0.1661</u> | <u>0.1428</u> | <u>0.2324</u> |
|         | LLMRec         | 0.1983        | 0.1535        | 0.1372        | 0.2257        |
|         | <b>LLM-Ins</b> | <b>0.2285</b> | <b>0.1747</b> | <b>0.1461</b> | <b>0.2368</b> |
|         | %Improv.       | 6.73%         | 5.18%         | 2.31%         | 1.90%         |
| Cold    | ChatGPT        | <u>0.3477</u> | <u>0.2079</u> | <u>0.1480</u> | <u>0.1556</u> |
|         | TALLRec        | 0.3352        | 0.1990        | 0.1374        | 0.1379        |
|         | LLMRec         | 0.3453        | 0.2076        | 0.1425        | 0.1508        |
|         | <b>LLM-Ins</b> | <b>0.3601</b> | <b>0.2126</b> | <b>0.1563</b> | <b>0.1566</b> |
|         | %Improv.       | 3.57%         | 2.27%         | 5.61%         | 0.64%         |
| Warm    | ChatGPT        | 0.3001        | 0.1933        | 0.3074        | 0.2921        |
|         | TALLRec        | <u>0.3102</u> | <u>0.2037</u> | <u>0.3172</u> | <u>0.3016</u> |
|         | LLMRec         | 0.2613        | 0.1645        | 0.3022        | 0.2902        |
|         | <b>LLM-Ins</b> | <b>0.3252</b> | <b>0.2156</b> | <b>0.3217</b> | <b>0.3060</b> |
|         | %Improv.       | 4.84%         | 5.84%         | 1.42%         | 1.46%         |

Experiment results when backbone = LightGCN

Comparison Experiment

Sequential recommendation systems (SRS) are often contaminated by noisy interactions.

Large Language Models (LLMs) has merged as a potential avenue to alleviate the noisy data. However, directly apply LLM to denoising task faces some **challenges**:

You are to analyze a list of item titles provided by a user. Your task is to identify any item(s) that do not align with the main interests reflected by the majority of the items. After identifying these noise items, suggest alternative items that better match the user's interests.

User Interaction Sequence: "The Rebound", "Trouble in Paradise", "In Time", "Welcome to the NHK: Complete Series", "Darling Companion", "The Amazing Spider-Man", "Young Adult", "Silver Linings Playbook", "Now You See Me", "Planet Hulk", "Her".

The user's interests are in movies and TV shows. The user's interests are not in books. The user's interests are not in comics. The user's interests are not in video games. The user's interests are not in music. The user's interests are not in anime. The user's interests are not in manga. The user's interests are not in webcomics. The user's interests are not in podcasts...

- **C1: Inadaptability of LLM:**

Direct application of pretrained LLMs may not be competent for the denoising task.

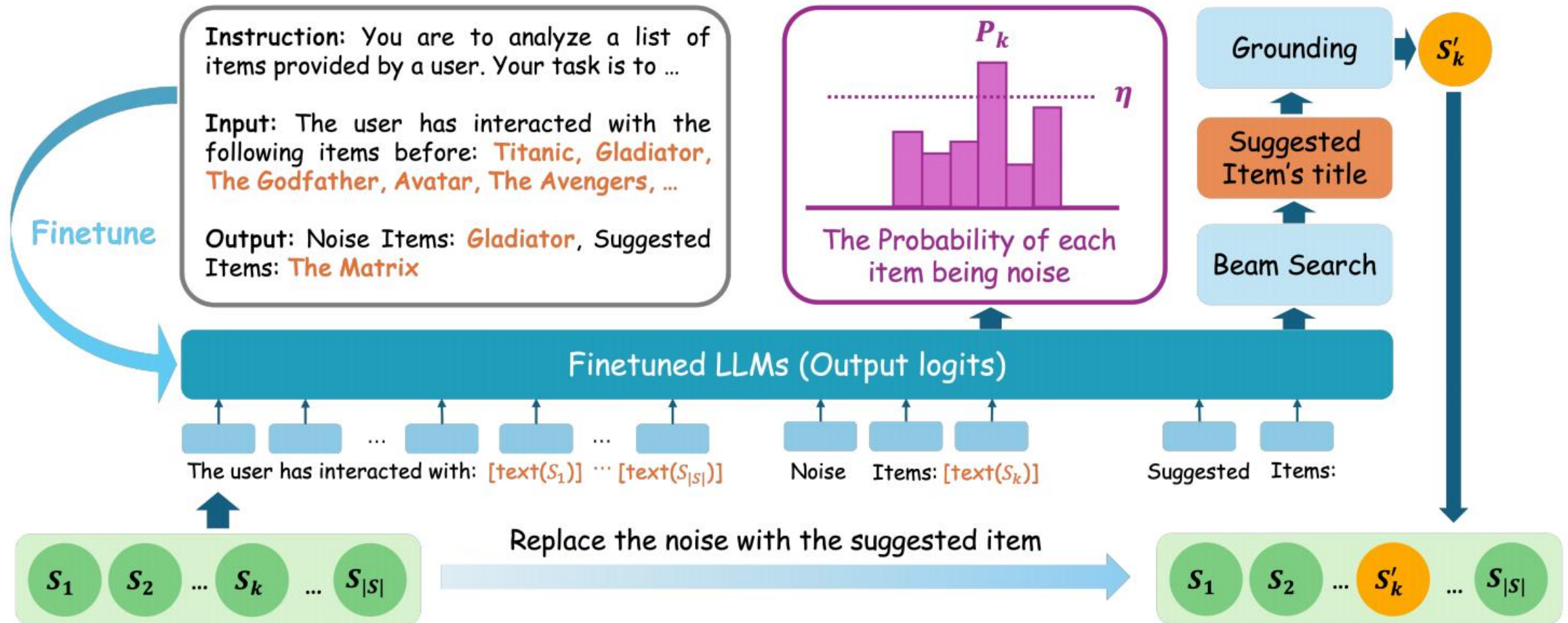
- **C2: Hallucination of LLM:**

The **reliability** of LLM's outputs remains questionable.



# Score-based: LLM4DSR

To address the above challenges, the authors proposed LLM4DSR, featuring with **self-supervised fine-tuning method** and an **uncertainty estimation mechanism** to fully exploit the capabilities of LLMs.





# Score-based: LLM4DSR

Specifically, for **C1: Inadaptability of LLM**, LLM4DSR constructed a new dataset and leveraged SFT to empower LLMs with denoising ability by using a “find & replace” task.

## Prompt for Self-Supervised Task

**Instruction:** You are to analyze a list of items provided by a user. Your task is to identify an item that do not align with the main interests reflected by the majority of the items. After identifying these noise items, suggest alternative items that better match the user’s interests.

**Input:** The user has interacted with the following items before:  $text(s_1), \dots, text(\hat{s}_t), \dots, text(\hat{s}_{|S|})$

**Output:** Noise Items:  $text(\hat{s}_t)$ , Suggested Items:  $text(s_t)$

## □ Training Objective:

By prompting the LLM using left instruction example, the authors construct a new binary prediction dataset and formulate the training objective as follows:

$$\max_{\Theta} \sum_{(x,z)} \sum_{i=1}^{|z|} \log (P_{\Phi} (z_i \mid x, z_{<i}))$$

However, there is still one limitation that current LLM-based denoiser can only address a single noise item per sequence. (**L1: Inadaptability for Multi-noise Scenario**)

# Score-based: LLM4DSR

To address **L1: Inadaptability for Multi-noise Scenario** & **C2: Hallucination of LLM**, the Uncertainty Estimation module is introduced.

## Prompt for Self-Supervised Task

**Instruction:** You are to analyze a list of items provided by a user. Your task is to identify an item that do not align with the main interests reflected by the majority of the items. After identifying these noise items, suggest alternative items that better match the user's interests.

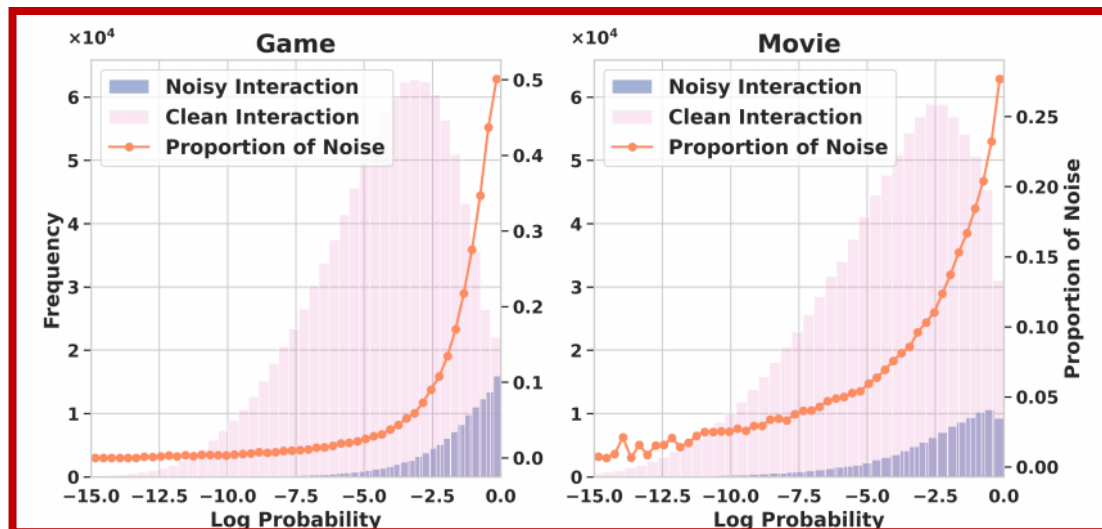
**Input:** The user has interacted with the following items before:  $text(s_1), \dots, text(\hat{s}_t), \dots, text(\hat{s}_{|S|})$

**Output:** Noise Items:  $text(\hat{s}_t)$ , Suggested Items:  $text(s_t)$

## □ Uncertainty Estimation:

Regarding the LLM provided probability after text "**Noise Items:**" as LLM's confidence, the probability values of all noisy items in the sequence can be obtained by the formula below:

$$\begin{aligned} P_{\text{noise}}(s_i|S) &= \prod_{j=1}^{|T_i|} P(T_{i,j}|x, z_{\text{noise}}, T_{i,<j}) \\ &= P(T_{i,1}|x, z_{\text{noise}})P(T_{i,2}, \dots, T_{i,|T_i|}|x, z_{\text{noise}}, T_{i,1}) \\ &= P(T_{i,1}|x, z_{\text{noise}}) \end{aligned}$$



# Score-based: LLM4DSR

For the experimental results, LLM4DSR achieves optimal results on **three raw public datasets** and **noise-added datasets**, validating the effectiveness of proposed methods.

| Backbone | Method  | Games         |               |               |               | Toy           |               |               |               | Movie         |               |               |               |
|----------|---------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
|          |         | NDCG@20       | NDCG@50       | HR@20         | HR@50         | NDCG@20       | NDCG@50       | HR@20         | HR@50         | NDCG@20       | NDCG@50       | HR@20         | HR@50         |
| SASRec   | None    | 0.0252        | 0.0314        | 0.0526        | 0.0838        | 0.0139        | 0.0178        | 0.0348        | 0.0542        | 0.0390        | 0.0455        | 0.0750        | 0.1084        |
|          | FMLP    | 0.0253        | 0.0315        | 0.0508        | 0.0824        | 0.0138        | 0.0173        | 0.0365        | 0.0547        | <u>0.0415</u> | <u>0.0484</u> | <u>0.0772</u> | <u>0.1120</u> |
|          | CL4SRec | <u>0.0278</u> | <u>0.0336</u> | <u>0.0556</u> | 0.0864        | <u>0.0151</u> | <u>0.0183</u> | <u>0.0394</u> | <u>0.0557</u> | 0.0405        | 0.0464        | 0.0770        | 0.1070        |
|          | HSD     | 0.0198        | 0.0250        | 0.0412        | 0.0678        | 0.0090        | 0.0127        | 0.0254        | 0.0444        | 0.0340        | 0.0407        | 0.0700        | 0.1042        |
|          | STEAM   | 0.0261        | 0.0333        | 0.0532        | <u>0.0900</u> | 0.0123        | 0.0161        | 0.0321        | 0.0513        | 0.0401        | 0.0476        | 0.0740        | 0.1118        |
|          | LLM4DSR | <b>0.0292</b> | <b>0.0357</b> | <b>0.0624</b> | <b>0.0956</b> | <b>0.0152</b> | <b>0.0200</b> | <b>0.0401</b> | <b>0.0642</b> | <b>0.0431</b> | <b>0.0517</b> | <b>0.0876</b> | <b>0.1314</b> |
| BERT4Rec | None    | 0.0266        | 0.0335        | 0.0534        | 0.0880        | 0.0150        | 0.0195        | 0.0346        | 0.0574        | 0.0485        | 0.0569        | 0.0826        | 0.1258        |
|          | FMLP    | 0.0281        | 0.0353        | 0.0598        | 0.0962        | 0.0147        | 0.0188        | 0.0330        | 0.0536        | 0.0478        | 0.0561        | 0.0818        | 0.1240        |
|          | CL4SRec | 0.0268        | 0.0341        | 0.0538        | 0.0906        | 0.0153        | 0.0196        | 0.0369        | <u>0.0586</u> | <u>0.0498</u> | <u>0.0579</u> | <u>0.0864</u> | <u>0.1276</u> |
|          | HSD     | 0.0288        | 0.0361        | <u>0.0600</u> | <u>0.0970</u> | 0.0152        | 0.0189        | 0.0340        | 0.0530        | 0.0390        | 0.0459        | 0.0682        | 0.1032        |
|          | STEAM   | <u>0.0294</u> | <u>0.0367</u> | 0.0562        | 0.0936        | <u>0.0182</u> | <u>0.0218</u> | <u>0.0388</u> | 0.0572        | 0.0472        | 0.0544        | 0.0836        | 0.1202        |
|          | LLM4DSR | <b>0.0316</b> | <b>0.0397</b> | <b>0.0644</b> | <b>0.1056</b> | <b>0.0183</b> | <b>0.0231</b> | <b>0.0401</b> | <b>0.0645</b> | <b>0.0509</b> | <b>0.0591</b> | <b>0.0890</b> | <b>0.1306</b> |
| LLaRA    | None    | 0.0128        | 0.0172        | 0.0260        | 0.0484        | 0.0085        | 0.0108        | 0.0188        | 0.0305        | <u>0.0117</u> | <u>0.0136</u> | <u>0.0198</u> | <u>0.0294</u> |
|          | HSD     | <u>0.0141</u> | <u>0.0184</u> | <u>0.0274</u> | <u>0.0488</u> | <u>0.0109</u> | <u>0.0141</u> | <u>0.0240</u> | <u>0.0403</u> | 0.0111        | 0.0132        | 0.0186        | 0.0294        |
|          | STEAM   | 0.0108        | 0.0146        | 0.0224        | 0.0416        | 0.0092        | 0.0120        | 0.0202        | 0.0348        | 0.0107        | 0.0127        | 0.0182        | 0.0286        |
|          | LLM4DSR | <b>0.0150</b> | <b>0.0193</b> | <b>0.0286</b> | <b>0.0504</b> | <b>0.0121</b> | <b>0.0162</b> | <b>0.0280</b> | <b>0.0482</b> | <b>0.0159</b> | <b>0.0186</b> | <b>0.0250</b> | <b>0.0390</b> |

Performance Comparison on Noise-added Datasets

Recent Studies have shown the rich semantic information can benefit the multi-interest modeling in Sequential Recommendation Systems (SRS).

However, there are still some problems when leveraging the semantic information to model user's behavioral information.

- **P1: Demand Conflict**

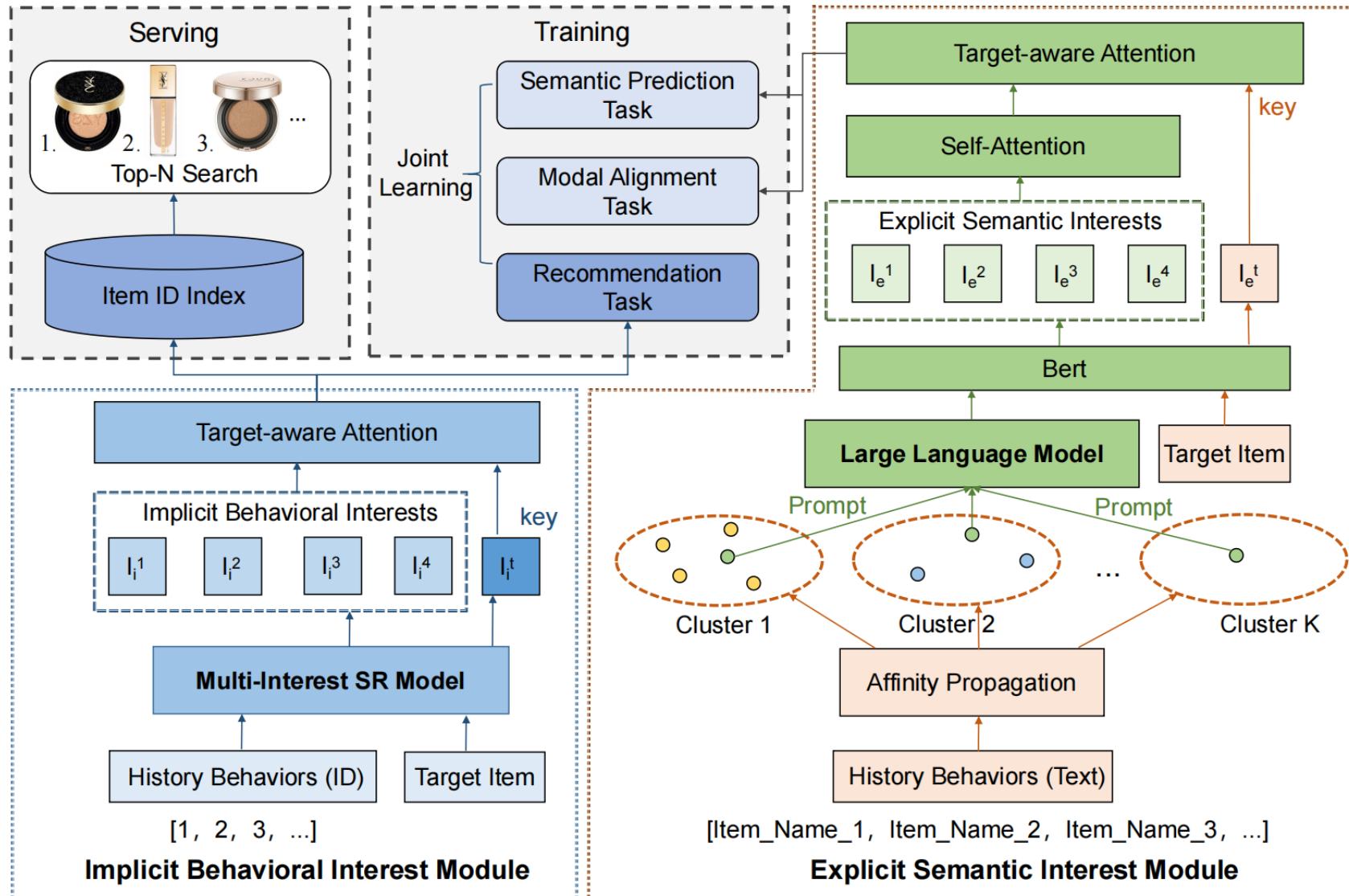
Traditional SRS has **high real-time requirements** while the fine-tuning and inference of LLM **requires a lot of time and computing resources**, leading to the demand conflict.

- **P2: Misunderstanding of User's Real Interests**

The semantic information from user/item content has **significant gap** with the behavioral information from well-trained recommendation model, leading to **misunderstanding of user's real interests**.

# Score-based: EIMF

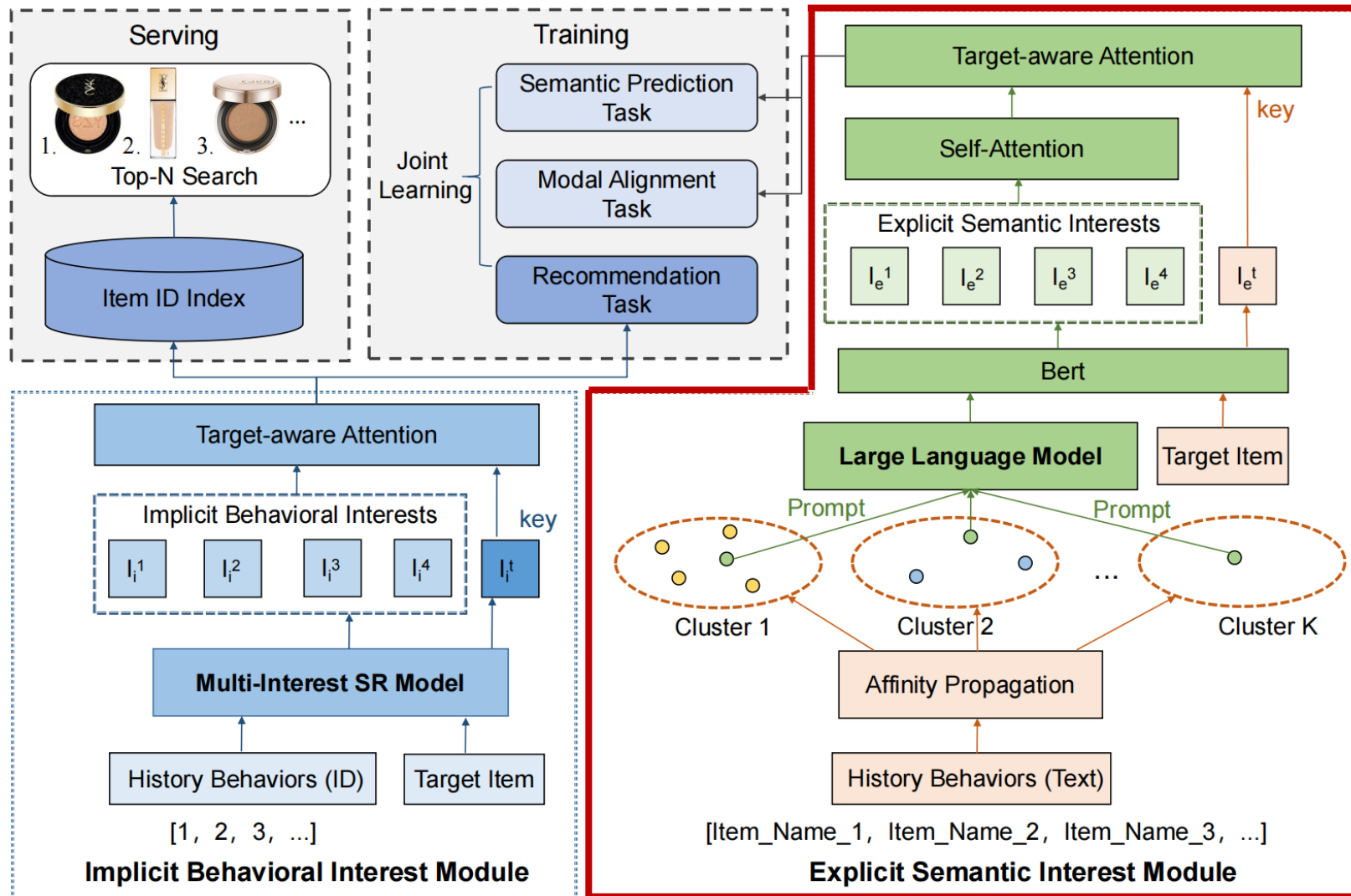
To address the above problems, the authors propose EIMF to adaptively combine traditional model with LLM.





# Score-based: EIMF

Specifically, to address the '**P1: Demand Conflict**', EIMF introduces **Explicit Behavioral Interest** to fully explore the LLM's capability while reducing the inference cost.



## □ Affinity Propagation & LLM Summarization:

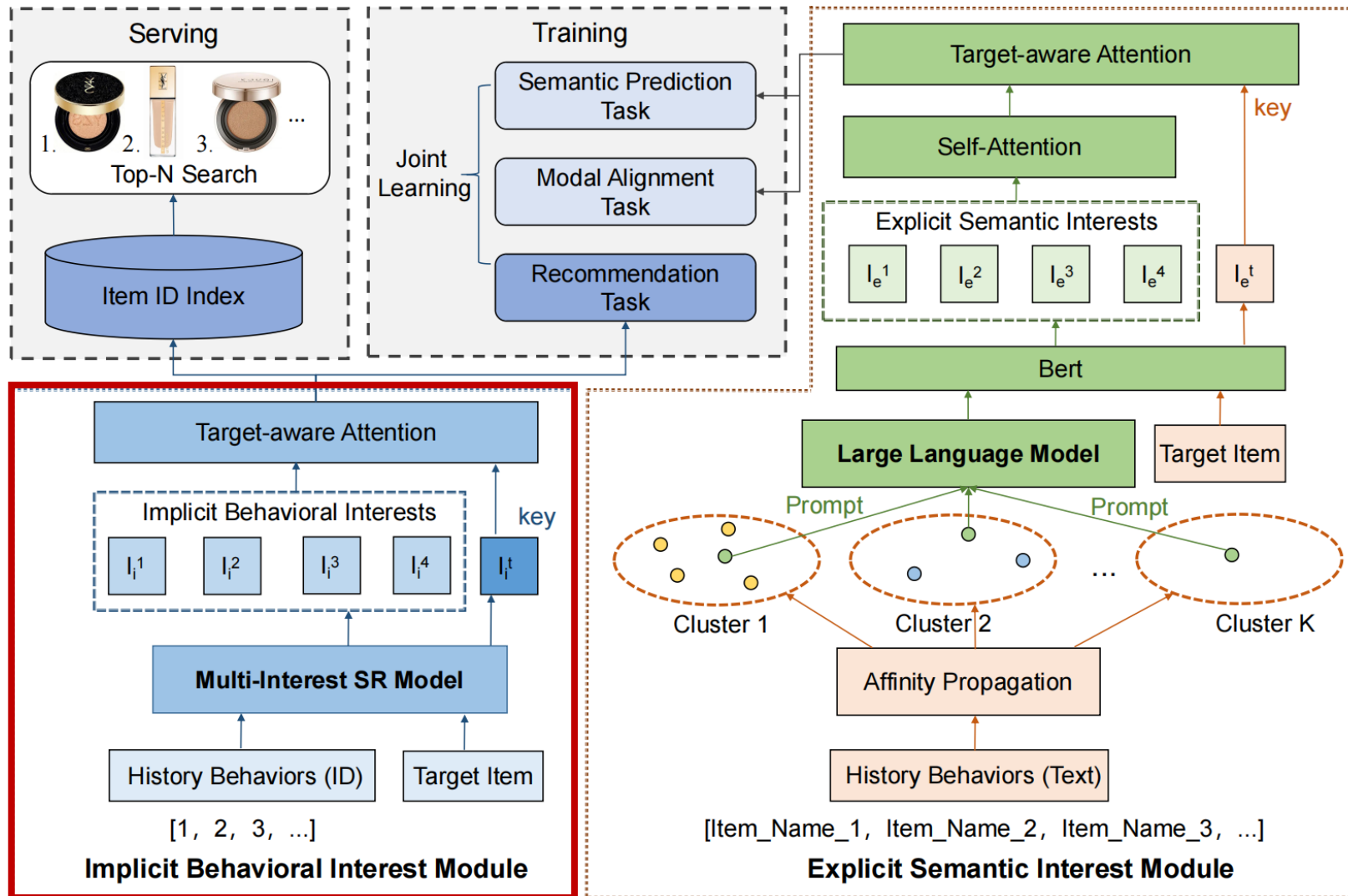
EIMF divide users into groups and utilize LLM as summarizer to reduce the inference cost.

## □ Interest Embedding

EIMF queries the semantic interest embedding by utilizing the target item as key, which can be used for alignment and semantic prediction.

# Score-based: EIMF

To address the '**P2: Misunderstanding of User's Real Interests**', EIMF combines the **Behavioral Interest** with the **Semantic Interest** by jointly learning three kinds of tasks.



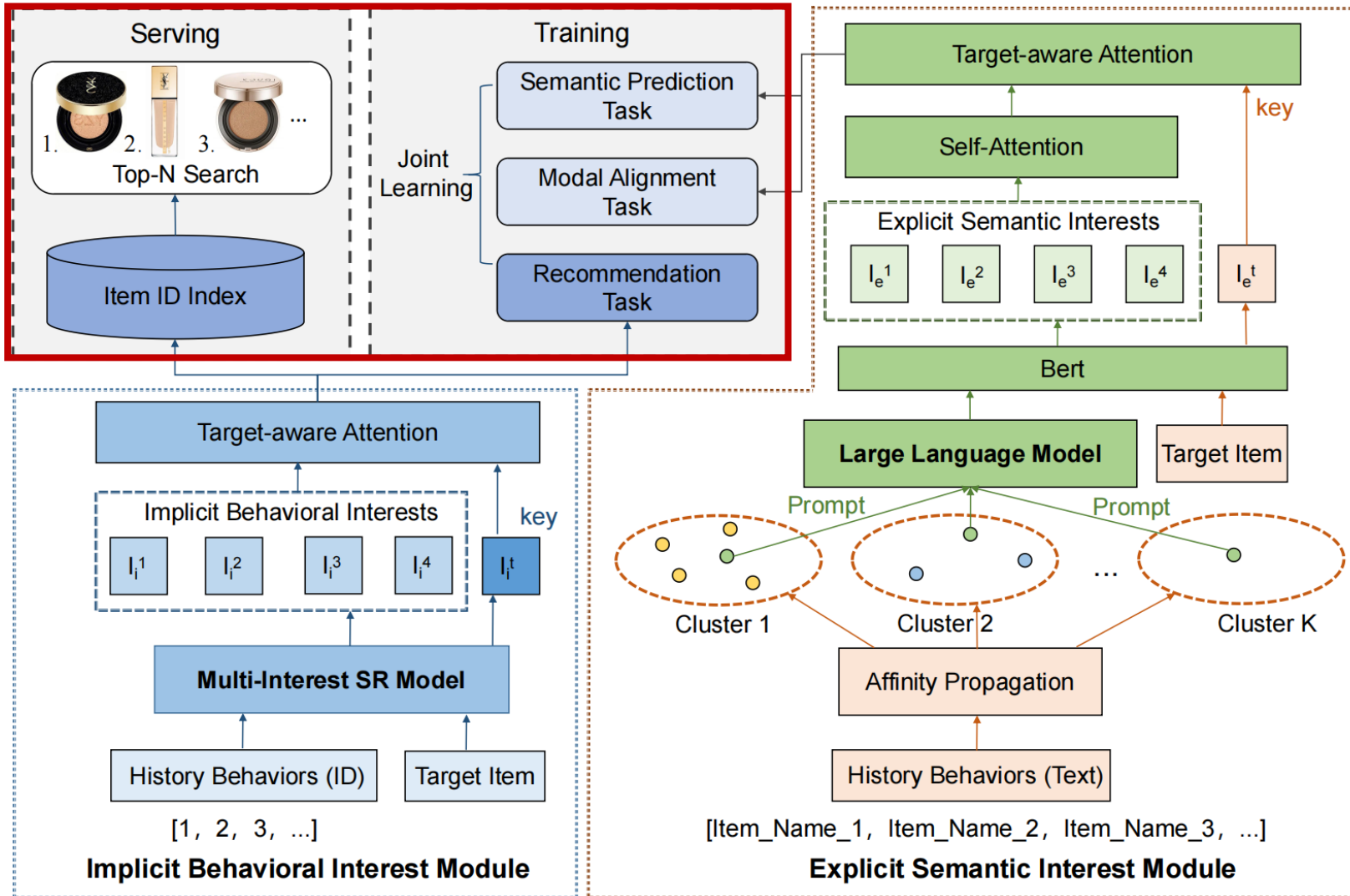
## □ Behavioral Interest:

Firstly, EIMF obtains the implicit behavioral interest by a SR model.

Then using similar **Target-aware Attention** network to acquire the behavioral embeddings to construct the loss.

# Score-based: EIMF

To address the '**P2: Misunderstanding of User's Real Interests**', EIMF combines the **Behavioral Interest** with the **Semantic Interest** by jointly learning three kinds of tasks.



## □ Joint Learning:

### - Semantic Prediction Task

Calculating the score between user's semantic interest and item text:

$$\mathcal{L}_S = \sum_{k=1}^n y_t^k \log(\hat{y}_t^k)$$

# Score-based: EIMF

To address the '**P2: Misunderstanding of User's Real Interests**', EIMF combines the **Behavioral Interest** with the **Semantic Interest** by jointly learning three kinds of tasks.

## Joint Learning:

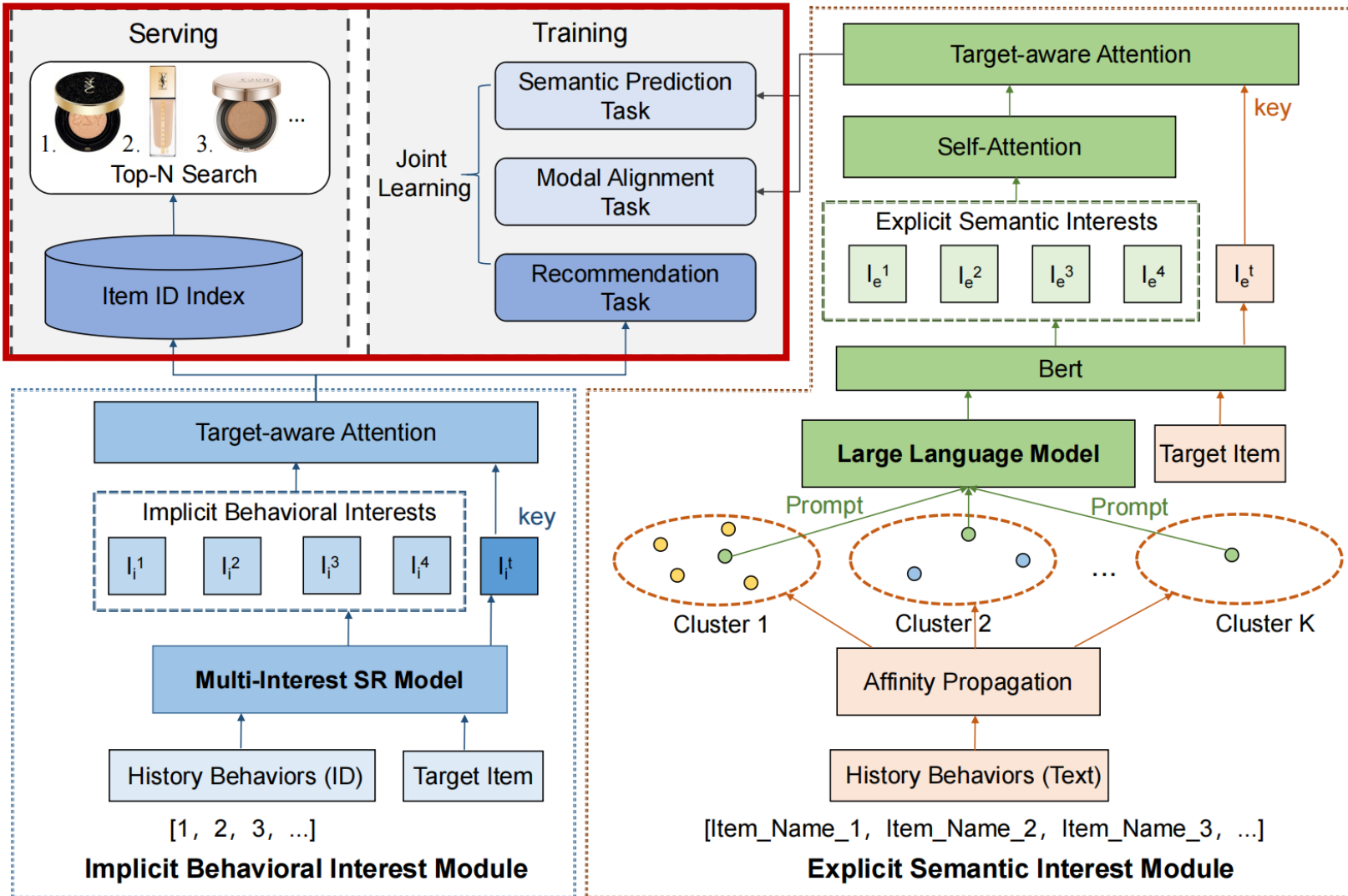
### - Modal Alignment Task

Building the **contrastive loss** to align the semantic embedding and behavioral embedding as:

$$CL(e_a, e_b) = -\frac{1}{N} \sum_{k=1}^N \log \left( \frac{\exp(\text{Sim}(e_a^k, e_b^k)/\tau)}{\sum_{j=1}^N \exp(\text{Sim}(e_a^k, e_b^j)/\tau)} \right),$$

$$\text{Cos}(e_a, e_b) = \frac{1}{N} \sum_{k=1}^N \left( 1 - \text{Sim}(e_a^k, e_b^k) \right),$$

$$\mathcal{L}_A = \alpha(CL(h_{\text{ex}}, h_{\text{im}}) + CL(e_t^T, e_i^T)) + \beta(\text{Cos}(h_{\text{ex}}, h_{\text{im}}) + \text{Cos}(e_t^T, e_i^T))$$



# Score-based: EIMF

To address the '**P2: Misunderstanding of User's Real Interests**', EIMF combines the **Behavioral Interest** with the **Semantic Interest** by jointly learning three kinds of tasks.

## Joint Learning:

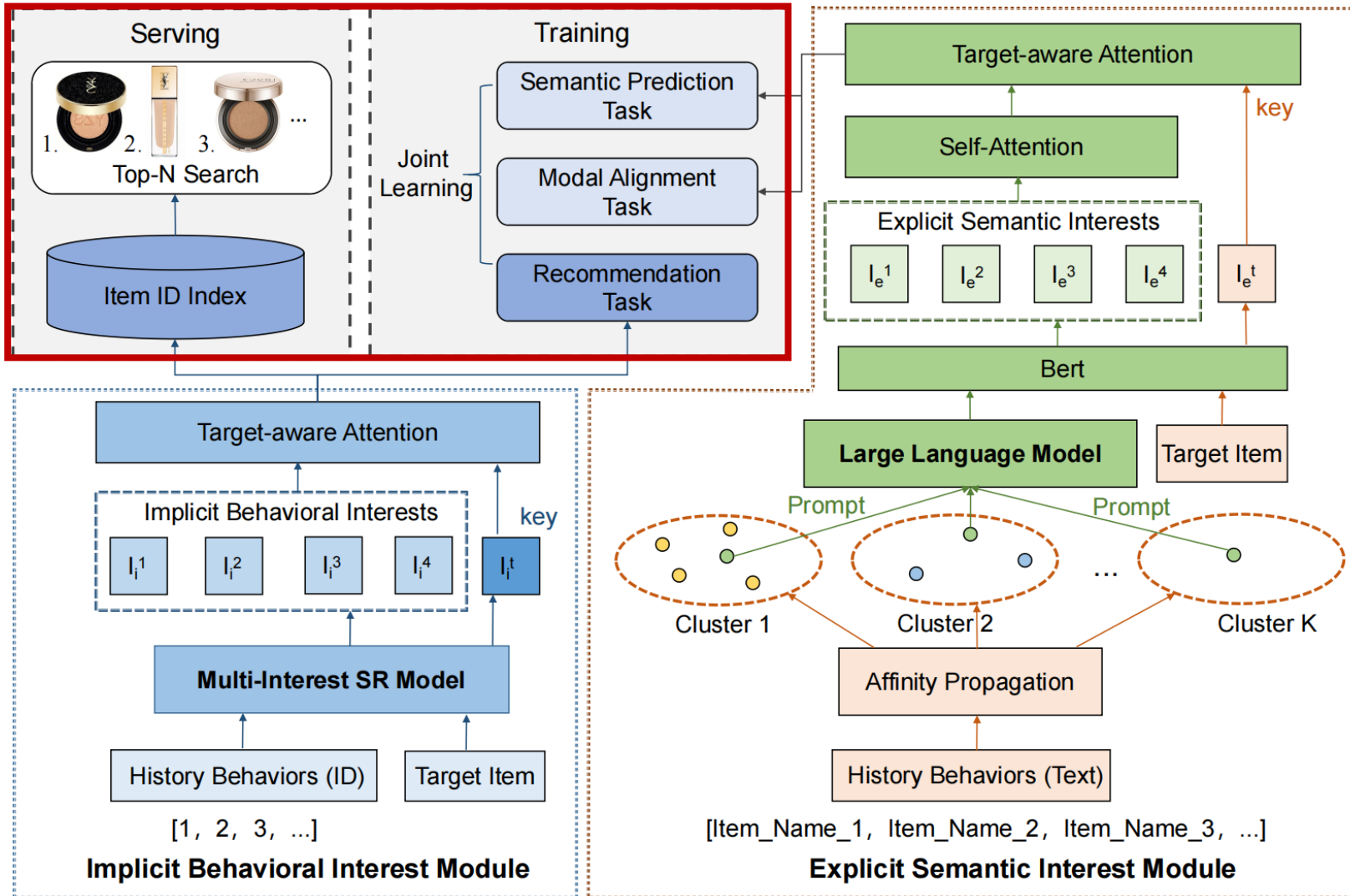
### - Recommendation Task

$$\hat{y}_i^k = \text{softmax}(h_{\text{ex}}^\top e_i^k)$$

$$\mathcal{L}_R = \sum_{k=1}^n y_i^k \log(\hat{y}_i^k)$$

Finally, the **whole loss** can be formulized as:

$$\mathcal{L} = \mathcal{L}_R + \gamma (\mathcal{L}_S + \mathcal{L}_A)$$





# Score-based: EIMF

The experimental results show that EIMF achieves remarkable results on **three public datasets** (two large and one small), presenting that EIMF's superiority.

| Dataset | Model     | Pop    | DNN    | GRU4Rec | SASRec        | Bert4Rec      | LLM2Bert4Rec | MIND   | ComiRec-SA | REMI          | EIMF(REMI)    | Improv.(%) |
|---------|-----------|--------|--------|---------|---------------|---------------|--------------|--------|------------|---------------|---------------|------------|
| Grocery | Recall@20 | 0.0729 | 0.1252 | 0.1387  | 0.1536        | 0.1544        | 0.1259       | 0.1445 | 0.1122     | <u>0.1617</u> | <b>0.1758</b> | +8.71      |
|         | Recall@50 | 0.1305 | 0.2044 | 0.2300  | 0.2508        | 0.2372        | 0.2080       | 0.2162 | 0.2076     | <u>0.2574</u> | <b>0.2704</b> | +5.05      |
|         | NDCG@20   | 0.0448 | 0.0804 | 0.0905  | 0.1041        | <b>0.1074</b> | 0.0793       | 0.0844 | 0.0706     | 0.0953        | 0.1025        | -4.56      |
|         | NDCG@50   | 0.0624 | 0.0969 | 0.1097  | <u>0.1139</u> | 0.1130        | 0.0954       | 0.0967 | 0.0918     | 0.1108        | <b>0.1181</b> | +3.69      |
|         | HR@20     | 0.1252 | 0.2076 | 0.2321  | <u>0.2586</u> | 0.2614        | 0.2164       | 0.2328 | 0.1851     | 0.2539        | <b>0.2750</b> | +6.34      |
|         | HR@50     | 0.2137 | 0.3227 | 0.3649  | 0.3750        | 0.3608        | 0.3294       | 0.3322 | 0.3220     | <u>0.3832</u> | <b>0.3955</b> | +3.21      |
| Beauty  | Recall@20 | 0.0452 | 0.1613 | 0.1388  | 0.1495        | 0.1430        | 0.1383       | 0.1563 | 0.1582     | <u>0.2189</u> | <b>0.2323</b> | +6.12      |
|         | Recall@50 | 0.0660 | 0.2361 | 0.2109  | 0.2220        | 0.2050        | 0.2053       | 0.2384 | 0.2594     | <u>0.3420</u> | <b>0.3636</b> | +6.31      |
|         | NDCG@20   | 0.0213 | 0.0886 | 0.0798  | 0.0854        | 0.0846        | 0.0788       | 0.0787 | 0.0854     | <u>0.1139</u> | <b>0.1204</b> | +5.71      |
|         | NDCG@50   | 0.0270 | 0.0932 | 0.0844  | 0.0879        | 0.0852        | 0.0823       | 0.0900 | 0.1004     | <u>0.1304</u> | <b>0.1348</b> | +3.37      |
|         | HR@20     | 0.0675 | 0.2401 | 0.2141  | 0.2320        | 0.2221        | 0.2181       | 0.2203 | 0.2325     | <u>0.3111</u> | <b>0.3268</b> | +5.04      |
|         | HR@50     | 0.0966 | 0.3299 | 0.2982  | 0.3129        | 0.2986        | 0.2919       | 0.3272 | 0.3545     | <u>0.4532</u> | <b>0.4802</b> | +5.95      |

## Performance comparison on two large datasets

| Model        | Recall@20     | NDCG@20       | HR@20         | Recall@50     | NDCG@50       | HR@50         |
|--------------|---------------|---------------|---------------|---------------|---------------|---------------|
| GRU4Rec      | 0.0800        | 0.0537        | 0.1384        | 0.1770        | 0.0809        | 0.2811        |
| SASRec       | <u>0.1152</u> | 0.0643        | 0.1792        | 0.1963        | 0.0805        | 0.2871        |
| Bert4Rec     | 0.1037        | 0.0635        | 0.1812        | 0.1920        | 0.0834        | 0.3116        |
| MIND         | 0.0915        | 0.0513        | 0.1466        | 0.1593        | 0.0706        | 0.2505        |
| ComiRec-SA   | 0.0788        | 0.0456        | 0.1202        | 0.1589        | 0.0653        | 0.2321        |
| REMI         | 0.1072        | 0.0637        | 0.1751        | 0.2018        | 0.0825        | 0.2973        |
| LLM2Bert4Rec | 0.1085        | <b>0.0660</b> | <b>0.1812</b> | <u>0.2117</u> | <b>0.0939</b> | <u>0.3360</u> |
| EIMF(REMI)   | <b>0.1222</b> | 0.0649        | 0.1772        | <b>0.2296</b> | 0.0930        | <b>0.3503</b> |
| Improv.(%)   | +6.08         | -1.66         | -2.20         | +8.45         | -0.95         | +4.25         |

## Performance comparison on one small dataset

# Score-based: EIMF

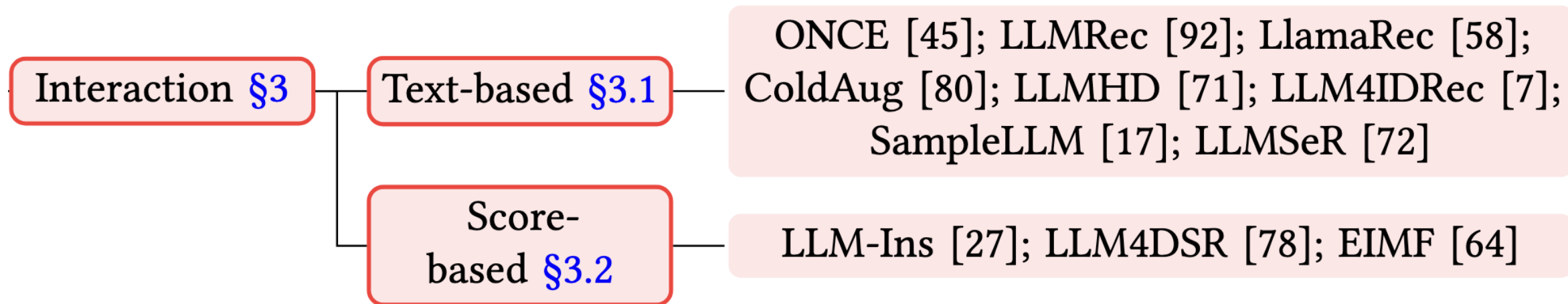
Moreover, the experiment performing on different backbones shows a significant improvement, further validating the effectiveness of proposed modules in EIMF.

| Dataset        | Beauty    |         |        | Office    |         |        |
|----------------|-----------|---------|--------|-----------|---------|--------|
| Model          | Recall@50 | NDCG@50 | HR@50  | Recall@50 | NDCG@50 | HR@50  |
| Bert4Rec       | 0.2050    | 0.0852  | 0.2986 | 0.1920    | 0.0834  | 0.3116 |
| EIMF(Bert4Rec) | 0.2155    | 0.0854  | 0.3071 | 0.2101    | 0.0909  | 0.3380 |
| Improv.(%)     | +5.12     | +0.23   | +2.84  | +9.42     | +8.99   | +8.47  |
| SASRec         | 0.2220    | 0.0879  | 0.3129 | 0.1964    | 0.0806  | 0.2872 |
| EIMF(SASRec)   | 0.2499    | 0.0999  | 0.3469 | 0.2088    | 0.0878  | 0.3198 |
| Improv.(%)     | +12.56    | +13.65  | +10.86 | +6.31     | +8.93   | +11.35 |
| MIND           | 0.2384    | 0.0900  | 0.3272 | 0.1594    | 0.0707  | 0.2505 |
| EIMF(MIND)     | 0.2518    | 0.0968  | 0.3424 | 0.2023    | 0.0793  | 0.2953 |
| Improv.(%)     | +5.62     | +7.55   | +4.64  | +26.91    | +12.16  | +17.88 |

Performance comparison on different backbones

# Summary

Here, we give an overview of the works on **interaction enhancement** for LLM Enhanced Sequential Recommendation (**LLMESR**).



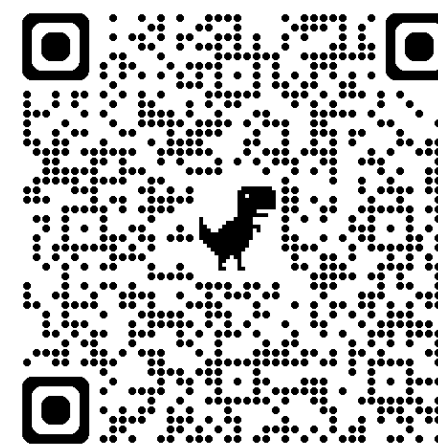
- **Text-based:** This kind of methods leverage LLM to give out the names of **pseudo-interacted items** in the user-item sequence as the augmentation.
- **Score-based:** The score-based methods utilize LLM to derive the **logits of the possible interactions** and generate the augmented items **by ranking**.



**KDD 25 LLMERS Tutorial  
Chat Group**



**AML Lab  
CityU**



**Tutorial Slides**

- [1] Large language model enhanced recommender systems: A Survey. <https://arxiv.org/abs/2412.13432>
- [2] Multi-Task Deep Recommendation Systems: A Survey. <https://arxiv.org/abs/2302.03525>
- [3] Multimodal Recommender Systems: A Survey. <https://arxiv.org/abs/2302.03883>

# Agenda

## 1 Introduction



Zijian Zhang



## 2 Knowledge Enhancement



Pengyue Jia



## 3 Interaction Enhancement



Ziwei Liu



## 4.1 Model Enhancement 1



Maolin Wang



## 4.2 Model Enhancement 2



Yuhao Wang



## 5 Conclusion

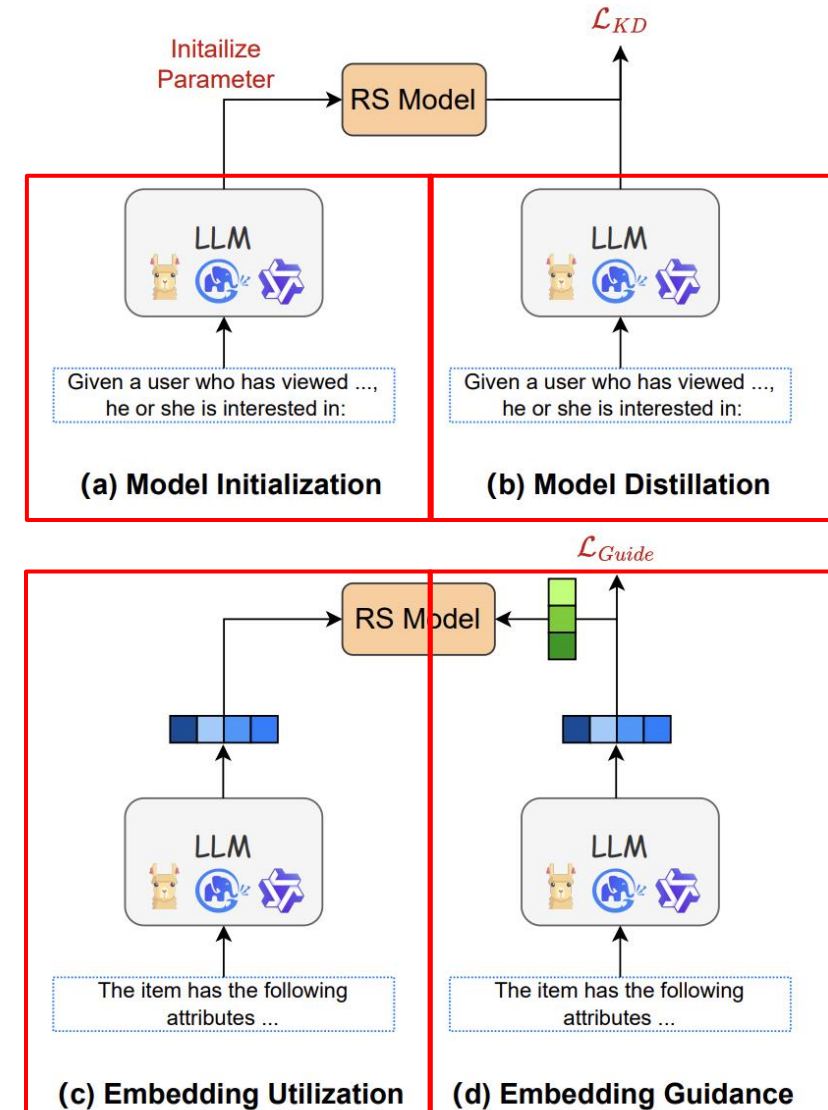


Yejing Wang



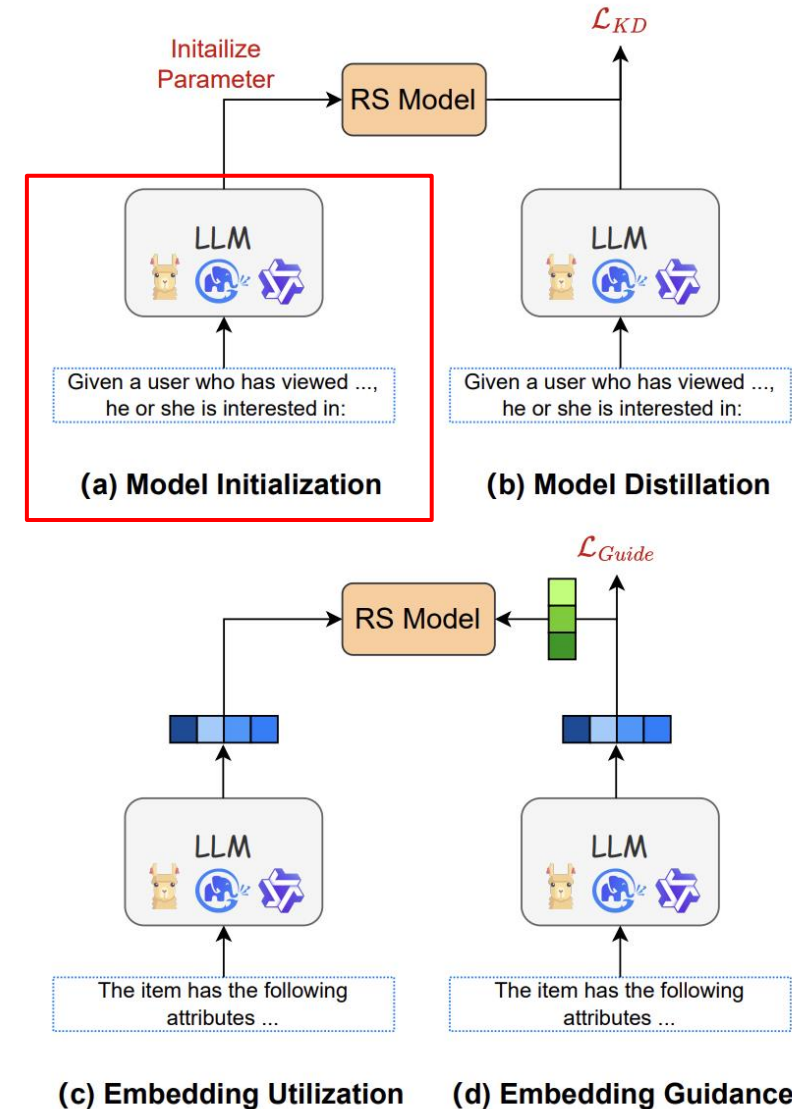
# Model Enhancement

- LLMs possess powerful semantic capabilities that can be directly integrated into recommendation systems to enhance models.
- Categories
  - Model Initialization
  - Model Distillation
  - Embedding Utilization
  - Embedding Guidance



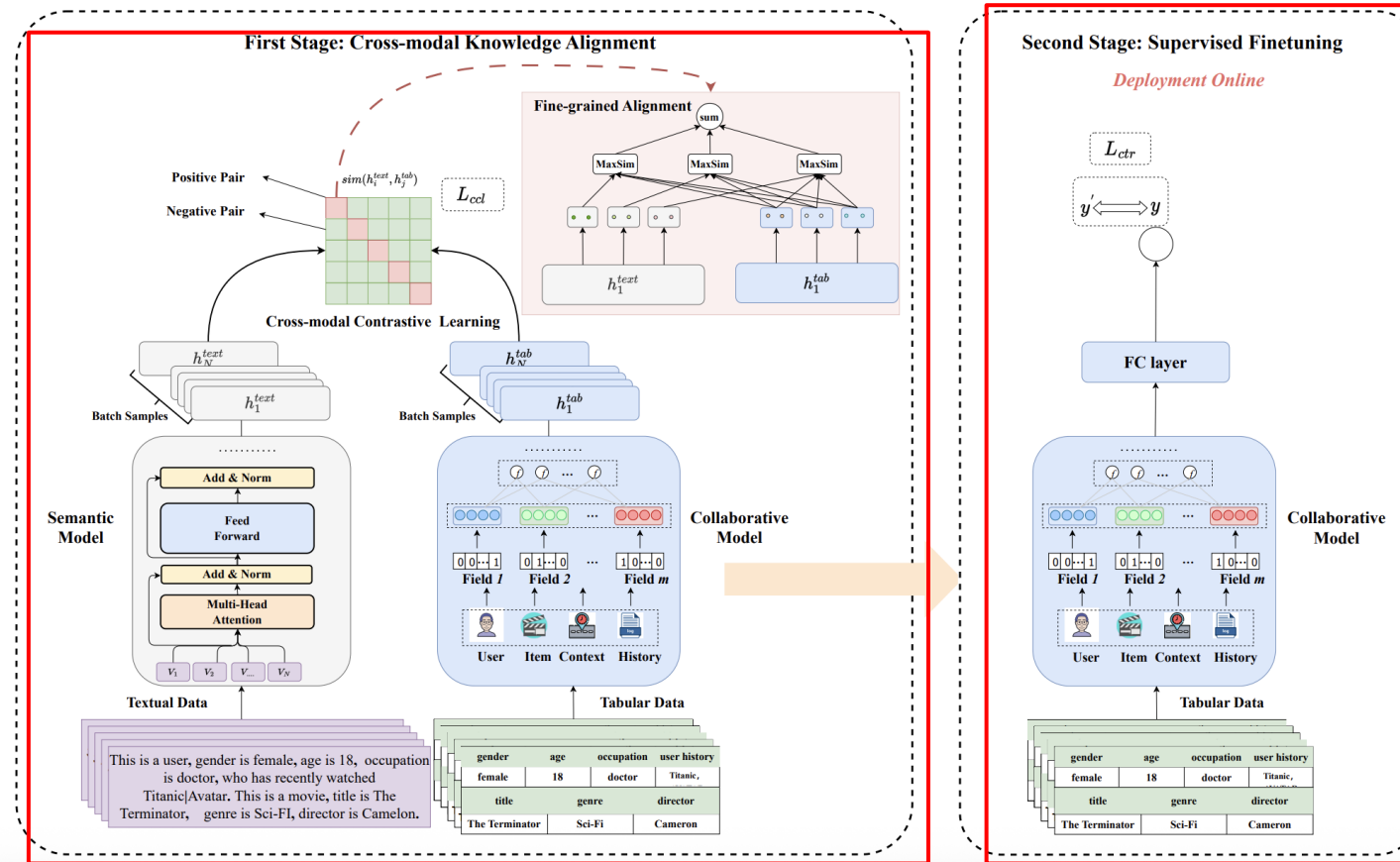
# Model Initialization

- This subcategory refers to utilizing LLM-derived semantics to initialize the weights of recommendation models before training begins.
- This approach accelerates model convergence while preserving semantic knowledge from LLM pretraining for downstream recommendation tasks.
- Categories
  - Whole
  - Embedding



- Background
  - CTR prediction is a core task in recommender systems, where traditional collaborative models capture user preferences through feature co-occurrence.
- Motivation
  - One-hot encoding discards semantic relationships, leading to poor performance in cold-start scenarios.
  - Direct use of Pre-trained Language Models is computationally expensive and fails to meet low-latency requirements for online inference.
- Key Stages
  - Cross-modal Knowledge Alignment
  - Supervised Fine-tuning

- CTRL proposes a two-stage framework to effectively integrate semantic knowledge into collaborative.
- Cross-modal Knowledge Alignment**
  - Tabular data feeds collaborative model, text prompts feed PLM. Contrastive learning aligns representations to distill semantic knowledge.
- Supervised Fine-tuning**
  - Enhanced collaborative model fine-tunes on CTR task. Only lightweight collaborative model deploys online, no PLM needed.



- CTRL significantly outperforms SOTA collaborative and semantic models on multiple public datasets and an industrial system, while maintaining high inference efficiency.

| Category             | Model     | MovieLens      |                |          | Amazon         |                |          | Alibaba        |                |          |
|----------------------|-----------|----------------|----------------|----------|----------------|----------------|----------|----------------|----------------|----------|
|                      |           | AUC            | Logloss        | RelaImpr | AUC            | Logloss        | RelaImpr | AUC            | Logloss        | RelaImpr |
| Collaborative Models | Wide&Deep | 0.8261         | 0.4248         | 3.52%    | 0.6968         | 0.4645         | 5.30%    | 0.6272         | 0.1943         | 5.19%    |
|                      | DeepFM    | 0.8268         | 0.4219         | 3.30%    | 0.6969         | 0.4645         | 5.33%    | 0.6280         | 0.1951         | 4.53%    |
|                      | DCN       | <u>0.8313</u>  | <u>0.4165</u>  | 1.90%    | 0.6999         | 0.4642         | 3.75%    | <u>0.6281</u>  | 0.1949         | 4.45%    |
|                      | PNN       | 0.8269         | 0.4220         | 3.27%    | 0.6979         | 0.4657         | 4.80%    | 0.6271         | 0.1956         | 5.27%    |
|                      | AutoInt   | 0.8290         | 0.4178         | 2.61%    | <u>0.7012</u>  | <u>0.4632</u>  | 3.08%    | 0.6279         | <u>0.1948</u>  | 4.61%    |
|                      | FiBiNet   | 0.8196         | 0.4188         | 5.63%    | 0.7003         | 0.4704         | 3.54%    | 0.6270         | 0.1951         | 5.35%    |
|                      | xDeepFM   | 0.8296         | 0.4178         | 2.43%    | 0.7009         | 0.4642         | 3.23%    | 0.6272         | 0.1959         | 5.19%    |
| Semantic Models      | P5        | 0.7583         | 0.4912         | 30.70%   | 0.6923         | 0.4608         | 7.85%    | 0.6034         | 0.3592         | 29.40%   |
|                      | CTR-BERT  | 0.7650         | 0.4944         | 27.40%   | 0.6934         | 0.4629         | 7.24%    | 0.6005         | 0.3620         | 33.13%   |
|                      | P-Tab     | 0.8031         | 0.4612         | 11.38%   | 0.6942         | 0.4625         | 6.80%    | 0.6112         | 0.3584         | 20.32%   |
| CTRL                 |           | <b>0.8376*</b> | <b>0.4025*</b> | -        | <b>0.7074*</b> | <b>0.4577*</b> | -        | <b>0.6338*</b> | <b>0.1890*</b> | -        |

- Practical Values:** CTRL achieves superior performance by fusing collaborative and semantic signals, maintains efficient inference, and is industrial-friendly with flexible model compatibility.



- Background
  - CTR prediction is a core task in recommender systems, where traditional ID-based models use one-hot encoding and PLMs use textual modality.
- Motivation
  - One-hot encoding discards semantic information, leading to poor performance in sparse scenarios.
  - PLMs struggle with field-wise collaborative signals and have high computational costs for online inference.
- Key Components
  - Modality Transformation
  - Modality Alignment Pretraining
  - Adaptive Finetuning

## • Modality Transformation

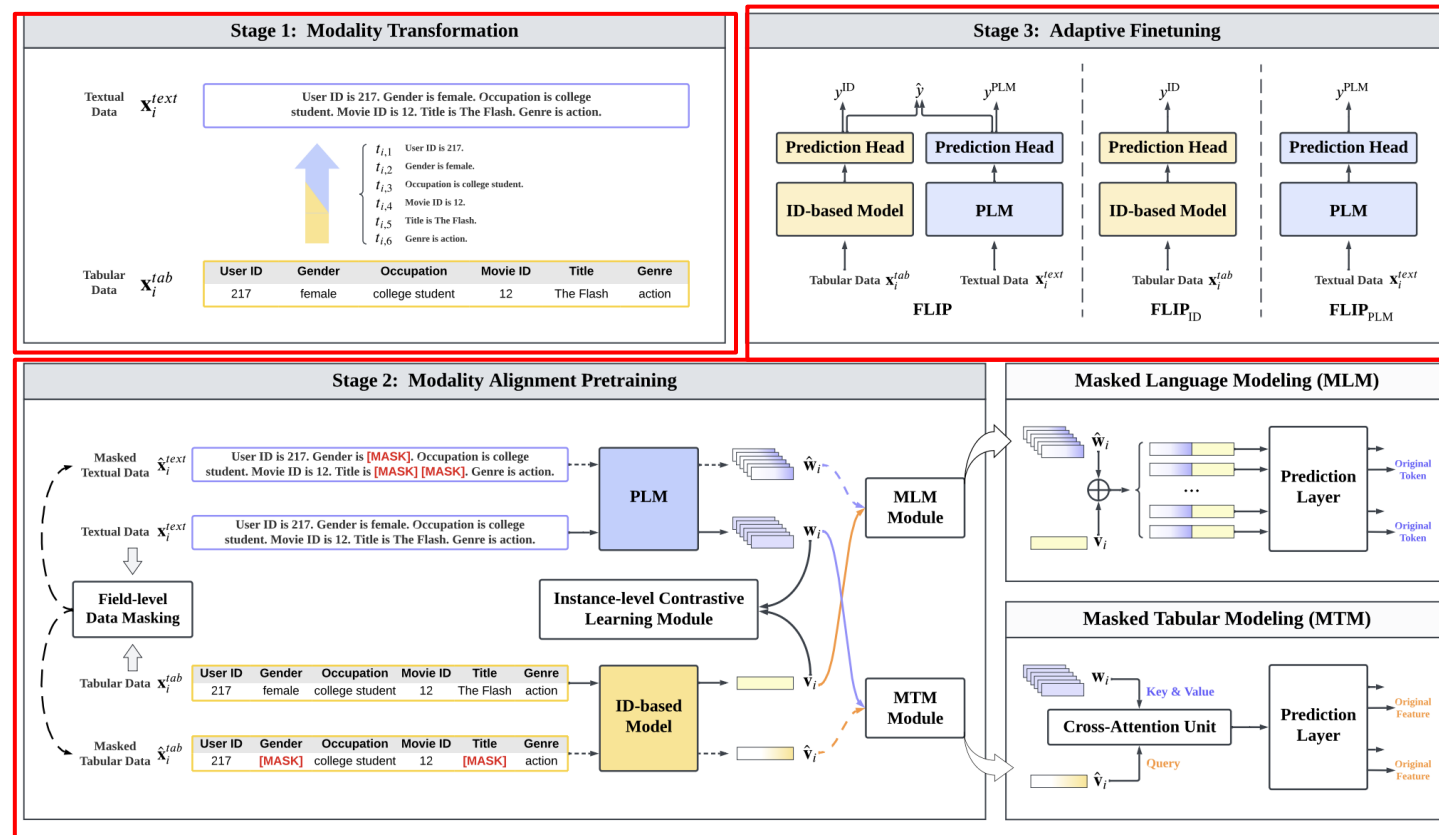
- Convert tabular features to textual sentences using simple templates, preserving semantic information in natural language format.

## • Modality Alignment Pretraining

- Field-level masking with MLM, MTM, and ICL for fine-grained cross-modal alignment.

## • Adaptive Finetuning

- Joint training with learnable weight  $\alpha$  combining both model outputs for CTR prediction.

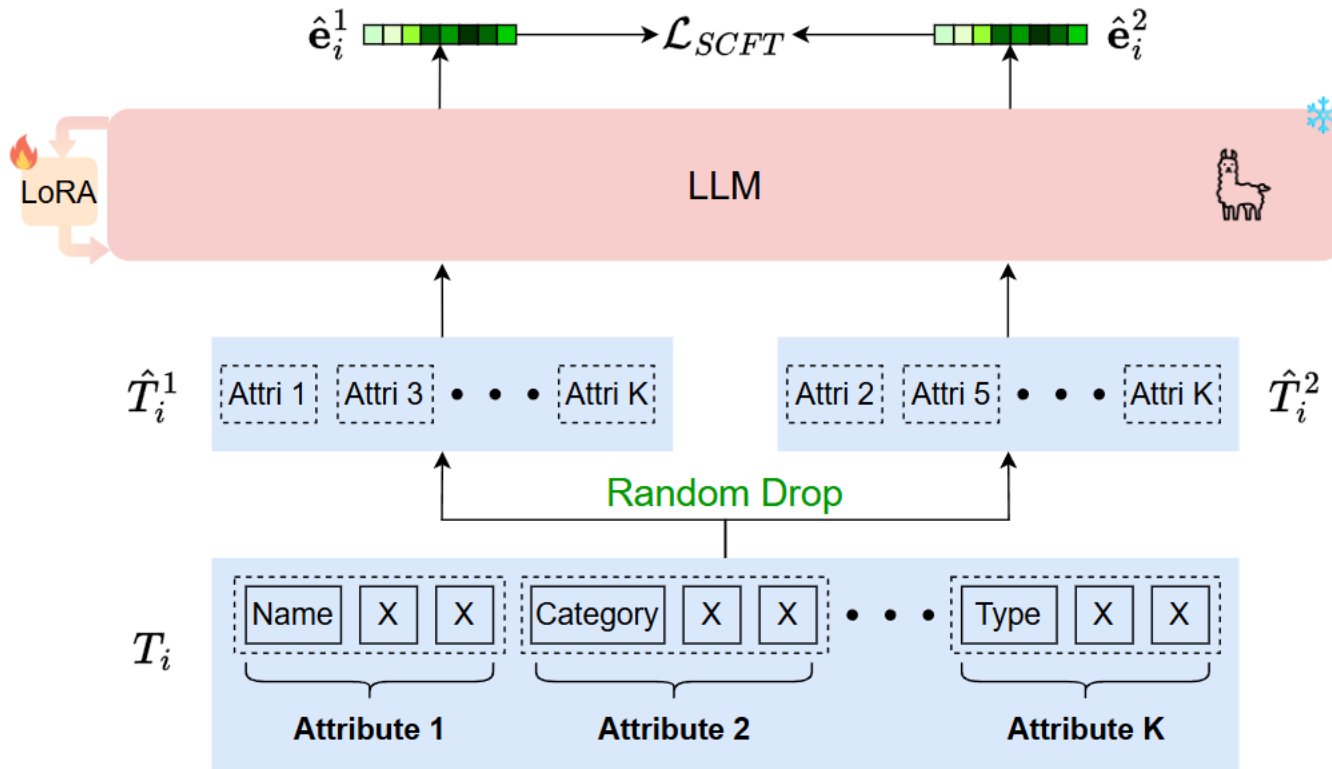


- **FLIP** significantly outperforms SOTA on three public datasets. It also shows strong compatibility with various ID models and PLMs.

| Model     |                            | MovieLens-1M   |                |          | BookCrossing   |                |          | GoodReads      |                |          |
|-----------|----------------------------|----------------|----------------|----------|----------------|----------------|----------|----------------|----------------|----------|
|           |                            | AUC            | Logloss        | Rel.Impr | AUC            | Logloss        | Rel.Impr | AUC            | Logloss        | Rel.Impr |
| ID-based  | AFM                        | 0.8449         | 0.3950         | 1.79%    | 0.7946         | 0.5116         | 1.06%    | 0.7630         | 0.5160         | 1.96%    |
|           | PNN                        | 0.8546         | 0.3946         | 0.63%    | 0.7956         | 0.5131         | 0.93%    | <u>0.7725</u>  | <u>0.5055</u>  | 0.70%    |
|           | Wide&Deep                  | 0.8509         | 0.3957         | 1.07%    | 0.7951         | 0.5116         | 1.00%    | 0.7684         | 0.5090         | 1.24%    |
|           | DCN                        | 0.8509         | 0.4056         | 1.07%    | <u>0.7957</u>  | 0.5108         | 0.92%    | 0.7693         | 0.5086         | 1.12%    |
|           | DeepFM                     | 0.8539         | 0.3905         | 0.71%    | 0.7947         | 0.5122         | 1.04%    | 0.7671         | 0.5138         | 1.41%    |
|           | xDeepFM                    | 0.8454         | 0.3934         | 1.72%    | 0.7953         | 0.5108         | 0.97%    | 0.7720         | 0.5079         | 0.77%    |
|           | AFN                        | 0.8525         | <u>0.3868</u>  | 0.88%    | 0.7932         | 0.5139         | 1.24%    | 0.7654         | 0.5118         | 1.64%    |
|           | AutoInt                    | 0.8509         | 0.4013         | 1.07%    | 0.7953         | 0.5118         | 0.97%    | 0.7716         | 0.5071         | 0.82%    |
|           | DCNv2                      | <u>0.8548</u>  | 0.3893         | 0.61%    | 0.7956         | <u>0.5103</u>  | 0.93%    | 0.7724         | 0.5057         | 0.72%    |
|           | FLIP <sub>ID</sub> (Ours)  | <b>0.8600*</b> | <b>0.3802*</b> | -        | <b>0.8030*</b> | <b>0.5043*</b> | -        | <b>0.7779*</b> | <b>0.5014*</b> | -        |
| PLM-based | CTR-BERT                   | 0.8304         | <u>0.4131</u>  | 1.88%    | 0.7795         | 0.5300         | 1.65%    | 0.7385         | 0.5316         | 1.23%    |
|           | P5                         | 0.8304         | 0.4173         | 1.88%    | 0.7801         | <u>0.5261</u>  | 1.58%    | 0.7365         | 0.5336         | 1.51%    |
|           | PTab                       | <u>0.8426</u>  | 0.4195         | 0.41%    | <u>0.7880</u>  | 0.5384         | 0.56%    | <u>0.7456</u>  | <u>0.5268</u>  | 0.27%    |
|           | FLIP <sub>PLM</sub> (Ours) | <b>0.8460*</b> | <b>0.4127*</b> | -        | <b>0.7924*</b> | <b>0.5304*</b> | -        | <b>0.7476*</b> | <b>0.5255*</b> | -        |
| ID+PLM    | CTRL                       | <u>0.8572</u>  | <u>0.3838</u>  | 0.57%    | 0.7985         | 0.5101         | 0.95%    | <u>0.7741</u>  | <u>0.5045</u>  | 0.59%    |
|           | MoRec                      | 0.8561         | 0.3896         | 0.70%    | <u>0.7990</u>  | <u>0.5087</u>  | 0.89%    | 0.7731         | 0.5085         | 0.72%    |
|           | FLIP (Ours)                | <b>0.8621*</b> | <b>0.3788*</b> | -        | <b>0.8061*</b> | <b>0.5004*</b> | -        | <b>0.7787*</b> | <b>0.5001*</b> | -        |

- **Practical Values:** FLIP's fine-grained alignment (joint MLM+MTM) outperforms instance-level methods, enables meaningful feature-level interactions, and enhances diverse ID-based models (DeepFM, AutoInt, DCNv2) and PLMs of varying sizes.

- Background
  - Sequential recommender systems (SRS) predict user preferences through interaction history using learned item embeddings
- Motivation
  - SRS models lack semantic understanding of textual item attributes.
  - LLMs have strong semantics but poor item-level distinction for recommendations.
- Key Stages
  - Supervised Contrastive Fine-Tuning (SCFT)
  - Recommendation Adaptation Training (RAT)

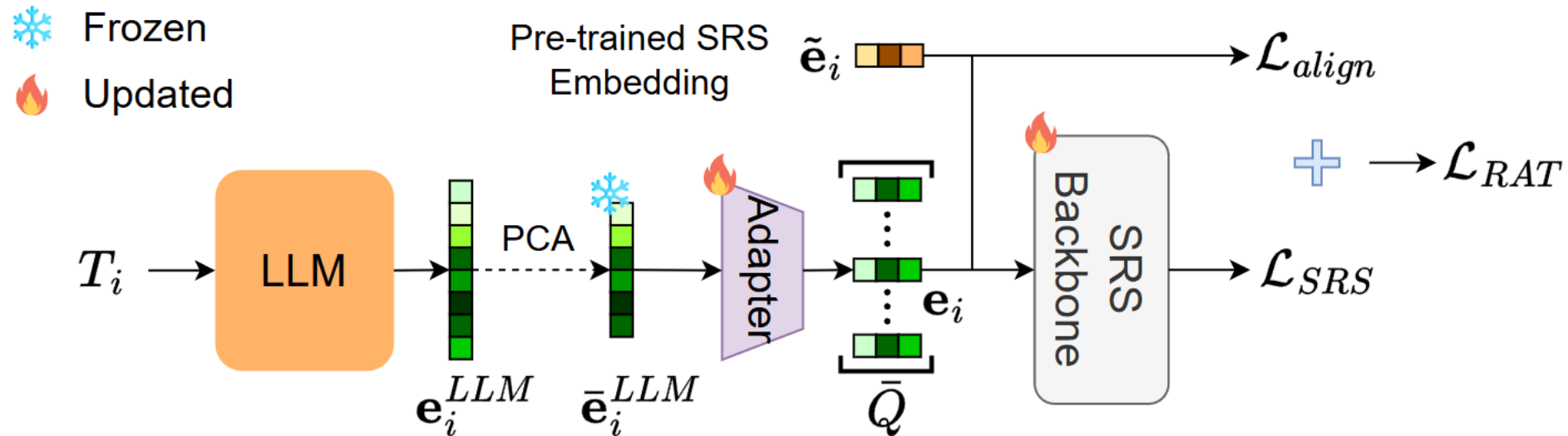


Stage 1: Supervised Contrastive Fine-Tuning (SCFT)

- **Build prompts for each item**
  - Includes instruction + attribute-value pairs.
- **Data Augmentation**
  - Randomly drop attributes to create two prompt views per item (as positive pairs).
- **Contrastive Learning**
  - Use contrastive loss to bring same-item embeddings closer, push different-item embeddings apart.
- **Outcome**
  - LLM learns to encode subtle attribute differences, providing stronger item representations.



## Stage 2: Recommendation Adaptation Training (RAT)



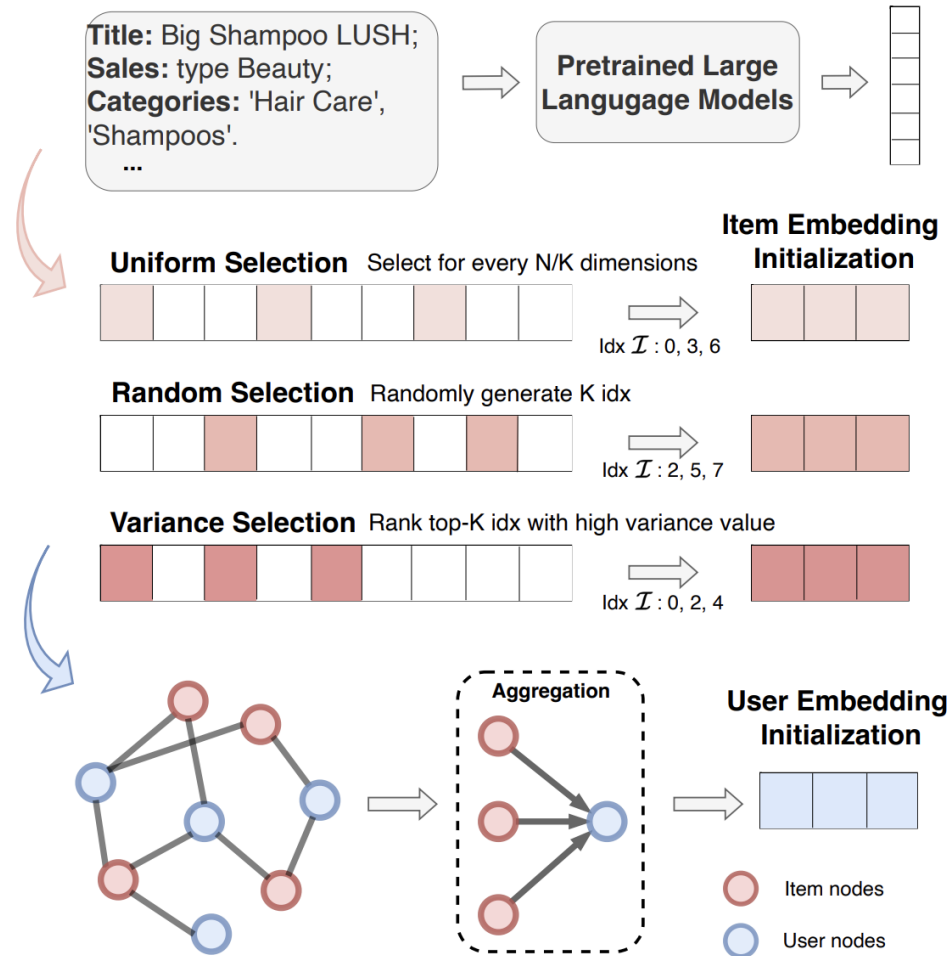
- **Embedding Transformation:** Use PCA to reduce dimensionality, then an adapter (two-layer MLP) to match the SRS size.
- **Adaptation:** Freeze LLM, train the adapter and SRS backbone together using SRS loss.
- **Collaborative Alignment:** Align adapter output with embeddings from a trained SRS, using contrastive loss to prevent overfitting.

- **LLMEmb** outperforms all baselines on Yelp, Amazon Beauty, and Fashion, especially improving recommendations for less popular (long-tail) items.

| Backbone Model |                 | Yelp           |                |                |                | Fashion        |                |                |                | Beauty         |                |                |                |
|----------------|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|                |                 | Overall        |                | Tail           |                | Overall        |                | Tail           |                | Overall        |                | Tail           |                |
|                |                 | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           |
| GRU4Rec        | - None          | 0.4879         | 0.2751         | 0.0171         | 0.0059         | 0.4798         | 0.3809         | 0.0257         | 0.0101         | 0.3683         | 0.2276         | 0.0796         | 0.0567         |
|                | - MELT          | <u>0.4985</u>  | <u>0.2825</u>  | 0.0201         | 0.0079         | 0.4884         | 0.3975         | 0.0291         | 0.0112         | 0.3702         | 0.2161         | 0.0009         | 0.0003         |
|                | - LLM2X         | 0.4872         | 0.2749         | 0.0201         | 0.0072         | 0.4881         | 0.4100         | 0.0264         | 0.0109         | 0.4151         | <u>0.2713</u>  | 0.0896         | 0.0637         |
|                | - SAID          | 0.4891         | 0.2764         | 0.0180         | 0.0062         | <u>0.4920</u>  | <u>0.4168</u>  | <u>0.0347</u>  | <u>0.0151</u>  | <u>0.4193</u>  | 0.2621         | <u>0.0936</u>  | <u>0.0661</u>  |
|                | - TSLRec        | 0.4528         | 0.2509         | <u>0.0255</u>  | <u>0.0095</u>  | 0.4814         | 0.4042         | 0.0149         | 0.0071         | 0.3119         | 0.1865         | 0.0750         | 0.0474         |
|                | - <b>LLMEmb</b> | <b>0.5270*</b> | <b>0.2980*</b> | <b>0.1116*</b> | <b>0.0471*</b> | <b>0.5062*</b> | <b>0.4329*</b> | <b>0.1046*</b> | <b>0.0477*</b> | <b>0.4445*</b> | <b>0.2726</b>  | <b>0.3183*</b> | <b>0.1793*</b> |
| Bert4Rec       | - None          | 0.5307         | 0.3035         | 0.0115         | 0.0044         | 0.4668         | 0.3613         | 0.0142         | 0.0067         | 0.3984         | 0.2367         | 0.0101         | 0.0038         |
|                | - MELT          | <u>0.6206</u>  | 0.3770         | 0.0429         | 0.0149         | 0.4897         | 0.3810         | 0.0059         | 0.0019         | 0.4716         | 0.2965         | 0.0709         | 0.0291         |
|                | - LLM2X         | 0.6199         | <u>0.3781</u>  | 0.0874         | 0.0330         | 0.5109         | <u>0.4159</u>  | 0.0377         | 0.0169         | 0.5029         | 0.3209         | 0.0927         | 0.0451         |
|                | - SAID          | 0.6156         | 0.3732         | <u>0.0973</u>  | <u>0.0382</u>  | <u>0.5135</u>  | 0.4124         | <u>0.0694</u>  | <u>0.0433</u>  | <u>0.5127</u>  | <u>0.3360</u>  | <u>0.1124</u>  | <u>0.0664</u>  |
|                | - TSLRec        | 0.6069         | 0.3680         | 0.0969         | <u>0.0388</u>  | 0.5078         | 0.4143         | 0.0418         | 0.0182         | 0.4936         | 0.3178         | 0.1013         | 0.0589         |
|                | - <b>LLMEmb</b> | <b>0.6294*</b> | <b>0.3881*</b> | <b>0.1876*</b> | <b>0.1094*</b> | <b>0.5244*</b> | <b>0.4238*</b> | <b>0.1485*</b> | <b>0.0764*</b> | <b>0.5247*</b> | <b>0.3485*</b> | <b>0.2430*</b> | <b>0.1224*</b> |
| SASRec         | - None          | 0.5940         | 0.3597         | 0.1142         | 0.0495         | 0.4956         | 0.4429         | 0.0454         | 0.0235         | 0.4388         | 0.3030         | 0.0870         | 0.0649         |
|                | - MELT          | 0.6257         | 0.3791         | 0.1015         | 0.0371         | 0.4875         | 0.4150         | 0.0368         | 0.0144         | 0.4334         | 0.2775         | 0.0460         | 0.0172         |
|                | - LLM2X         | <u>0.6415</u>  | <u>0.3997</u>  | <u>0.1760</u>  | <u>0.0789</u>  | 0.5210         | 0.4486         | 0.0768         | 0.0473         | 0.5043         | 0.3319         | <u>0.1608</u>  | <u>0.0940</u>  |
|                | - SAID          | 0.6277         | 0.3841         | 0.1548         | 0.0669         | <u>0.5316</u>  | <u>0.4619</u>  | <u>0.0901</u>  | <u>0.0540</u>  | <u>0.5097</u>  | 0.3343         | 0.1549         | 0.0906         |
|                | - TSLRec        | 0.6152         | 0.3795         | 0.1383         | 0.0620         | 0.5125         | 0.4594         | 0.0652         | 0.0382         | 0.4977         | <u>0.3366</u>  | 0.1211         | 0.0789         |
|                | - <b>LLMEmb</b> | <b>0.6647*</b> | <b>0.4113*</b> | <b>0.2951*</b> | <b>0.1456*</b> | <b>0.5521*</b> | <b>0.4730*</b> | <b>0.1513*</b> | <b>0.0826*</b> | <b>0.5277*</b> | <b>0.3460*</b> | <b>0.4194*</b> | <b>0.2595*</b> |

- **Practical Values:** LLMEmb's SCFT+RAT framework bridges semantic understanding with collaborative filtering, enables efficient precomputed embeddings, and enhances sequential recommenders without runtime overhead.

- Background
  - Collaborative filtering models predict user preferences through interaction patterns but face cold-start and embedding collapse challenges.
- Motivation
  - CF models capture interaction patterns but lack semantic understanding.
  - LLMs provide rich semantics but struggle with item-level distinctions.
- Key Stages
  - Selective Sampling
  - Efficient Initialization



- **LLMInit** bridges the LLM-CF embedding gap using selective initialization with sampling strategies to transfer knowledge from large LLM embeddings to lightweight CF models.
- **Selective Sampling**
  - Extract K-dimensional embeddings from LLM representations using variance-based and random selection strategies.
- **Efficient Initialization**
  - Initialize CF models with LLM embeddings to inherit semantic knowledge while maintaining scalability.

- **LLMInit** significantly outperforms baseline CF models across all datasets (Beauty, Toys-Games, Tools-Home, Office-Products).

| Method              | Beauty                 |                        | Toys-Games             |                        | Tools-Home             |                        | Office-Products        |                        |
|---------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
|                     | Recall@10              | NDCG@10                | Recall@10              | NDCG@10                | Recall@10              | NDCG@10                | Recall@10              | NDCG@10                |
| <b>LightGCN</b> [7] | 0.0910                 | 0.0432                 | 0.0775                 | 0.0360                 | 0.0574                 | 0.0283                 | 0.0745                 | 0.0365                 |
| +LLMInit-Rand       | 0.0960 (+5.5%)         | 0.0467 (+8.1%)         | 0.0805 (+3.9%)         | 0.0387 (+7.50%)        | 0.0612 (+6.6%)         | 0.0313 (+10.6%)        | 0.0773 (+3.8%)         | 0.0387 (+6.0%)         |
| +LLMInit-Uni        | 0.1006 (+10.6%)        | 0.0469 (+8.6%)         | 0.0806 (+4.0%)         | 0.0388 (+7.8%)         | 0.0633 (+10.3%)        | <b>0.0319</b> (+12.7%) | 0.0791 (+6.2%)         | 0.0395 (+8.2%)         |
| +LLMInit-Var        | <b>0.1019</b> (+12.0%) | <b>0.0485</b> (+12.3%) | <b>0.0808</b> (+4.3%)  | <b>0.0389</b> (+8.1%)  | <b>0.0633</b> (+10.3%) | 0.0317 (+12.0%)        | <b>0.0816</b> (+9.5%)  | <b>0.0414</b> (+13.4%) |
| <b>SGL</b> [23]     | 0.1017                 | 0.0474                 | 0.0832                 | 0.0380                 | 0.0580                 | 0.0284                 | 0.0669                 | 0.0297                 |
| +LLMInit-Rand       | 0.1069 (+5.1%)         | 0.0520 (+9.7%)         | 0.0885 (+6.4%)         | 0.0418 (+10.0%)        | <b>0.0692</b> (+19.3%) | 0.0337 (+18.7%)        | <b>0.0810</b> (+21.1%) | <b>0.0426</b> (+43.4%) |
| +LLMInit-Uni        | 0.1101 (+8.3%)         | 0.0513 (+8.2%)         | 0.0920 (+10.6%)        | 0.0424 (+11.6%)        | 0.0676 (+16.6%)        | 0.0333 (+17.3%)        | 0.0773 (+15.6%)        | 0.0350 (+17.9%)        |
| +LLMInit-Var        | <b>0.1106</b> (+8.8%)  | <b>0.0530</b> (+11.8%) | <b>0.0927</b> (+11.4%) | <b>0.0427</b> (+12.4%) | 0.0686 (+18.3%)        | <b>0.0339</b> (+19.4%) | 0.0794 (+18.7%)        | 0.0421 (+41.8%)        |
| <b>SGCL</b> [32]    | 0.1027                 | 0.0499                 | 0.0828                 | 0.0382                 | 0.0585                 | 0.0294                 | 0.0647                 | 0.0298                 |
| +LLMInit-Rand       | 0.1094 (+6.5%)         | 0.0512 (+2.6%)         | 0.0929 (+12.2%)        | 0.0418 (+9.4%)         | <b>0.0651</b> (+11.3%) | 0.0326 (+10.9%)        | 0.0770 (+19.0%)        | 0.0365 (+22.5%)        |
| +LLMInit-Uni        | <b>0.1115</b> (+8.6%)  | 0.0513 (+2.8%)         | 0.0923 (+11.5%)        | <b>0.0422</b> (+10.5%) | 0.0650 (+11.1%)        | 0.0327 (+11.2%)        | 0.0742 (+14.7%)        | 0.0354 (+18.8%)        |
| +LLMInit-Var        | 0.1104 (+7.5%)         | <b>0.0522</b> (+4.6%)  | <b>0.0941</b> (+13.6%) | 0.0421 (+10.2%)        | 0.0646 (+10.4%)        | <b>0.0327</b> (+11.2%) | <b>0.0776</b> (+19.9%) | <b>0.0366</b> (+22.8%) |

- **Practical Values:** LLMInit provides a "free lunch" approach for enhancing CF models with LLM semantic knowledge while maintaining high efficiency and scalability for real-world deployment.



- Background
  - Real-world LLM deployment in recommendation is hindered by the Domain Gap and Catastrophic Forgetting.
- Motivation
  - Domain Gap: Mismatch between LLM's open-world knowledge and recommendations' collaborative knowledge.
  - Catastrophic Forgetting: Fine-tuning LLMs to their full extent causes them to forget pre-trained knowledge.
- Key Components
  - Content-Embedding Generation (CEG)
  - Preference Comprehension (PCH)

# Embedding - LEARN

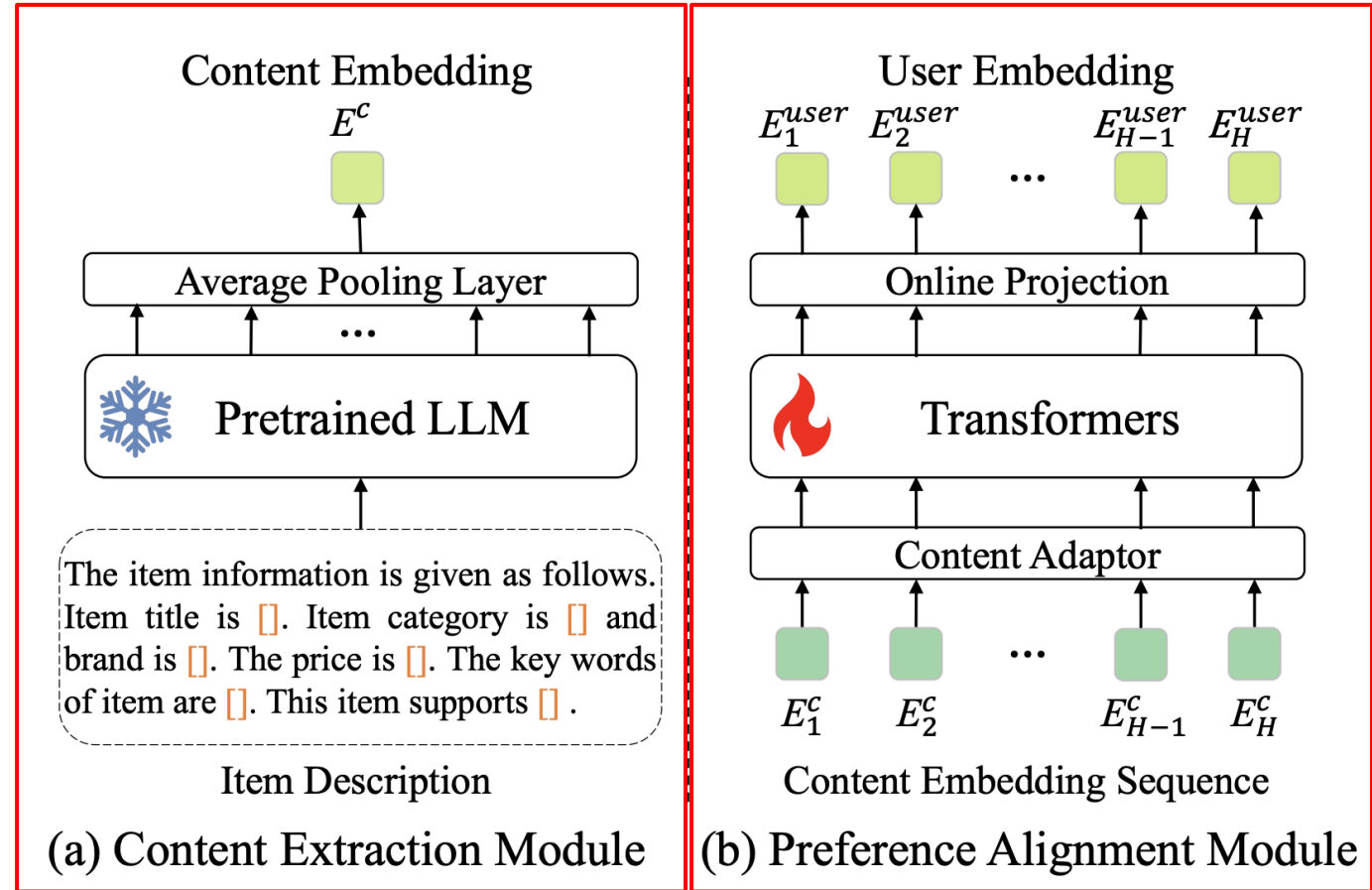
**LEARN** is a twin-tower framework that adapts LLM knowledge by freezing the LLM and using self-supervised training.

- **Content-Embedding Generation**

- Uses a frozen LLM as an item encoder to convert text into rich content embeddings. Freezing is key to preventing catastrophic forgetting.

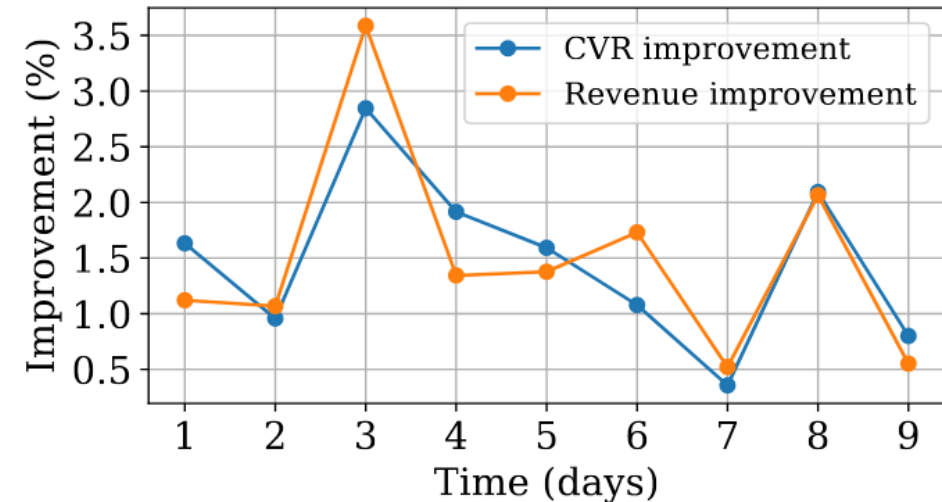
- **Preference Comprehension**

- A lightweight Transformer learns user preferences from item embeddings via contrastive learning.



- **LEARN** achieves SOTA performance on large-scale offline datasets and delivers significant business impact in real-world online A/B tests on Kuaishou.

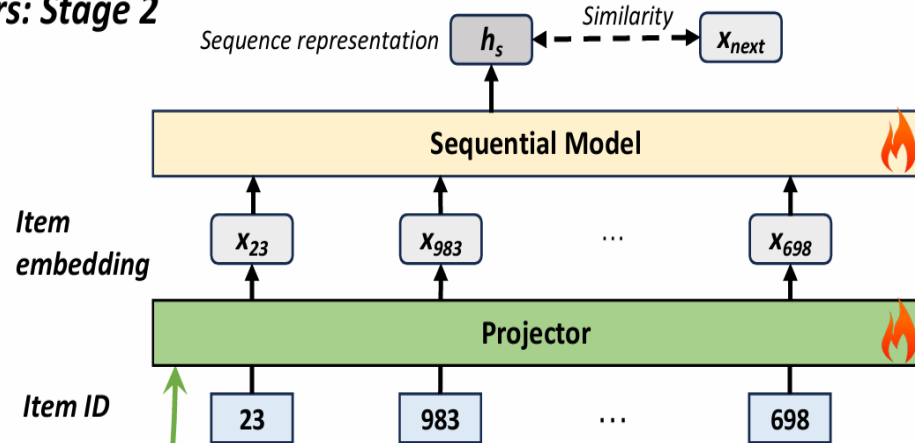
| Method       | H@10                    | H@50                    | H@200                   |
|--------------|-------------------------|-------------------------|-------------------------|
| SASRec [20]  | 0.0306                  | 0.0754                  | 0.1431                  |
| HSTU [43]    | <b>0.0416</b> (+35.95%) | 0.0957 (+26.92%)        | 0.1735 (+21.24%)        |
| LEARN (Ours) | 0.0407 (+33.01%)        | <b>0.0979</b> (+29.84%) | <b>0.1874</b> (+30.96%) |
| –            | N@10                    | N@50                    | N@200                   |
| SASRec [20]  | 0.0164                  | 0.0260                  | 0.0362                  |
| HSTU [43]    | <b>0.0227</b> (+38.41%) | 0.0344 (+32.31%)        | 0.0461 (+27.35%)        |
| LEARN (Ours) | 0.0224 (+36.59%)        | <b>0.0371</b> (+42.69%) | <b>0.0483</b> (+33.43%) |



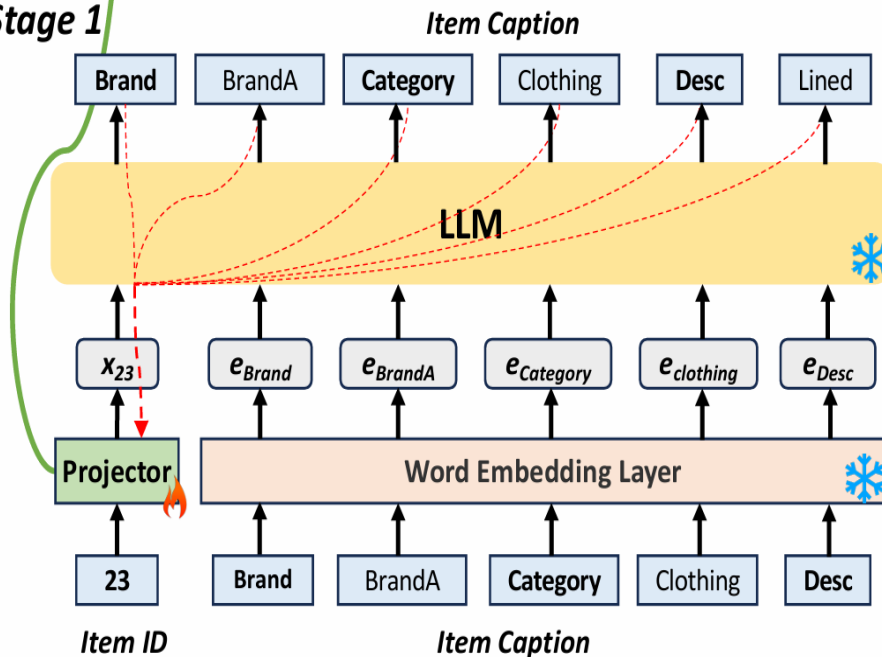
- **Practical Values:** LEARN is the first LLM-enhanced solution successfully deployed and monetized in a large-scale industrial recommender system, setting a new standard for practical application.

- Background
  - LLMs enhance sequential recommendation models with their generalization ability and knowledge base.
- Motivation
  - LLM embeddings may lose fine-grained item information
  - Long token sequences cause efficiency issues
- Key Stages
  - Semantically Aligned Embedding Learning
  - Model-agnostic Sequential Recommender Training

## Ours: Stage 2



## Ours: Stage 1



SAID learns item embeddings aligned with LLM text descriptions, usable with lightweight sequential models.

- **Semantically Aligned Embedding Learning**
  - SAID learns to generate an embedding for each item by leveraging the projector module and an on-the-shelf LLM.
- **Model-agnostic Sequential Recommender Training**
  - The embeddings acquired during the first stage are utilized as initial features of the items, which are then inputted into a downstream model for sequential recommendation.



# Embedding - SAID

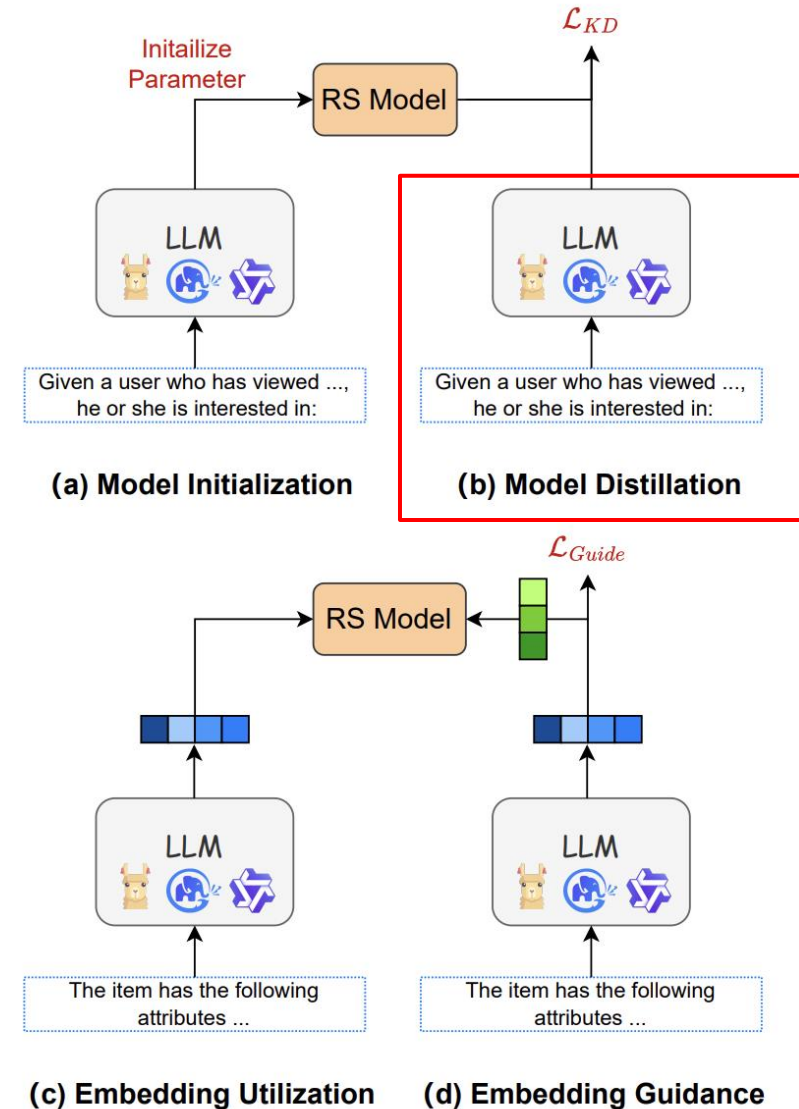
- Experiments conducted on public datasets above and Alipay's online advertising deployment justify the efficiency and efficacy of SAID.

| Dataset     | Metrics   | GRU4Rec |               |               |         | SASRec        |               |               |         |
|-------------|-----------|---------|---------------|---------------|---------|---------------|---------------|---------------|---------|
|             |           | Random  | LH            | SAID          | Improv. | Random        | LH            | SAID          | Improv. |
| Scientific  | NDCG@10   | 0.0896  | <u>0.0958</u> | <b>0.1021</b> | 6.6%    | 0.0869        | <u>0.0992</u> | <b>0.1050</b> | 5.8%    |
|             | Recall@10 | 0.1037  | <u>0.1317</u> | <b>0.1321</b> | 0.3%    | 0.1088        | <u>0.1313</u> | <b>0.1353</b> | 3.0%    |
|             | MRR       | 0.0877  | <u>0.0913</u> | <b>0.0977</b> | 7.0%    | 0.0825        | <u>0.0945</u> | <b>0.1005</b> | 6.3%    |
| Instruments | NDCG@10   | 0.0760  | <u>0.0774</u> | <b>0.0862</b> | 11.4%   | 0.0768        | <u>0.0837</u> | <b>0.0928</b> | 10.9%   |
|             | Recall@10 | 0.0910  | <u>0.1085</u> | <b>0.1122</b> | 3.4%    | 0.0920        | <u>0.1158</u> | <b>0.1211</b> | 4.6%    |
|             | MRR       | 0.0741  | <u>0.0741</u> | <b>0.0832</b> | 12.3%   | 0.0750        | <u>0.0802</u> | <b>0.0896</b> | 11.7%   |
| Arts        | NDCG@10   | 0.0878  | <u>0.1057</u> | <b>0.1180</b> | 11.6%   | <u>0.1017</u> | 0.0995        | <b>0.1090</b> | 7.2%    |
|             | Recall@10 | 0.0978  | <b>0.1441</b> | 0.1413        | -       | 0.1302        | <u>0.1433</u> | <b>0.1487</b> | 3.8%    |
|             | MRR       | 0.0866  | <u>0.0993</u> | <b>0.1145</b> | 15.3%   | <u>0.0951</u> | 0.0910        | <b>0.1017</b> | 6.9%    |
| Office      | NDCG@10   | 0.1127  | <u>0.1136</u> | <b>0.1202</b> | 5.8%    | 0.1084        | <u>0.1134</u> | <b>0.1208</b> | 6.5%    |
|             | Recall@10 | 0.1285  | <u>0.1374</u> | <b>0.1440</b> | 4.8%    | 0.1265        | <u>0.1377</u> | <b>0.1450</b> | 5.3%    |
|             | MRR       | 0.1097  | <u>0.1098</u> | <b>0.1160</b> | 5.6%    | 0.1047        | <u>0.1092</u> | <b>0.1165</b> | 6.7%    |
| Games       | NDCG@10   | 0.0641  | <u>0.0748</u> | <b>0.0785</b> | 4.9%    | 0.0673        | <u>0.0752</u> | <b>0.0812</b> | 8.0%    |
|             | Recall@10 | 0.0877  | <b>0.1221</b> | 0.1173        | -       | 0.0936        | <b>0.1221</b> | 0.1204        | -       |
|             | MRR       | 0.0617  | <u>0.0694</u> | <b>0.0741</b> | 6.8%    | 0.0647        | <u>0.0696</u> | <b>0.0770</b> | 10.6%   |
| Pet         | NDCG@10   | 0.0854  | <u>0.0925</u> | <b>0.0960</b> | 3.8%    | 0.0878        | <u>0.0881</u> | <b>0.0951</b> | 7.9%    |
|             | Recall@10 | 0.0945  | <u>0.1120</u> | <b>0.1152</b> | 2.9%    | 0.0978        | <u>0.1062</u> | <b>0.1129</b> | 6.3%    |
|             | MRR       | 0.0843  | <u>0.0902</u> | <b>0.0934</b> | 3.5%    | <u>0.0866</u> | 0.0859        | <b>0.0929</b> | 7.3%    |

- Practical Values:** SAID significantly enhances inference efficiency and recommendation accuracy, making large language models practically applicable in industrial recommendation scenarios.

# Model Distillation

- This subcategory refers to compressing LLM knowledge into smaller RS models through distillation techniques.
- This approach transfers LLM capabilities to lightweight models while maintaining recommendation quality for efficient deployment.
- Categories
  - Feature-based
  - Response-based



- Background
  - Medication recommendation provides prescription suggestions to doctors. LLMs have two advantages: medical semantic understanding and cold-start capabilities.
- Motivation
  - Complex medication names cause LLMs to output recommendations not in drug databases, leading to failures.
  - Resource-intensive LLM inference challenges resource-constrained medical institutions in using LLMs for drug recommendations.
- Key Stages
  - Improve for LLMs
  - Distilling LLMs

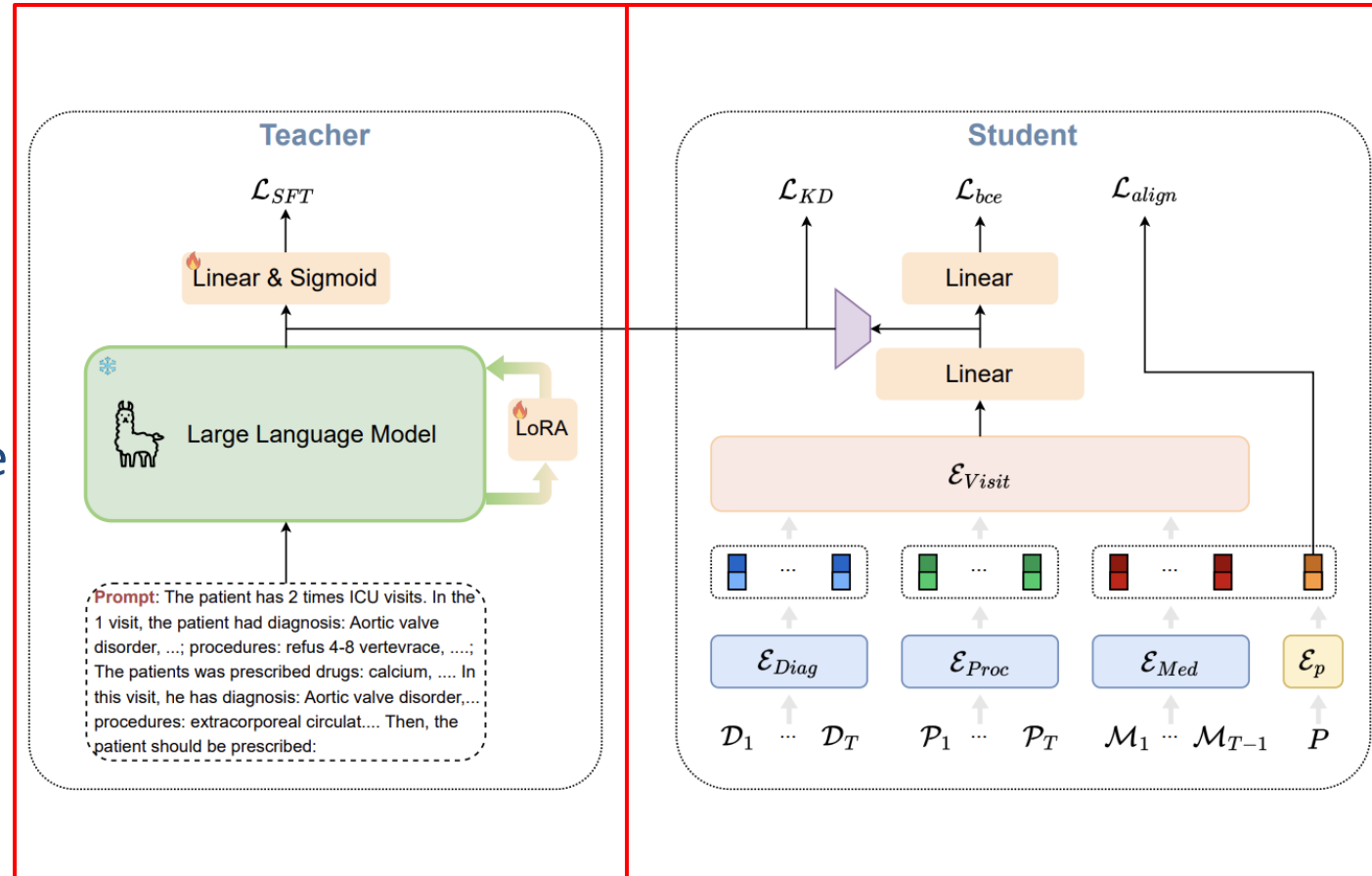
**LEADER** improved the recommendation performance of LLMs by enhancing the output layer.

- **Improve for LLMs**

- Replace the original text generation head layer of the large model with a new classification layer, and fine-tune the model using BCE loss.

- **Distilling LLMs**

- Design the distillation process where the hidden states from the final layer of the LLMs are used as guidance to distill semantic understanding into the designed smaller model.



- **LEADER** significantly outperforms baseline. The distilled smaller model demonstrates excellent performance in both overall and cold-start scenarios.

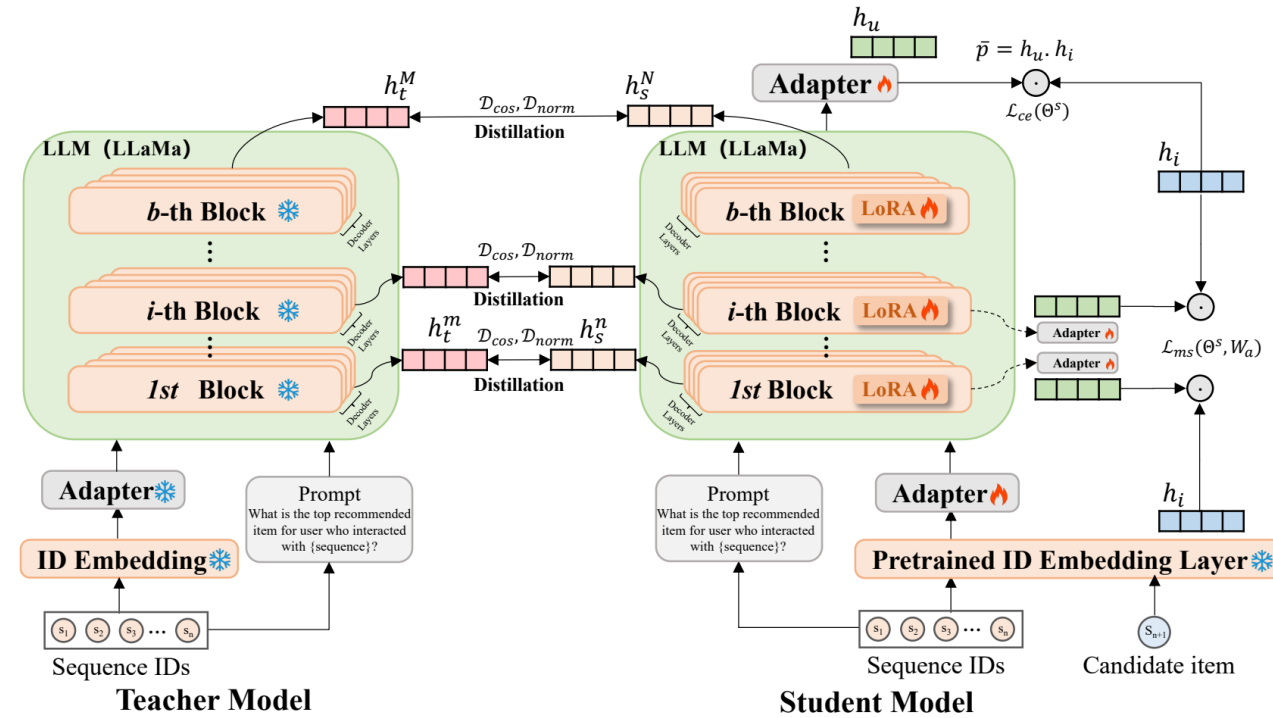
| Model            | Overall                                |                                        |                                        | Multi-visit                            |                                        |                                        | Single-visit                           |                                        |                                        |
|------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|----------------------------------------|
|                  | PRAUC                                  | Jaccard                                | F1                                     | PRAUC                                  | Jaccard                                | F1                                     | PRAUC                                  | Jaccard                                | F1                                     |
| RETAIN           | 0.7513 $\pm$ 0.0025                    | 0.4943 $\pm$ 0.0023                    | 0.6516 $\pm$ 0.0022                    | 0.7580 $\pm$ 0.0020                    | 0.5106 $\pm$ 0.0023                    | 0.6674 $\pm$ 0.0022                    | 0.7337 $\pm$ 0.0067                    | 0.4811 $\pm$ 0.0053                    | 0.6403 $\pm$ 0.0049                    |
| G-Bert           | -                                      | -                                      | -                                      | 0.6904 $\pm$ 0.0017                    | 0.4578 $\pm$ 0.0019                    | 0.6186 $\pm$ 0.0018                    | -                                      | -                                      | -                                      |
| GAMENet          | 0.7605 $\pm$ 0.0011                    | 0.5024 $\pm$ 0.0010                    | 0.6595 $\pm$ 0.0008                    | 0.7638 $\pm$ 0.0023                    | 0.5070 $\pm$ 0.0028                    | 0.6635 $\pm$ 0.0025                    | 0.7451 $\pm$ 0.0053                    | 0.4840 $\pm$ 0.0038                    | 0.6442 $\pm$ 0.0036                    |
| SafeDrug         | 0.7582 $\pm$ 0.0020                    | 0.5054 $\pm$ 0.0024                    | 0.6621 $\pm$ 0.0021                    | 0.7623 $\pm$ 0.0029                    | 0.5095 $\pm$ 0.0027                    | 0.6658 $\pm$ 0.0024                    | 0.7416 $\pm$ 0.0044                    | 0.4900 $\pm$ 0.0043                    | 0.6481 $\pm$ 0.0042                    |
| MICRON           | -                                      | -                                      | -                                      | 0.7651 $\pm$ 0.0027                    | 0.5110 $\pm$ 0.0025                    | 0.6741 $\pm$ 0.0023                    | -                                      | -                                      | -                                      |
| COGNet           | -                                      | -                                      | -                                      | 0.7771 $\pm$ 0.0019                    | 0.5275 $\pm$ 0.0021                    | 0.6805 $\pm$ 0.0019                    | -                                      | -                                      | -                                      |
| REFINE           | -                                      | -                                      | -                                      | 0.7791 $\pm$ 0.0017                    | 0.5235 $\pm$ 0.0018                    | 0.6794 $\pm$ 0.0017                    | -                                      | -                                      | -                                      |
| <b>LEADER(T)</b> | <b>0.7816 <math>\pm</math> 0.0015*</b> | <b>0.5391 <math>\pm</math> 0.0015*</b> | <b>0.6921 <math>\pm</math> 0.0014*</b> | <b>0.7854 <math>\pm</math> 0.0015*</b> | <b>0.5450 <math>\pm</math> 0.0021*</b> | <b>0.6971 <math>\pm</math> 0.0018*</b> | 0.7590 $\pm$ 0.0046*                   | <b>0.5090 <math>\pm</math> 0.0044*</b> | <b>0.6668 <math>\pm</math> 0.0041*</b> |
| <b>LEADER(S)</b> | <b>0.7795 <math>\pm</math> 0.0025*</b> | <b>0.5175 <math>\pm</math> 0.0022*</b> | <b>0.6737 <math>\pm</math> 0.0019*</b> | <b>0.7830 <math>\pm</math> 0.0019</b>  | 0.5208 $\pm$ 0.0020                    | 0.6768 $\pm$ 0.0017                    | <b>0.7631 <math>\pm</math> 0.0056*</b> | <b>0.5038 <math>\pm</math> 0.0062*</b> | <b>0.6614 <math>\pm</math> 0.0057*</b> |

- **Practical Values:** LEADER can be used for recommending prescriptions for patients who first visit the hospital (cold-start scenarios).



- Background
  - Large Language Models (LLMs) have demonstrated exceptional performance in Sequential Recommendation (SR).
- Motivation
  - Intermediate layers of LLMs are largely redundant for SR tasks.
  - The massive computational and deployment costs make LLMs impractical for industrial applications.
- Key Stages
  - Layer-wise Feature Distillation
  - Dual-Loss Feature Alignment
  - Auxiliary Supervision

- **Layer-wise Feature Distillation**
  - Aligns intermediate hidden states to transfer rich feature-level knowledge, not just mimic final outputs.
- **Dual-Loss Feature Alignment**
  - Employs a dual-loss mechanism to match features in both direction and magnitude.
- **Auxiliary Supervision**
  - Applies early, task-specific supervision to shallow layers, enabling efficient learning in a compact model.



- SLMRec achieves state-of-the-art performance while reducing computational costs.

| Model                 | Cloth        |              |              |              | Movie        |              |              |              | Rank        |
|-----------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|-------------|
|                       | HR@1         | HR@5         | NDCG@5       | MRR          | HR@1         | HR@5         | NDCG@5       | MRR          |             |
| Caser                 | 9.66         | 15.18        | 12.66        | 13.03        | 4.27         | 14.96        | 9.57         | 10.36        | 13.50       |
| GRU4Rec               | 13.79        | 15.46        | 14.64        | 15.15        | 10.56        | 19.47        | 15.11        | 15.46        | 9.25        |
| BERT4Rec              | 13.60        | 14.66        | 14.14        | 14.59        | 9.68         | 14.91        | 12.40        | 12.74        | 11.63       |
| SASRec                | 13.08        | 16.94        | 15.01        | 15.76        | 5.57         | 16.80        | 11.17        | 12.08        | 11.63       |
| HGN                   | 15.96        | 18.70        | 17.30        | 18.27        | 7.54         | 19.20        | 13.42        | 14.73        | 6.50        |
| LightSANs             | 14.12        | 20.32        | 17.30        | 16.86        | 6.08         | 17.54        | 11.81        | 12.82        | 8.00        |
| S <sup>3</sup> -Rec   | 14.10        | 18.67        | 16.10        | 16.95        | 7.75         | 20.39        | 15.69        | 14.34        | 7.50        |
| DuoRec                | 13.06        | 18.29        | 15.79        | 15.42        | 10.07        | 20.37        | 17.96        | 16.61        | 7.88        |
| MAERec                | 13.29        | 18.35        | 15.68        | 16.13        | 8.89         | 20.24        | 16.03        | 15.28        | 8.38        |
| Open-P5               | 14.13        | 17.68        | 17.02        | -            | 12.66        | 21.98        | 17.13        | -            | 5.67        |
| E4SRec                | <b>16.71</b> | <b>19.45</b> | <b>18.09</b> | <b>18.77</b> | <b>14.74</b> | <b>23.79</b> | <b>19.45</b> | <b>19.74</b> | <b>1.75</b> |
| E4SRec <sub>8</sub>   | 15.30        | 18.54        | 16.91        | 17.60        | 13.32        | 22.49        | 17.99        | 18.46        | 4.00        |
| E4SRec <sub>4</sub>   | 14.58        | 18.05        | 16.32        | 17.01        | 11.80        | 21.54        | 16.73        | 17.20        | 5.75        |
| SLMRec <sub>4←8</sub> | <b>16.69</b> | <b>19.47</b> | <b>18.07</b> | <b>18.74</b> | <b>15.29</b> | <b>24.25</b> | <b>19.90</b> | <b>20.36</b> | <b>1.50</b> |

- **Practical Values:** SLMRec offers the industry a path to harness the power of LLMs in recommendation systems in an economical, efficient, and high-performance manner.

- Background
  - LLM-based recommenders face serious inference inefficiency issues, posing substantial challenges to their practical applications.
- Motivation
  - Teacher Knowledge Reliability: The teacher's knowledge isn't always reliable.
  - Model Capacity Gap: The student struggles to assimilate teacher knowledge.
  - Semantic Space Divergence: Different "languages" make direct alignment counterproductive.
- Key Stages
  - Importance-aware Ranking Distillation
  - Collaborative Embedding Distillation

# Response-based - DLLM2Rec

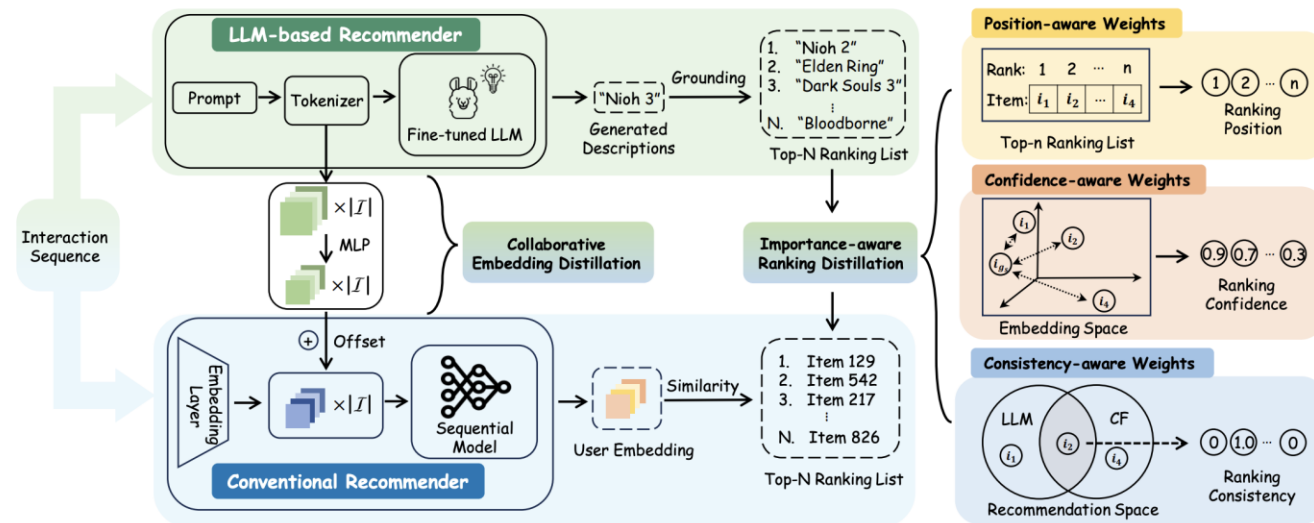
**DLLM2Rec** is designed to effectively distill knowledge from LLM-based recommenders to conventional recommenders.

- **Importance-aware Ranking Distillation**

- DLLM2Rec introduces importance weights to selectively learn reliable knowledge rather than blindly mimicking teacher rankings.

- **Collaborative Embedding Distillation**

- DLLM2Rec performs collaborative fusion to bridge semantic space divergence between teacher and student.





- **DLLM2Rec's** lightweight student model not only achieves a million-fold speedup in inference but also outperforms the massive LLM teacher on several datasets.

| Backbone | Model     | Games         |               | MovieLens     |               | Toys          |               |
|----------|-----------|---------------|---------------|---------------|---------------|---------------|---------------|
|          |           | HR@20         | NDCG@20       | HR@20         | NDCG@20       | HR@20         | NDCG@20       |
| Teacher  | BIGRec    | 0.0532        | 0.0341        | 0.0541        | 0.0370        | 0.0420        | 0.0207        |
|          | +None     | 0.0305        | 0.0150        | 0.0608        | 0.0236        | 0.0172        | 0.0081        |
|          | +Hint     | 0.0284        | 0.0120        | 0.0646        | 0.0240        | 0.0128        | 0.0058        |
|          | +HTD      | 0.0299        | 0.0128        | 0.0578        | 0.0229        | 0.0155        | 0.0062        |
|          | +RD       | 0.0398        | 0.0177        | 0.0667        | 0.0254        | 0.0157        | 0.0076        |
| GRU4Rec  | +CD       | 0.0306        | 0.0149        | 0.0699        | 0.0256        | 0.0126        | 0.0052        |
|          | +RRD      | 0.0359        | 0.0163        | 0.0657        | 0.0243        | 0.0215        | 0.0097        |
|          | +DCD      | 0.0427        | 0.0190        | 0.0666        | 0.0263        | 0.0262        | 0.0114        |
|          | +UnKD     | 0.0370        | 0.0170        | 0.0607        | 0.0226        | 0.0235        | 0.0114        |
|          | KAR       | 0.0307        | 0.0149        | 0.0603        | 0.0229        | 0.0184        | 0.0079        |
|          | LLM-CF    | 0.0393        | 0.0174        | 0.0677        | 0.0246        | 0.0132        | 0.0058        |
|          | +DLLM2Rec | <b>0.0446</b> | <b>0.0205</b> | <b>0.0815</b> | <b>0.0308</b> | <b>0.0281</b> | <b>0.0118</b> |
|          |           | <i>Gain.S</i> | +46.17%       | +36.94%       | +34.05%       | +30.43%       | +63.88%       |
|          |           | <i>Gain.B</i> | +4.56%        | +7.64%        | +16.60%       | +16.80%       | +7.40%        |

| Dataset   | Model       | HR@20   | NDCG@20 | Inference time       |
|-----------|-------------|---------|---------|----------------------|
| Games     | BIGRec      | 0.0532  | 0.0341  | $2.3 \times 10^4$ s  |
|           | DLLM2Rec    | 0.0751  | 0.0331  | 1.8s                 |
|           | <i>Gain</i> | +37.41% | -2.99%  | $+1.3 \times 10^6$ % |
| MovieLens | BIGRec      | 0.0541  | 0.0370  | $1.8 \times 10^4$ s  |
|           | DLLM2Rec    | 0.1063  | 0.0437  | 1.7s                 |
|           | <i>Gain</i> | +96.49% | +18.18% | $+1.1 \times 10^6$ % |
| Toys      | BIGRec      | 0.0420  | 0.0207  | $1.1 \times 10^4$ s  |
|           | DLLM2Rec    | 0.0463  | 0.0225  | 1.6s                 |
|           | <i>Gain</i> | +10.24% | +8.70%  | $+6.8 \times 10^5$ % |

- **Practical Values:** DLLM2Rec provides the industry with a path to integrate the powerful capabilities of large language models into existing recommendation systems in an economical, efficient, and feasible manner.

# Agenda

## 1 Introduction



Zijian Zhang



## 2 Knowledge Enhancement



Pengyue Jia



## 3 Interaction Enhancement



Ziwei Liu



## 4.1 Model Enhancement 1



Maolin Wang



## 4.2 Model Enhancement 2



Yuhao Wang



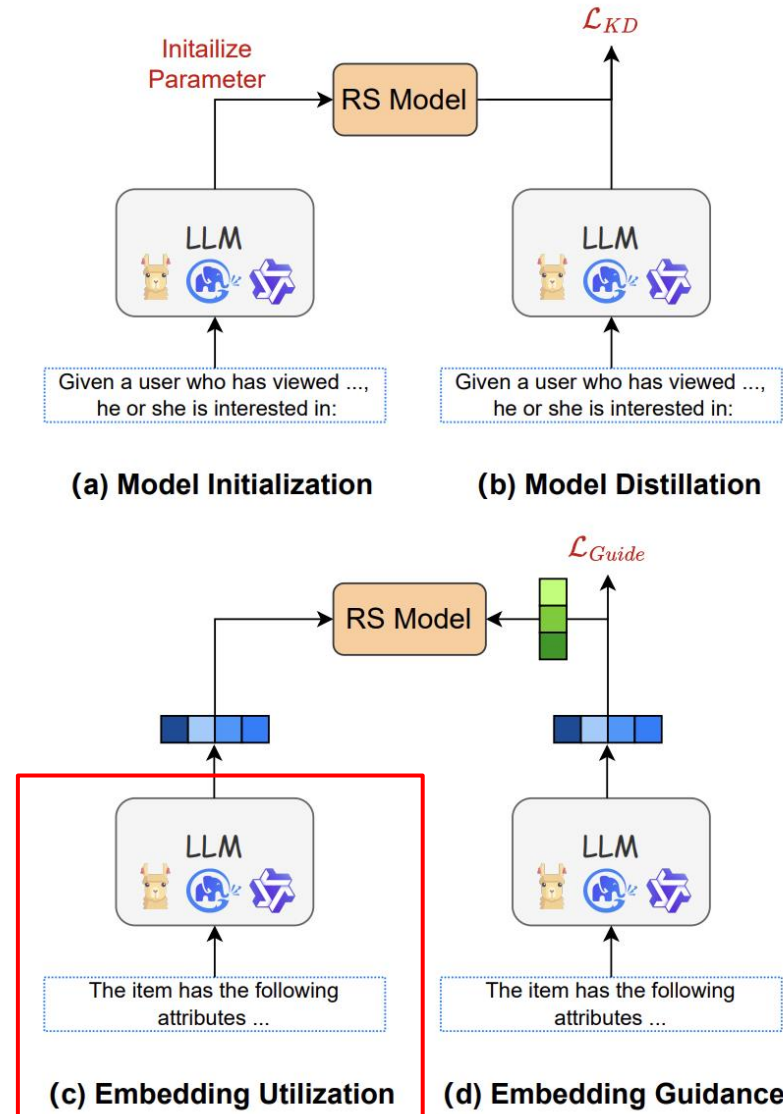
## 5 Conclusion



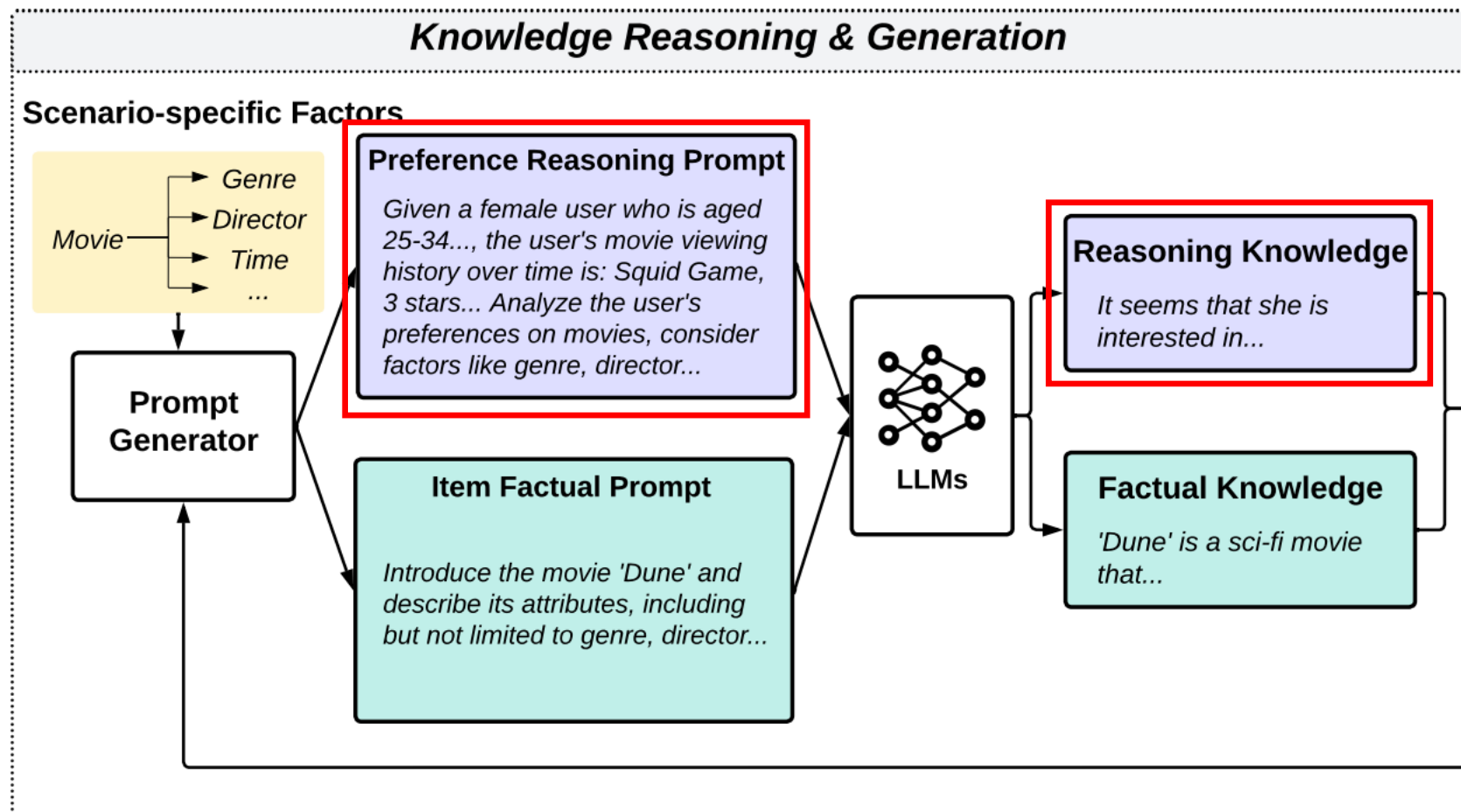
Yejing Wang

# Embedding Utilization

- This subcategory refers to utilizing the embeddings derived from LLM to enhance traditional RS as a semantic supplement.
- This approach tackles the deficiency that textual outputs are often difficult and inefficient to be integrated into RS directly.
- Categories
  - User Only
  - Item Only
  - User & Item



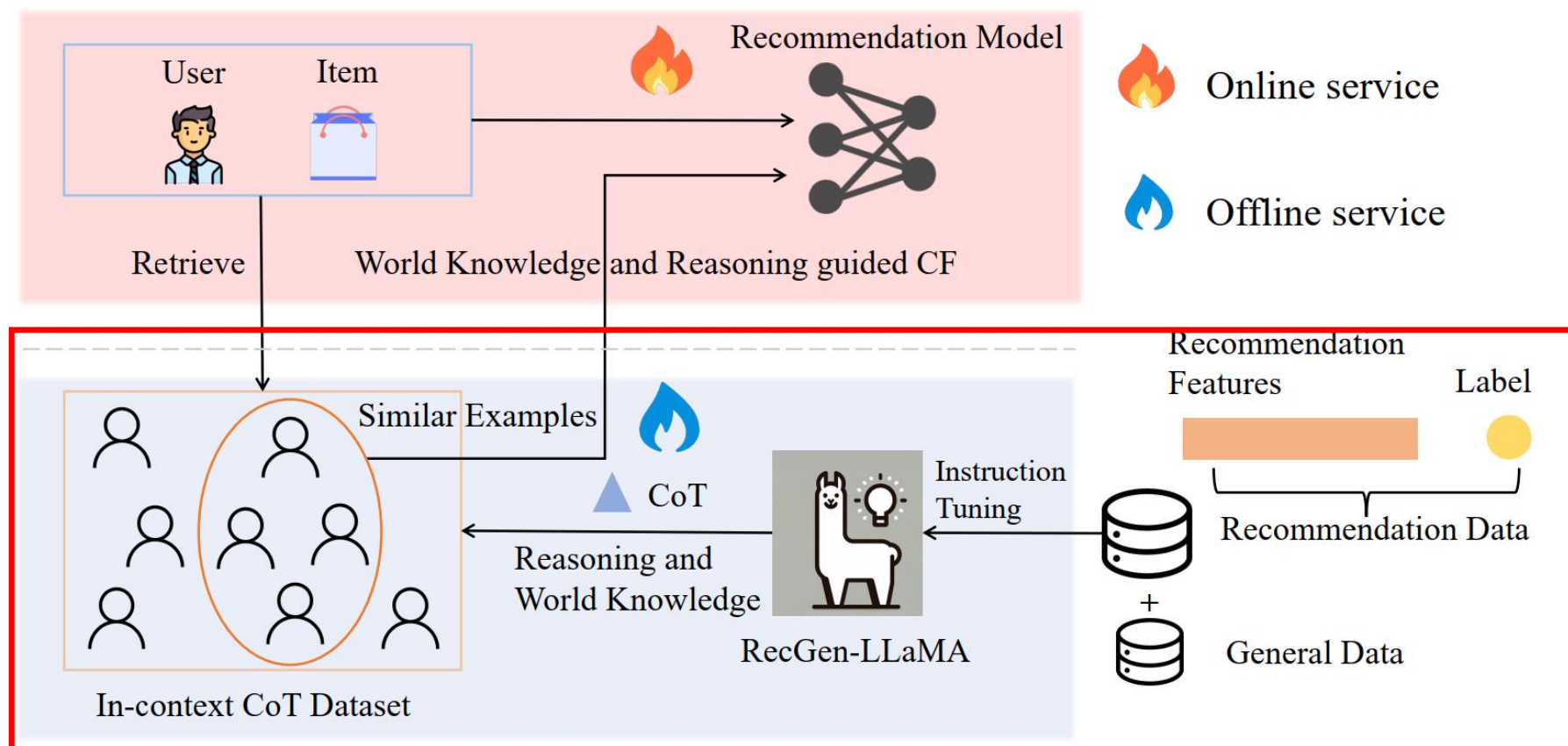
- Reasoning about user preferences given **user profile** and **interactions**





## Offline service

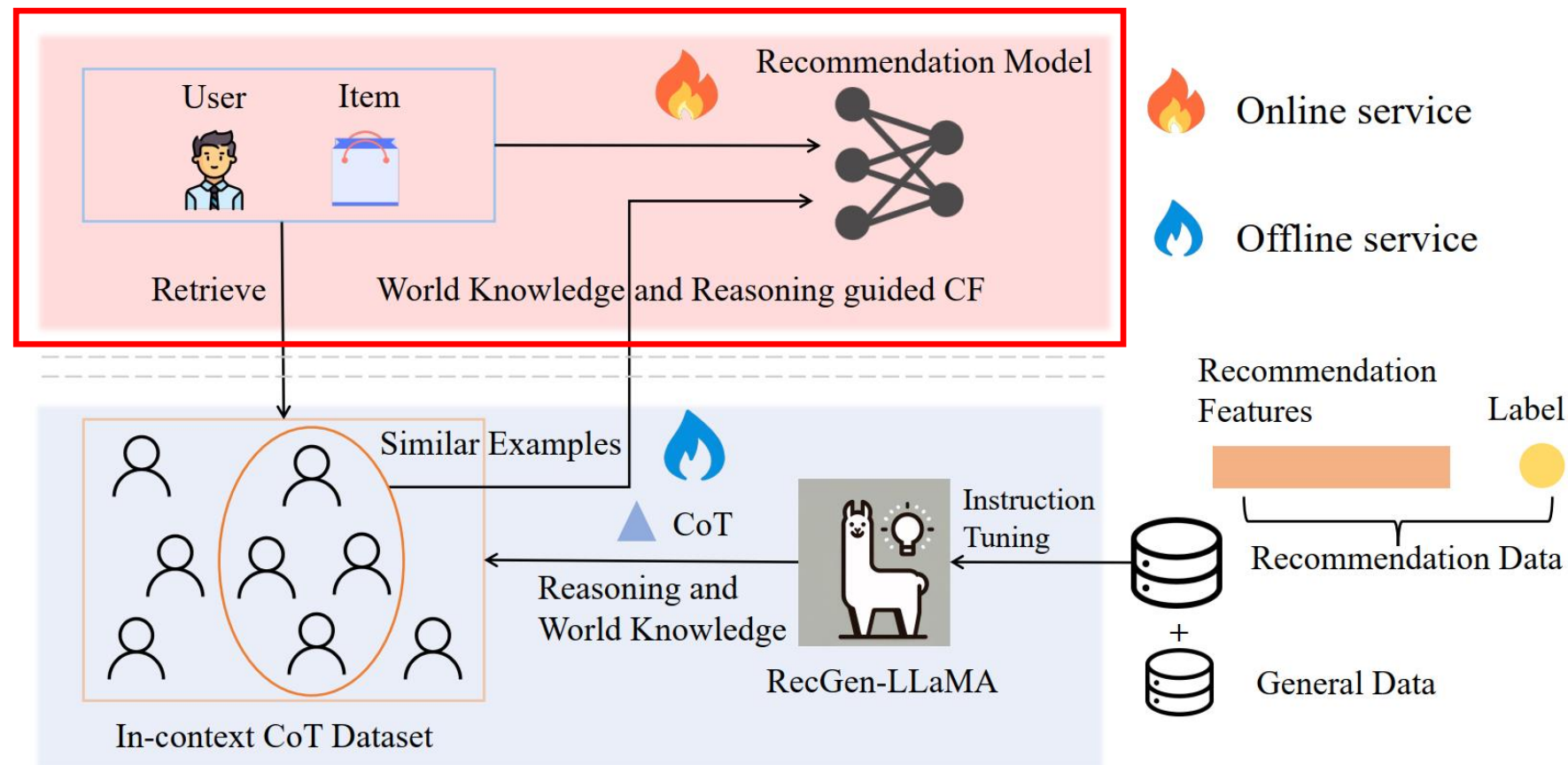
- Chain of Thought (CoT) reasoning
- In-context CoT dataset construction





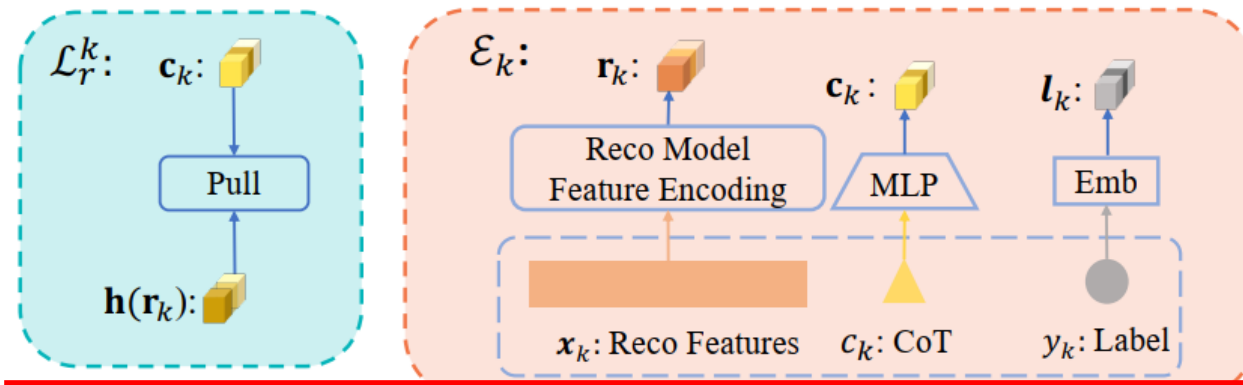
## • Online service

- In-context CoT examples Retrieval
- World knowledge & reasoning guided CF
- Feature enhanced RS



## CoT Prompt

- Generating CoT of the user's decision making on the target item



### Chain of Thought Prompt

<<SYS>> As an AI model developed for analyzing consumer behavior, your task is to generate a chain of thought that considers the following points:

1. Utilize the user's interaction history and review comments to summarize their profiles.
2. Introduce the target new item and detail its features precisely. In addition, integrate information about items related to the current target new item ...
3. Contemplate the alignment between the user's profile and the features of the target item.
4. Reflect on the user's potential desire for diversity in their purchases.

Your output should be a clear and logical chain of thought... Ensure your analysis is impartial... Focus should be on understanding factors that influence the user's decision-making regarding the target item. </SYS>>

Please generate a chain of thought based on the user's ... considering how these might relate to their interest in the target new item.

{Recommendation Features}

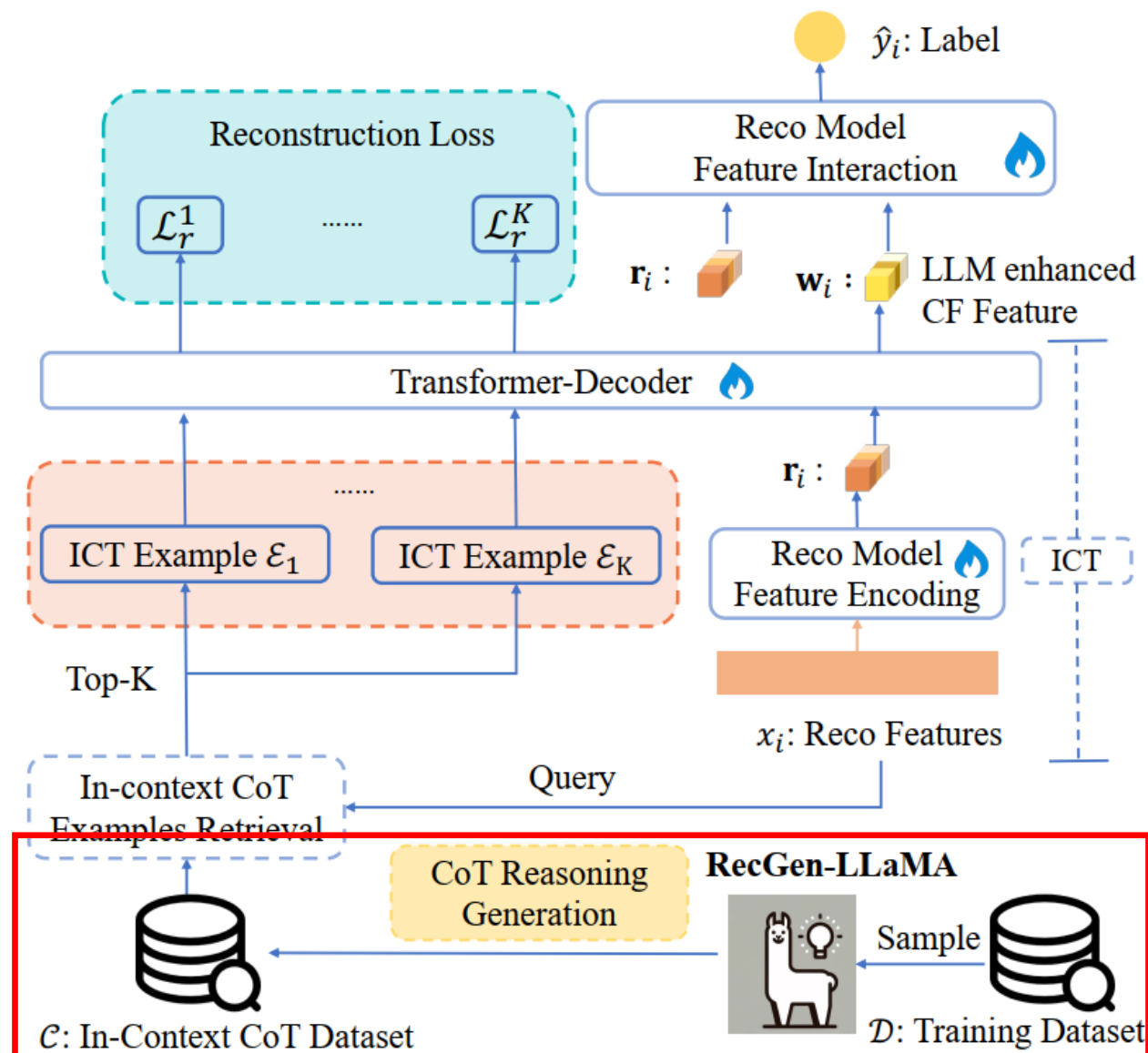
User's decision-making: The user {Ground-Truth Label} the target new item.

Let's think step by step and develop the chain of thought for the above considerations.

Commence with the chain of thought immediately:

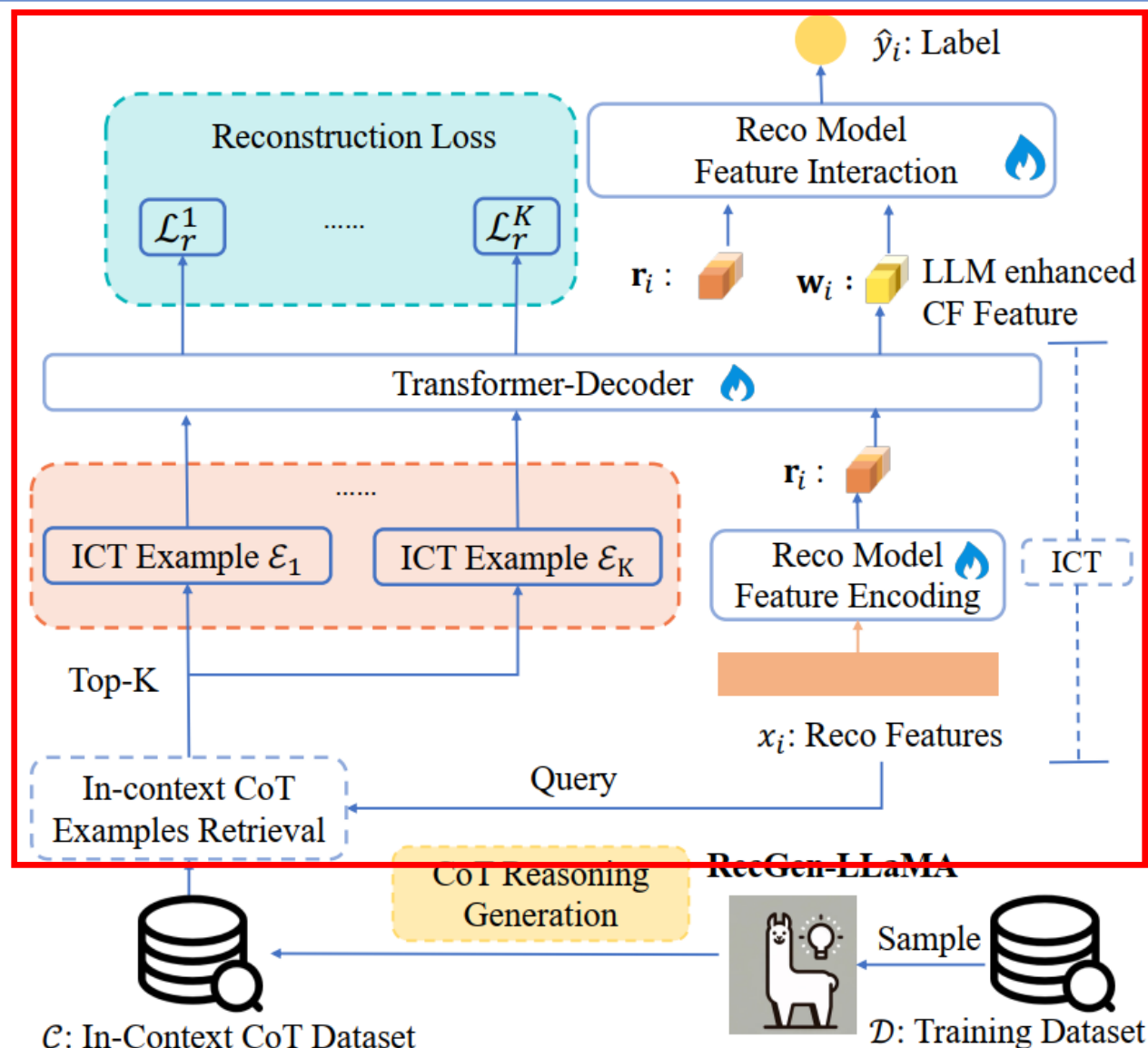
## Offline Service

- Instruction tuning on LLaMA2
- RecGen-LLaMA generating CoT forming ICT dataset



## • Online Service

- Embedding-based retrieval forming ICT examples
- Learning world-knowledge and reasoning guided CF feature

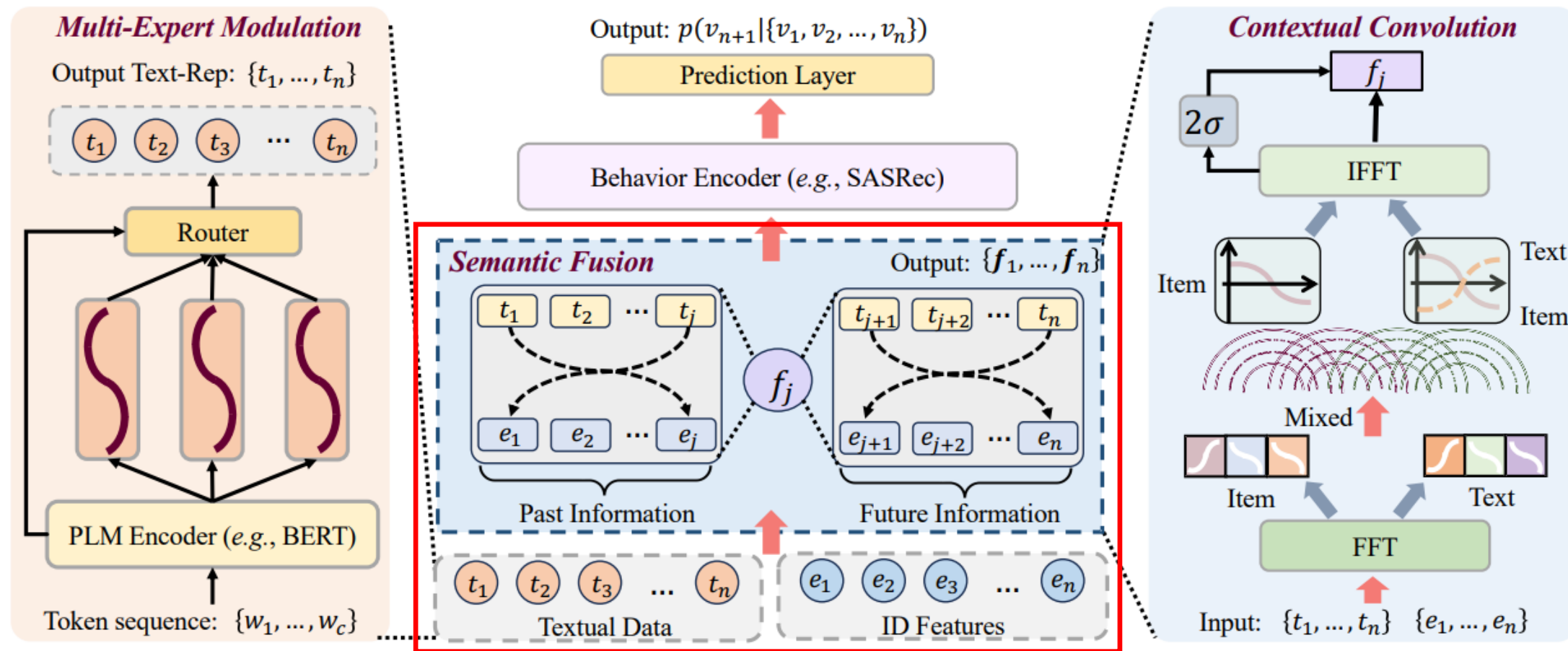


# Overall performance

| Backbone | Framework | Sports         |                |               | Beauty         |                |               | Toys           |                |               |
|----------|-----------|----------------|----------------|---------------|----------------|----------------|---------------|----------------|----------------|---------------|
|          |           | AUC↑           | Logloss↓       | RelaImpr↑     | AUC↑           | Logloss↓       | RelaImpr↑     | AUC↑           | Logloss↓       | RelaImpr↑     |
| DeepFM   | None      | 0.7990         | 0.5471         | 0.000%        | 0.7853         | 0.5545         | 0.000%        | 0.7681         | 0.5770         | 0.000%        |
|          | KD        | <u>0.8043</u>  | <u>0.5404</u>  | 1.773%        | <u>0.7959</u>  | <u>0.5442</u>  | 3.715%        | <u>0.7713</u>  | <u>0.5716</u>  | 1.194%        |
|          | KAR       | 0.7991         | 0.5469         | 0.033%        | 0.7870         | 0.5546         | 0.596%        | 0.7698         | 0.5718         | 0.634%        |
|          | LLM-CF    | <b>0.8137*</b> | <b>0.5306*</b> | <b>4.916%</b> | <b>0.8044*</b> | <b>0.5366*</b> | <b>6.695%</b> | <b>0.7881*</b> | <b>0.5581*</b> | <b>7.460%</b> |
| xDeepFM  | None      | 0.8158         | 0.5318         | 0.000%        | 0.8065         | 0.5359         | 0.000%        | 0.7836         | 0.5589         | 0.000%        |
|          | KD        | <u>0.8169</u>  | 0.5298         | 0.348%        | <u>0.8104</u>  | 0.5345         | 1.272%        | 0.7865         | 0.5553         | 1.023%        |
|          | KAR       | 0.8161         | 0.5279         | 0.094%        | 0.8101         | 0.5315         | 1.175%        | 0.7898         | 0.5529         | 2.186%        |
|          | LLM-CF    | <b>0.8196*</b> | <b>0.5248*</b> | <b>1.203%</b> | <b>0.8113</b>  | <b>0.5311</b>  | <b>1.566%</b> | <b>0.7947*</b> | <b>0.5473*</b> | <b>3.914%</b> |
| AutoInt  | None      | 0.8003         | 0.5444         | 0.000%        | 0.7949         | 0.5469         | 0.00%         | 0.7630         | 0.5770         | 0.000%        |
|          | KD        | 0.8012         | 0.5439         | 0.300%        | <u>0.7961</u>  | <u>0.5444</u>  | 0.407%        | 0.7635         | 0.5770         | 0.190%        |
|          | KAR       | 0.8039         | <b>0.5390</b>  | 1.199%        | 0.7939         | 0.5476         | -0.339%       | 0.7683         | 0.5741         | 2.015%        |
|          | LLM-CF    | <b>0.8088*</b> | 0.5391         | <b>2.831%</b> | <b>0.8090*</b> | <b>0.5321*</b> | <b>4.781%</b> | <b>0.7754*</b> | <b>0.5685*</b> | <b>4.714%</b> |
| DCNv1    | None      | 0.8023         | 0.5442         | 0.000%        | 0.8146         | 0.5255         | 0.000%        | 0.7621         | 0.5831         | 0.000%        |
|          | KD        | <u>0.8040</u>  | <u>0.5441</u>  | 0.562%        | 0.8147         | 0.5286         | 0.031%        | <u>0.7652</u>  | 0.5847         | 1.183%        |
|          | KAR       | 0.8024         | 0.5469         | 0.033%        | 0.8165         | 0.5229         | 0.604%        | 0.7651         | 0.5821         | 1.144%        |
|          | LLM-CF    | <b>0.8092*</b> | <b>0.5368*</b> | <b>2.282%</b> | <b>0.8182*</b> | <b>0.5216*</b> | <b>1.144%</b> | <b>0.7702*</b> | <b>0.5745*</b> | <b>3.090%</b> |
| DCNv2    | None      | 0.8110         | 0.5331         | 0.000%        | 0.8028         | 0.5378         | 0.00%         | 0.7774         | 0.5650         | 0.000%        |
|          | KD        | <u>0.8112</u>  | <u>0.5320</u>  | 0.064%        | <b>0.8057</b>  | <b>0.5343</b>  | <b>0.958%</b> | <b>0.7827</b>  | <b>0.5609</b>  | <b>1.911%</b> |
|          | KAR       | 0.8087         | 0.5363         | -0.739%       | 0.8003         | 0.5404         | -0.825%       | 0.7759         | 0.5662         | -0.541%       |
|          | LLM-CF    | <b>0.8131*</b> | <b>0.5307*</b> | <b>0.675%</b> | 0.8033         | 0.5372         | 0.165%        | 0.7812         | 0.5619         | 1.370%        |
| DIN      | None      | 0.7986         | 0.5519         | 0.000%        | 0.7861         | 0.5613         | 0.000%        | 0.7586         | 0.5885         | 0.000%        |
|          | KD        | <u>0.8023</u>  | <u>0.5422</u>  | 1.239%        | <u>0.7934</u>  | <u>0.5518</u>  | 2.551%        | <u>0.7652</u>  | <u>0.5847</u>  | 2.552%        |
|          | KAR       | 0.7971         | 0.5525         | -0.502%       | 0.7861         | 0.5604         | 0.000%        | 0.7620         | 0.5874         | 1.315%        |
|          | LLM-CF    | <b>0.8089*</b> | <b>0.5374*</b> | <b>3.449%</b> | <b>0.7967*</b> | <b>0.5492*</b> | <b>3.705%</b> | <b>0.7783*</b> | <b>0.5699*</b> | <b>7.618%</b> |

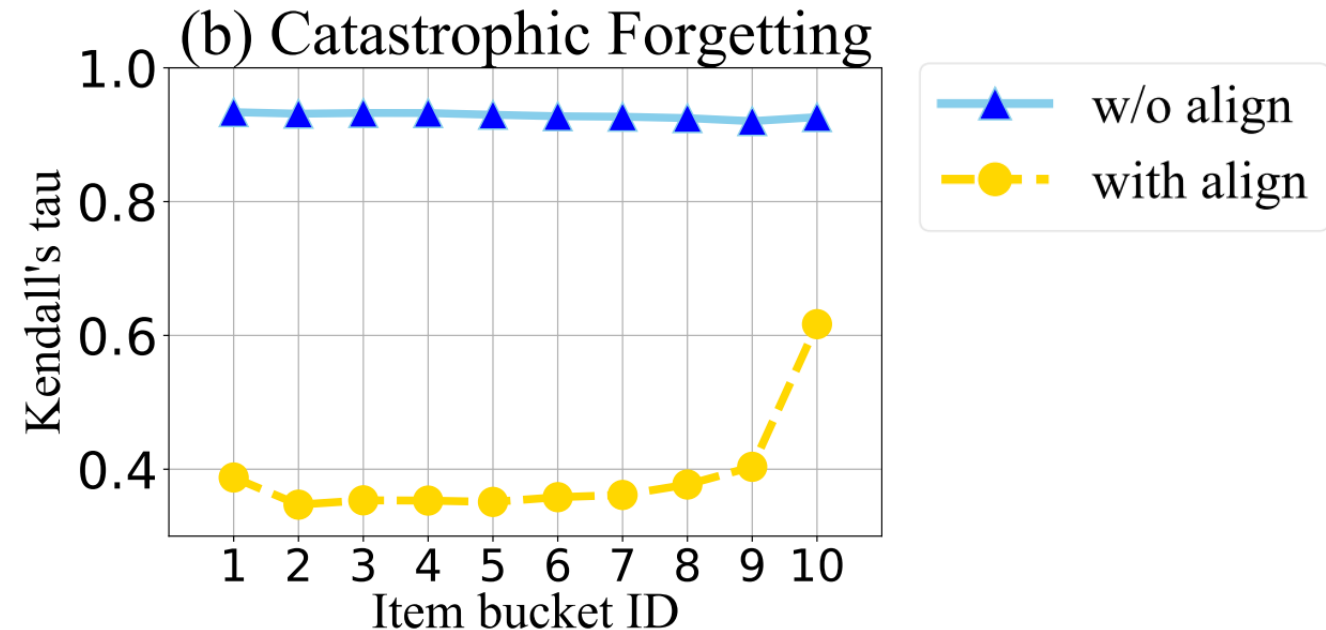
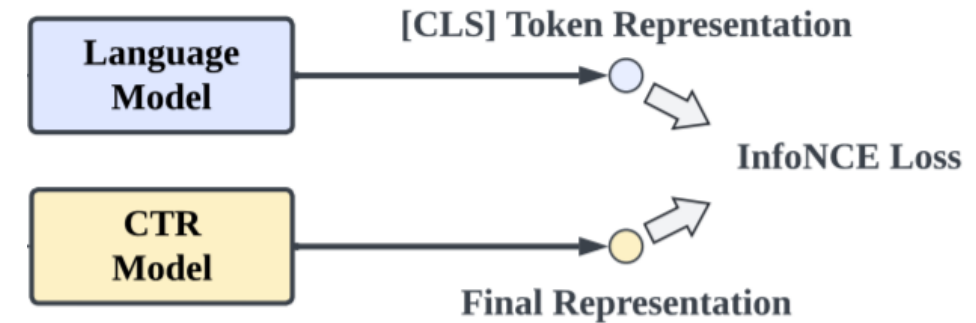
- LLM-CF achieves improvement on different CTR backbones.





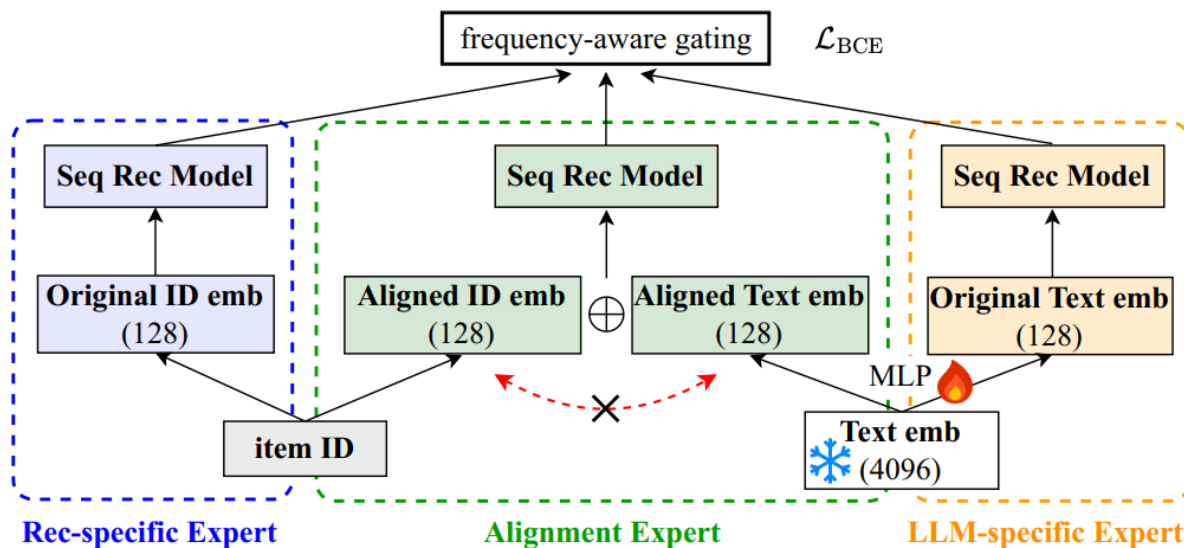
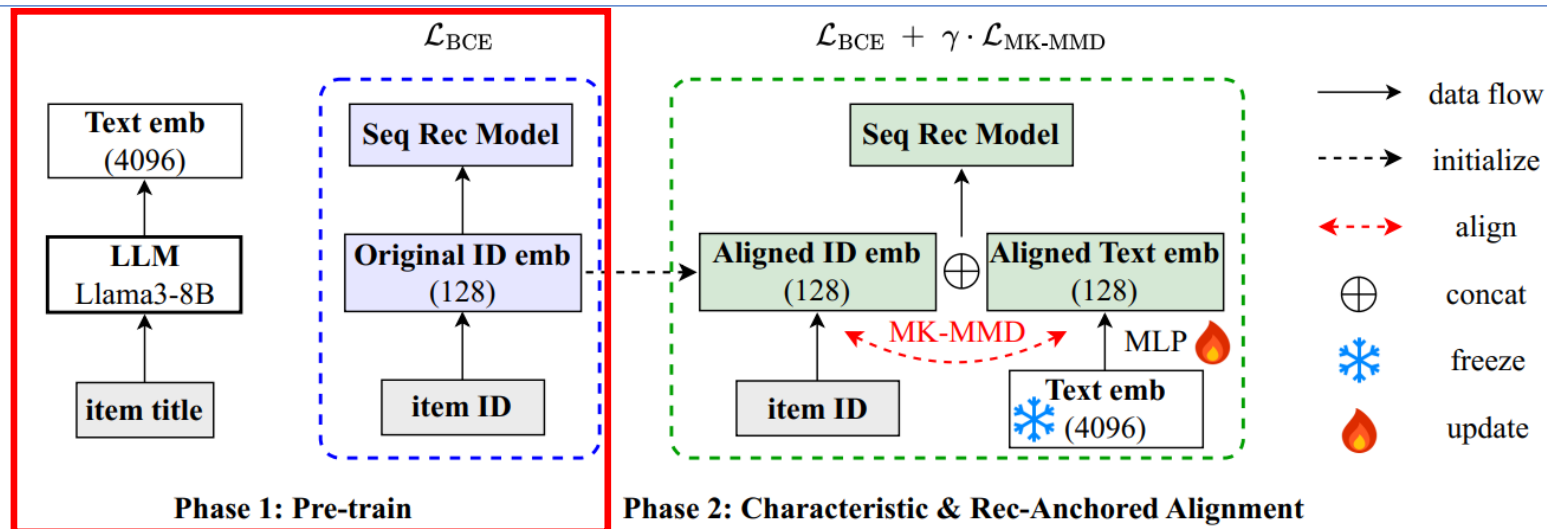
- TedRec applies Fourier Transform to conduct semantic fusion in the frequency domain.

- **Challenges for Alignment + SFT**
  - **Inability to Capture All Statistics of Data Distribution:** Contrastive learning with common cosine kernel is not optimal
  - **Catastrophic Forgetting in Alignment:** With only one set of collaborative embeddings, the alignment leads to catastrophic forgetting on the collaborative embeddings



- **Pre-train**

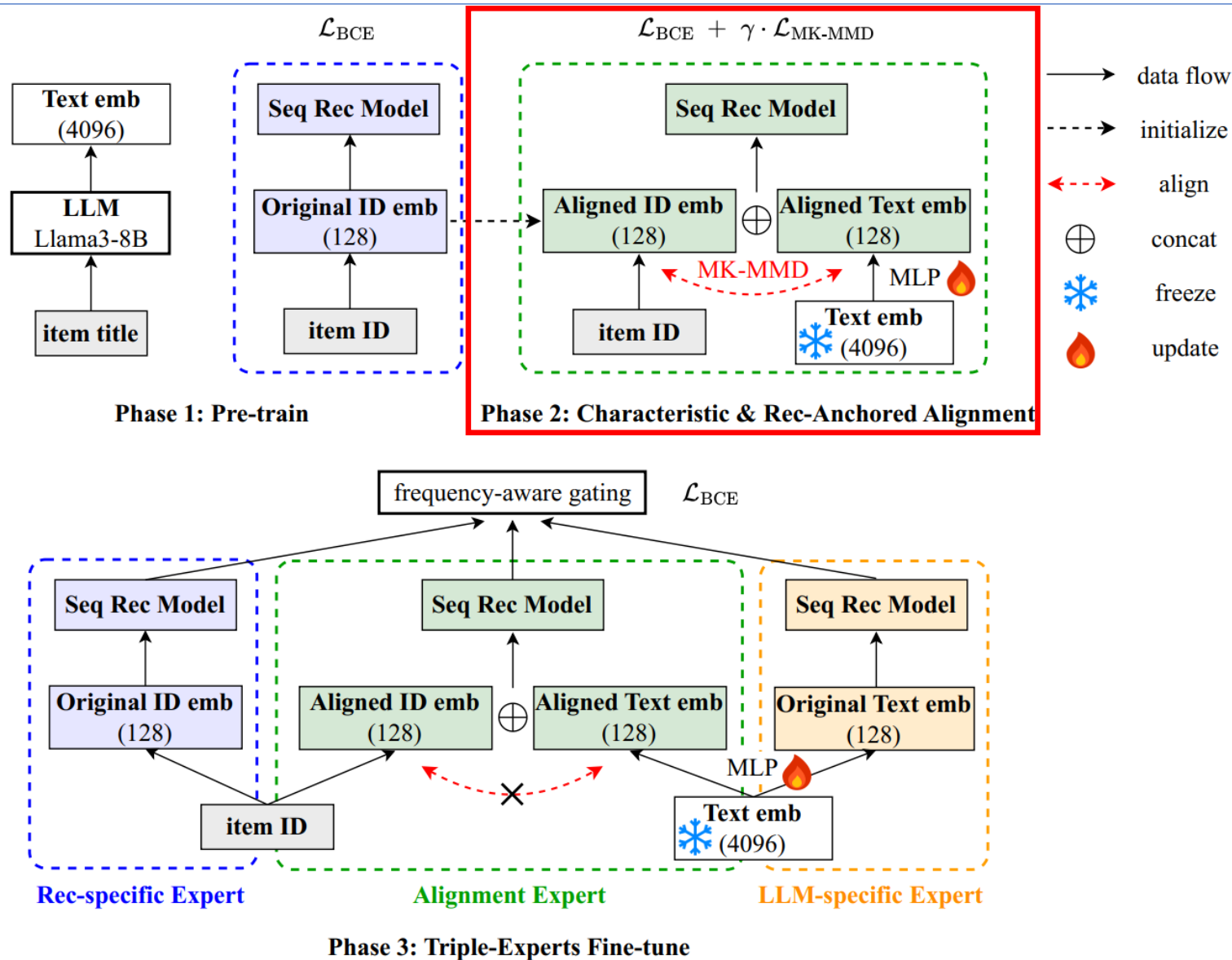
- Text & Collaborative emb



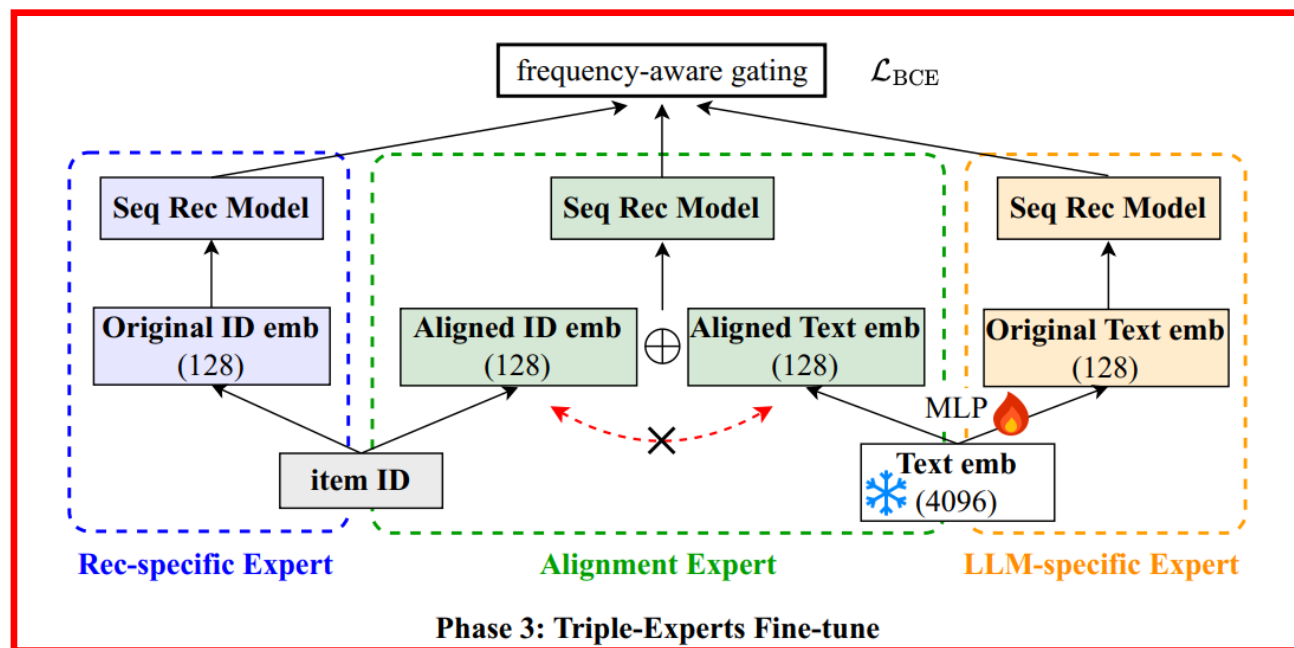
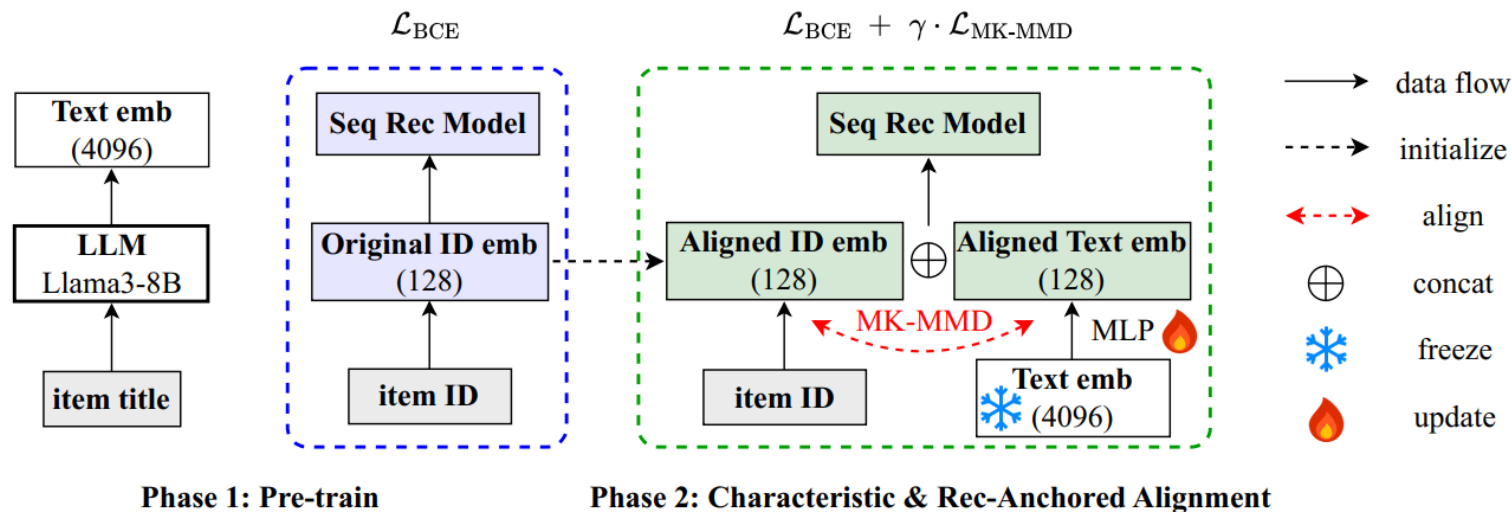
Phase 3: Triple-Experts Fine-tune

## Align

- Recommendation as anchor
- Multi-kernel maximum mean discrepancy as alignment loss based on characteristic kernel



- **Disentangle**
  - Triple-Experts & Multi-Emb

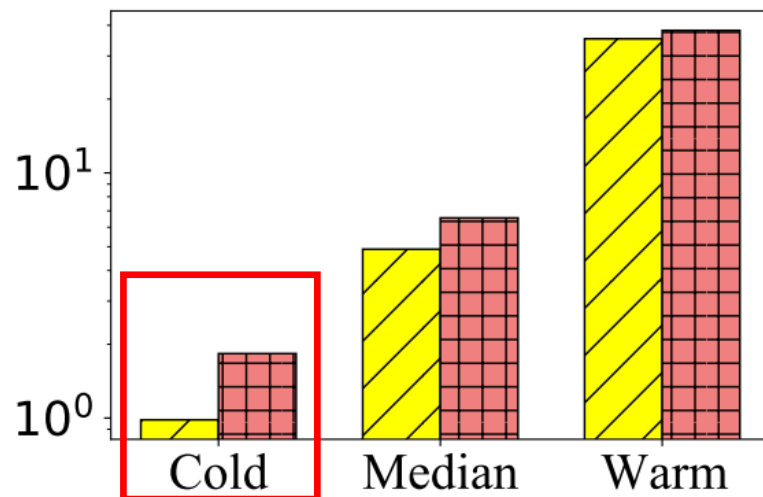




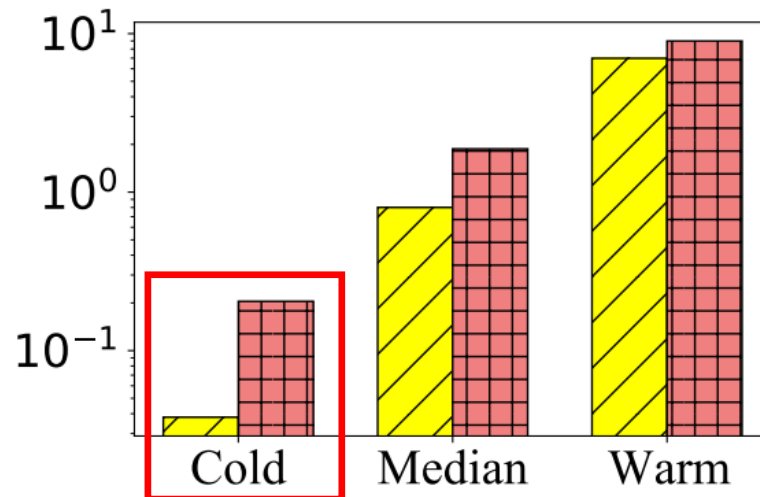
| Datasets     | Metric  | SASRec  | Hybrid  | MoRec   | CTRL    | MAKE    | DisCo   | SMEM           | Ours                       | Impr.  |
|--------------|---------|---------|---------|---------|---------|---------|---------|----------------|----------------------------|--------|
| MIND         | HR@10   | 16.8437 | 17.3385 | 14.1235 | 18.1318 | 16.4317 | 17.3367 | <u>18.4319</u> | <b>18.6703<sup>★</sup></b> | 10.84% |
|              | nDCG@10 | 9.0520  | 9.3321  | 7.7090  | 9.7996  | 8.9684  | 9.4131  | <u>10.0003</u> | <b>10.1515<sup>★</sup></b> | 12.15% |
| Electronics  | HR@10   | 1.6754  | 1.5673  | 0.9562  | 1.9468  | 0.9495  | 1.8148  | <u>2.3492</u>  | <b>2.4804<sup>★</sup></b>  | 48.05% |
|              | nDCG@10 | 0.7938  | 0.8385  | 0.4721  | 0.9370  | 0.4611  | 0.9025  | <u>1.4287</u>  | <b>1.5373<sup>★</sup></b>  | 93.67% |
| Prime Pantry | HR@10   | 2.6338  | 3.1000  | 3.1699  | 2.8048  | 3.3408  | 2.8358  | <u>3.4341</u>  | <b>3.8303<sup>★</sup></b>  | 45.43% |
|              | nDCG@10 | 1.2926  | 1.5606  | 1.6013  | 1.3700  | 1.6379  | 1.3451  | <u>1.7440</u>  | <b>1.9104<sup>★</sup></b>  | 47.80% |

- PAD surpasses all baselines and achieves significant improvement.

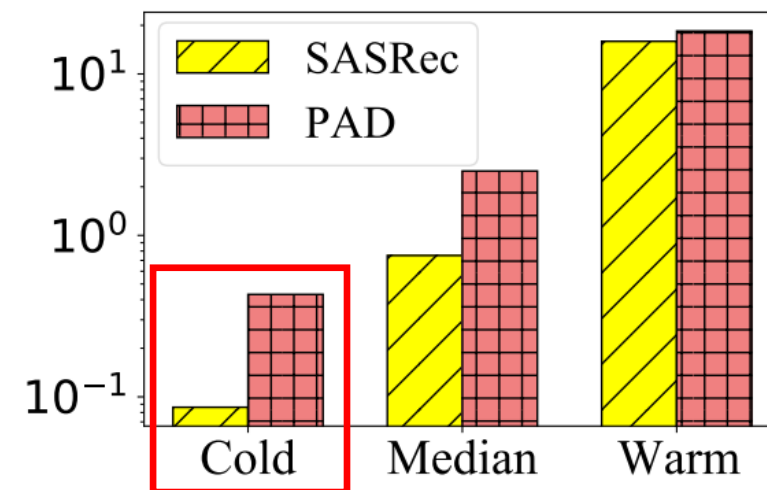
MIND



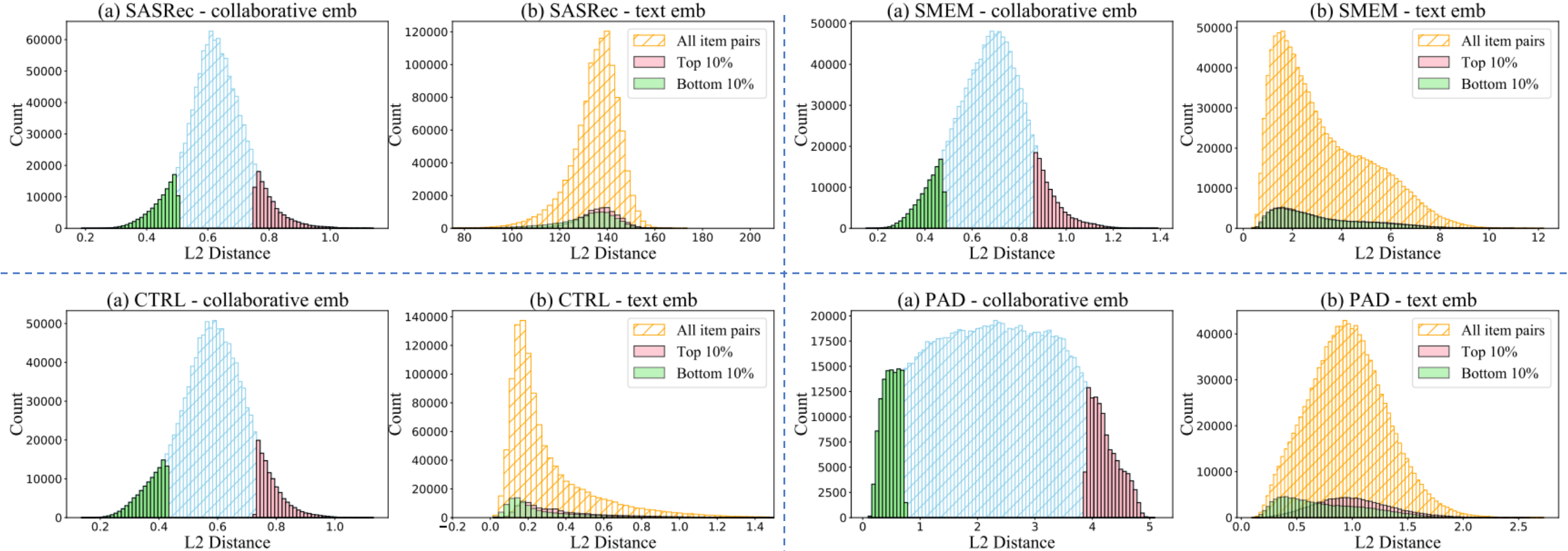
Electronics



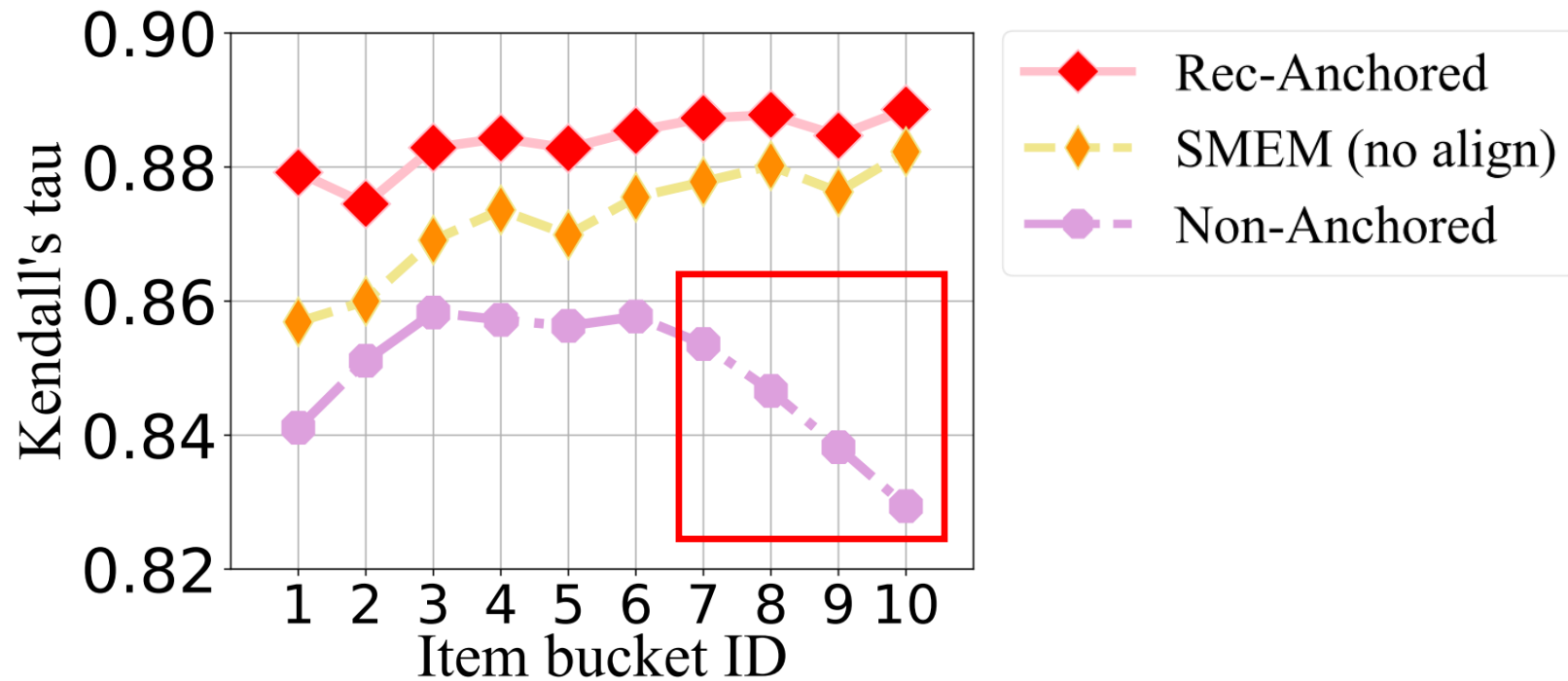
Prime Pantry



- PAD can mitigate cold-start problem with LLM knowledge.



- PAD greatly improves the performance on the cold items with better alignment of textual embeddings towards the collaborative space.



- The proposed Rec-Anchored Alignment loss avoids the catastrophic forgetting of collaborative embeddings and succeeds in aligning textual embedding towards the collaborative space.

| Datasets     | GRU4Rec |         |         |         |        |         | Caser  |         |        |         |         |         |
|--------------|---------|---------|---------|---------|--------|---------|--------|---------|--------|---------|---------|---------|
|              | Origin  |         | +PAD    |         | Impr.  |         | Origin |         | +PAD   |         | Impr.   |         |
|              | HR@10   | nDCG@10 | HR@10   | nDCG@10 | HR@10  | nDCG@10 | HR@10  | nDCG@10 | HR@10  | nDCG@10 | HR@10   | nDCG@10 |
| MIND         | 13.3969 | 6.9098  | 16.5762 | 8.8475  | 23.73% | 28.04%  | 8.6879 | 3.7981  | 9.0569 | 4.2696  | 4.25%   | 12.41%  |
| Electronics  | 0.8437  | 0.4104  | 1.1016  | 0.5898  | 30.57% | 43.73%  | 0.5746 | 0.2355  | 0.6351 | 0.2501  | 10.53%  | 6.20%   |
| Prime Pantry | 1.7248  | 0.8227  | 2.5484  | 1.2307  | 47.75% | 49.59%  | 0.6526 | 0.4136  | 1.4063 | 0.5507  | 115.49% | 33.16%  |

- GRU4Rec and Caser as sequential recommendation backbones
- PAD acts as model-agnostic enhancement paradigm

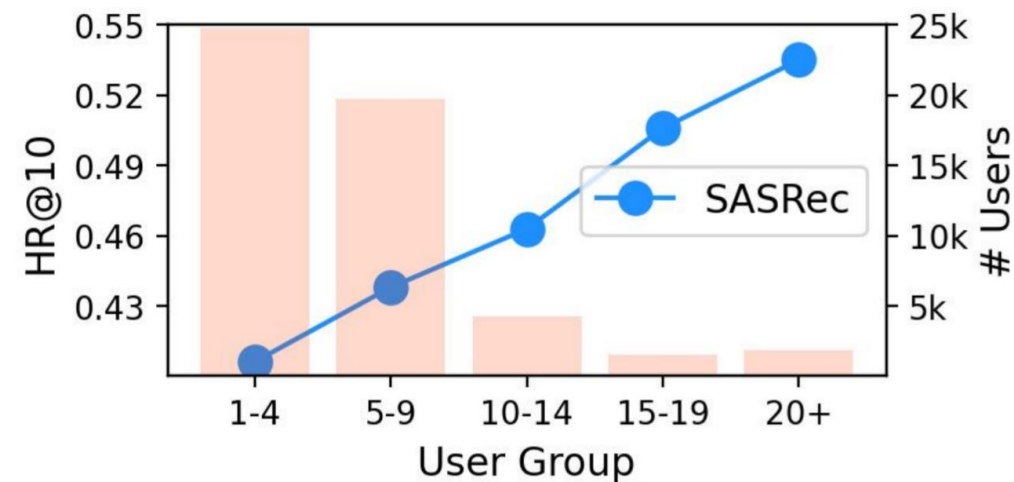


## • Long-tail Challenges for SR

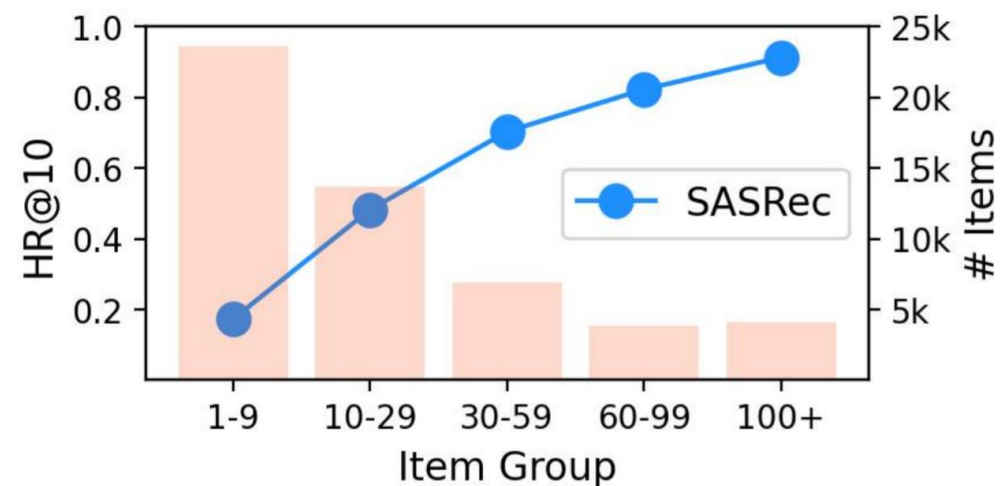
- **Long-tail User Challenge:** The majority of users receive less than optimal recommendation services  
→ Poor experience for **new users**

- **Long-tail Item Challenge:** Current SRS models often underperform those unpopular items  
→ Less profits for **small sellers**

- **LLMs are promising to address the long-tail challenges from semantic view!**

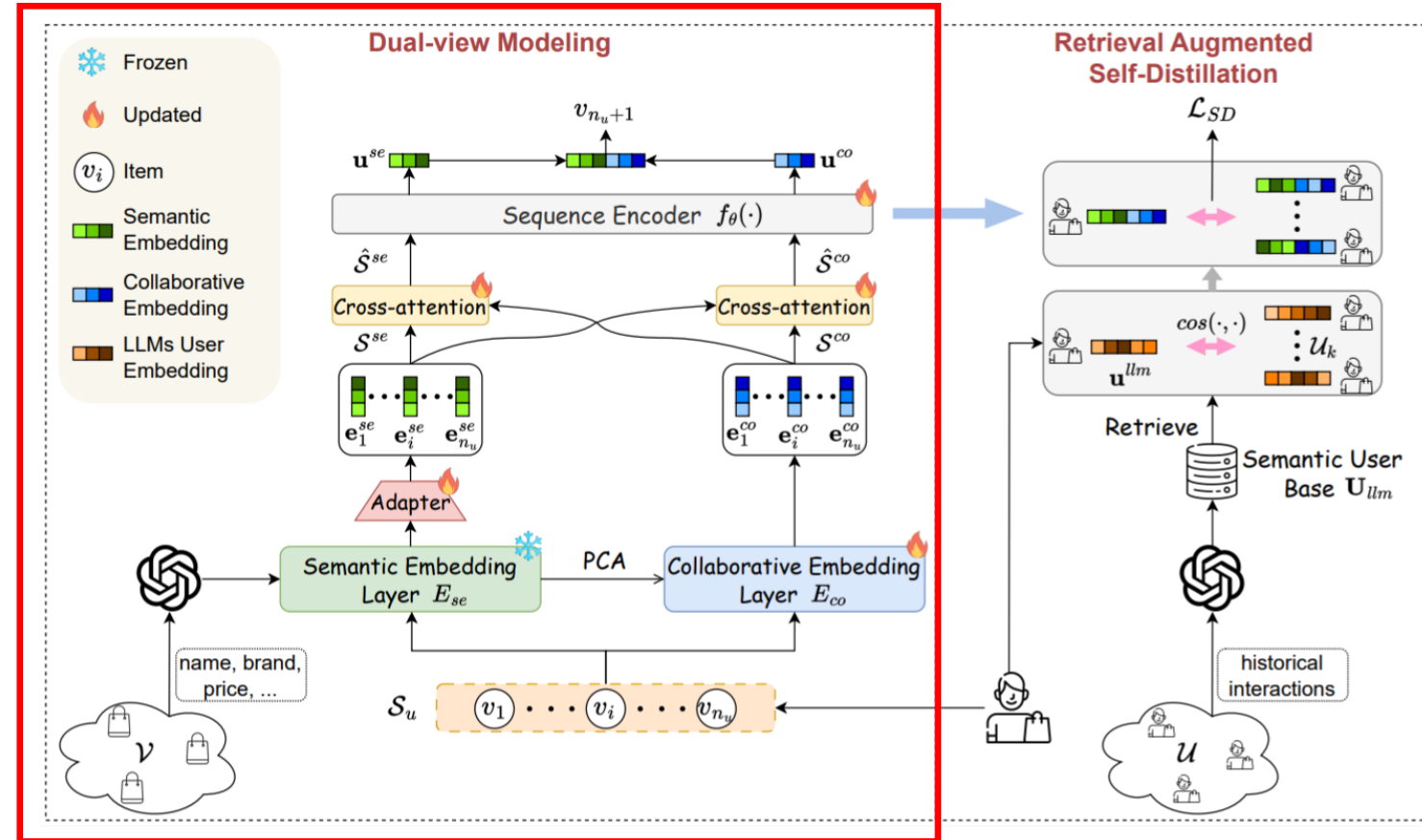


(a) Long-tail User Challenge

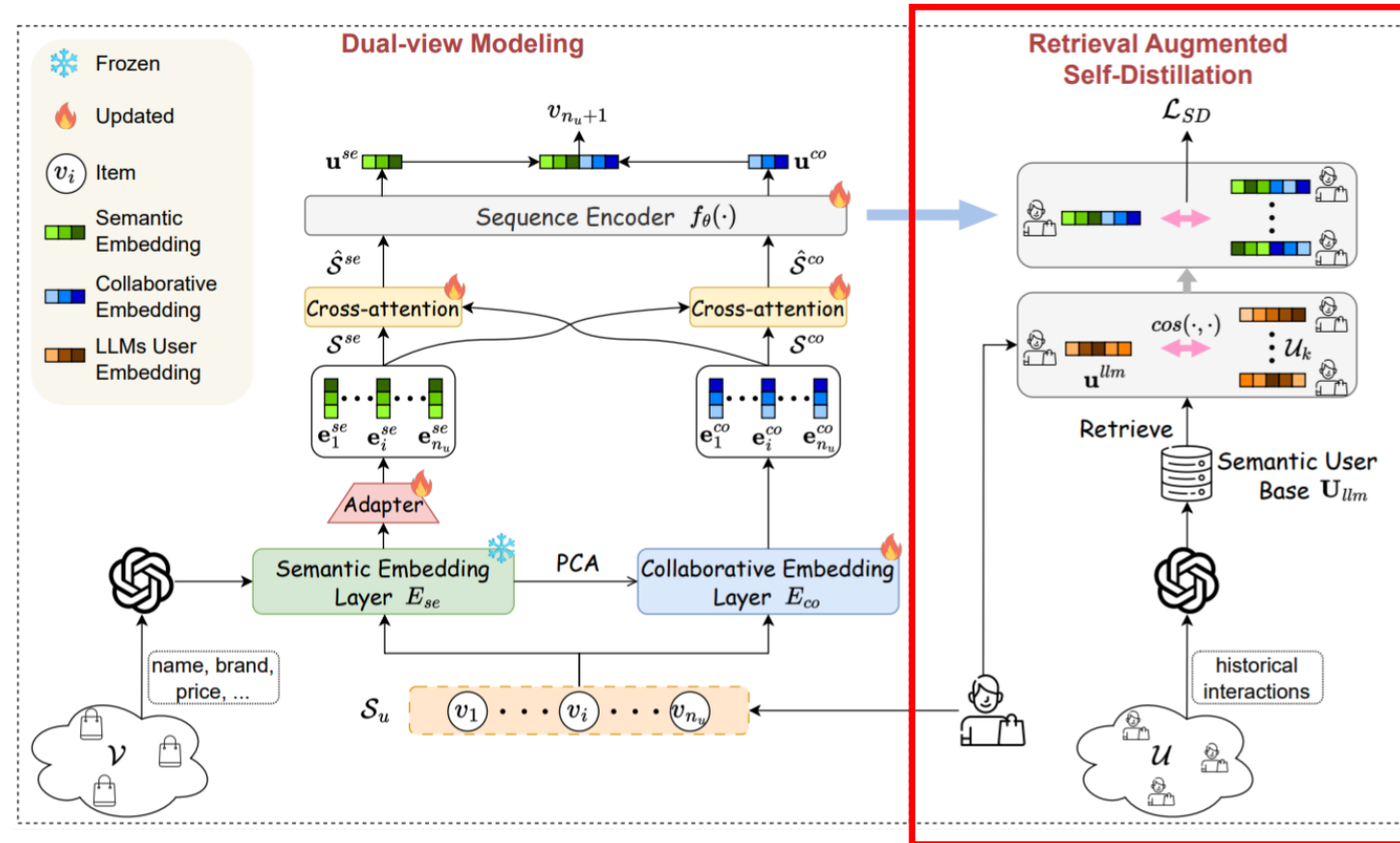


(b) Long-tail Item Challenge

- **Dual-view Modeling:** Consisting of semantic-view modeling and collaborative-view modeling to address the long-tail item issue



- Retrieval Augmented Self-Distillation:**  
Enhancing the SRS model to address the long-tail user problem



| Dataset | Model     | Overall |         | Tail Item |         | Head Item |         | Tail User |         | Head User |         |
|---------|-----------|---------|---------|-----------|---------|-----------|---------|-----------|---------|-----------|---------|
|         |           | H@10    | N@10    | H@10      | N@10    | H@10      | N@10    | H@10      | N@10    | H@10      | N@10    |
| Yelp    | GRU4Rec   | 0.4879  | 0.2751  | 0.0171    | 0.0059  | 0.6265    | 0.3544  | 0.4919    | 0.2777  | 0.4726    | 0.2653  |
|         | - CITIES  | 0.4898  | 0.2749  | 0.0134    | 0.0051  | 0.6301    | 0.3543  | 0.4936    | 0.2783  | 0.4756    | 0.2618  |
|         | - MELT    | 0.4985  | 0.2825  | 0.0201    | 0.0079  | 0.6393    | 0.3633  | 0.5046    | 0.2865  | 0.4750    | 0.2671  |
|         | - RLMRec  | 0.4886  | 0.2777  | 0.0188    | 0.0067  | 0.6269    | 0.3574  | 0.4920    | 0.2804  | 0.4756    | 0.2671  |
|         | - LLMInit | 0.4872  | 0.2749  | 0.0201    | 0.0072  | 0.6246    | 0.3537  | 0.4908    | 0.2775  | 0.4732    | 0.2647  |
|         | - LLM-ESR | 0.5724* | 0.3413* | 0.0763*   | 0.0318* | 0.7184*   | 0.4324* | 0.5782*   | 0.3456* | 0.5501*   | 0.3247* |
|         | Bert4Rec  | 0.5307  | 0.3035  | 0.0115    | 0.0044  | 0.6836    | 0.3916  | 0.5325    | 0.3047  | 0.5241    | 0.2988  |
|         | - CITIES  | 0.5249  | 0.3015  | 0.0041    | 0.0014  | 0.6783    | 0.3899  | 0.5274    | 0.3032  | 0.5155    | 0.2954  |
|         | - MELT    | 0.6206  | 0.3770  | 0.0429    | 0.0149  | 0.7907    | 0.4836  | 0.6210    | 0.3780  | 0.6191    | 0.3733  |
|         | - RLMRec  | 0.5306  | 0.3039  | 0.0104    | 0.0040  | 0.6938    | 0.3922  | 0.5351    | 0.3065  | 0.5137    | 0.2936  |
|         | - LLMInit | 0.6199  | 0.3781  | 0.0874    | 0.0330  | 0.7766    | 0.4797  | 0.6204    | 0.3796  | 0.6178    | 0.3723  |
|         | - LLM-ESR | 0.6623* | 0.4222* | 0.1227*   | 0.0500* | 0.8212*   | 0.5318* | 0.6637*   | 0.4247* | 0.6571*   | 0.4127* |
|         | SASRec    | 0.5940  | 0.3597  | 0.1142    | 0.0495  | 0.7353    | 0.4511  | 0.5893    | 0.3578  | 0.6122    | 0.3672  |
|         | - CITIES  | 0.5828  | 0.3540  | 0.1532    | 0.0700  | 0.7093    | 0.4376  | 0.5785    | 0.3511  | 0.5994    | 0.3649  |
|         | - MELT    | 0.6257  | 0.3791  | 0.1015    | 0.0371  | 0.7801    | 0.4799  | 0.6246    | 0.3804  | 0.6299    | 0.3744  |
|         | - RLMRec  | 0.5990  | 0.3623  | 0.0953    | 0.0412  | 0.7474    | 0.4568  | 0.5966    | 0.3613  | 0.6084    | 0.3658  |
|         | - LLMInit | 0.6415  | 0.3997  | 0.1760    | 0.0789  | 0.7785    | 0.4941  | 0.6403    | 0.4010  | 0.6462    | 0.3948  |
|         | - LLM-ESR | 0.6673* | 0.4208* | 0.1893*   | 0.0845* | 0.8080*   | 0.5199* | 0.6685*   | 0.4229* | 0.6627*   | 0.4128* |

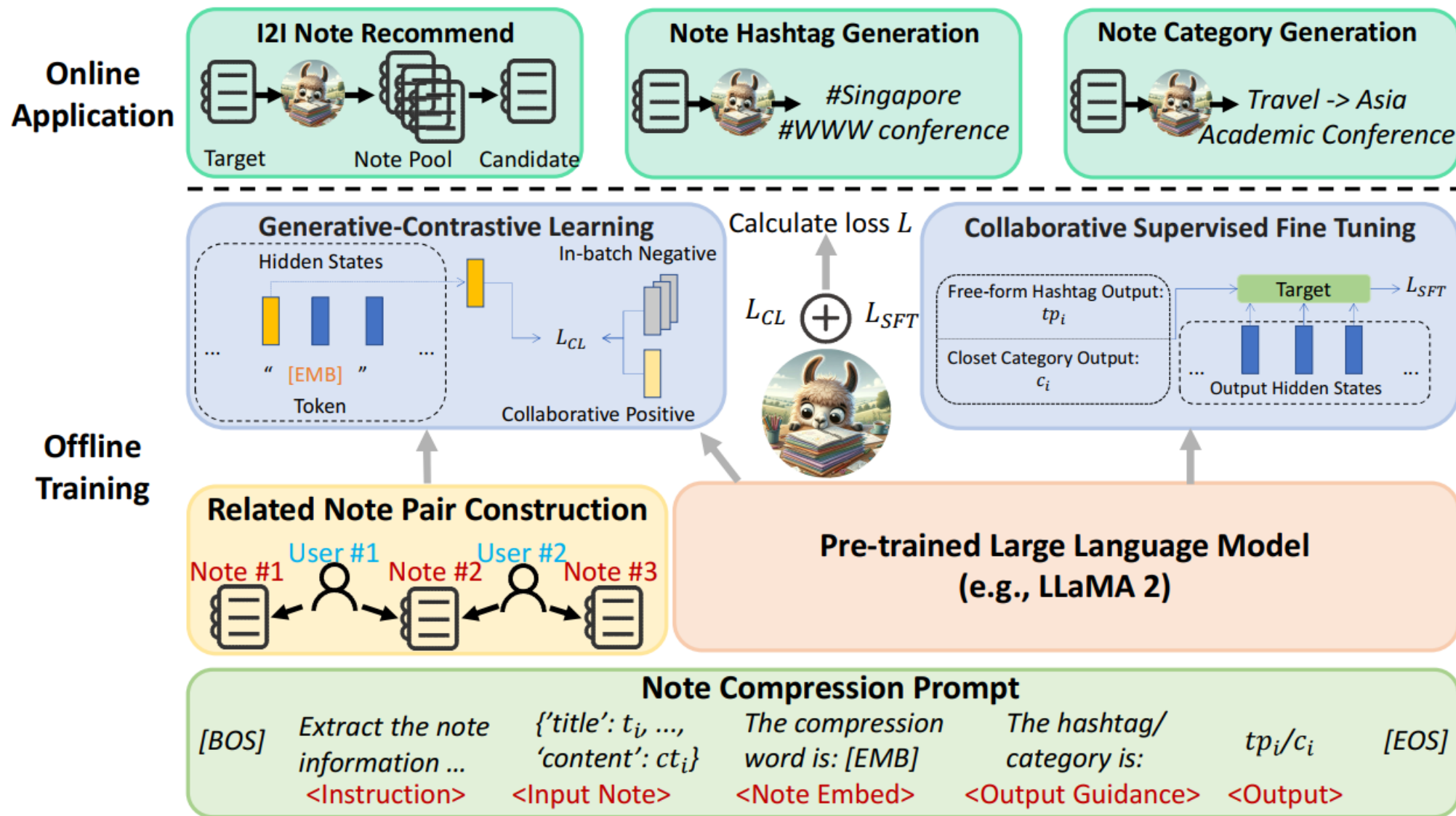
- LLM-ESR leads the overall performance, which indicates better enhancing effects.

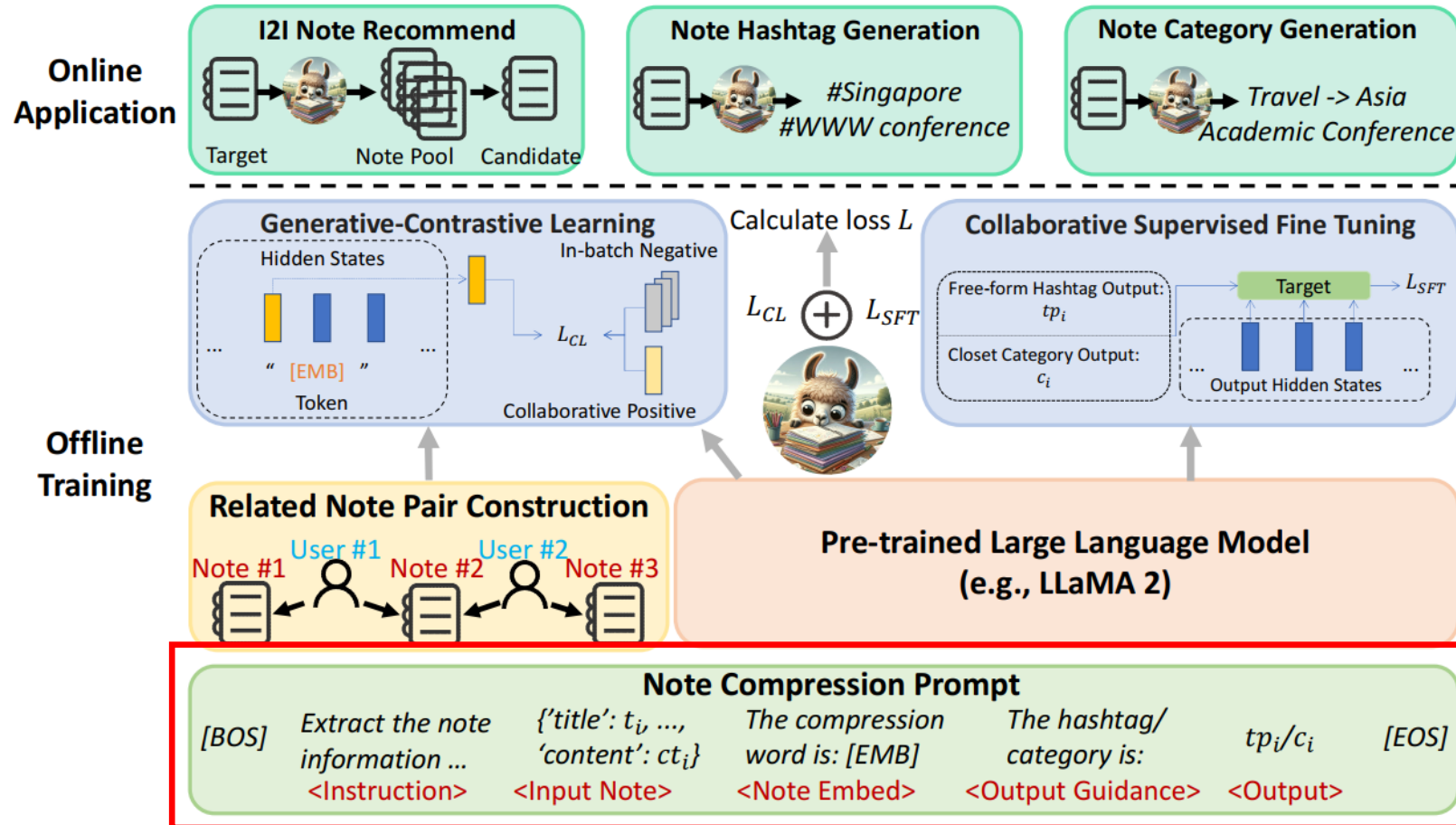


| Dataset | Model     | Overall        |                | Tail Item      |                | Head Item      |                | Tail User      |                | Head User      |                |
|---------|-----------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|         |           | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           | H@10           | N@10           |
| Yelp    | GRU4Rec   | 0.4879         | 0.2751         | 0.0171         | 0.0059         | 0.6265         | 0.3544         | 0.4919         | 0.2777         | 0.4726         | 0.2653         |
|         | - CITIES  | 0.4898         | 0.2749         | 0.0134         | 0.0051         | 0.6301         | 0.3543         | 0.4936         | 0.2783         | 0.4756         | 0.2618         |
|         | - MELT    | 0.4985         | 0.2825         | 0.0201         | 0.0079         | 0.6393         | 0.3633         | 0.5046         | 0.2865         | 0.4750         | 0.2671         |
|         | - RLMRec  | 0.4886         | 0.2777         | 0.0188         | 0.0067         | 0.6269         | 0.3574         | 0.4920         | 0.2804         | 0.4756         | 0.2671         |
|         | - LLMInit | 0.4872         | 0.2749         | 0.0201         | 0.0072         | 0.6246         | 0.3537         | 0.4908         | 0.2775         | 0.4732         | 0.2647         |
|         | - LLM-ESR | <b>0.5724*</b> | <b>0.3413*</b> | <b>0.0763*</b> | <b>0.0318*</b> | <b>0.7184*</b> | <b>0.4324*</b> | <b>0.5782*</b> | <b>0.3456*</b> | <b>0.5501*</b> | <b>0.3247*</b> |
|         | Bert4Rec  | 0.5307         | 0.3035         | 0.0115         | 0.0044         | 0.6836         | 0.3916         | 0.5325         | 0.3047         | 0.5241         | 0.2988         |
|         | - CITIES  | 0.5249         | 0.3015         | 0.0041         | 0.0014         | 0.6783         | 0.3899         | 0.5274         | 0.3032         | 0.5155         | 0.2954         |
|         | - MELT    | 0.6206         | 0.3770         | 0.0429         | 0.0149         | 0.7907         | 0.4836         | 0.6210         | 0.3780         | 0.6191         | 0.3733         |
|         | - RLMRec  | 0.5306         | 0.3039         | 0.0104         | 0.0040         | 0.6938         | 0.3922         | 0.5351         | 0.3065         | 0.5137         | 0.2936         |
|         | - LLMInit | 0.6199         | 0.3781         | 0.0874         | 0.0330         | 0.7766         | 0.4797         | 0.6204         | 0.3796         | 0.6178         | 0.3723         |
|         | - LLM-ESR | <b>0.6623*</b> | <b>0.4222*</b> | <b>0.1227*</b> | <b>0.0500*</b> | <b>0.8212*</b> | <b>0.5318*</b> | <b>0.6637*</b> | <b>0.4247*</b> | <b>0.6571*</b> | <b>0.4127*</b> |
|         | SASRec    | 0.5940         | 0.3597         | 0.1142         | 0.0495         | 0.7353         | 0.4511         | 0.5893         | 0.3578         | 0.6122         | 0.3672         |
|         | - CITIES  | 0.5828         | 0.3540         | 0.1532         | 0.0700         | 0.7093         | 0.4376         | 0.5785         | 0.3511         | 0.5994         | 0.3649         |
|         | - MELT    | 0.6257         | 0.3791         | 0.1015         | 0.0371         | 0.7801         | 0.4799         | 0.6246         | 0.3804         | 0.6299         | 0.3744         |
|         | - RLMRec  | 0.5990         | 0.3623         | 0.0953         | 0.0412         | 0.7474         | 0.4568         | 0.5966         | 0.3613         | 0.6084         | 0.3658         |
|         | - LLMInit | 0.6415         | 0.3997         | 0.1760         | 0.0789         | 0.7785         | 0.4941         | 0.6403         | 0.4010         | 0.6462         | 0.3948         |
|         | - LLM-ESR | <b>0.6673*</b> | <b>0.4208*</b> | <b>0.1893*</b> | <b>0.0845*</b> | <b>0.8080*</b> | <b>0.5199*</b> | <b>0.6685*</b> | <b>0.4229*</b> | <b>0.6627*</b> | <b>0.4128*</b> |

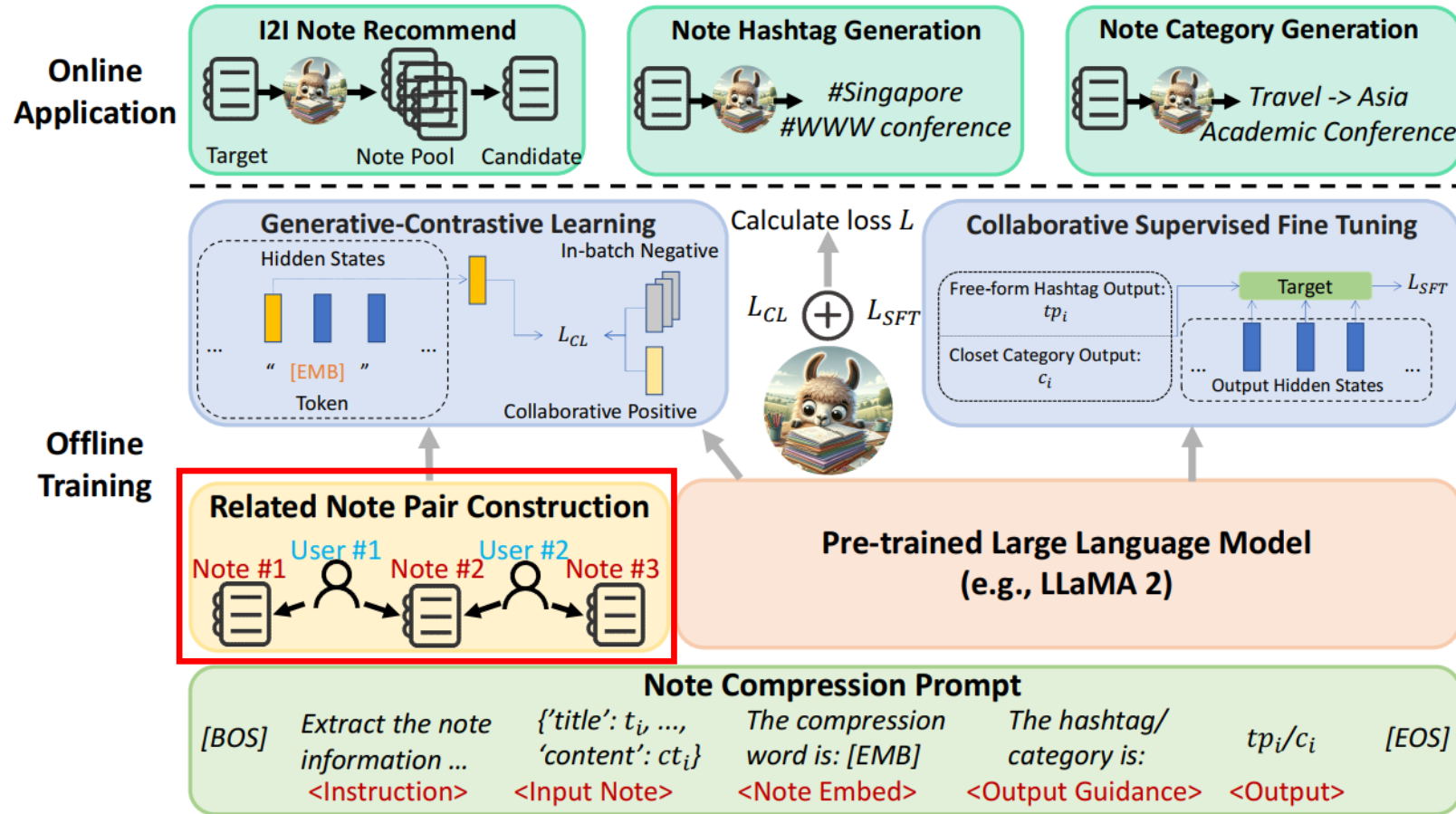
- LLM-ESR achieves the best on tail and popular item group.
- LLM-ESR can augment the tail user group better.





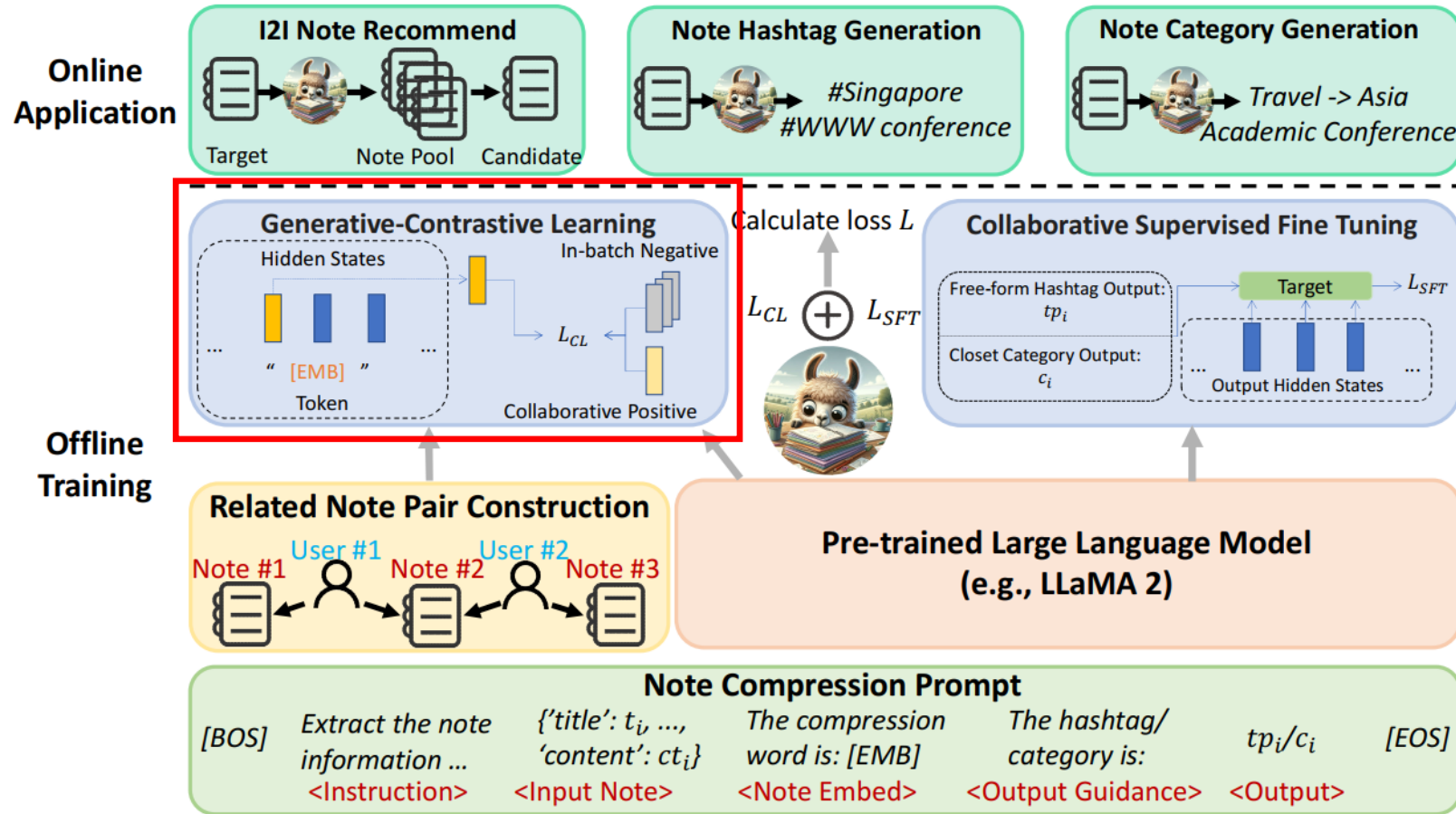


- **Note Compression Prompt**
  - Compressing the note content
  - Generating hashtags/categories

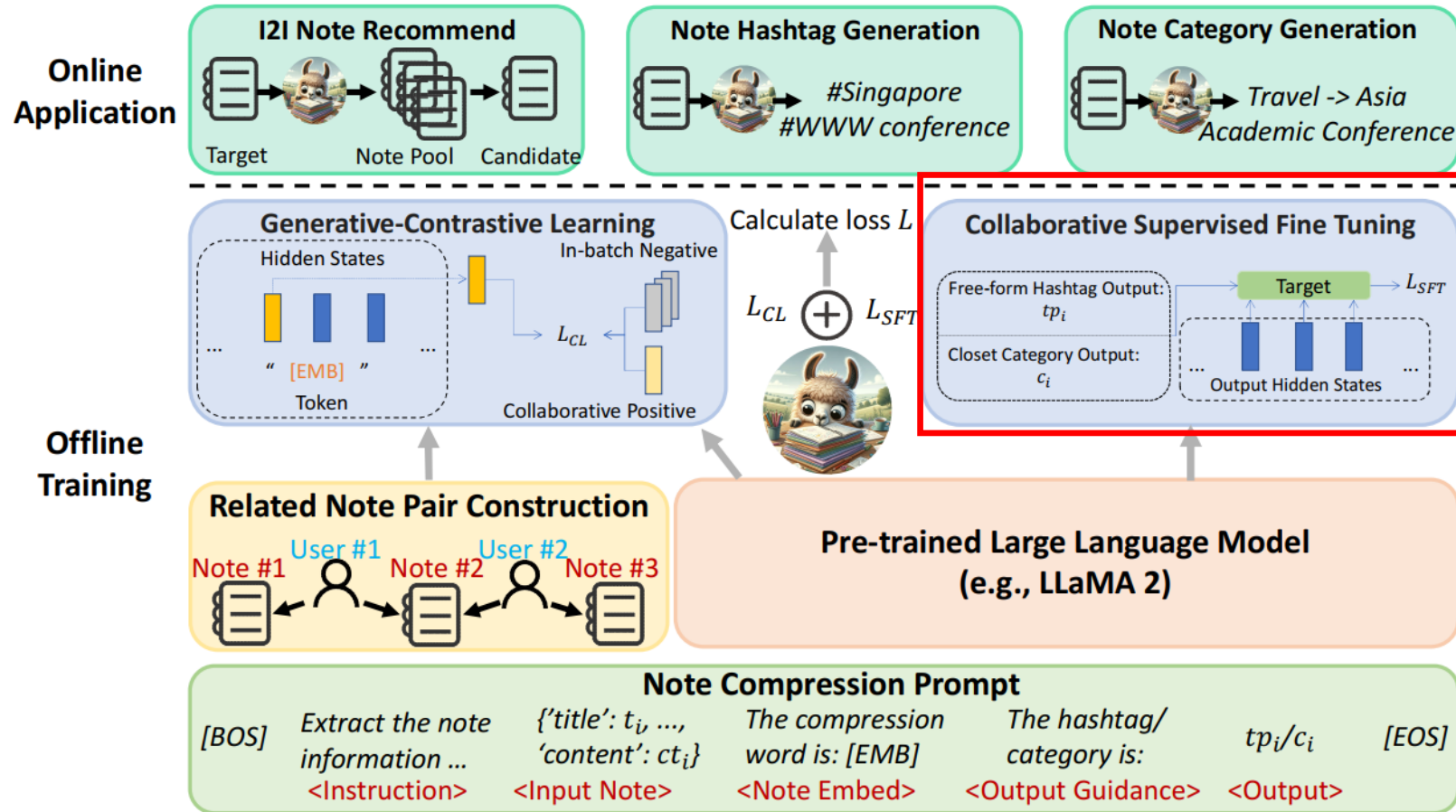


## • Constructing related note pairs

- Counting co-occurrence scores
- Selecting notes with highest scores as related notes



- **Generative-Contrastive Learning**
  - Identifying related notes from in-batch negatives

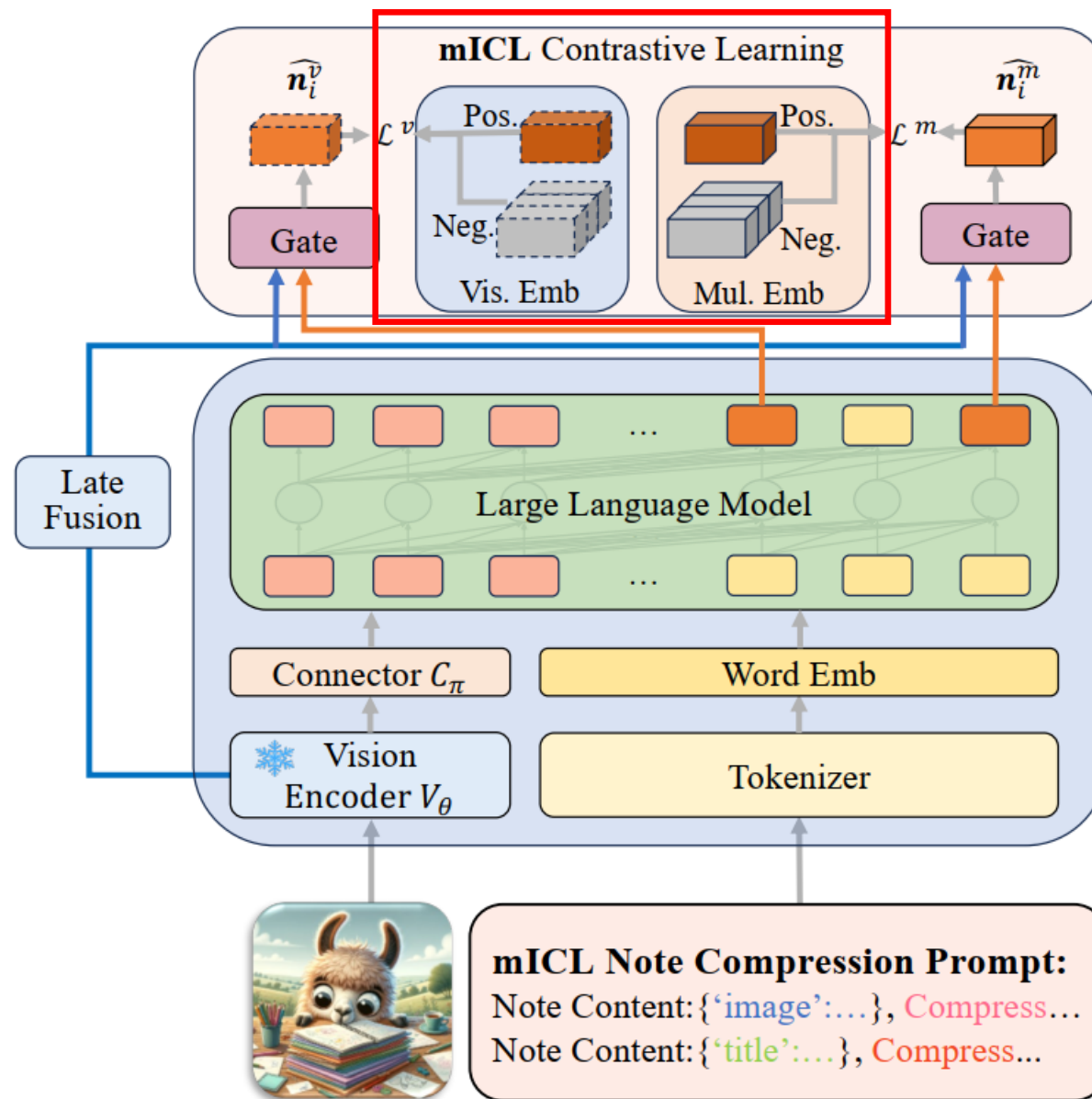


- **Collaborative Supervised Fine-tuning**
  - Generating hashtags/categories for each note



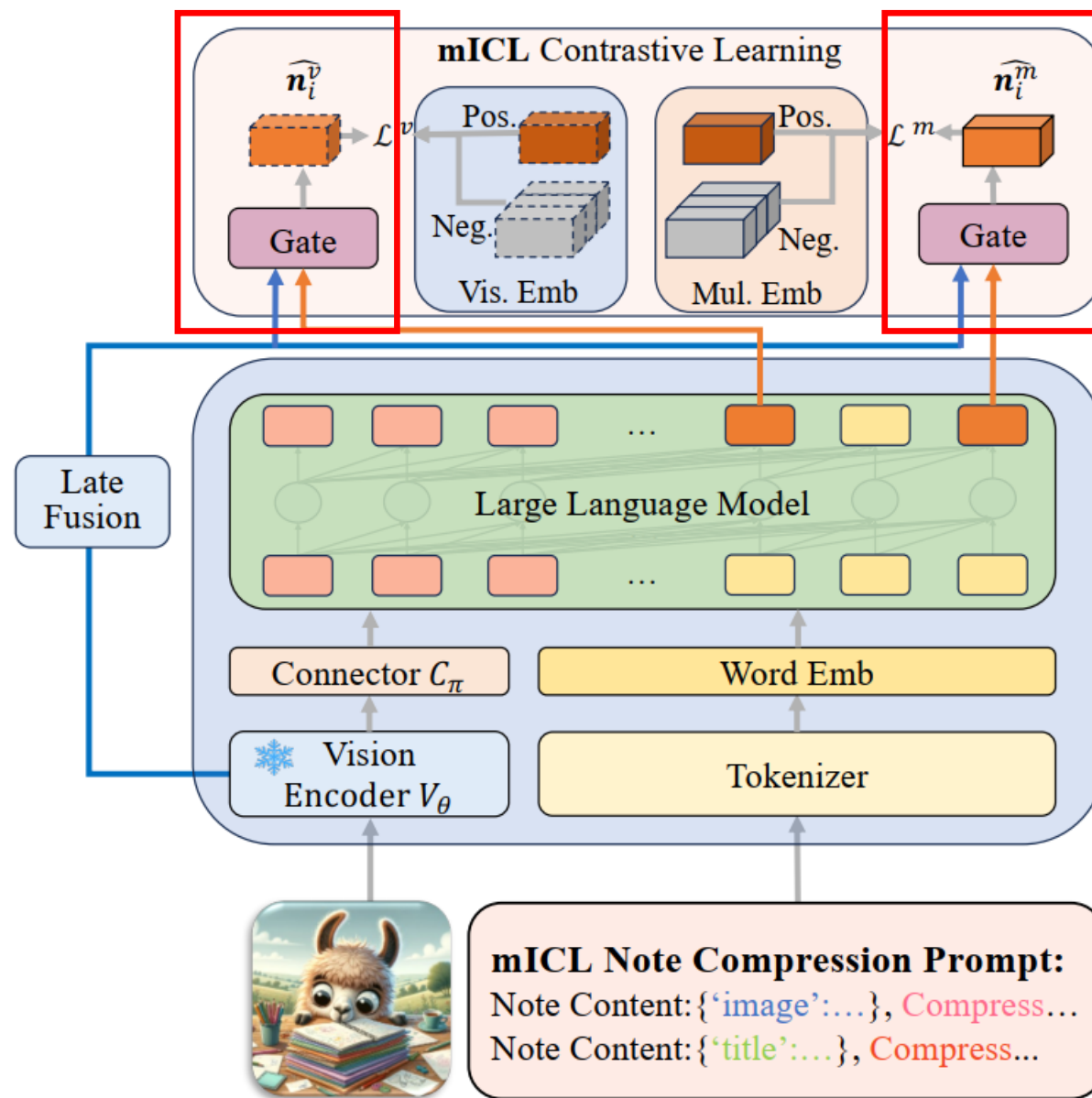
## Multimodal In-Context Learning

- Separating multimodal content into visual & textual components
- Compressing into modality-compressed words



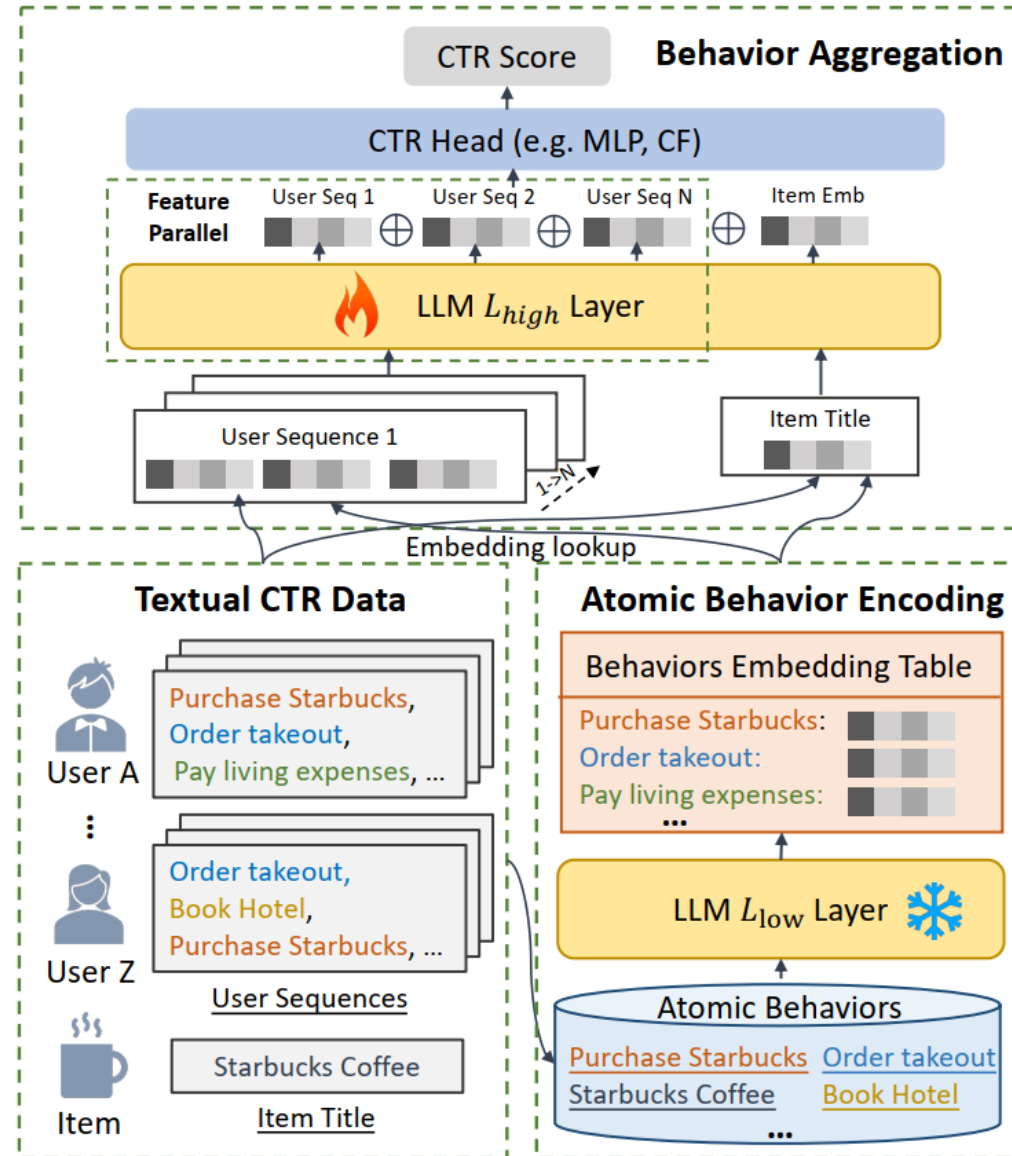
## Late Fusion

- Multimodal Gating mechanism



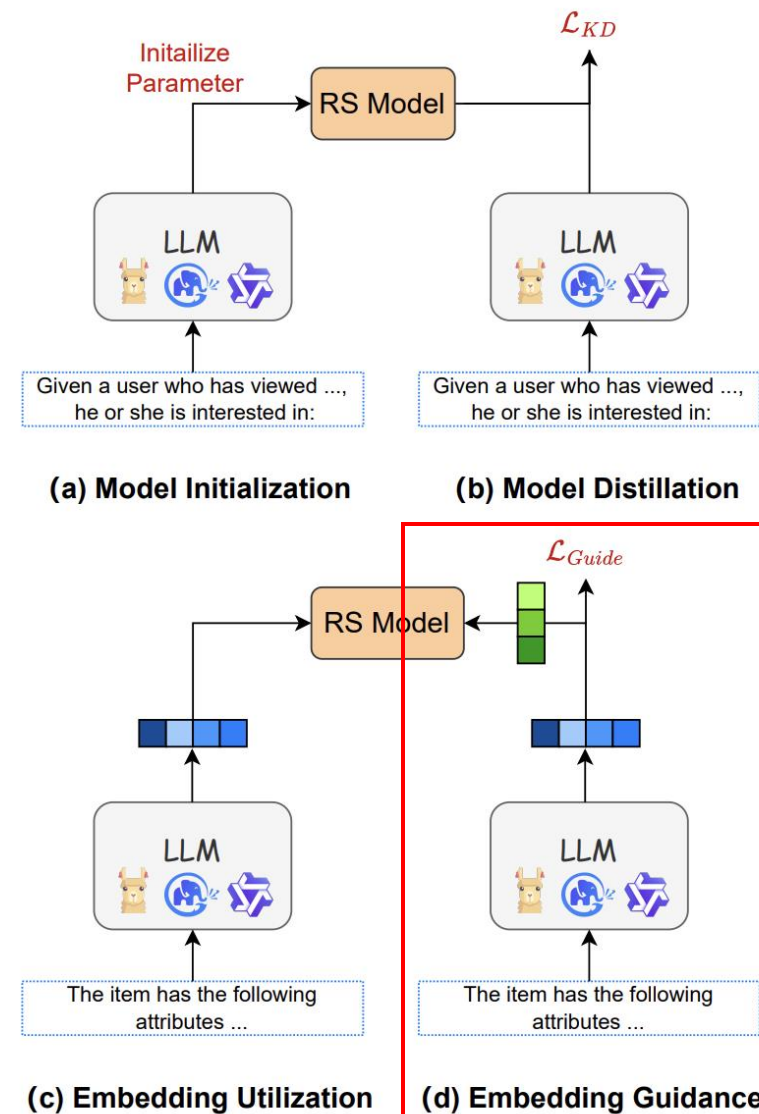
## • Hierarchical Encoding

- Lower layers encoding item into atomic representation
- Higher layers generating high-level user representations



# Embedding Guidance

- This subcategory refers to only using the LLM embeddings as the guidance for training or parameter synthesis.
- Categories
  - User Only
  - User & Item



## • Multi-Scenario Modeling

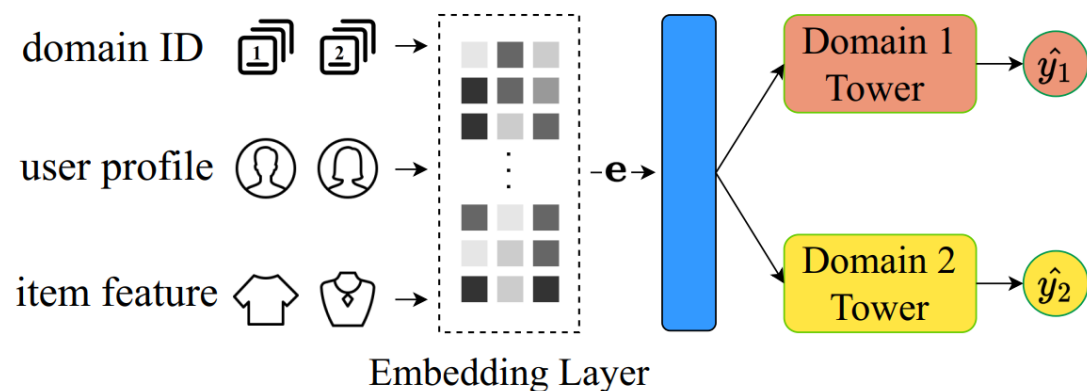
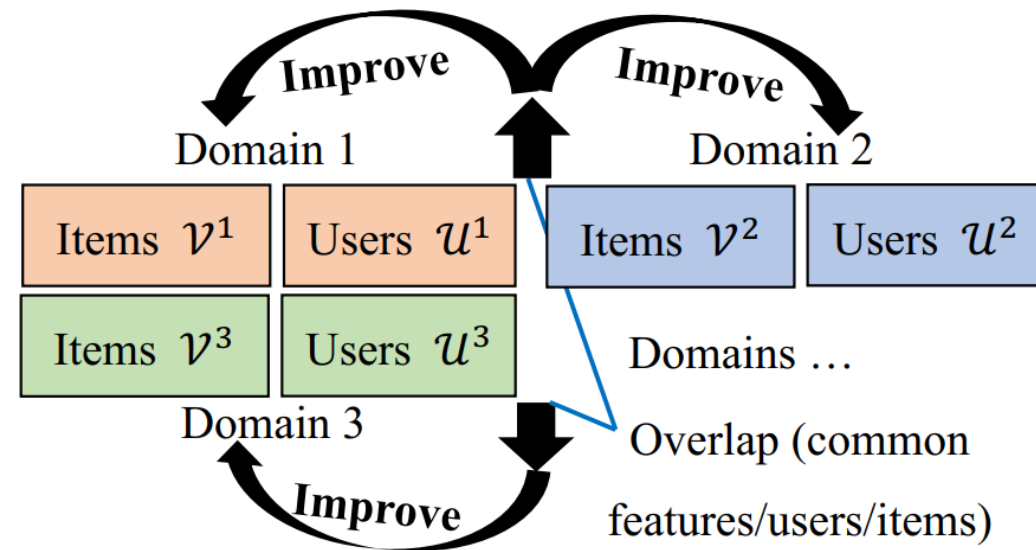
- Tackling data sparsity
- Reducing computation cost
- High efficiency

## • Challenges

- Insufficient scenario knowledge
- Ignoring cross-scenario preferences

## • Motivation

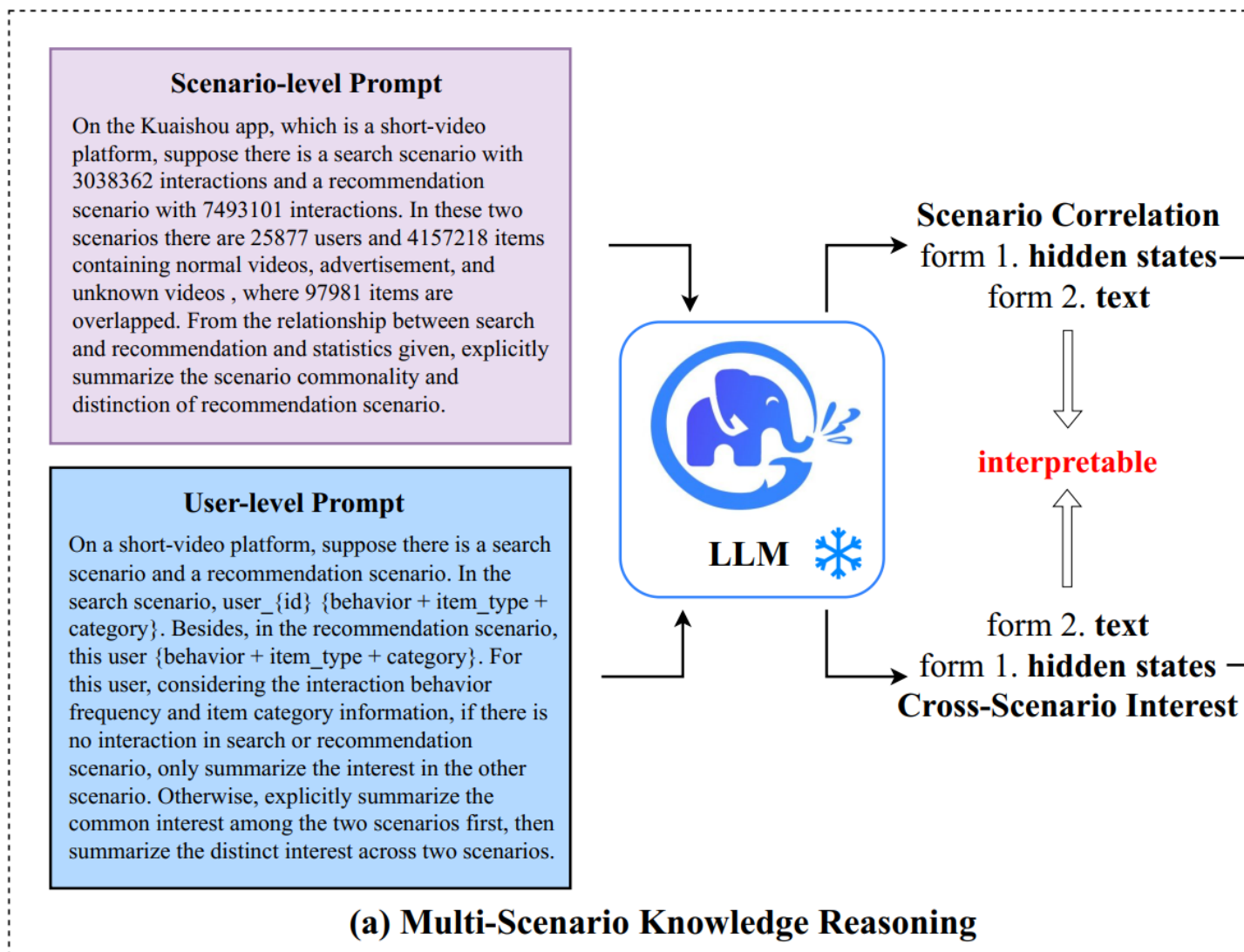
- Incorporating LLM to improve conventional recommender system
- LLM as reasoner + encoder





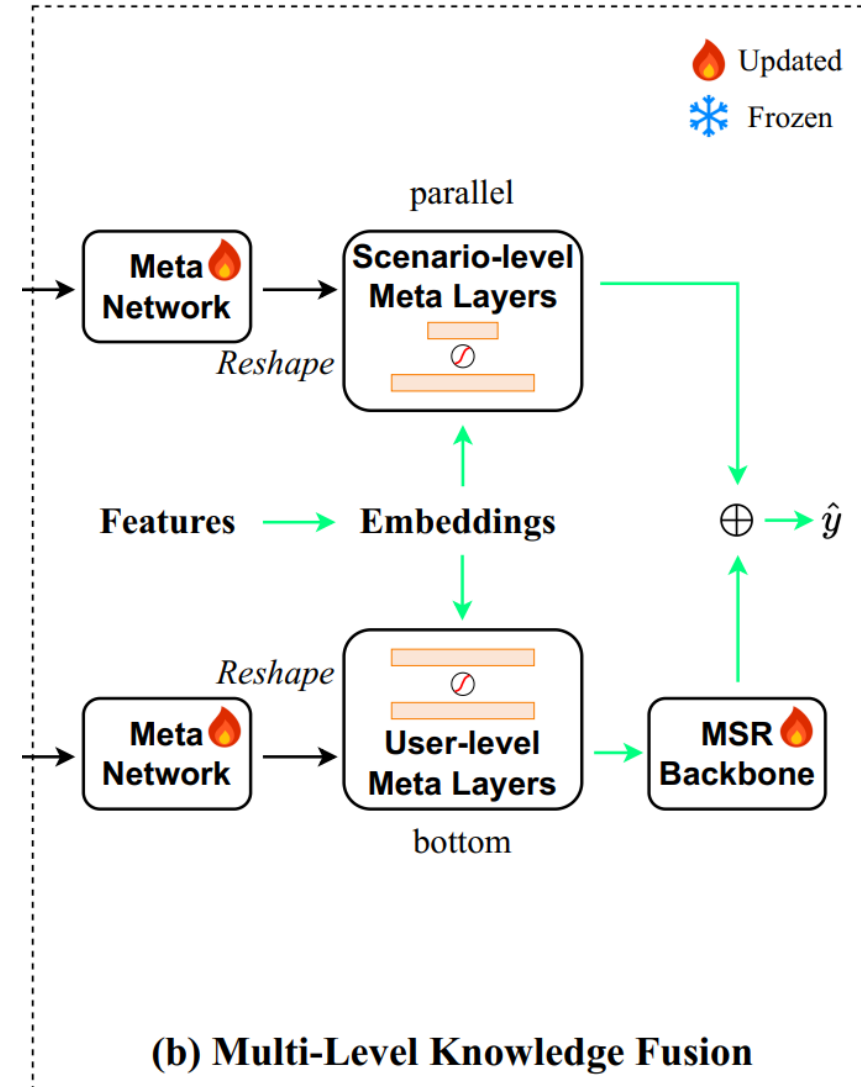
- **Multi-scenario reasoning**

- Scenario-level prompt
- User-level prompt



## • Multi-level Fusion

- Meta-networks generating meta layers
- Hierarchical bottom + parallel structure



- LLM4MSR achieves an increase of 1.5%, 1%, and 40% in AUC on three datasets
  - Domain correlation & Personalized interest + Adaptive meta network

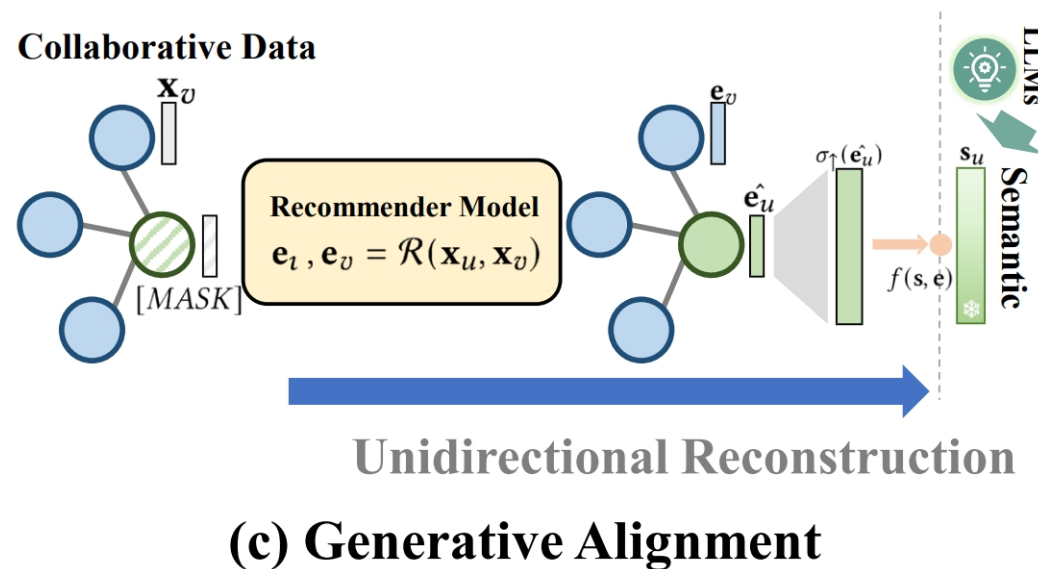
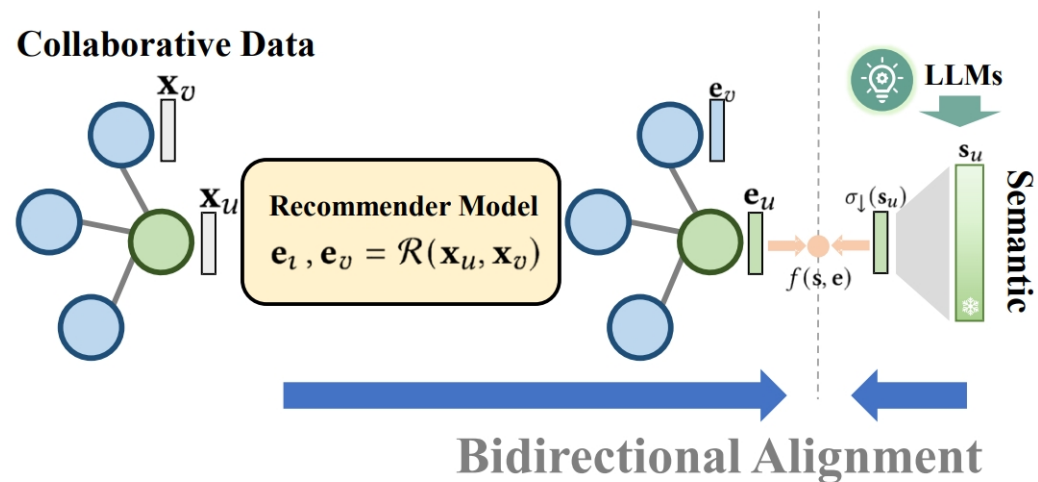
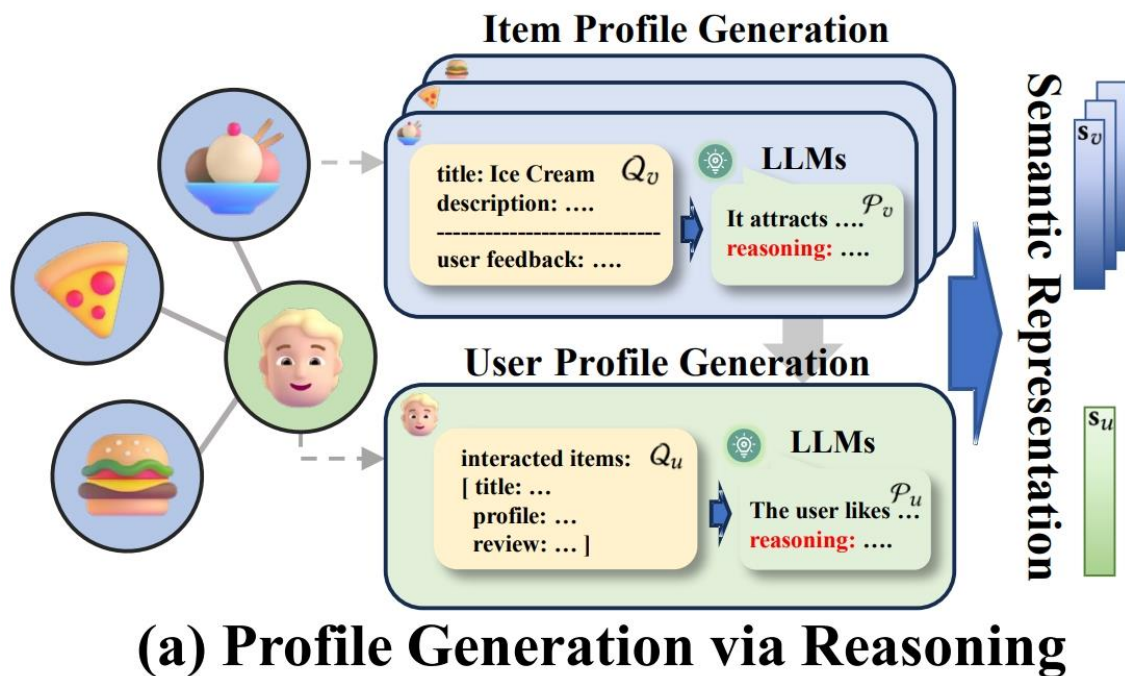
| Backbones          | KuaiSAR-small |         | KuaiSAR |         | Amazon  |         |         |
|--------------------|---------------|---------|---------|---------|---------|---------|---------|
|                    | Rec           | Search  | Rec     | Search  | Rec#1   | Rec#2   | Rec#3   |
| STAR               | 0.7225        | 0.6089  | 0.7387  | 0.6268  | 0.6156  | 0.6320  | 0.6225  |
| STAR_DN            | 0.7241        | 0.6116  | 0.7404  | 0.6270  | 0.6306  | 0.6350  | 0.6355  |
| STAR_EP            | 0.7230        | 0.6082  | 0.7388  | 0.6266  | 0.6211  | 0.6302  | 0.6378  |
| STAR_ours          | 0.7276★       | 0.6181★ | 0.7408★ | 0.6332★ | 0.8720★ | 0.6364★ | 0.7543★ |
| OMoE               | 0.7241        | 0.6163  | 0.7394  | 0.6310  | 0.5995  | 0.6043  | 0.6252  |
| OMoE_DN            | 0.7249        | 0.6183  | 0.7402  | 0.6319  | 0.6027  | 0.6171  | 0.6289  |
| OMoE_EP            | 0.7246        | 0.6179  | 0.7392  | 0.6330  | 0.6143  | 0.6176  | 0.6312  |
| OMoE_ours          | 0.7265★       | 0.6186★ | 0.7413★ | 0.6332★ | 0.8182★ | 0.6180★ | 0.6823★ |
| MMoE               | 0.7235        | 0.6150  | 0.7392  | 0.6313  | 0.5907  | 0.5999  | 0.6032  |
| MMoE_DN            | 0.7246        | 0.6164  | 0.7394  | 0.6330  | 0.6310  | 0.6201  | 0.6371  |
| MMoE_EP            | 0.7245        | 0.6178  | 0.7397  | 0.6326  | 0.6218  | 0.6221  | 0.6315  |
| MMoE_ours          | 0.7264★       | 0.6166★ | 0.7410★ | 0.6341★ | 0.7860★ | 0.6222★ | 0.6854★ |
| PLE                | 0.7249        | 0.6149  | 0.7396  | 0.6313  | 0.6059  | 0.6127  | 0.5995  |
| PLE_DN             | 0.7258        | 0.6165  | 0.7400  | 0.6333  | 0.6066  | 0.6195  | 0.6162  |
| PLE_EP             | 0.7252        | 0.6184  | 0.7402  | 0.6339  | 0.6173  | 0.6229  | 0.6263  |
| PLE_ours           | 0.7269★       | 0.6184★ | 0.7408★ | 0.6343★ | 0.8265★ | 0.6250★ | 0.7074★ |
| AITM               | 0.7236        | 0.6154  | 0.7398  | 0.6329  | 0.6039  | 0.6126  | 0.6046  |
| AITM_DN            | 0.7243        | 0.6172  | 0.7398  | 0.6318  | 0.6057  | 0.6151  | 0.6166  |
| AITM_EP            | 0.7237        | 0.6177  | 0.7389  | 0.6337  | 0.6064  | 0.6163  | 0.6238  |
| AITM_ours          | 0.7273★       | 0.6178★ | 0.7407★ | 0.6349★ | 0.8196★ | 0.6178★ | 0.7089★ |
| Shared Bottom      | 0.7228        | 0.6169  | 0.7389  | 0.6321  | 0.6073  | 0.6077  | 0.6268  |
| Shared Bottom_DN   | 0.7250        | 0.6176  | 0.7396  | 0.6323  | 0.6081  | 0.6253  | 0.6294  |
| Shared Bottom_EP   | 0.7243        | 0.6182  | 0.7392  | 0.6331  | 0.6268  | 0.6174  | 0.6233  |
| Shared Bottom_ours | 0.7269★       | 0.6182★ | 0.7400★ | 0.6341★ | 0.8323★ | 0.6262★ | 0.7182★ |

- LLM4MSR enhances various MSR backbone models, showing great **compatibility** and **deployability**

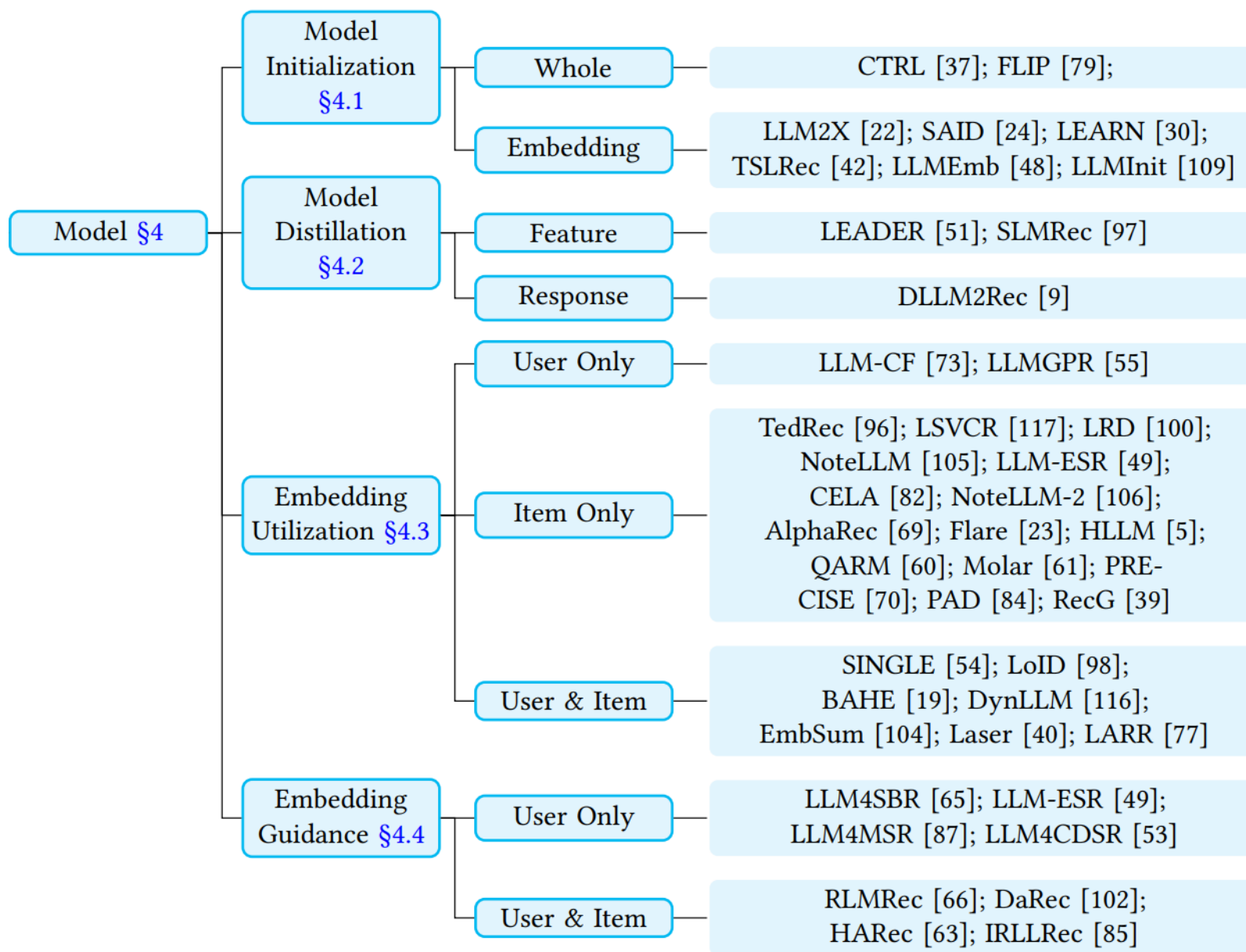
| Backbones                 | KuaiSAR-small  |                | KuaiSAR        |                | Amazon         |                |                |
|---------------------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
|                           | Rec            | Search         | Rec            | Search         | Rec#1          | Rec#2          | Rec#3          |
| STAR                      | 0.7225         | 0.6089         | 0.7387         | 0.6268         | 0.6156         | 0.6320         | 0.6225         |
| STAR_DN                   | 0.7241         | 0.6116         | 0.7404         | 0.6270         | 0.6306         | 0.6350         | 0.6355         |
| STAR_EP                   | 0.7230         | 0.6082         | 0.7388         | 0.6266         | 0.6211         | 0.6302         | 0.6378         |
| <b>STAR_ours</b>          | <b>0.7276★</b> | <b>0.6181★</b> | <b>0.7408★</b> | <b>0.6332★</b> | <b>0.8720★</b> | <b>0.6364★</b> | <b>0.7543★</b> |
| OMoE                      | 0.7241         | 0.6163         | 0.7394         | 0.6310         | 0.5995         | 0.6043         | 0.6252         |
| OMoE_DN                   | 0.7249         | 0.6183         | 0.7402         | 0.6319         | 0.6027         | 0.6171         | 0.6289         |
| OMoE_EP                   | 0.7246         | 0.6179         | 0.7392         | 0.6330         | 0.6143         | 0.6176         | 0.6312         |
| <b>OMoE_ours</b>          | <b>0.7265★</b> | <b>0.6186★</b> | <b>0.7413★</b> | <b>0.6332★</b> | <b>0.8182★</b> | <b>0.6180★</b> | <b>0.6823★</b> |
| MMoE                      | 0.7235         | 0.6150         | 0.7392         | 0.6313         | 0.5907         | 0.5999         | 0.6032         |
| MMoE_DN                   | 0.7246         | 0.6164         | 0.7394         | 0.6330         | 0.6310         | 0.6201         | 0.6371         |
| MMoE_EP                   | 0.7245         | 0.6178         | 0.7397         | 0.6326         | 0.6218         | 0.6221         | 0.6315         |
| <b>MMoE_ours</b>          | <b>0.7264★</b> | <b>0.6166★</b> | <b>0.7410★</b> | <b>0.6341★</b> | <b>0.7860★</b> | <b>0.6222★</b> | <b>0.6854★</b> |
| PLE                       | 0.7249         | 0.6149         | 0.7396         | 0.6313         | 0.6059         | 0.6127         | 0.5995         |
| PLE_DN                    | 0.7258         | 0.6165         | 0.7400         | 0.6333         | 0.6066         | 0.6195         | 0.6162         |
| PLE_EP                    | 0.7252         | 0.6184         | 0.7402         | 0.6339         | 0.6173         | 0.6229         | 0.6263         |
| <b>PLE_ours</b>           | <b>0.7269★</b> | <b>0.6184★</b> | <b>0.7408★</b> | <b>0.6343★</b> | <b>0.8265★</b> | <b>0.6250★</b> | <b>0.7074★</b> |
| AITM                      | 0.7236         | 0.6154         | 0.7398         | 0.6329         | 0.6039         | 0.6126         | 0.6046         |
| AITM_DN                   | 0.7243         | 0.6172         | 0.7398         | 0.6318         | 0.6057         | 0.6151         | 0.6166         |
| AITM_EP                   | 0.7237         | 0.6177         | 0.7389         | 0.6337         | 0.6064         | 0.6163         | 0.6238         |
| <b>AITM_ours</b>          | <b>0.7273★</b> | <b>0.6178★</b> | <b>0.7407★</b> | <b>0.6349★</b> | <b>0.8196★</b> | <b>0.6178★</b> | <b>0.7089★</b> |
| Shared Bottom             | 0.7228         | 0.6169         | 0.7389         | 0.6321         | 0.6073         | 0.6077         | 0.6268         |
| Shared Bottom_DN          | 0.7250         | 0.6176         | 0.7396         | 0.6323         | 0.6081         | 0.6253         | 0.6294         |
| Shared Bottom_EP          | 0.7243         | 0.6182         | 0.7392         | 0.6331         | 0.6268         | 0.6174         | 0.6233         |
| <b>Shared Bottom_ours</b> | <b>0.7269★</b> | <b>0.6182★</b> | <b>0.7400★</b> | <b>0.6341★</b> | <b>0.8323★</b> | <b>0.6262★</b> | <b>0.7182★</b> |



- **Generating profile**
- **Extra alignment loss**







# Agenda

## 1 Introduction



Zijian Zhang



## 2 Knowledge Enhancement



Pengyue Jia



## 3 Interaction Enhancement



Ziwei Liu



## 4.1 Model Enhancement 1



Maolin Wang



## 4.2 Model Enhancement 2



Yuhao Wang



## 5 Conclusion

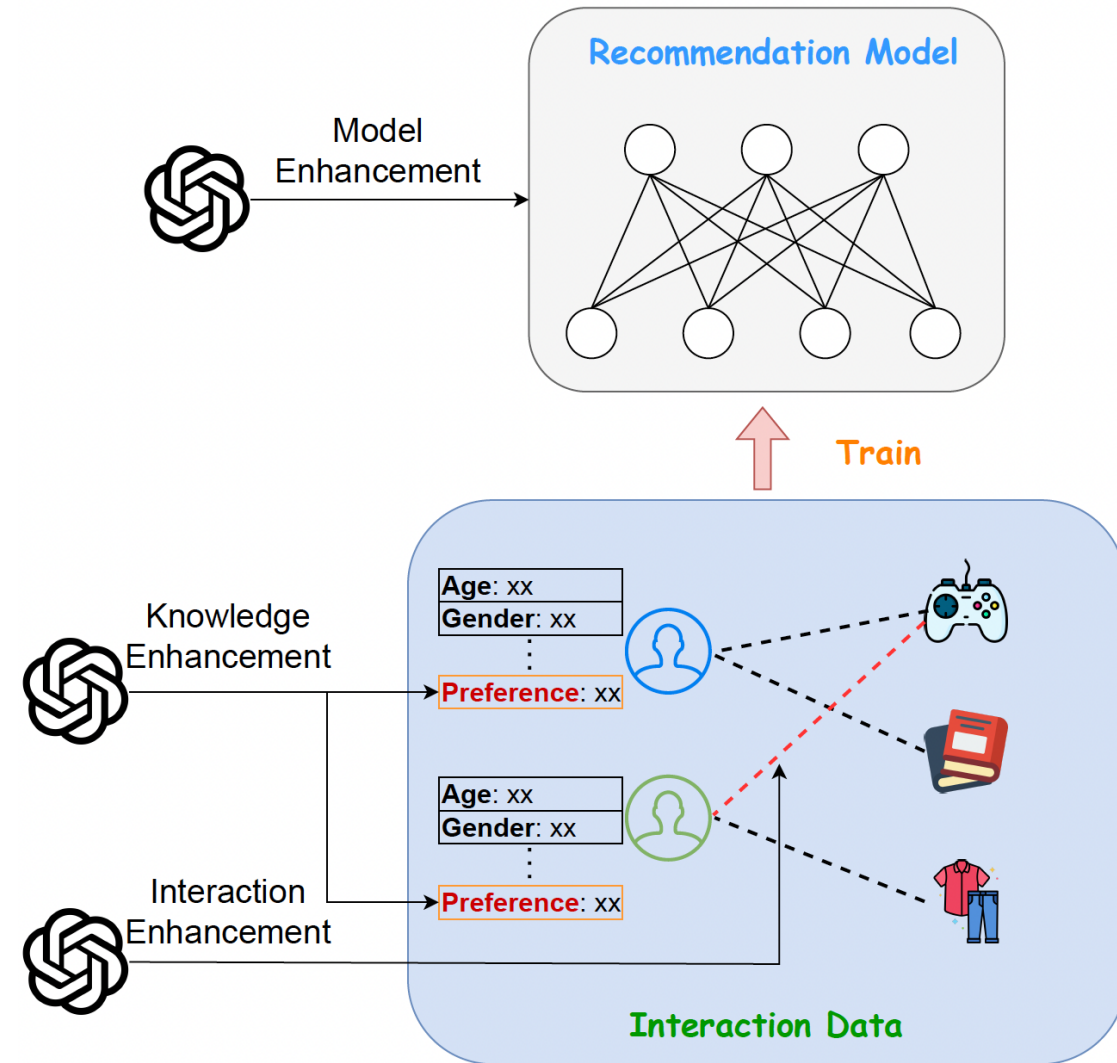


Yejing Wang

# Summary

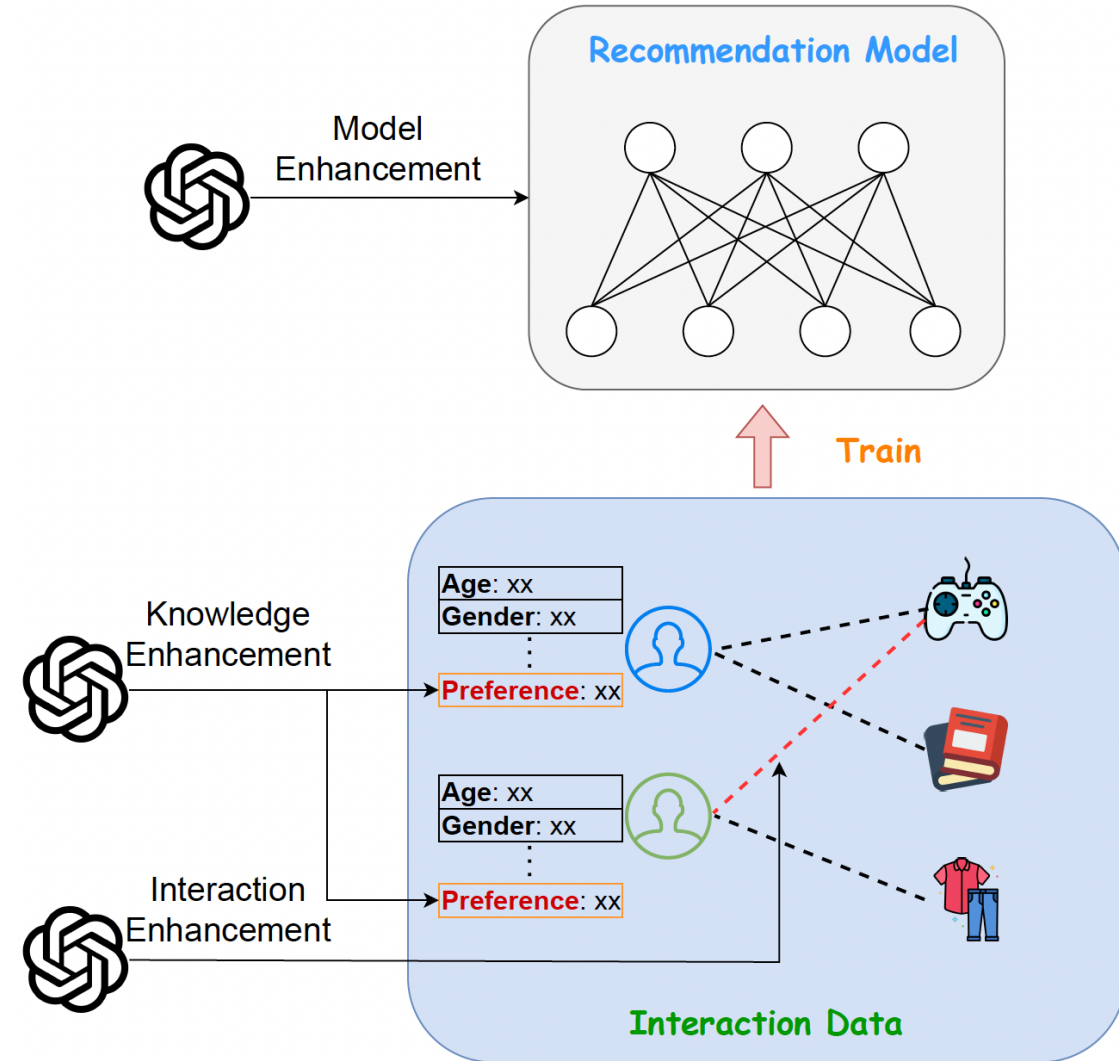
## LLM-enhanced RS (LLMERS)

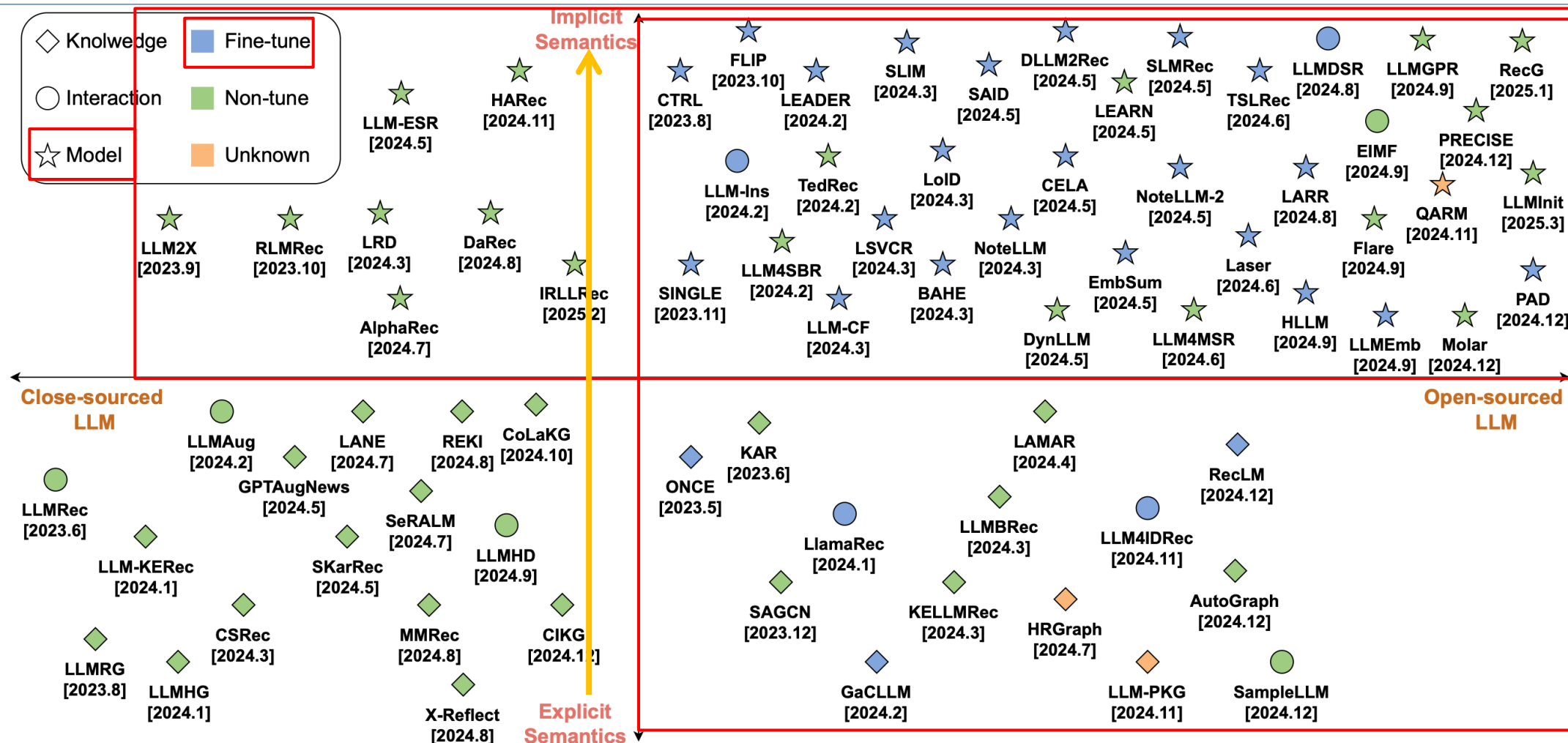
- Knowledge enhancement
  - World Knowledge, Reasoning
  - Explicit semantic
- Interaction enhancement
  - Generation
  - Implicit/Explicit semantic
- Model enhancement
  - Representation
  - Implicit semantic



## LLM-enhanced RS (LLMERS)

- Knowledge enhancement
  - Encoding the knowledge (LLM)
  - Converting to representations (Small LM)
  - Dual encoding & storage/update issues
- Interaction enhancement
  - Independent of the inference
  - Highly scalable
- Model enhancement
  - LLM representations
  - (Small) pre-encoding & storage issues

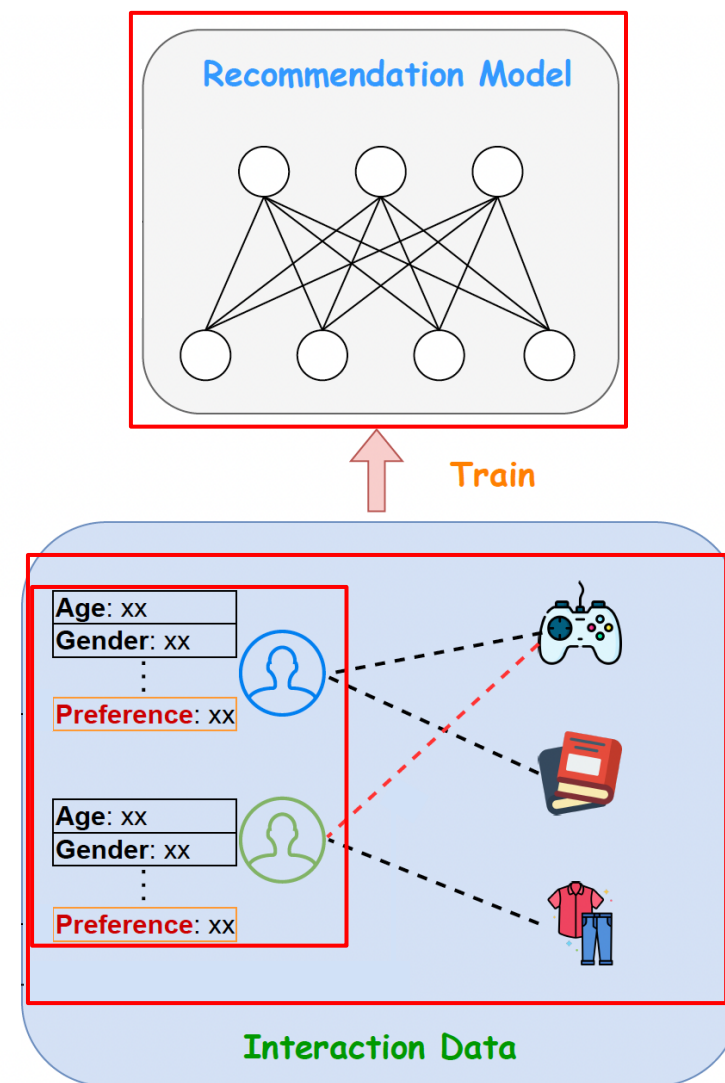




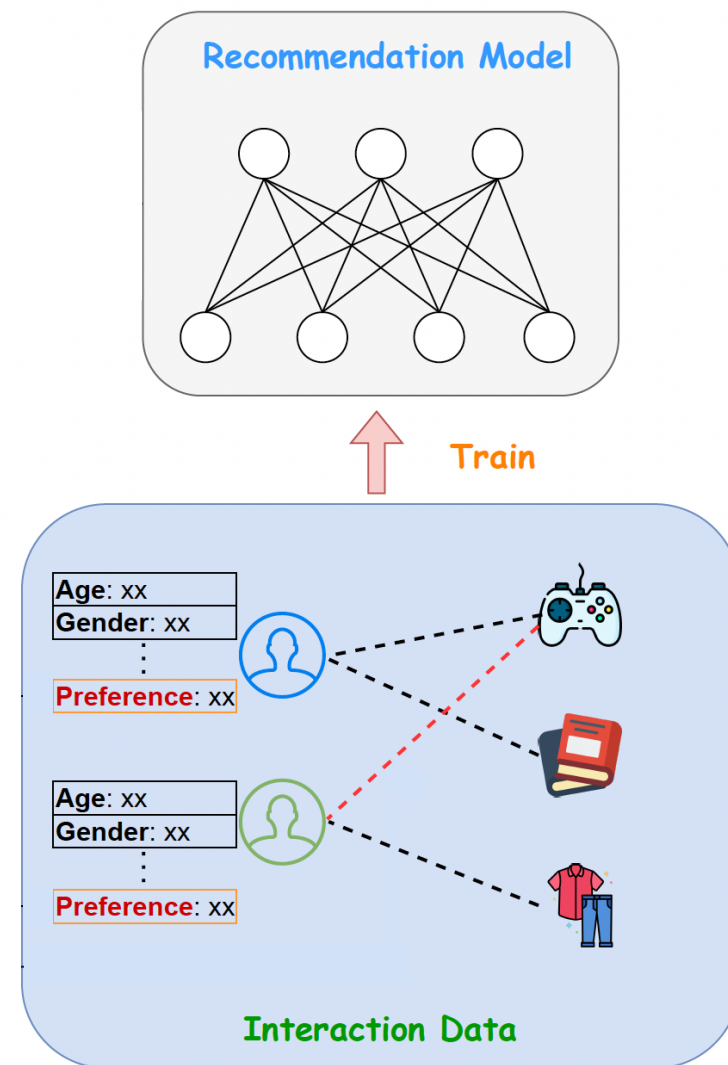
- Semantics: **explicit** -> **implicit**, **model enhancement** solutions are dominant
- **Fine-tuned open-source LLMs** are more popular for better adaptability



- Extended to more recommendation tasks
  - Focused on collaborative filtering and sequential recommendation
  - Potential in other tasks like multi-task, cross-domain...
- Support for multimodal/text-free RS
  - Multimodal input: MLLM -> adaptive representation
    - E.g., phone case vs. dresses on E-commerce platforms
  - Text-free RS: tabular data comprehension
- User-centric enhancement
  - Relying on item description/representations -> lengthy prompt/behavior-unaware integration
  - Item-free user enhancement, behavior-aware integration



- Scalability
  - More efficient than “LLM as RS” solutions
  - Huge computational & storage burden for extremely numerous items, especially **knowledge & model enhancement** -> Limited ROI
- Explainability
  - Not addressed by LLM
  - Generating explanations while enhancing RS
- Evaluations
  - Only evaluated combined with traditional RS
  - Lack of benchmarks and metrics for LLM **enhancement**



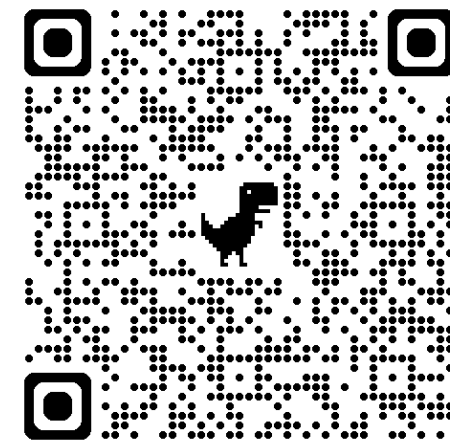
# Thanks for Listening!



KDD 25 LLMERS Tutorial  
Chat Group



AML Lab  
CityU



Tutorial Slides

- [1] Large language model enhanced recommender systems: A Survey. <https://arxiv.org/abs/2412.13432>
- [2] Multi-Task Deep Recommendation Systems: A Survey. <https://arxiv.org/abs/2302.03525>
- [3] Multimodal Recommender Systems: A Survey. <https://arxiv.org/abs/2302.03883>